

Spectrum Power 5
UIOS

5.30

Reference Guide

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NOTE

For your own safety, observe the warnings and safety instructions contained in this document, if available.

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Preface

Notes on Safety

This manual is not a complete index of all safety measures required for operation of the equipment (module or device). However, it comprises important information that must be followed for personal safety, as well as to avoid material damage. Information is highlighted and illustrated as follows according to the degree of danger:



WARNING

WARNING means that death or severe injury **may** result if the measures specified are not taken.

- ✧ Comply with all instructions, in order to avoid death or severe injuries.



CAUTION

CAUTION means that medium-severe or slight injuries **can** occur if the specified measures are not taken.

- ✧ Comply with all instructions, in order to avoid moderate or minor injuries.



NOTE

Important information about the product, product handling or a certain section of the documentation which must be given particular attention.

Qualified Electrical Engineering Personnel

Only qualified electrical engineering personnel may commission and operate the equipment (module, device) described in this document. Qualified electrical engineering personnel in the sense of this manual are people who can demonstrate technical qualifications as electrical technicians. These persons may commission, isolate, ground and label devices, systems and circuits according to the standards of safety engineering. Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Proper Use

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Scope

This document provides specific and detailed information on how to use a particular product or product component.

**NOTE**

Note that the screenshots used in this document contain sample data which may not be available in some systems.

Typical Users

This document is designed for users that are already familiar with operational and technical aspects of power generation and power transmission and distribution as well as product concepts:

- **System Engineers**
The System Engineer is able to install and to customize the system. He needs deep knowledge about the internal structures and processes of the network control system.
- **Operators**
The Operator is controlling the network or the plants with the system therefore he needs to know how the network control system has to be operated.
- **Data Engineers**
The Data Engineer translates the real network of a utility or the generation plants to the data model of the network control system. He also maintains the data model.
- **Technical Specialists**
The Technical Specialist is responsible for a special technical area which is related to the network control system. This can be the planning department, which plans changes in the network and uses network analyses applications of the network control system or the RTU department, which has to connect the RTU to the network control system. The Technical Specialist has to know how specific functions of the network control system work in detail.

Structure of the Manual

- **Introduction:**
Basic information about the component
- **Workflow/Operations:**
Main use cases, workflows and operational steps for optimal use of the component
- **References:**
Reference and description of user interfaces and dialogs

Documentation Conventions

Example	Meaning
File > Open	Selecting the menu item Open on the File menu
F1	Pressing the F1 function key on the keyboard
Control+o	Holding down the Ctrl key and pressing the letter o on the keyboard
Control+Shift+o	Holding down the Ctrl key and the Shift key and pressing the letter o on the keyboard
Left-click	Clicking the left mouse button
Right-click	Clicking the right mouse button
Shift-click	Holding down the Shift key on the keyboard and clicking the left mouse button

Questions?

This document is part of a set of manuals that describes the complete product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these products, please contact your Siemens representative.

Open Source Software

Among other things, this product contains Open Source Software that was developed by third parties. The Open Source Software included with this product and the relevant Open Source Software license agreement can be found in Readme_OSS. The Open Source Software program is protected by copyright. You are permitted to use the Open Source Software according to the Open Source Software license agreement. Should contradictions occur between the Open Source Software license conditions and the Siemens license conditions applicable to the product, the Open Source Software license agreement with respect to the Open Source Software shall prevail. The Open Source Software is provided free of charge. If stipulated by the Open Source Software license agreement, the source text of the software is available until the end of the third year after purchasing the product. Shipping charges will apply separately. We are liable for the product including the Open Source Software according to the license agreement for the product. Any liability arising from the use of the Open Source Software that exceeds the intended program flow of the product, as well as liability for defects that were caused by changes made to the Open Source Software, is excluded. If the product was modified, we cannot provide technical support.

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Among other things, this product contains Commercial Software that was developed by third parties and requires the placement of a logo or trademark legends, or other acknowledgements:



[javalogo, 1, --_--]

Java is a trademark of Oracle.

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1 UIOS

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1.1 Introduction

The Spectrum Power 5 **User Interface** enables you to monitor and control the power system.

User interfaces are:

- Power 5 Application Toolbar (PAT)
- Single-line diagrams
- Runtime Explorer
- Summaries
- Trends And Charts display



NOTE

Applications have user interfaces of their own, web user interfaces and administrator user interfaces.



NOTE

The properties of the object are shown as a tool tip (help) on moving the mouse over the object.



NOTE

A Worldmap, which is displayed in the Web user interface and is modified through an activation, must be reopened manually on the Web user interface to force the reload of the activated version of the worldmap in the Web user interface.

Warning Indication

In case of an alarm, you are notified on several Spectrum Power 5 user interfaces:

- The PAT
- The **Alarm Summary**
- The **Runtime Explorer**
- **Single-line Diagram**

Flashing icons, special coloring, and text messages indicate alarms. Optionally, loudspeakers audibly raise alarms. Another possible, alarm indicator is wireless alarming using SMS or voice mail.

Operating Tasks

The Spectrum Power 5 user interface enables you to perform several tasks:

- Viewing and acknowledging alarms
- Marking and tagging
- Viewing and updating values
- Supervisory control
- Entering comments and notes
- Generating reports

2 Power 5 Application Toolbar

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2.1 Introduction

The Power 5 Application Toolbar (PAT) is the central display of Spectrum Power 5. It provides access to Spectrum Power 5 applications, tools, and displays. In addition, the PAT presents system information like the currently active environment (real-time, study context, or test mode), the actual date, and time.

For example, it provides central alarming indicators.

**NOTE**

The PAT is located at the top border of the display and cannot be moved due to safety reasons. Other windows cannot cover the PAT.

**NOTE**

Worldmap AOR's and security handling to open worldmaps only take the user AORs into consideration, but not the console AOR's.

2.2 Operational Steps

2.2.1 Logging on Spectrum Power 5

The goal of this task is to log on to the system.



NOTE

After logon, you can access Spectrum Power 5 applications and functions according to the predefined access rights.

Log-on Process

- Click **Logon Logoff** on the PAT.

The **Power 5 Logon** dialog opens.

- Enter the access data into the following text boxes:
 - **SP5 User Name**
 - **Password**
- Select the **Domain Name** (if necessary) from the corresponding list box.
- Click **OK**.



NOTE

The **Cancel** button closes the respective dialog without applying the changes.



NOTE

If a new user logs on to the system at shift change, the former user on this operator console is logged off automatically.

2.2.2 Logging Off Spectrum Power 5

The goal of this task is to log off the system.

Log-off Process

- Click **Logon Logoff** on the PAT.
The **Power 5 Logon** information dialog opens.
- Click **Log Off**.

Shift Change Process

- Click **Logon Logoff** on the PAT.
The **Power 5 Logon** information dialog opens.
- Click **Log On**.
The **Power 5 Logon** dialog opens. The new user can log on to the console, the currently logged in user is logged off automatically.

2.2.3 Calling Displays and Summaries

The goal of this task is to call displays and summaries using PAT.

Calling a Single-line Diagram using PAT

For additional information on how to call a single-line diagram, refer to [4.2.1 Calling a Single-line Diagram](#).

Calling the Runtime Explorer using PAT

For additional information on how to call the Runtime Explorer, refer to [5.2.1 Calling the Runtime Explorer](#).

Calling a Summary Using PAT

For additional information on how to call a summary, refer to [8.2.1 Displaying a Summary](#).

Using the Display Callup

For additional information on how to use **Display Callup**, refer to [2.2.4 Using the Display Callup](#).

Using Application Buttons

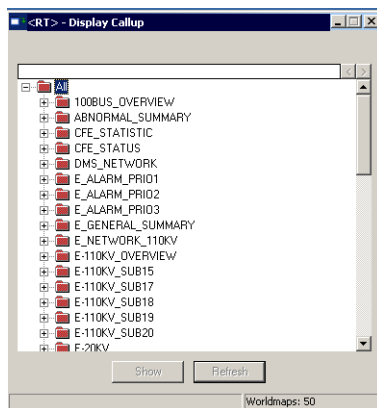
For additional information on how to use application buttons, refer to [4.2.9 Calling Application Buttons](#).

2.2.4 Using the Display Callup

The goal of the task is to search and open a display with the **Display Callup** dialog.

Calling the Display Callup

- Click  on the PAT to open the **Display Callup**.



[rguios02-301109-01.tif, 1, en_US]

Figure 2-1 Display Callup Dialog

The **Display Callup** dialog opens, showing a tree with all available displays (single-line diagrams).

Searching by Name

- Enter the search string in the text box of the **Display Callup** dialog to search in the displayed tree and click **Enter**.
The search results are displayed in the tree, the number of items is displayed on the left side on the bottom of the dialog.
- Select the desired view.
- Double-click or click the button **Show**.
The selected view is displayed.
- Use the arrows to the right to switch to the next result.

Searching the Tree

- Navigate through the tree using the + and - characters to expand or to collapse the tree.
- Select the desired view.
- Double-click or click the button **Show**.

The selected view is displayed.



NOTE

The **Refresh** button is only displayed, if a display is added or removed by another user.



NOTE

More than one **Display Callup** dialogs can be displayed. If a desktop layout is saved and invoked again, the **Display Callup** dialogs are displayed again, ready for use.

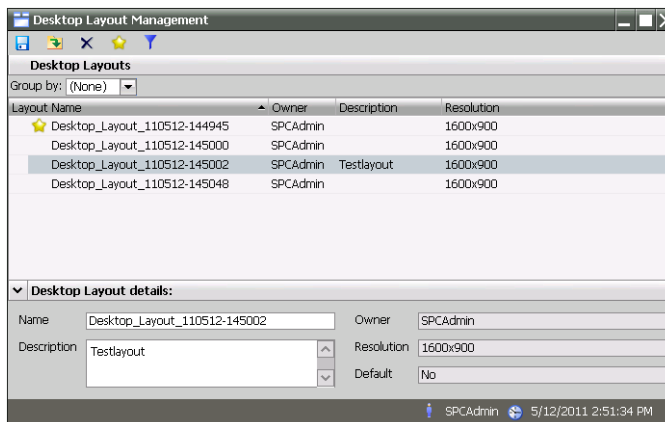
2.2.5 Managing Desktop Layout

2.2.5.1 Starting the Desktop Layout Management User Interface

The goal of this task is to start the **Desktop Layout Management** user interface.

To start the **Desktop Layout Management** UI, proceed as follows:

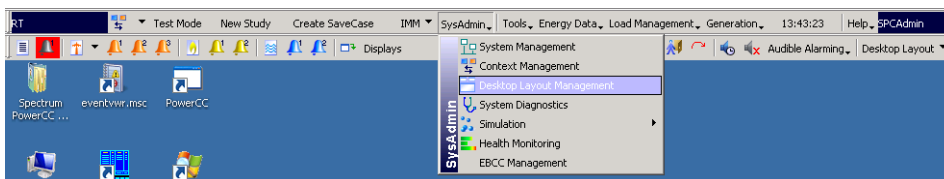
- From the **Power 5 Application Toolbar (PAT)**, select **SysAdmin -> Desktop Layout Management**.
The **Desktop Layout Management** UI opens.



[rgdlm021-300511-01.tif, 1, en_US]

Figure 2-2 Desktop Layout Management

- Alternatively, from **PAT**, select **Desktop Layout -> Desktop Layout Management**.



[rgdlm023-300511-01.tif, 1, en_US]

Figure 2-3 Selecting Desktop Layout Management

The **Desktop Layout Management** UI opens.

2.2.5.2 Saving a Desktop Layout

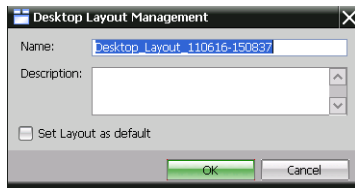
The goal of this task is to save the current desktop layout of the currently logged-in user.

To save the desktop layout, proceed as follows:

- From the **PAT**, select **SysAdmin -> Desktop Layout Management**.
The **Desktop Layout Management** UI opens.

- Click .

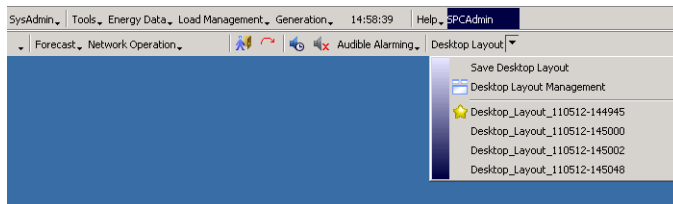
The **Desktop Layout Management** dialog opens.



[rgdlm001-170611-01.tif, 1, en_US]

Figure 2-4 Desktop Layout Management Dialog

- Enter the details in the fields and click **OK**.
- Alternatively, from **PAT**, select **Desktop Layout -> Save Desktop Layout**.
Enter the details in the **Desktop Layout Management** dialog and click **OK**.
The **Desktop Layout** is saved.



[rguia001-080611-01.tif, 1, en_US]

Figure 2-5 Saving Desktop Layout



NOTE

To save the desktop layout as default, check the **Use as default** check box in the **Desktop Layout Management** dialog.

2.2.5.3 Loading a Desktop Layout

The goal of this task is to load the saved desktop.

To load the desktop layout, proceed as follows:

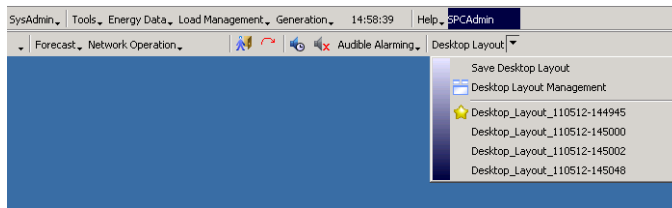
- From the **PAT**, select **SysAdmin -> Desktop Layout Management**.
The **Desktop Layout Management** UI opens.

- Select the required desktop layout.

- Click .

The **Desktop Layout** is loaded.

- Alternatively, from **PAT**, select **Desktop Layout** -> click on the required **Desktop Layout**.
For example, Desktop_Layout_110512_145000.



[rguio001-080611-01.tif, 1, en_US]

Figure 2-6 Loading a Desktop Layout



NOTE

If the resolution of the layout is bigger than the screen, the layout cannot be loaded. A yellow exclamation mark is displayed in the list beside the resolution of the layout.

2.2.5.4 Saving a Desktop Layout as Default

The goal of this task is to set a saved desktop layout as the default layout of the currently logged-in user.
To save the desktop layout as default layout, proceed as follows:

- From the **PAT**, select **SysAdmin** -> **Desktop Layout Management**.
The **Desktop Layout Management** UI opens.
- Select the required desktop layout to set as default.
- Click  from the tool bar.
The desktop layout is set as default.

2.2.5.5 Deleting a Desktop Layout

The goal of this task is to delete a saved desktop layout of the currently logged-in user.
To delete the desktop layout, proceed as follows:

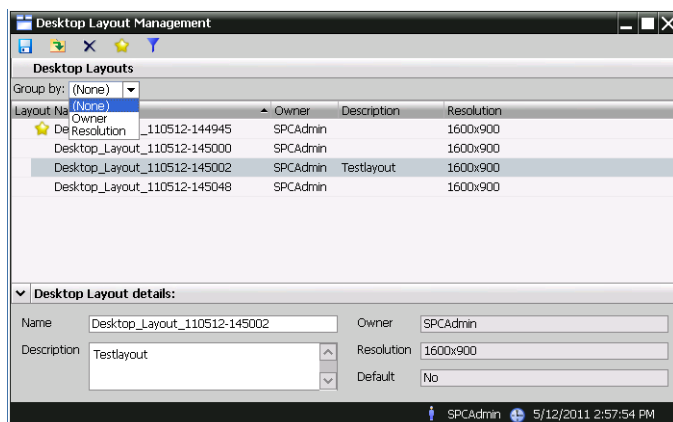
- From the **PAT**, select **SysAdmin** -> **Desktop Layout Management**.
The **Desktop Layout Management** UI opens.
- Select the required desktop layout.
- Click .
The selected desktop layout is deleted.

2.2.5.6 Grouping the Desktop Layouts

The goal of this task is to group the columns in the desktop layout.
To group the columns in desktop layout, proceed as follows:

- From the **PAT**, select **SysAdmin** -> **Desktop Layout Management**.
The **Desktop Layout Management** UI opens.

- Click the **Group** by drop-down menu and select an entry.
For example, **Owner**.



[rgdlm024-300511-01.tif, 1, en_US]

Figure 2-7 Grouping



NOTE

By default the grouping is Owner, if the logged user is not an admin user.

2.2.5.7 Enable/Disable the Inline Filter

The goal of this task is to enable/disable the columns in the desktop layout.

To enable/disable the columns in desktop layout, proceed as follows:

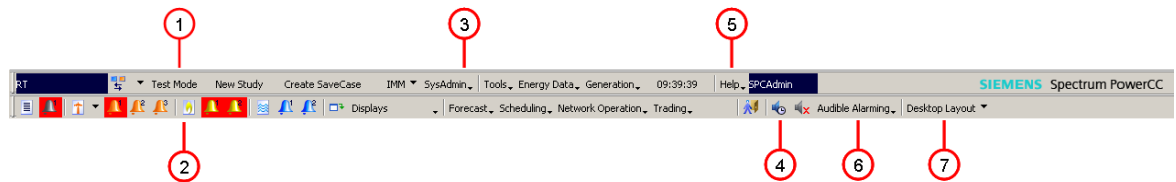
- From the **PAT**, select **SysAdmin** -> **Desktop Layout Management**.
The **Desktop Layout Management** UI opens.
- Click  to enable/disable the columns.

The desktop layout is grouped based on the selected column.

2.3 User Interface Descriptions

2.3.1 Power 5 Application Toolbar

The **Power 5 Application Toolbar (PAT)** is subdivided in several areas of operations.



[rguios17-031209-01.tif, 1, en_US]

Figure 2-8 Structure of the PAT

- | | |
|-----|---|
| (1) | Context management |
| (2) | Alarming area |
| (3) | Buttons and menus |
| (4) | Icons for Silence , Alarm Inhibit and Logon/Logoff |
| (5) | General information |
| (6) | Audible Alarming |
| (7) | Desktop layout management |

Context Management

This area displays the currently selected context. Furthermore, it contains buttons and menus that allow you

- to create new studies
- to switch between real-time context, test mode and study context
- to create and load save cases

For additional information on context management, refer to Technical Specification System Software (SYSW).

Alarming Area

The alarm area contains buttons for calling up different summaries. Flashing buttons of alarm summaries indicate an unacknowledged alarm in the respective summary.

For additional information on the alarming area, refer to [7.2.1 Handling an Incoming Alarm](#).

Buttons and Menus

Buttons and menus provide access to Spectrum Power 5 tools and displays.

For example, Runtime Explorer, single-line diagrams, and Display Callup.

Icons

These icons provide a quick access for alarm handling and log on log off.

General Information

Read-only display fields provide general system information.

For example, date and time, current context, and current user.

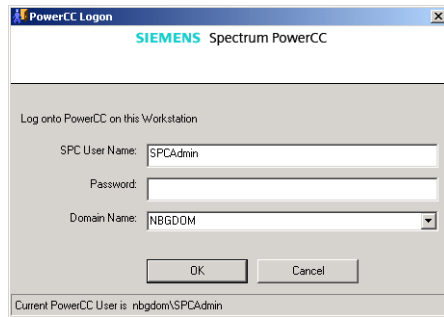
Desktop Layout Management

The **Desktop Layout** menu provides items for saving or loading desktop layouts.

2.3.2 Power 5 Logon Dialog

In the **Power 5 Logon** dialog, enter the access data to log on to the system.
The log on process is part of the console system startup:

- Each logon of a user is recorded in the **General Summary**.
- Predefined access rights and areas of responsibility are active after logon.



[scaap002-190209-01.tif, 1, en_US]

Figure 2-9 Power 5 Logon Dialog

Text Boxes:

- **SP5 User Name**
Enter the SP5 user name.
- **Password**
Enter the password.

List Box:

- **Domain Name**
Select the domain name.

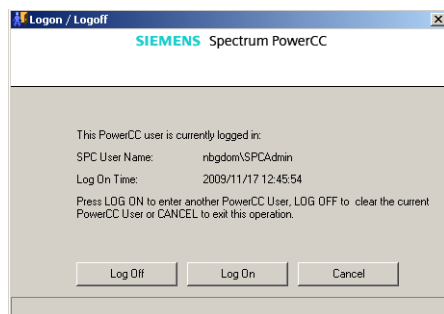


NOTE

The **Cancel** button always closes the respective dialog without applying the changes.

2.3.3 Power 5 Logon Information Dialog

The **Power 5 Logon** information dialog displays the logon status.



[rguios16-021209-01.tif, 1, en_US]

Figure 2-10 Power 5 Logon Information Dialog

2.3.4 Power 5 Display Callup

The Display Callup Dialog

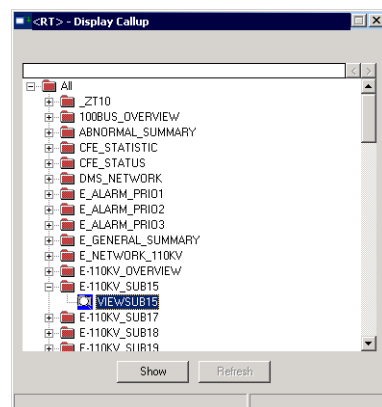
The Power 5 **Display Callup** dialog provides displays in a tree view.

For example, Single-line diagrams and Summaries.

The hierarchy is:

- Group of single-line diagrams
- Single-line diagrams
- Views

Single-line diagrams and views are only displayed, when you have valid access rights.



[scaap016-190209-01.tif, 1, en_US]

Figure 2-11 Display Callup Dialog

- List box over the tree view:
Enter the search string.
- Buttons < and > :
Select the previous or next match of the search string in the tree view.
- Button **Show**:
Click this button to display the view. The button is active when a view is selected.
You can also use drag and drop to display a view.
- Button **Refresh**:
The button is active when a display is added to or removed after IMM activation.

2.3.5 Desktop Layout Management

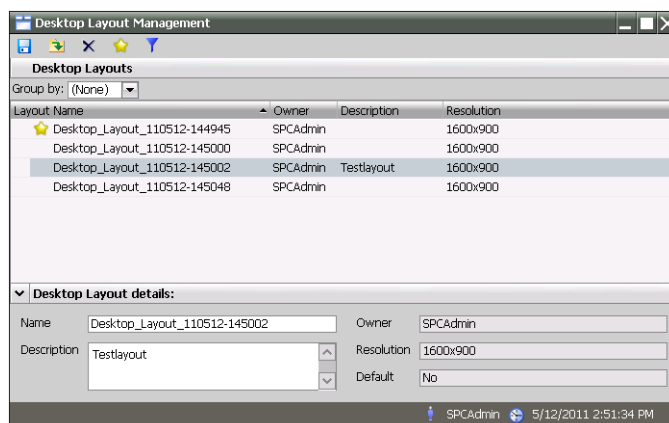
The **Desktop Layout Management** user interface allows you to perform the following operations:

- View the desktop layouts
- Create new desktop layout
- Load the selected desktop layout
- Delete the selected desktop layout
- Enable/Disable the inline filtering in the desktop layout table
- Set the desktop layout as default

PAT **Desktop Layout Management** menu displays entries for storing a new layout (**Save**), starting the DLM UI, and all layouts of the current user that fit to the screen.

Desktop Layout Management UI consists of 3 sections:

- Toolbar
- Desktop Layouts
- Desktop Layout details



[rgdlm021-300511-01.tif, 1, en_US]

Figure 2-12 Desktop Layout Management

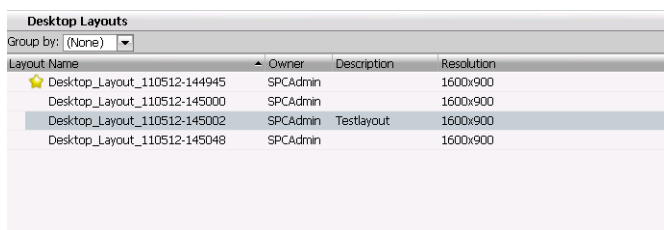
Toolbar

A toolbar consists of set of icons, which are described in the following table:

Icon	Description
	Saves the current desktop layout
	Loads the selected layout into the current context
	Deletes the selected layout
	Sets the selected desktop layout as default layout for the current user
	Enables or disables the inline filtering in the desktop layout table

Desktop Layouts

The Desktop Layouts section gives an overview about the available layouts.



[rgdlm025-300511-01.tif, 1, en_US]

Figure 2-13 Desktop Layouts Section

Desktop layout details

For each desktop layout (row) selected in the **Desktop Layout** section, additional information is displayed in **Desktop Layout Details** section.

Desktop Layout details:

Name	Desktop_Layout_110512-145002	Owner	SPCAAdmin
Description	Testlayout	Resolution	1600x900
		Default	No

SPCAAdmin 5/12/2011 2:51:34 PM

[rgdlm026-300511-01.tif, 1, en_US]

Figure 2-14 Desktop Layout Details Section



NOTE

The **Desktop Layout Management** menu from PAT displays the desktop layouts of the current user.

Inline Filter

Desktop Layouts supports an inline or auto filter for faster filtering of simple filter conditions. Toolbar has a toggle button called show inline filter. Clicking this result in a new section that provides the inline filter conditions as shown in the figure.

Desktop Layout Management

Group by: (None)

Layout Name	Owner	Description	Resolution
(None)	(None)	(None)	(None)
★ Desktop_Layout_110512-144945	SPCAAdmin		1600x900
Desktop_Layout_110512-145000	SPCAAdmin		1600x900
Desktop_Layout_110512-145002	SPCAAdmin	Testlayout	1600x900
Desktop_Layout_110512-145048	SPCAAdmin		1600x900

Desktop Layout details:

Name	Desktop_Layout_110512-145002	Owner	SPCAAdmin
Description	Testlayout	Resolution	1600x900
		Default	No

SPCAAdmin 5/12/2011 2:52:43 PM

[rgdlm022-300511-01.tif, 1, en_US]

Figure 2-15 Inline Filter

Click  to activate the filter function. When a filter is activated on a desirable column, data shown in the **Desktop Layouts** is automatically filtered.

3 Context Management

3.1	Operational Steps	34
3.2	User Interface Description	56

3.1 Operational Steps

3.1.1 Using Real-time Context

The goal of this task is to start using Real-time context.

To start using Real-time context, proceed as follows:

- Startup the system for real-time context.
- Open the desired application using the PAT list boxes.
- Perform required processes.



NOTE

For additional information on operations in the real-time context, refer to Reference Guides for the corresponding applications.

3.1.2 Using Study Context



NOTE

For additional information on study context functionality, refer to Technical Specification System Software.

Starting Study Context and Check Switching Procedures

The goal of this task is to start study context and check switching procedures.

- Click **New Study** on PAT to start the context type.

The PAT displays the button .

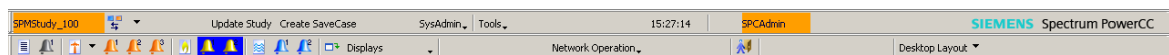
- Click  to switchover to the study context.

In the study context, a different color is used for the context and user label in the PAT, window headers of single-line diagrams, alarm lists, and SPM displays.



NOTE

Orange is the default color for study context, but this color can be customized. The type of study context (TNA, DNA, or SPM) started depends on the role assigned to you.



[rgsysw25-200110-01.tif, 1, en_US]

Figure 3-1 PAT Displaying Study Context

As long as the study context is selected the context indicator displays the context name.

For example, SPMStudy_100.



NOTE

For additional information on study context types, refer to **PAT Configuration** in Date Modelling Guide Spectrum Power 5.

- After switching to the study context, the PAT displays **Update Study** button.
- Click **Update Study** to update the dynamic data.
This step is optional. It is required only if actual latest dynamic values are used in the study context.
- Perform switching procedures.
- Save the context state as save case, if required.

**NOTE**

For additional information on operations of the switching procedures, refer to Reference Guide UIOS.

Switching Between Real-time and Study Contexts

After the study context is started, switchover from study to real-time context using the toggle button  in the PAT. Switchover to the real-time context implies that the started study context continues to run on background.

Stopping Study Context

There are 2 ways to stop a running study context:

- It is stopped automatically after logoff.
This way it is ensured that you do not forget to stop their study context, which can result in affecting the performance of the system.
- Manually, using the **Context Management User Interface**.
After study context is stopped on the local machine, the **Context Management User Interface** overview assigns the stopping state to this study context (during process stopping) and then assigns the stopped state.

3.1.3 Using Test Mode

**NOTE**

For additional information on test mode functionality, refer to Technical Specification System Software.

Starting Test Mode and Checking Telemetry Test

The goal of this task is to start test mode and check telemetry.

**NOTE**

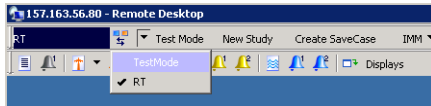
It is a precondition to have adequate permissions and to select a telemetered value of a device with **Object-in-Test** marker applied in the real-time context. After switching to the test mode, the **Object-in-Test** marker cannot be applied. Additionally, you must know the SCADA operations.

For additional information on SCADA operations, refer to Reference Guide UIOS.

Perform following work steps:

- Select the object of interest in the real-time context.
For example, RTU.
The telemetered value remains unchanged in the real-time context as long as this marker is set. Data transmission in the real-time context is interrupted. In test mode, you can change this data and perform the telemetry test.

- Switchover from the real-time context to the test mode using the PAT button.
 - On the PAT, click **Test Mode**. You are attached to the **Test Mode**.
If the attachment process is completed, the entry **TestMode** in the list box is activated.
 - Click on **TestMode** in the list box to switch to the **Test Mode**.



[syswr052-010210-01.tif, 1, en_US]

Figure 3-2 Using PAT, Attach to Test Mode

- The PAT in test mode is displayed and labeled in another color. Orange is the default color for test mode, but this color can be customized. The test mode provides all data copied from real-time context except those markers with **Object-in-Test** marker.



[syswr053-010210-01.tif, 1, en_US]

Figure 3-3 PAT Displaying Test Mode

- In the test mode, select the device to test and check the communication functionality.
- Perform the intended communication test using the SCADA control functions.
The measurement values are continuously updated in the test mode. In the real time context, the values remain unchanged and thus differ from the values in test mode.
- Switch back to the real-time context after successful test and delete the set **Object-in-Test** marker. The object blocked before is reintegrated in real-time processes. Its telemetered value at system controlling and monitoring changes again.



NOTE

The **Object-in-Test** marker has to be set in the real-time context before initializing the test mode. Otherwise the telemetry test cannot be performed in the test mode. Without set **Object-in-Test** marker, the test mode is a read-only copy of the real-time context.

Switching Between Real-time Context and Test Mode

The goal of this task is to switch between real-time context and test mode.

After finishing the telemetry tests within test mode, switchover from test-mode to real-time context using the toggle button .



[syswr055-010210-01.tif, 1, en_US]

Figure 3-4 On PAT, Switching Back to Real-time Context

Stopping Test Mode

The test mode can be stopped manually by using the **Context Management User Interface**.

3.1.4 Using PAT Buttons

3.1.4.1 Overview

In this section, the PAT buttons are used to create a save case, a study, and update an existing study.



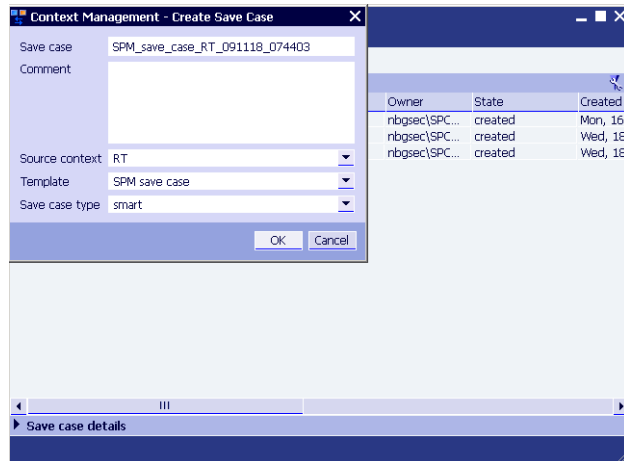
[rgsys025-210110-01.tif, 1, en_US]

Figure 3-5 PAT Overview

3.1.4.2 Creating Save Cases

The goal of this task is to create a save case.

- Click **Create Savecase** to create a save case.



[rgsys001-201109-01.tif, 1, en_US]

Figure 3-6 Create Save Case

- For the new save case, define the following data:
 - Save case: The system assigns the name automatically. It is editable by any single-context user.
 - Comment: Optionally, you can write any text here.
 - Source context: This context is fixed with exception of context-management administrator.
 - Template: It is fixed with exception of multi-context user and context-management administrator.
 - Save case type: It is optional. The type is smart by default.
- Click **OK**.
The new save case is created.

The status of the newly created savecase is viewed in **Context Management**.



NOTE

The save case name cannot be changed after the save case is created.
The save case name has to be unique and certain characters are not allowed in save case names. The restrictions are verified in the dialog.



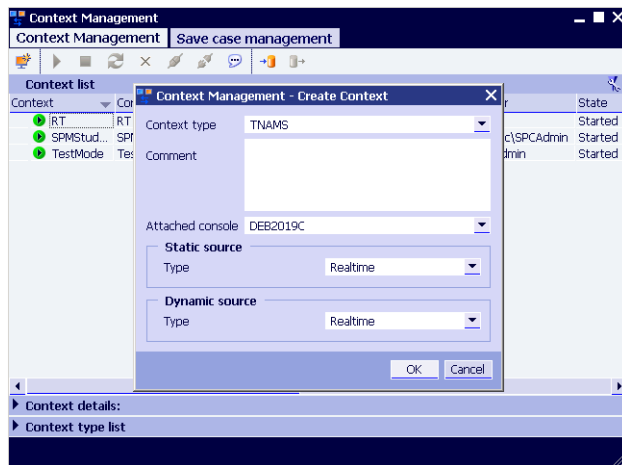
NOTE

The selection of template for the save case is only possible, when the current context is RT.

3.1.4.3 Creating Study Context

The goal of this task is to create a study context.

- Click **New Study** to create a new study context.



[rgsys002-201109-01.tif, 1, en_US]

Figure 3-7 Create New Study Context

- For new study context, define the following data:
 - Context type: The system assigns the context type automatically. It is fixed with exception of multi-context user and context-management administrator.
 - Comment: Optionally, you can write any text here.
 - Attached console: The system assigns the console name automatically. This context is fixed with exception of context-management administrator.
 - Static source: Click the list box to select the source. RT is the default static source.
 - Dynamic source: Click the list box to select the initialization type for dynamic data of the new context. RT is the default dynamic source. It is optional.

Available sources:

- **Real-Time** - The initial values for the study are taken from the current real time situation.
- **Normal** - The initial values for the study are defined by the normal state of all digital measurements.
- **HIS** - The values are taken at a certain time in the past from the archive.
- **Save Case** - The initial values are taken from a previously stored save case.
- **Context** - The values are taken from the current situation in another context.
For example, study.

- Click **OK**
The new study context is created.

Status of creation is observed within **Context Management User Interface**.



NOTE

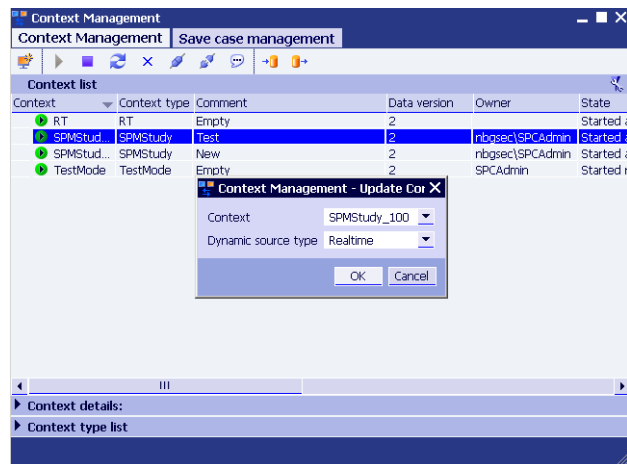
- In order to stop and delete a context, you can use the **Context Management User Interface**.
- One user can create several studies on one user interface. But those studies must be of the same type.
- The creation of a study is only possible if a free study slot is available. Those study slots are defined during configuration per study type.

3.1.4.4 Updating Study Context

The goal of this task is to update a study context.

The dynamic data of a running study can be changed using Update.

- Select the existing and running context from the context list to be updated.
- Click .



[rgsys004-201109-01.tif, 1, en_US]

Figure 3-8 Update Study Context

- For updating the study context, define the following data:
 - Context: Used as source for the context initializing. This is optional and the default is the current study context. It is fixed within single context with exception of multi-context user and context-management administrator. Multi-context user can select any own study context. Context-management administrator can select any running study context.
 - Dynamic source type: Select the dynamic source from the available list box.

Available sources:

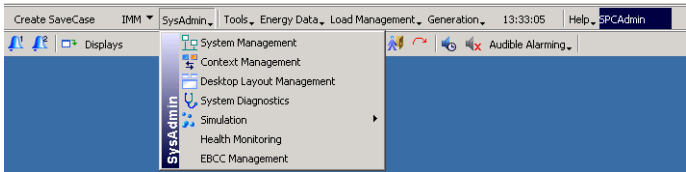
- **Normal state**
The initial values for the study are defined by the normal state of all digital measurements.
- **HIS restore**
The values are taken at a certain time in the past from the archive.
- **Save case**
The initial values are taken from a previously stored save case.
- **Realtime**
- **Context**
The values are taken from the current situation in another context.
For example, study.
- Click **OK**.
The context is updated.

3.1.5 Using Context Management UI

3.1.5.1 Calling-up Context Management UI Displays

To view the **Context Management** user interface displays, proceed as follows:

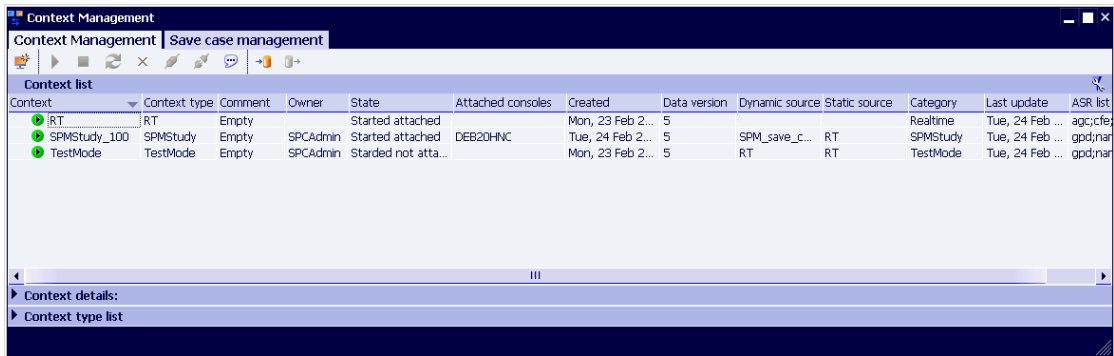
- Select **SysAdmin** from the PAT.
The Drop-down menu is displayed.



[rgsys007-060511-01.tif, 1, en_US]

Figure 3-9 Calling-up Context Management

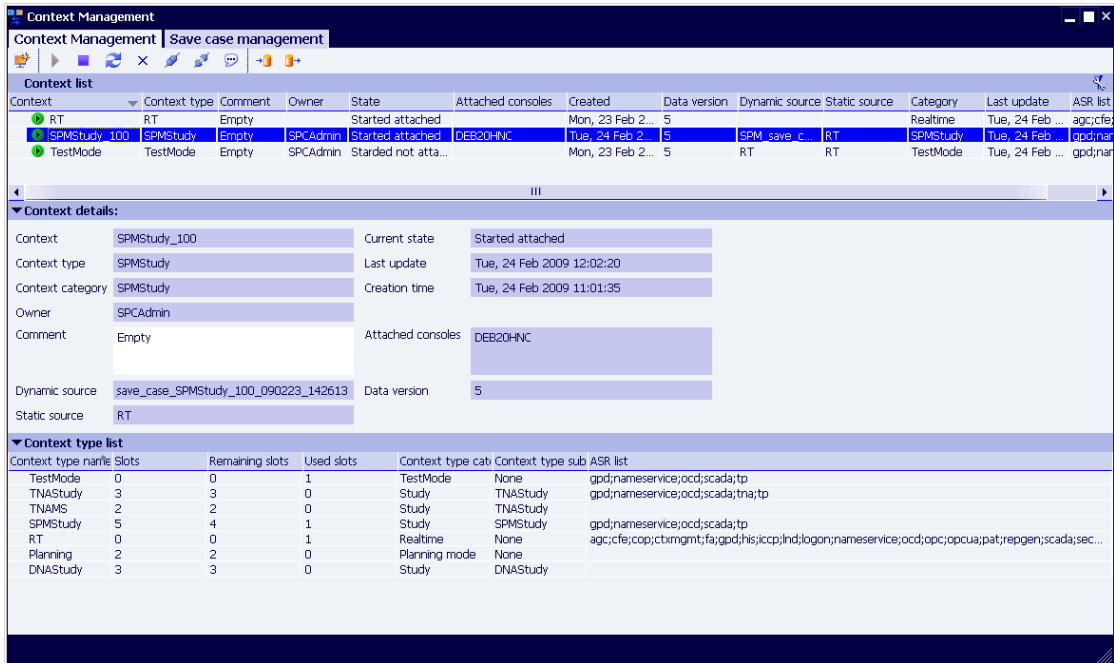
- Select **Context Management**.
The **Context Management** dialog opens.
The **Context List** shows already defined context data. This user interface allows you to manage both, contexts and save cases.



[syscm002-250209-01.tif, 1, en_US]

Figure 3-10 Overview of Available Contexts

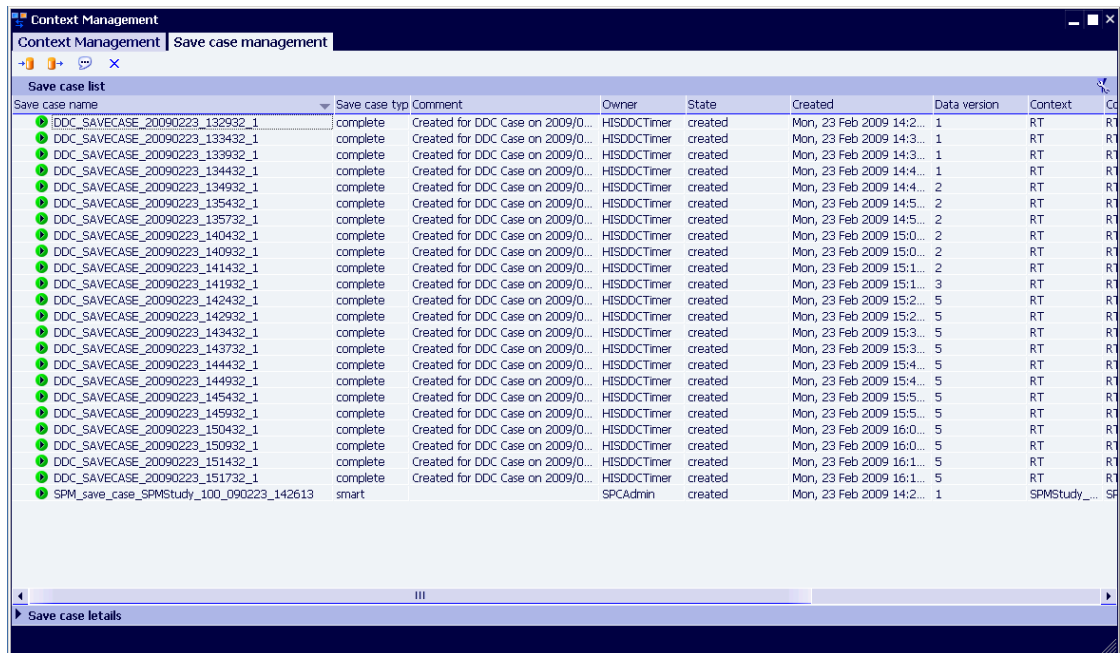
- Click **Context details** or **Context type list** to display or hide the detailed data for the selected context.



[syscm003-250209-01.tif, 1, en_US]

Figure 3-11 Detail Data for Selected Context

- Click **Save case management** to display the corresponding user interface.



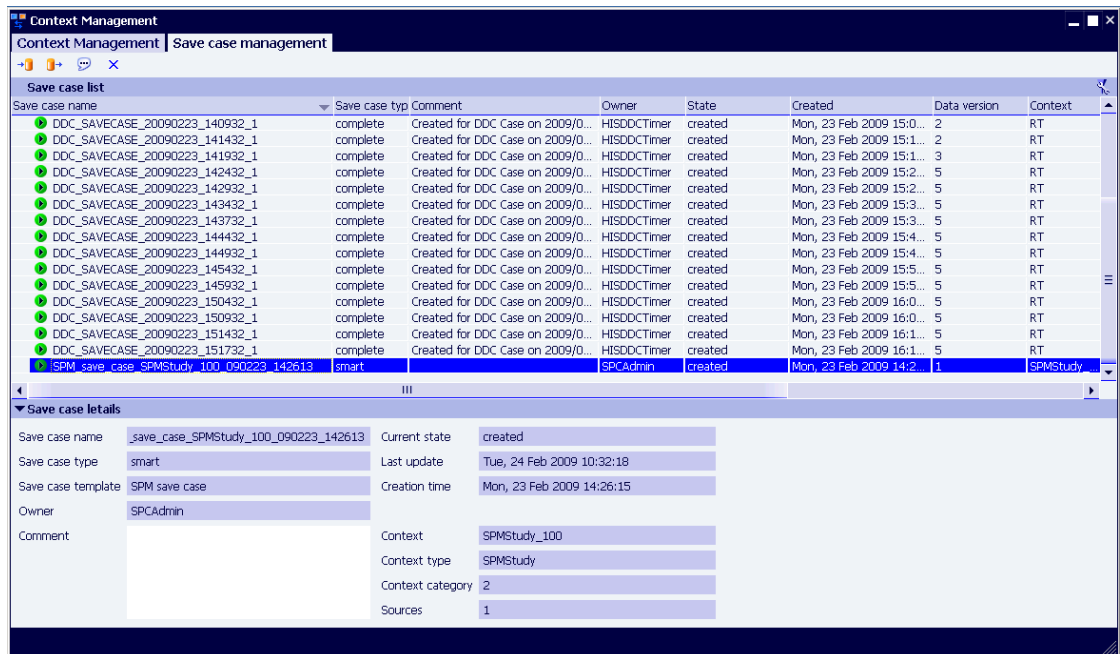
The screenshot shows the 'Context Management' application window with the 'Save case management' tab selected. It displays a table of save cases with columns: Save case name, Save case type, Comment, Owner, State, Created, Data version, and Context. The table lists 24 cases, mostly of type 'complete' created by 'HISDOCTimer', and one of type 'smart' created by 'SPCAdmin'.

Save case name	Save case type	Comment	Owner	State	Created	Data version	Context
DDC_SAVECASE_20090223_132932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:2...	1	RT
DDC_SAVECASE_20090223_139432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:3...	1	RT
DDC_SAVECASE_20090223_139932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:3...	1	RT
DDC_SAVECASE_20090223_134432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:4...	1	RT
DDC_SAVECASE_20090223_134932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:4...	2	RT
DDC_SAVECASE_20090223_135432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:5...	2	RT
DDC_SAVECASE_20090223_135732_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 14:5...	2	RT
DDC_SAVECASE_20090223_140432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:0...	2	RT
DDC_SAVECASE_20090223_140932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:0...	2	RT
DDC_SAVECASE_20090223_141432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:1...	2	RT
DDC_SAVECASE_20090223_141932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:1...	3	RT
DDC_SAVECASE_20090223_142432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:2...	5	RT
DDC_SAVECASE_20090223_142932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:2...	5	RT
DDC_SAVECASE_20090223_143432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:3...	5	RT
DDC_SAVECASE_20090223_143732_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:3...	5	RT
DDC_SAVECASE_20090223_144432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:4...	5	RT
DDC_SAVECASE_20090223_144932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:4...	5	RT
DDC_SAVECASE_20090223_145432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:5...	5	RT
DDC_SAVECASE_20090223_145932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:5...	5	RT
DDC_SAVECASE_20090223_150432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:0...	5	RT
DDC_SAVECASE_20090223_150932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:0...	5	RT
DDC_SAVECASE_20090223_151432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:1...	5	RT
DDC_SAVECASE_20090223_151732_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:1...	5	RT
SPM_save_case_SPMStudy_100_090223_142613	smart	Created for DDC Case on 2009/0...	SPCAdmin	created	Mon, 23 Feb 2009 14:2...	1	SPMStudy_...

[syscm012-250209-01.tif, 1, en_US]

Figure 3-12 Overview of Available Save Cases

- Click **Save case details** to display or hide the detailed data for the selected save case.



The screenshot shows the 'Context Management' application window with the 'Save case management' tab selected. The 'Save case list' table is visible, and the 'Save case details' section is expanded for the selected case 'SPM_save_case_SPMStudy_100_090223_142613'.

Save case name	Save case type	Comment	Owner	State	Created	Data version	Context
DDC_SAVECASE_20090223_140932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:0...	2	RT
DDC_SAVECASE_20090223_141432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:1...	2	RT
DDC_SAVECASE_20090223_141932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:1...	3	RT
DDC_SAVECASE_20090223_142432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:2...	5	RT
DDC_SAVECASE_20090223_142932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:2...	5	RT
DDC_SAVECASE_20090223_143432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:3...	5	RT
DDC_SAVECASE_20090223_143732_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:3...	5	RT
DDC_SAVECASE_20090223_144432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:4...	5	RT
DDC_SAVECASE_20090223_144932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:4...	5	RT
DDC_SAVECASE_20090223_145432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:5...	5	RT
DDC_SAVECASE_20090223_145932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 15:5...	5	RT
DDC_SAVECASE_20090223_150432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:0...	5	RT
DDC_SAVECASE_20090223_150932_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:0...	5	RT
DDC_SAVECASE_20090223_151432_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:1...	5	RT
DDC_SAVECASE_20090223_151732_1	complete	Created for DDC Case on 2009/0...	HISDOCTimer	created	Mon, 23 Feb 2009 16:1...	5	RT
SPM_save_case_SPMStudy_100_090223_142613	smart	Created for DDC Case on 2009/0...	SPCAdmin	created	Mon, 23 Feb 2009 14:2...	1	SPMStudy

Save case details	
Save case name	_save_case_SPMStudy_100_090223_142613
Save case type	smart
Save case template	SPM save case
Owner	SPCAdmin
Comment	
Current state	created
Last update	Tue, 24 Feb 2009 10:32:18
Creation time	Mon, 23 Feb 2009 14:26:15
Context	SPMStudy_100
Context type	SPMStudy
Context category	2
Sources	1

[syscm013-250209-01.tif, 1, en_US]

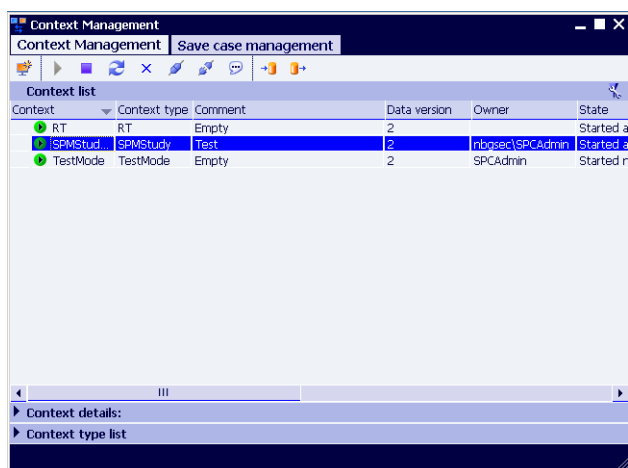
Figure 3-13 Detail Data for Selected Save Case

3.1.5.2 Context Management

Creating Study Context

The goal of this task is to create a study context.

- Select **SysAdmin > Context Management** from the PAT tool bar.

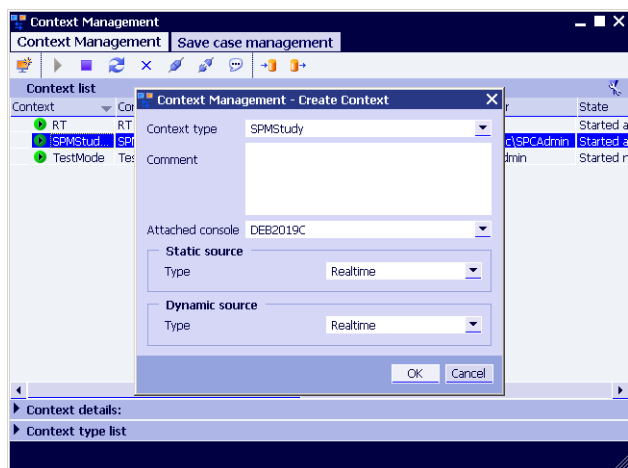


[rgsys005-201109-01.tif, 1, en_US]

Figure 3-14 Context Management Tab

Select **Context Management** tab.

- Click .



[rgsys007-201109-01.tif, 1, en_US]

Figure 3-15 Creating Study Context

- For new study context, define the following data:
 - Context type: The system assigns the context type automatically. It is fixed with exception of multi-context user and context-management administrator.
 - Comment: Optionally, you can write any text here.
 - Attached console: The system assigns the console name automatically. This context is fixed with exception of context-management administrator.
 - Static source: Click the list box to select the source. RT is the default static source. It is optional.
 - Dynamic source: Click the list box to select the initialization type for dynamic data of the new context. RT is the default dynamic source. It is optional.

Available dynamic data sources are:

- Normal state**
The initial values for the study are defined by the normal state of all digital measurements.
- HIS restore**
The values are taken at a certain time in the past from the archive.
- Save case**
The initial values are taken from a previously stored save case.
- Realtime**
- Context**
The values are taken from the current situation in another context.
For example, SPMStudy_100.



NOTE

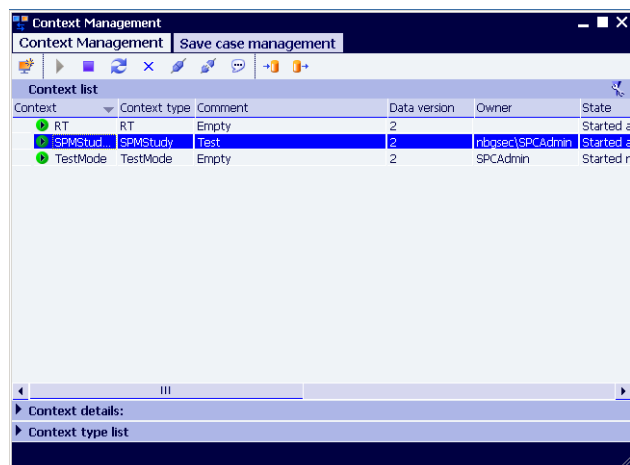
The retrieval of TNE values is not supported.

- Click **OK**.
- The study context is created.

Starting and Stopping Study Context

The goal of this task is to start and stop a study context.

- Select **SysAdmin > Context Management** from the PAT tool bar.



[rgsys005-201109-01.tif, 1, en_US]

Figure 3-16 Context Management

- Select the context, which you want to start.

- Click  to start the study context.
- Click  to stop the study context.
- Alternatively, right-click the selected context, that must be started or stopped. Select the option **Start** or **Stop**.



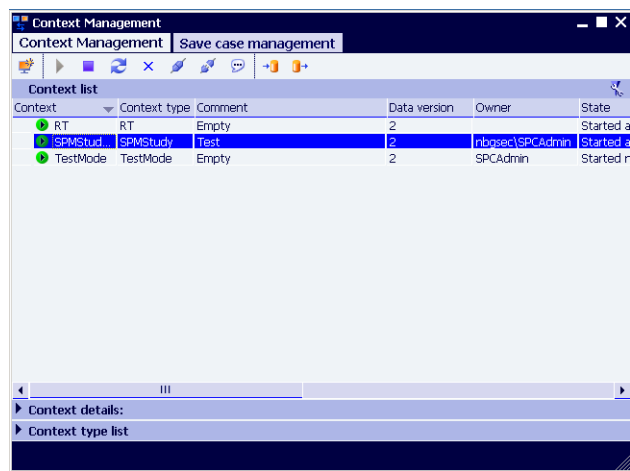
NOTE

From user perspective, **Stopping Study Context** is the same function as **Deleting Study Context**.

Updating Study Context

The goal of this task is to update a study context.

- Select **SysAdmin > Context Management** from the PAT tool bar.

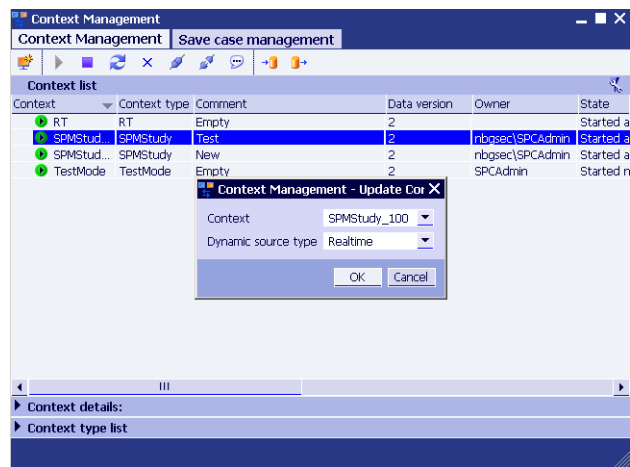


[rgsys005-201109-01.tif, 1, en_US]

Figure 3-17 Context Management

This is possible from **Context Management View** only.

- Select the context.
- Click .



[rgsys011-201109-01.tif, 1, en_US]

Figure 3-18 Update Study Context

- For updating the study context, define the following data:
 - Context: Used as source for the context initializing. This is optional and the default is the current study context. It is fixed within single context with exception of multi-context user and context-management administrator. Multi-context user can select any own study context. Context-management administrator can select any running study context.
 - Dynamic source type: Select the dynamic source from the available list box.

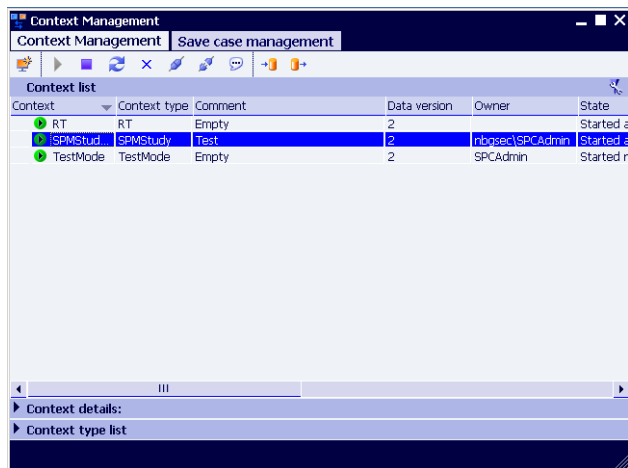
Available sources are:

- **Normal state**
The initial values for the study are defined by the normal state of all digital measurements.
- **HIS restore**
The values are taken at a certain time in the past from the archive.
- **Save case**
The initial values are taken from a previously stored save case.
- **Realtime**
- **Context**
The values are taken from the current situation in another context.
For example, SPMStudy_100.

Deleting Study Context

The goal of this task is to delete the study context that is no longer needed.

- Select **SysAdmin > Context Management** from the PAT tool bar.



[rgsys005-201109-01.tif, 1, en_US]

Figure 3-19 Context Management

It is possible to load save cases from **Context Management View** or **Save Case Management View**.

- Select the context.
- Click .
The study context is deleted.
- Alternatively, right-click on the selected context that has to be deleted.
The context menu appears.
- Select **Delete**.
The message that deleting process is successfully completed, appears at the bottom.

For a single-context, only 1 study is allowed. If current context is deleted, it switches automatically to RT. For a multi-context user, multiple studies are allowed of the same type. If current context is deleted, it switches

automatically to RT. For an context-management administrator, multiple studies are allowed of the same type. If current context is deleted, it switches automatically to RT.



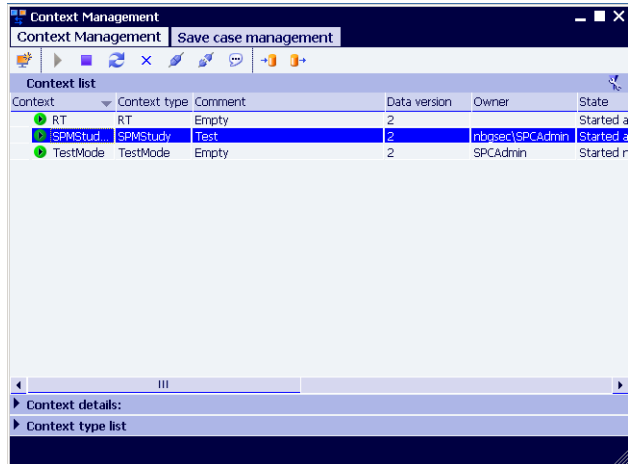
NOTE

- TestMode cannot be deleted by any user.
- From user perspective **Deleting Study Context** is the same function as **Stopping Study Context**.

Attaching and Detaching Study Context

The goal of this task is to attach and detach a study context. It is possible only in RT.

- Select **SysAdmin > Context Management** from the PAT tool bar.

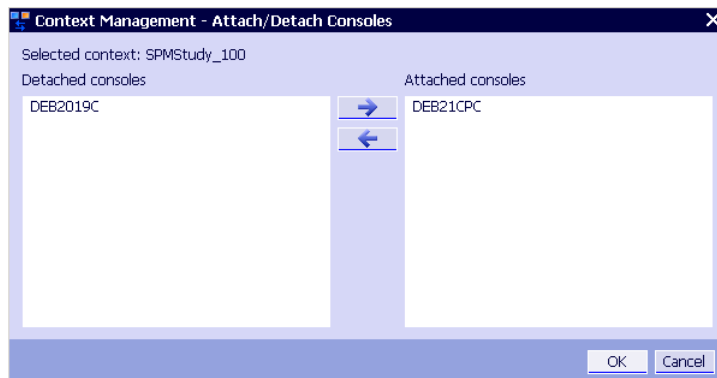


[rgsys005-201109-01.tif, 1, en_US]

Figure 3-20 Context Management

- Select the context, which you want to attach to your console.
- Click to attach the context.
- Click to detach the context.

If the console to be attached or detached is not unique, a separate dialog for specifying the console opens.



[rgsys023-251109-01.tif, 1, en_US]

Figure 3-21 Configuration Dialog to Attach or Detach Consoles

- To attach the console, select an available console from the detached console list on the left and click the arrow-to-right button between the console lists.
The selected console is replaced from detached console list to the attached console list.

- Click **OK** to confirm.
- To detach the console, select the attached console from the console list on the right and click the arrow-to-left button between the console lists.
The selected console is replaced from attached console list to the detached console list.
- Click **OK** to confirm.



NOTE

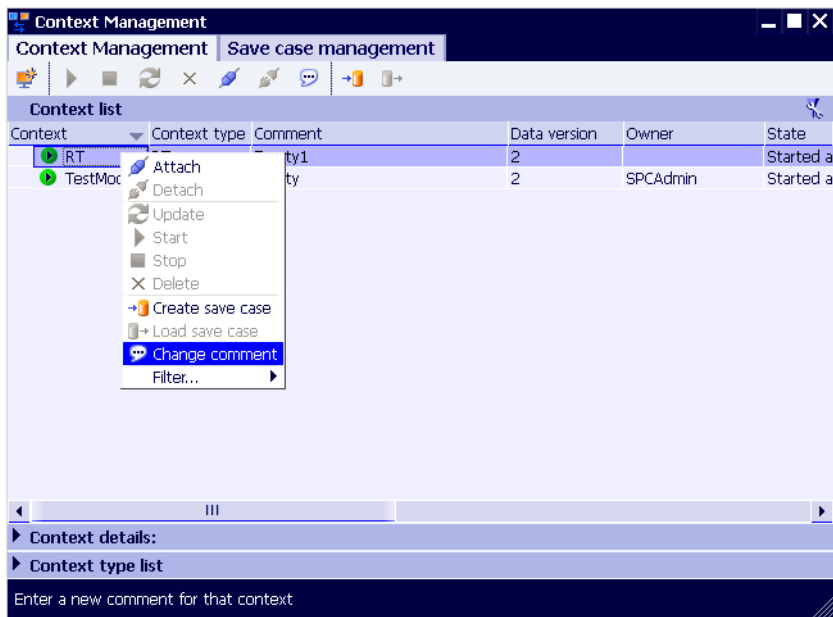
- In general, you can only detach the context you own and only attach a context to your console.
- An administrator can detach any context and attach a context to any console.

Entering New Comment for Selected Context

The goal of this task is to enter a new comment to a selected context.

- Select the context to which you can enter a comment.
 - Click  .

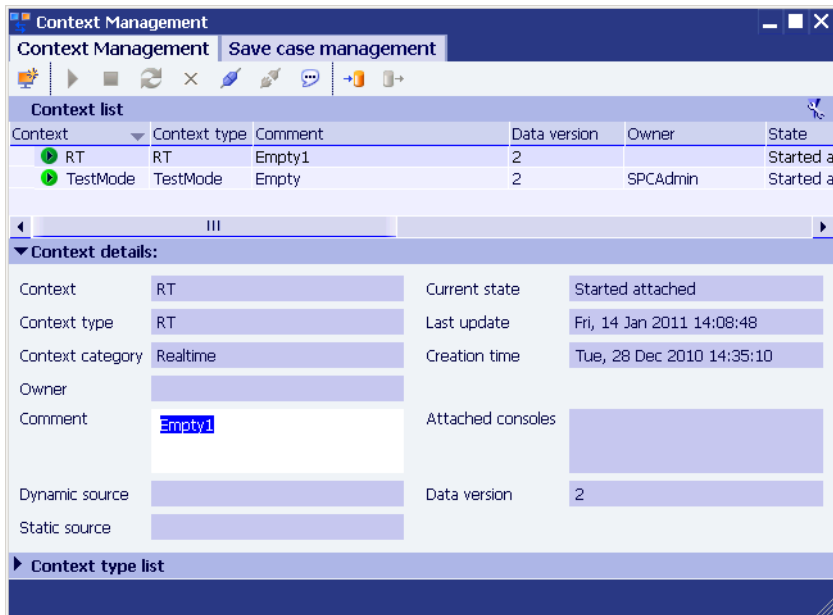
- Alternatively, select **Change comment** from the context menu.



[rgsysw03-140111-01.tif, 1, en_US]

Figure 3-22 Context Management - Change Comment

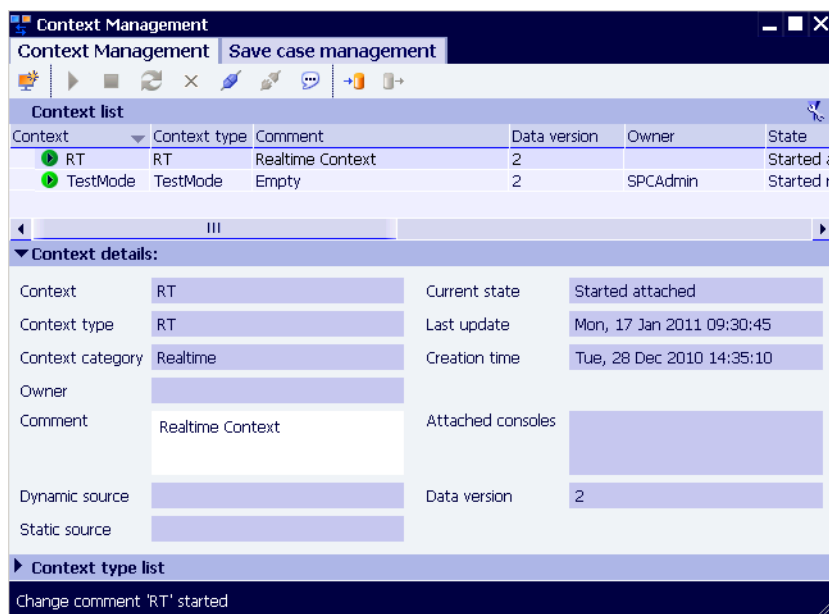
The **Context details** view of the selected context opens and the mouse pointer is placed in the **Comment** data field.



[rgsysw04-140111-01.tif, 1, en_US]

Figure 3-23 Entering New Comment for Selected Context with Completed Message

- Enter a comment into the data field.
- Click another data field, to assume the entered comment and display in the context list.



[rgsysw05-170111-01.tif, 1, en_US]

Figure 3-24 Updated Comment for Selected Context

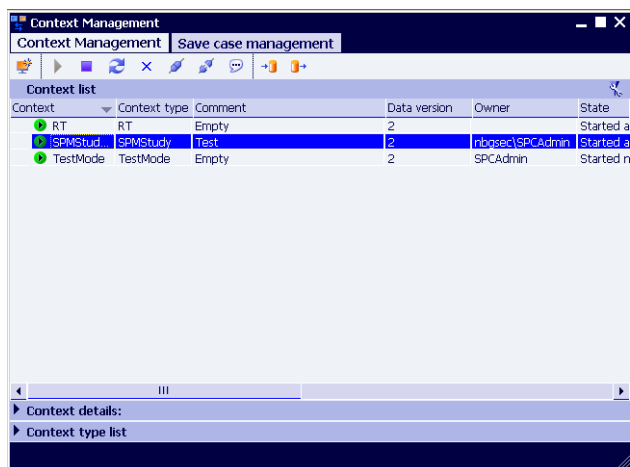
**NOTE**

The comment is only stored, when another field is selected.

3.1.5.3 Save Case Management**Creating Save Cases**

The goal of this task is to create a save case.

- Select **SysAdmin > Context Management** from the PAT tool bar.



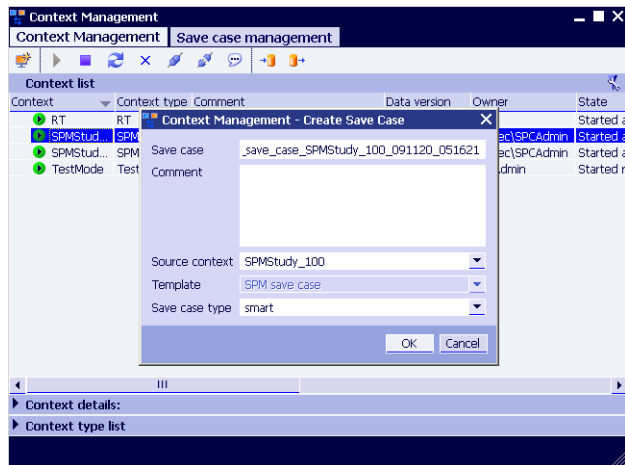
[rgsys005-201109-01.tif, 1, en_US]

Figure 3-25 Context Management

It is possible to create save cases from **Context Management** view or **Save Case Management** view.

- Select a context.

- Click .



[rgsys016-201109-01.tif, 1, en_US]

Figure 3-26 Create Save Case

- For the new save case, define the following data:
 - Save case: The system assigns the name automatically. It is editable by any single-context user.
 - Comment: Optionally, you can write any text here.
 - Source context: This context is fixed with exception of multi-context user and context-management administrator.
 - Template: It is fixed with exception of multi-context user and context-management administrator.
 - Save case type: It is optional. The type is smart by default.
- Click **OK**.
The status of the newly created savecase is viewed in **Context Management**.



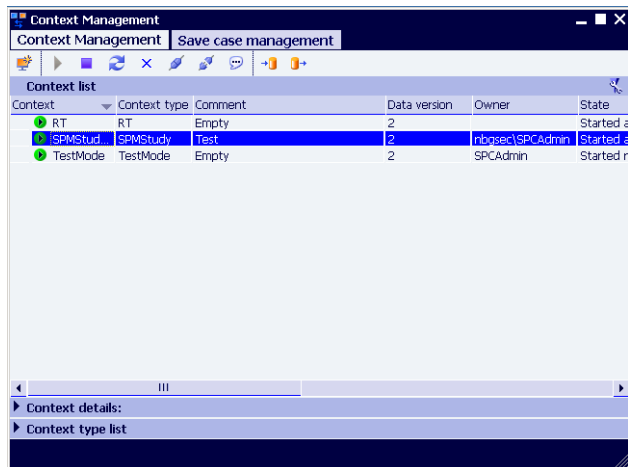
NOTE

- The process of creating a save case from the **Save Case Management** view is the same. There is no difference in **Create SaveCase** dialog from **Save Case** view.
- Creation of save cases from **Test Mode** is not supported.

Loading Save Cases

The goal of this task is to load a save case.

- Select **SysAdmin > Context Management** from the PAT tool bar.

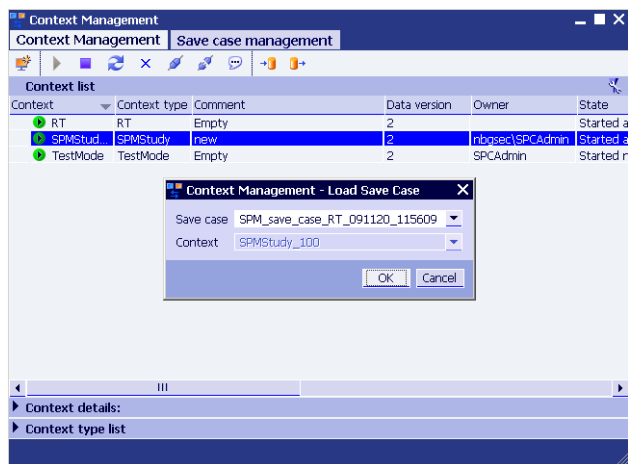


[rgsys005-201109-01.tif, 1, en_US]

Figure 3-27 Context Management

It is possible to load save cases from **Context Management** view or **Save Case Management** view.

- Select the context.
- Click .



[rgsys018-201109-01.tif, 1, en_US]

Figure 3-28 Loading Save Case

- For loading the save case, define the following data:
 - Save case: Select a save case from the list.
 - Context: This context is fixed with exception of context-management administrator.
- Click **OK**.

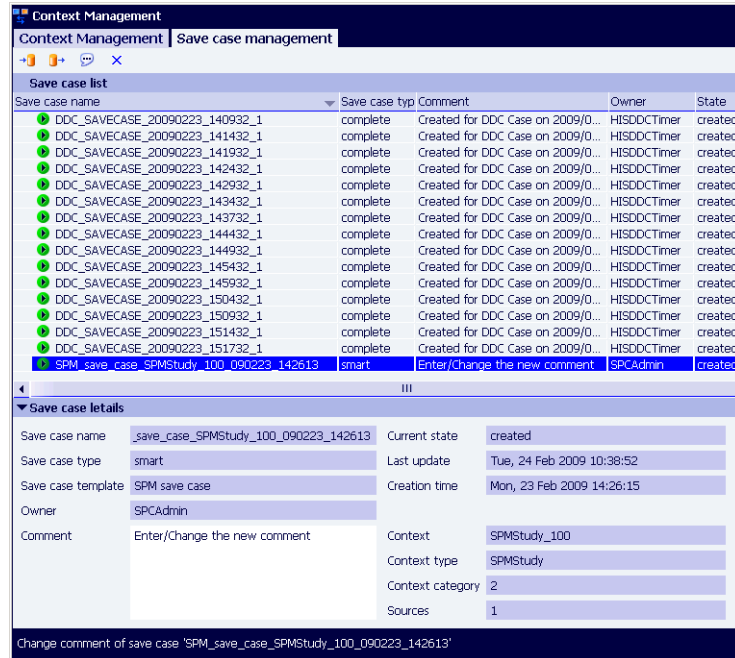
Entering New Comment for Selected Save Case

The goal of this task is to enter new comment for a save case.

- Select the save case to which a you want to enter a comment.
- Click .

- Click **Save case details**.

The **Save case details** view of the selected save case opens and the mouse pointer is placed in the comment data field.



[syscm042-260209-01.tif, 1, en_US]

Figure 3-29 Entering New Comment for Selected Save Case

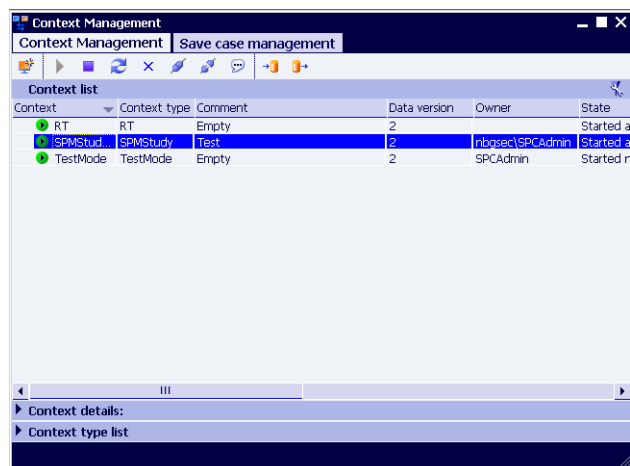
- Enter a comment into the text box.
- Click another data field, to assume the entered comment and display in the save case list.

Deleting Save Cases

The goal of this task is to delete a save case.

You can delete created save cases that are no longer needed.

- Select **SysAdmin > Context Management** from the PAT tool bar.



[rgsys005-201109-01.tif, 1, en_US]

Figure 3-30 Context Management

This is possible from **Save case management** tab only.

- Select the context.

- Click  .
The save case is deleted.
- Alternatively, right-click on the selected save case that has to be deleted.
The Context menu appears.
- Select **Delete**.

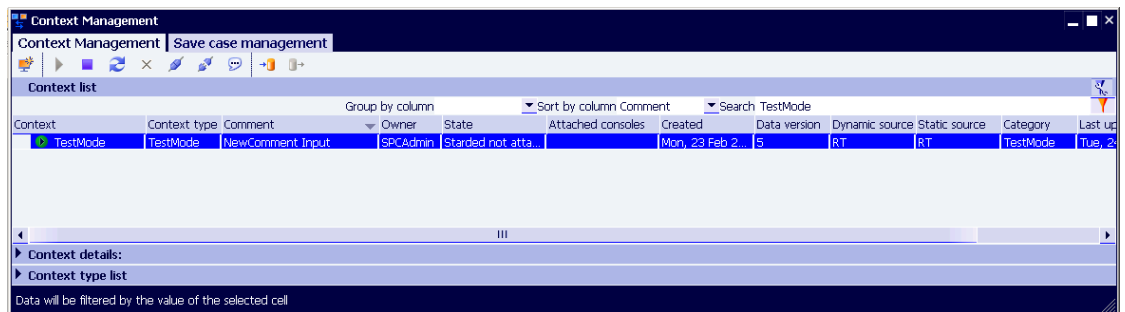
Any user can delete their own save cases. Context-management administrator can comment or delete any save case.

3.1.5.4 Filtering, Sorting, and Grouping

The goal of this task is to filter, sort, and group contexts.

The filter options are performed using context menu options with right-click for a selected context.

- Select a context from the context list.
- Open the context menu.

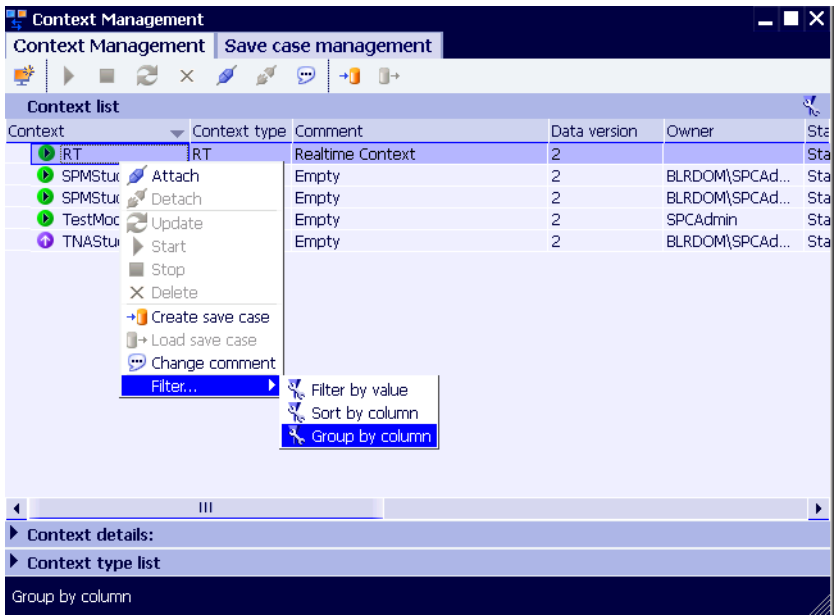


[syscm030-260209-01.tif, 1, en_US]

Figure 3-31 Filtering Using Filter Criteria Search

- Select **Filter > Filter By Value**
- Enter a text string into the **Search** filter box for target filtering by value.
For example, **Test Mode**, for specified **Context Type Name** filtering.
Only the filtered option is shown in the context list.
- Select a context from the context list.
- Open the context menu.

- Select **Filter > Group by column**.

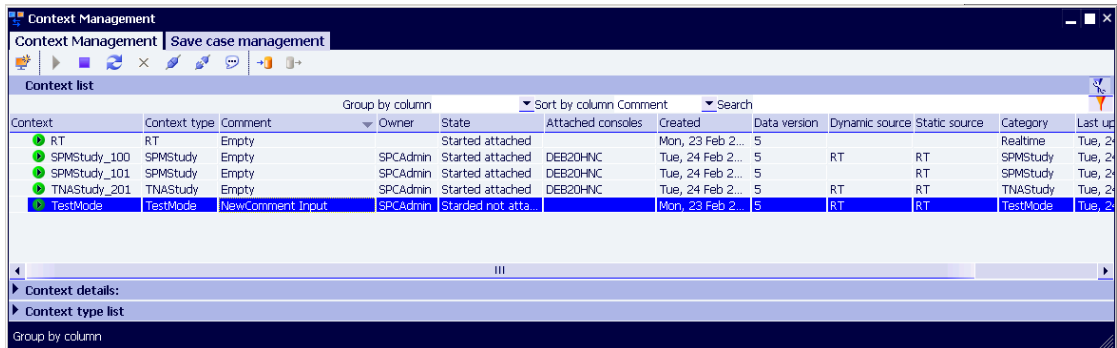


[rgsysw06-170111-01.tif, 1, en_US]

Figure 3-32 Context Menu - Group by Column

Contexts are sorted and listed in ascending order.

- Click the column header for descending order listing.



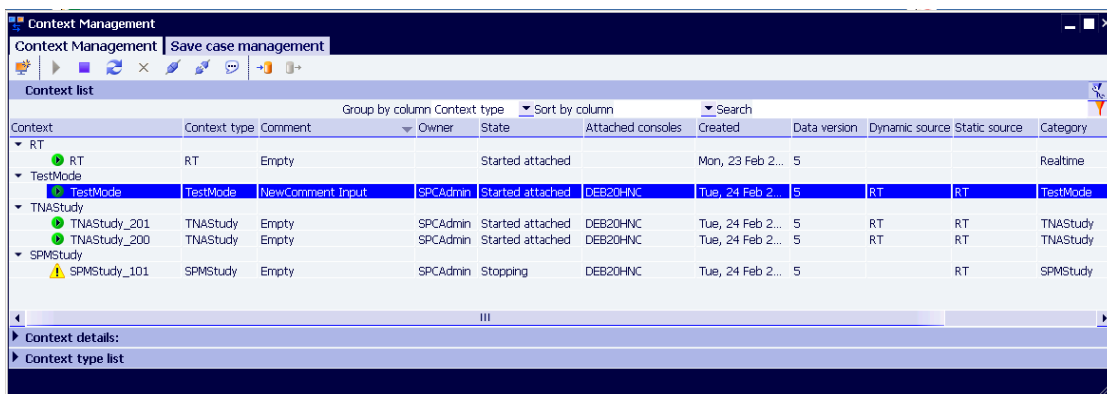
[syscm031-260209-01.tif, 1, en_US]

Figure 3-33 Example of Sorting Contexts Using Filter Criteria Sort by Column

- Select a context from the context list.
- Open the context menu

- Select **Filter > Group by column**.

Contexts are grouped and listed using the filter criteria **Group by column**.



The screenshot shows the 'Context Management' application window with the 'Save case management' tab selected. The 'Context list' is displayed, showing a table of contexts grouped by column. The table has columns: Context, Context type, Comment, Owner, State, Attached consoles, Created, Data version, Dynamic source, Static source, and Category. The contexts are grouped into four main categories: RT, TestMode, TNASTudy, and SPMStudy. The 'RT' group contains one context. The 'TestMode' group contains one context. The 'TNASTudy' group contains two contexts. The 'SPMStudy' group contains one context. The table is sorted by 'Created' date, and the 'Group by column' is set to 'Context type'.

Context	Context type	Comment	Owner	State	Attached consoles	Created	Data version	Dynamic source	Static source	Category
RT	RT	Empty		Started attached		Mon, 23 Feb 2...	5			Realtime
TestMode	TestMode	NewComment Input	SPCAdmin	Started attached	DEB20HNC	Tue, 24 Feb 2...	5	RT	RT	TestMode
TNASTudy_201	TNASTudy	Empty	SPCAdmin	Started attached	DEB20HNC	Tue, 24 Feb 2...	5	RT	RT	TNASTudy
TNASTudy_200	TNASTudy	Empty	SPCAdmin	Started attached	DEB20HNC	Tue, 24 Feb 2...	5	RT	RT	TNASTudy
SPMStudy_101	SPMStudy	Empty	SPCAdmin	Stopping	DEB20HNC	Tue, 24 Feb 2...	5		RT	SPMStudy

[syscm032-260209-01.tif, 1, en_US]

Figure 3-34 Example of Filtered Groups



NOTE

Filtering, Sorting, and Grouping can also be done in the **Save Case Management** view.

3.2 User Interface Description

3.2.1 Overview

Context Management is an application of the Spectrum Power 5 base system. This application runs on the RTS server. It operates in a warm-standby mode so that there is only 1 leading context manager at any point. In this way, the standby can start providing context management services if the master fails.

Context Management provides these features:

- Stores information of which contexts or save cases exist
 - What is the current state of existing contexts or save cases
 - Coordinates all the context management function
- If a study context must be started, the context management application finds out on which computers of the Spectrum Power 5 system it must be started. It issues corresponding commands to the involved SSR instances on these computers and monitors the progress of this operation. The context management **Application Suite Repository (ASR)** is updated to reflect all changes.

The **Context Management** application has no specific user interface (UI). It implements an interface (the context management API) that allows applications (like the context management UI and others) to read the current state of all contexts. Additionally, system applications can invoke system wide context operations.

For example, start, stop, and join of a context.

The Spectrum Power 5 operational environment is divided into several independent contexts. A context is a users specific view on a set of resources (variable information or data) that models the working environment (workspace) and its behavior.

The PAT distinguishes between the following contexts:

- Real-time context (RT)
- Study context
- Test Mode context



NOTE

The configured functional scope can vary between different context types.

3.2.2 User Roles Using Context Management

Context Management Configuration

In order to simplify operation for specific users, the behavior of the context management is dependent on role of the user.

The following context management roles are defined:

Single-context User

- It has only 1 context type beside real time (RT) and test mode (configured).
For example, **SPMStudy**.
- Only 1 configured context type can be started at the same time (only 1 instance for study context).
- Context type defines implicitly the **SaveCase** template.

Multi-context User

- It has 1 to n context types beside real time (RT) and test mode (configured).
For example, **SPMStudy**, **DNASStudy**.
- Can have more than 1 started context (even of same type).

- 1 context type (1 to n) configured as default.
- Configured context types define implicitly the available **SaveCase** templates.

Context-management Administrator

Has all permissions for **Context Management UI**.

3.2.3 Real-time Context (RT)

The real-time context satisfies the operational, planning, testing, and other activities of the Spectrum Power 5 system.

After system startup, the real-time context runs as per default setting.

The real-time context supports the actual real-time operation and control on current telemeter data. It provides direct operator interaction with the external world. Only this context allows control commands to the external world. It is also used to deliver control commands to field equipment.

The RT context is the only context in the system that provides redundancy.



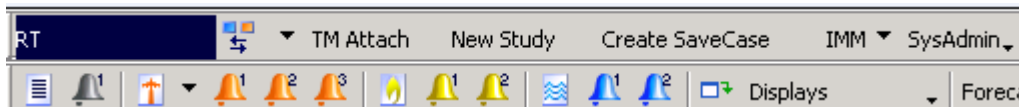
NOTE

For additional information on RT operations, refer to the specific Reference Guide for the corresponding application.

3.2.4 Study Context

A study context is used to support the operators during different network analysis and operational activities. This action is performed independently or in parallel to their (online) network monitoring, supervisory control, and management activities.

The PAT provides the necessary buttons to start or stop the study context and to move a console from study to RT and back.



[syscm015-271108-03.tif, 1, en_US]

Figure 3-35 PAT Displaying Study Context Buttons

- **New Study:** To create a study context.
- To switch from realtime to study and back to realtime
- **Update Study:** To update dynamic data in the study context.
Toggle button appears on the PAT after a study is loaded.

After switchover to the study context, the type of study context (TNA, DNA, or SPM) is started. Operations in study context depend on the user role assigned to the currently logged-on user.

After study context is started, the PAT changes colored background of the context indication from blue to orange.

As long as the study context is selected the context indicator displays the context name via the toggle button



The user can switch at any time back to the real time context. If an unacknowledged alarm exists in real-time context it is flashing in a customized color.

For example, SPMStudy_100.

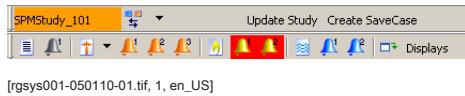
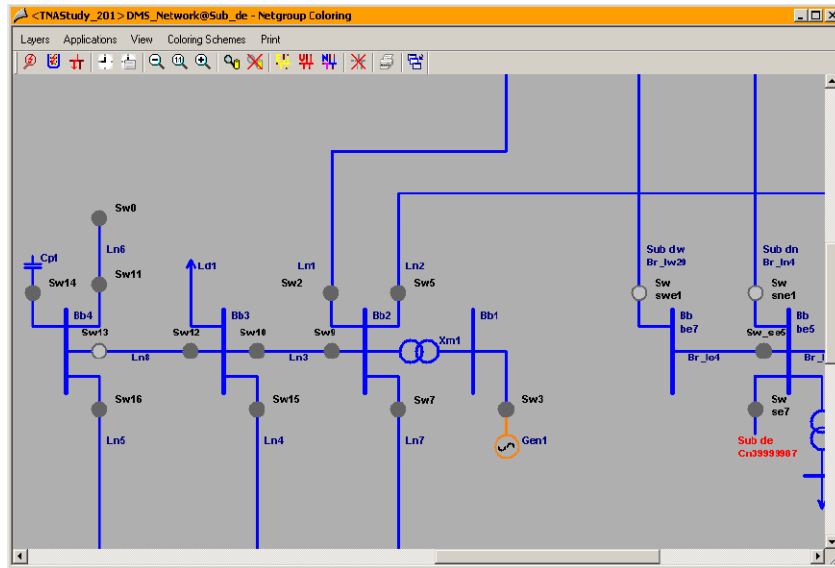


Figure 3-36 PAT Displaying Study Context

To comply with security constraints controls the instrumentation on the PAT is automatically adapted whenever the context is changed or when another user logs on. If a control or instrumentation item is not available in the selected context or for the current user it is not displayed. To ensure that users are aware of the current context, every user interface in the system clearly identifies the context being viewed. Configurable background colors for alarm lists, station single-line diagrams, and the switching procedure management accomplish this action.



[syscm016-271108-01.tif, 1, en_US]

Figure 3-37 Example of User Display in Study Context

The possibility to switchover from study to real-time context and in reversed order allows users to react immediately to an incoming alarm or any critical situation in the real network. Hence, whenever required, you can switch back to the real-time context.

Important remarks:

- If an object-test-marker is set for a measurement, a test run of a switching procedure in study mode fails. This presetting is important because the same switching procedure also fails in RT.
- If a switching procedure is changed or programmed in study mode, it is possible to revoke object-in-test markers to force a test run of the procedure. Object-in-test markers can be obstacle in study mode.



NOTE

For additional information on study context operations, refer to [3.1.2 Using Study Context](#).

3.2.5 Test Mode

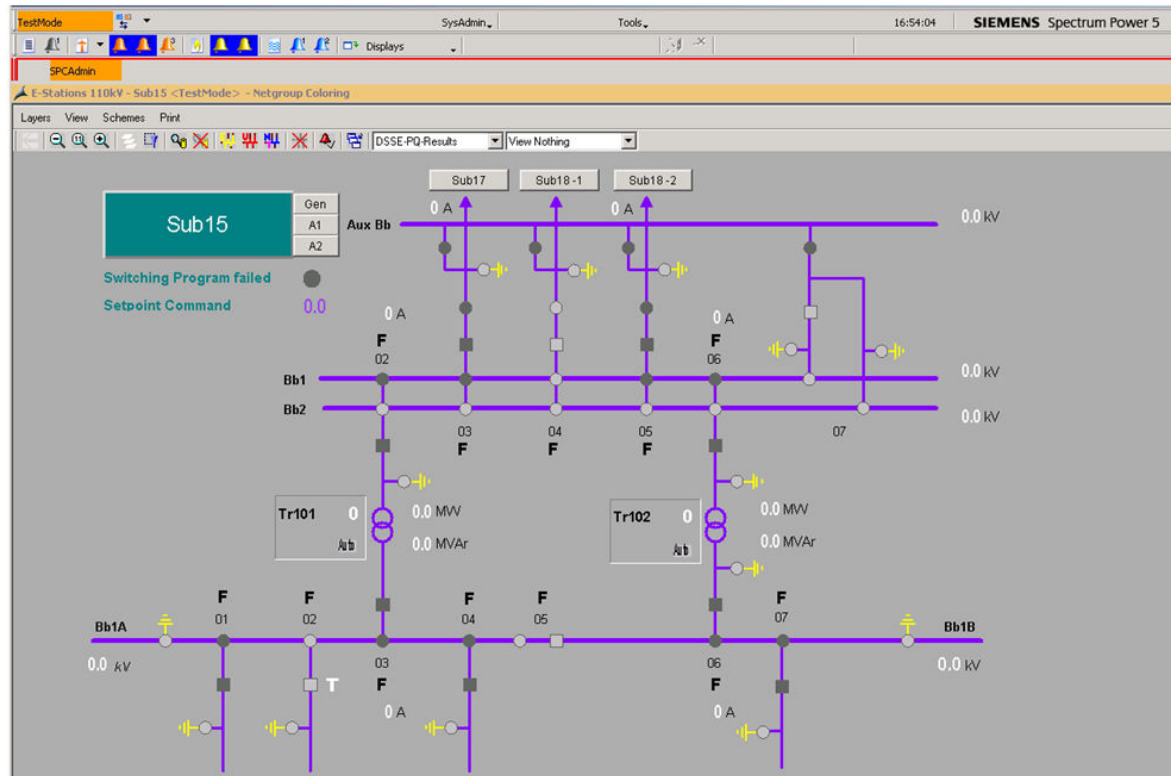
The purpose of the test mode consists of the fact that users can perform telemetry tests for selected measurements. Precondition is that these measurements must have set in the real-time context the Object-In-Test marker. These tests in the test mode are not visible in the RT context. The test mode is started on user request and runs parallel to and independent of the RT context.

The behavior in the test mode is distinct from RT or study context. There is only 1 system context for test mode. Test mode does not have console specific contexts.

Measurement controlling is limited to the corresponding environment. This means that a real-time operator is not able to control a measurement that is in test mode. And in reversed order a test mode operator is unable to control a measurement that is not in test mode. Thus, you are not guided to perform an operation that is not allowed.

Individual measurements are switched to the test mode state. In the RT context, these measurements remain unchanged. In the test mode, the test mode operator can change these measurements and can perform the telemetry test. After the Object-in-Test marker is removed the measurement in the test mode context is consequently read-only again. The measurement is updated in the RT context with the current telemetry value and has its normal real-time behavior (RT shows changes again).

After switchover to the test mode depending on the user role assigned to the currently logged-on user, the test mode UI is opened.



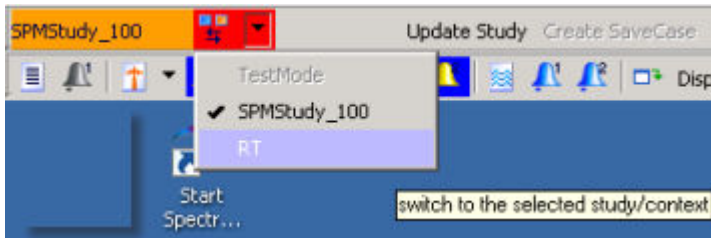
[syscom017-270209-01.tif, 2, en_US]

Figure 3-38 Example of User Display in Test Mode

- The test mode is a separated context, displays different value set, provides visual depiction of test mode using a color scheme.
- Test mode does not provide function to update or control a measurement that is not in test mode.
- Test mode does not provide the function to set or reset an object-in-test marker.
- RT context does not provide function to update or control a measurement that is in test mode.
- The test mode context is not kept in synchronization with activations occurring in the RT context.

For this reason, it is necessary to stop and explicitly start the test mode server package. This action requires that a visual indication of the different domain structure between test mode context and RT is given. It also requires distinguishing between the operations to switch to test mode versus to restart test mode.

- To switch from RT context to the test Mode, select the entry Test mode in the drop-down list of the  in the PAT.
PAT shows context label with colored background. The toggle button is labeled with **Switch to RT**.



[syscm018-270209-02.tif, 1, en_US]

Figure 3-39 PAT Button to Switch Back from Test Mode to RT

- To switch back from test Mode to real-time context, click on the toggle button .
If an unacknowledged alarm exists in real-time context, it flashes in red.



NOTE

Context Management allows you to manage contexts. It also monitors and controls the details of the test mode.

3.2.6 Save Case Management Using PAT Buttons

The save case functionality allows you to store static and dynamic data of the selected context. This dataset can later be used to run the system using this data instead of current data. Thus, special process situations can be reproduced for further analysis.

A save case can be created via the corresponding PAT button.

3.2.7 Context Management UI

3.2.7.1 Context Management View

Screen-specific Visual Elements

On invoking PAT, the context management UI is displayed. It allows you to manage both, contexts and save cases.



NOTE

The context menu entries is following the rules described below:

- In case of multi selection, the context menu will contain those entries that are common for all selected objects.
 - Entries are grayed, when the operator is missing the required authorization or when the corresponding action is not allowed.
-

Context	Context type	Comment	Owner	State	Attached consoles	Created	Data version	Dynamic source	Static source	Category	Last update	ASR list
RT	RT	Empty		Started attached		Mon, 23 Feb 2...	5			Realtime	Tue, 24 Feb ...	agc;cf...
SPMStudy_100	SPMStudy	Empty	SPCAdmin	Started attached	DEB20HNC	Tue, 24 Feb 2...	5	SPM save c...	RT	SPMStudy	Tue, 24 Feb ...	gpd;nar
TestMode	TestMode	Empty	SPCAdmin	Starded not atta...		Mon, 23 Feb 2...	5	RT	RT	TestMode	Tue, 24 Feb ...	gpd;nar

Context type name	Slots	Remaining slots	Used slots	Context type cat	Context type sub	ASR list
TestMode	0	0	1	TestMode	None	gpd;nameservice;ocd;scada;tp
TNASTudy	3	3	0	Study	TNASTudy	gpd;nameservice;ocd;scada;tna;tp
TNAMS	2	2	0	Study	TNASTudy	
SPMStudy	5	4	1	Study	SPMStudy	gpd;nameservice;ocd;scada;tp
RT	0	0	1	Realtime	None	agc;cf;cop;ctxmgmt;fa;gpd;his;iccp;ld;logon;nameservice;ocd;opc;opcua;pat;repgen;scada;sec...
Planning	2	2	0	Planning mode	None	
DNASTudy	3	3	0	Study	DNASTudy	

[syscm003-250209-01.tif, 1, en_US]

Figure 3-40 Context Management View

The context management display shows:

- Context List

This list shows the significant attributes of already defined context data. Only important user hints are specified. On the right is displayed the button for filter settings.

- Context: Defines the name of created contexts.
This column also includes RT and Test mode context, which cannot be deleted from the context list. Several users can be attached to the test mode. Only 1 user can be attached to the study context.
- Context type: Specifies the context type based on the system available context type.
- Comment: Can be edit with a free text.
- Owner: Defines the user account.
- State: Defines the context state.
For example, stopped, started, or started attached.
- Attached Consoles: Specifies the name of the attached console.
- Created: Specifies the time (as format day, date hour) when the context was created.
For example, Thu, 12 Aug 2010 07:41:17.
- Data version: Specifies the version of the RT context when this context was created.
If this version differs from the actual RT context version a visual indication occurs. The context name is automatically displayed with yellow background. In this manner, you can decide whether for the own analysis it is better to delete the created context and to recreate it again. Context deletion followed with context recreation allows you to operate again with actual RT data.
- Dynamic Source: Dynamic data source is selected from the available list box.
Available dynamic sources: RT, Normal State, Save Case (includes all types), HIS Restore Data, and Context.
Distinctions in comparison to RT selection refers to additional specific data fields, that is defined during context creation.
Option Normal State: No additional input required. Default configurations of RT settings are requested.
Option Save Case: Name of the save case (data field) is required, available list box can be used.
Option HIS Restore Data:
 - **HIS Restore Date/Time** can be defined using the calendar.
 - **Alarm Summary prefetch in minutes** number of minutes input required to define the time for alarm receiving and recording in alarm lists.
- Static Source: Source of the static data of the context.
Selecting the option Complete Save Case, the save case name must be additionally specified.
- Category: Is based on context type
For example, the category Study is assigned to the context types SPM Study, TNA Study, and DNA Study.
- Last Update: Time (as format day, date hour) of last data update.
For example, Thu, 20 Aug 2010 17:10:10.
- ASR list: Name of required ASR.

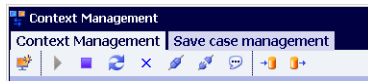
- Context details: Click the arrow left on the list name to hide or to display the context details.

This list shows the attributes for the selected context from the context list. These attributes are also listed in the context list.

- Context Type list: Click the arrow left on the list name to inhibit or to display the attributes of this list.
 - Context Type Name: List of all available system context types
 - Slots: Number of available slots for context access (subdivided into used and remaining slots)
 - Remaining Slots: Number of unused and still available slots (difference of slots and used slots)
 - Used Slots: Number of used slots
 - Context Type Category: Assigned category to the context type.
For example, Study to SPM Study.
 - Context Type Sub Category.
 - ASR List: Name of the required ASR.
It lists the ASRs for the particular context type.
For example, for test mode, name service and SCADA are used.

Context Management Functions

The context management functions can be accessed using the icons listed on top:



[syscm050-270209-01.tif, 1, en_US]

Figure 3-41 Available Icons for Context Functions

These icons listed from left to right are:

- Create context.
- Start, stop, update, delete, attach, detach context, enter or change context comment.
- Create save case out of the selected context and load save case into a running study.
The save case icons are also available in the **Save Case Management** tab.

Click  on the right in the second filter row to open filter settings.

If you set filter options, the user interface also displays the available filter criteria.



[rguio010-290610-02.tif, 1, en_US]

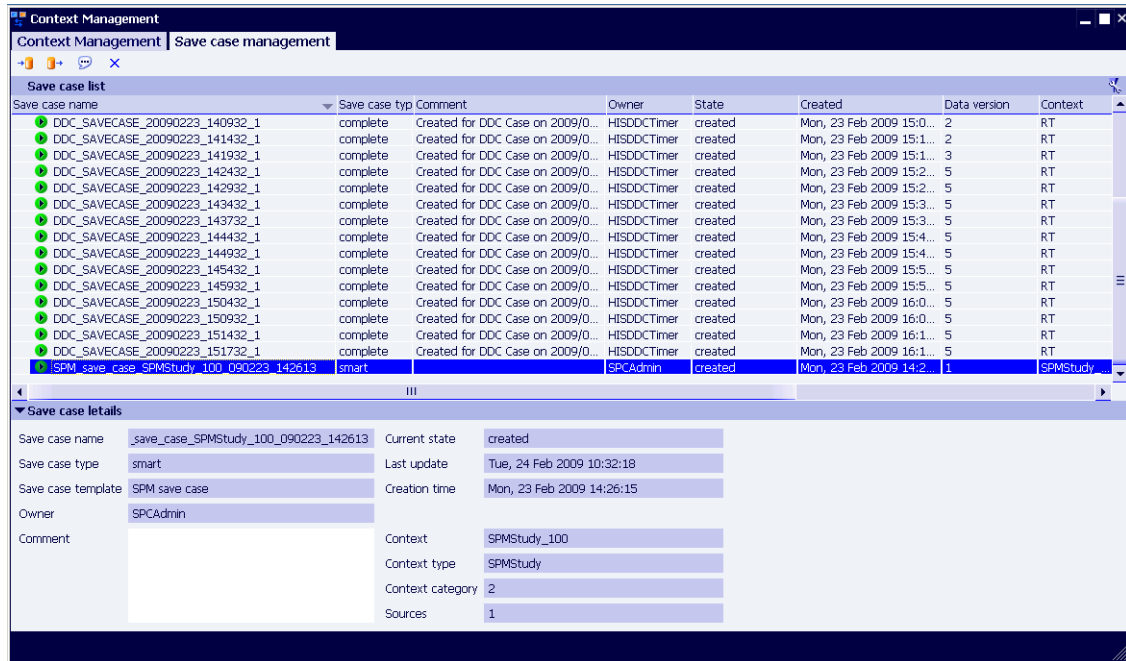
Figure 3-42 Displaying Filter Criteria and Set Filter Button

- Group by column: Provided list box can be used.
- Sort by column: Provided list box can be used.
- Search: Is depending on the entered string.

3.2.7.2 Save Case Management View

Screen-specific Visual Elements

On clicking the **Save Case Management** tab the corresponding user interface is displayed.



[syscm013-250209-01.tif, 1, en_US]

Figure 3-43 Save Case Management View

The save case management display shows:

- **Save Case List**
This list shows the significant attributes of already defined save cases. Only important user hints are specified.
 - **Save Case Name:** Defines the name of created save cases for study contexts.
 - **Save Case Type:** Specifies the save case type whether partial, smart, or complete
 - **Comment:** Can be edit with a free text
 - **Owner:** Defines the user account
 - **State:** Defines the save case state.
For example, created.
 - **Created:** Specifies the time (as format day, date hour) when the save case has been created.
 - **Data version:** Specifies the version of the RT context when the save case was created.
If this version differs from the actual RT context version a visual indication occurs.
For example, the save case name is automatically displayed with yellow background. In this manner, you can decide whether for the own analysis it is better to delete the created save case and to recreate it again. Save case deletion followed with save case recreation allows you to operate again with actual RT data.
 - **Context:** Refers to the context name of the created save case.
For example, SPMStudy_100.
 - **Context Type:** Refers to the context type of the created save case.
For example, SPMStudy, or RT.
 - **Context Category:** Identifier of the context category.
For example, 1 for RT and 2 for Study.
 - **Template:** Name of the applied template during save case creation.
 - **Last Update:** Time (as format day, date hour) of last data update.
 - **ASR List:** Name of required ASR.

- **Save Case Details:** Click the arrow left on the list name to hide or to display the save case details. This list shows the attributes for the selected save case from the save case list. These attributes are also listed in the save case list and not explained again. The attribute Data Sources coincide with the attribute Data Version shown in the save case list and also not explained again.

Save Case Management Functions

The save case management functions can be accessed using the icons listed on top:



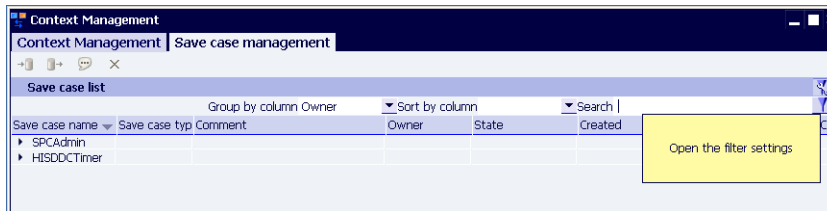
[syscm052-270209-01.tif, 1, en_US]

Figure 3-44 Available Icons for Save Case Functions

These icons listed from left to right are:

- Create a new save case out of a context
- Load save case into a running study
- Enter/change save case comment
- Delete selected save case

Click the set filter button on the right in the second filter row to open filter settings. If users set filter options, the user interface also displays the available filter criteria.



[syscm053-270209-01.tif, 1, en_US]

Figure 3-45 Displaying Filter Criteria and Set Filter Button

- **Group by column:** Provided list box can be used.
- **Sort by column:** Provided list box can be used.
- **Search:** Is depending on the entered string

4 Single-line Diagram

4.1	Introduction	68
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4.1 Introduction

A single-line diagram is a graphical presentation of the network topology. Single-line diagrams help you to visualize the complex technical assembly of the networks and to control measurements and devices.

The main features are:

- Marking and tagging
- Supervisory control
- Manual updating
- Acknowledging alarms
- Interactive tracing
- Schemes



NOTE

A tooltip occurs when the mouse pointer is next to a device in the single-line diagram.

Schemes

The operator needs information about network energization, network grouping, and other information represented by color code.

Spectrum Power 5 provides multiple schemes for different purposes:

- Network energization
- Voltage levels
- Network groups
- Looped/parallel network components
- Multi or single phase



[rguios18-031209-01.tif, 2, en_US]

Figure 4-1 Selecting a Scheme on the Toolbar of the Single Line Diagram

4.2 Operational Steps

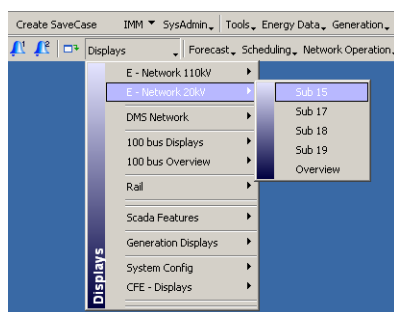
4.2.1 Calling a Single-line Diagram

There are alternative ways to open a single-line diagram:

- Call using the PAT.
- Call using a context menu.
- Call using an application button within a single-line diagram or the PAT.
- Call using the Display Callup.

Calling the Single-line Diagram Using PAT

- Click **Displays**



[rguios04-301109-01.tif, 1, en_US]

Figure 4-2 Calling a Diagram Using PAT

- Select the desired single-line diagram from the **Displays** menu.
For example, **E - Network 20kV > Sub15**.
- Double-click or use drag and drop to open the single-line diagram.

Calling the Single-line Diagram Using Context Menu

- Right-click an entry in a summary, the Runtime Explorer, or another single-line diagram.
For example, the **Alarm Summary**.
- Select **Display Callup** from the context menu.
The attached single-line diagram opens immediately.
Alternatively, if more than 1 display is attached to the selected object, the **Display Search Results** dialog opens and you can select 1 of the results.

For additional information on search results, refer to [4.3.6 Display Search Results Dialog](#).

Calling the Single-line Diagram Using Display Callup

You can call the single-line diagram using a **Display Callup**.

For additional information on how to call single-line diagrams using **Display Callup**, refer to [2.2.4 Using the Display Callup](#).

Calling the Single-line Diagram Using an Application Button

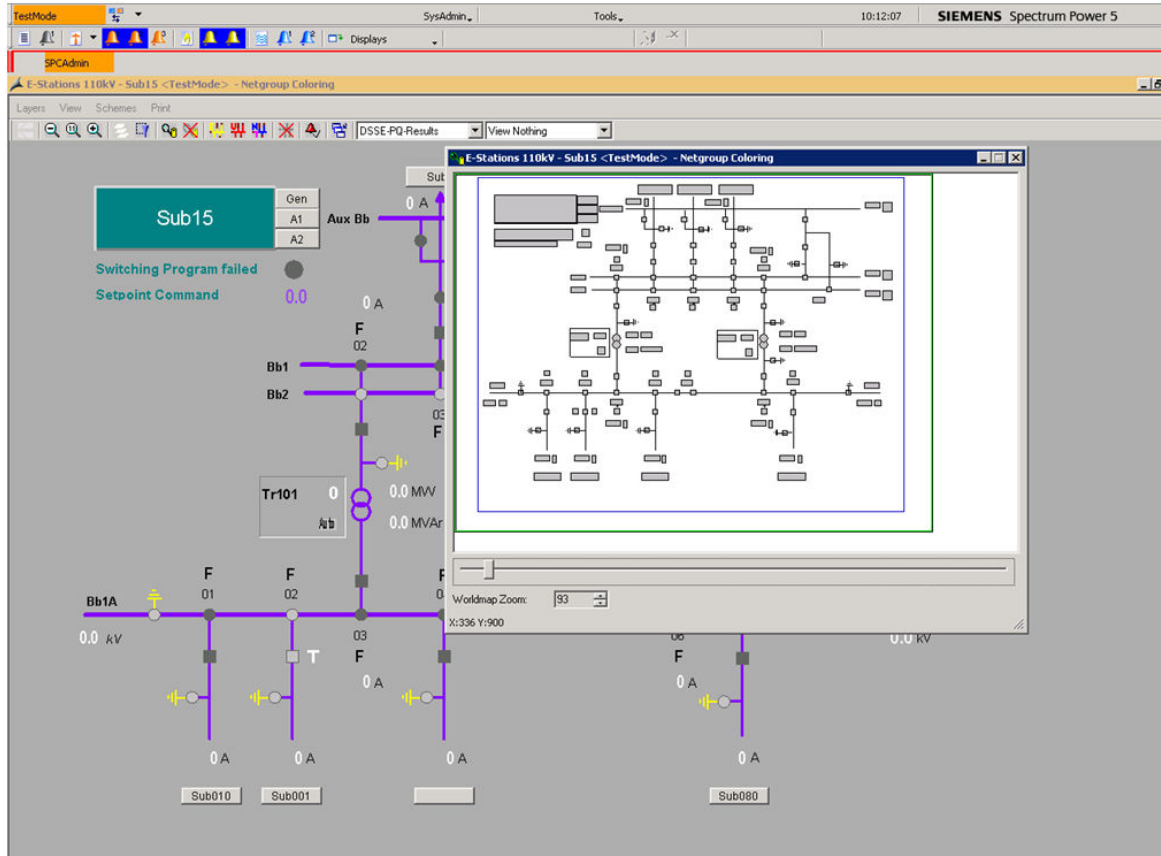
- Click the appropriate button to display the single-line diagram, on a single-line diagram or the PAT.
The single-line diagram opens.
For additional information on application buttons, refer to [4.2.9 Calling Application Buttons](#).

4.2.2 Navigating Through a Single-line Diagram

The goal of this task is to display a specific segment of the single-line diagram by using the panning functionality.

Navigating Through the Single-line Diagram

- On the menu bar of the single-line diagram, select **View > Open Navigator**.



[sc_SLDNavigatorWindow, 1, en_US]

Figure 4-3 Single-line Diagram and Navigator Window

The navigator window opens. The green rectangle represents the displayed segment in the associated main window.

- To scroll to a different part of the single-line diagram, move the green rectangle in the navigator window by using drag and drop.
- To enlarge or reduce the size of the rectangle, press and hold the **CTRL** key and use the scroll wheel.
- To zoom the display of the navigator window, use the slider control.

Using the Sliders of the Single-line Diagram

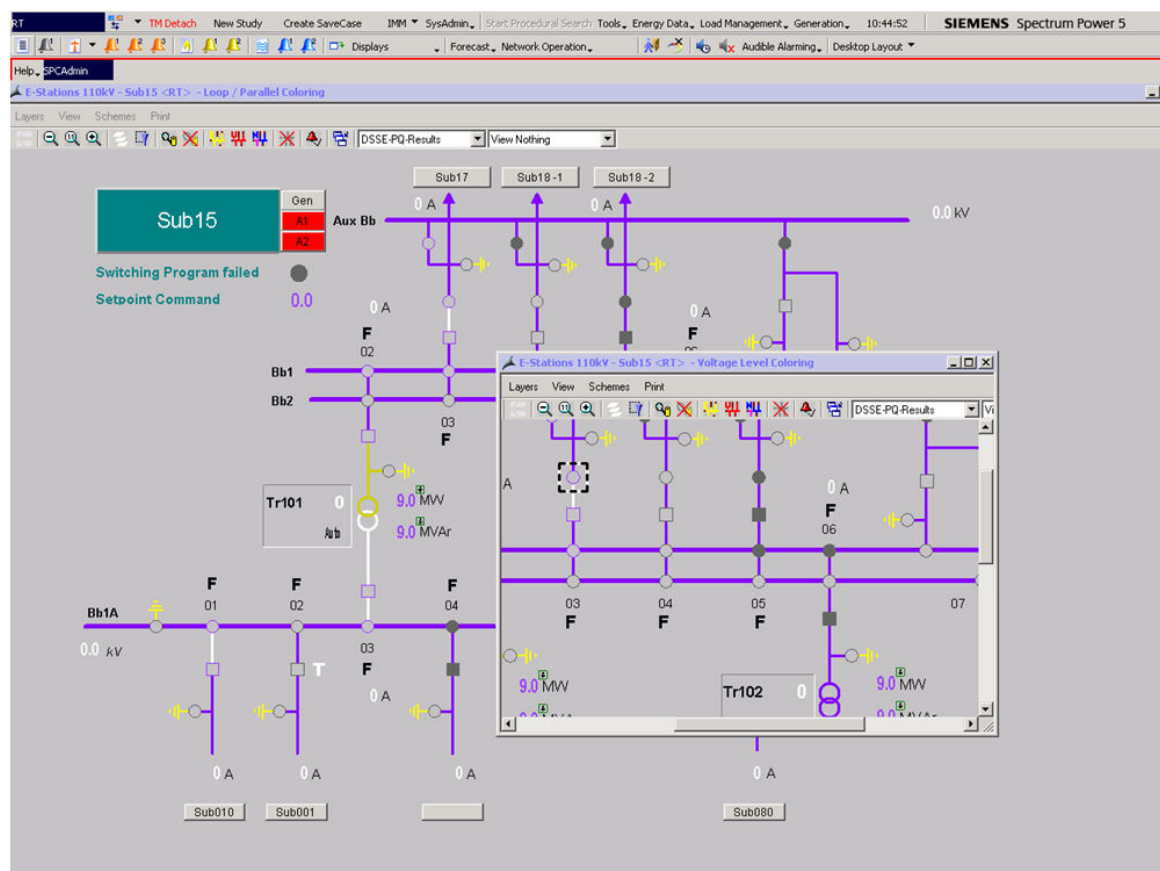
- You can also use the slider on the right and on the bottom to navigate through the single-line diagram.

4.2.3 Zooming a Single-line Diagram

The goal of this task is to zoom the display of the single-line diagram.

Zooming

- Zooming of the single-line diagram:
 - To perform smooth zooming, press the **CTRL** key and simultaneously use the mouse scroll.
 - To perform rectangle zooming, press the **CTRL** key and simultaneously left-click.
 - Step zooming by 3 dedicated buttons in the toolbar (in, out, original size).
 - When zooming in, not only do objects become larger, but a new, more detailed, plane with new information becomes visible. The reverse occurs when zooming out. This decluttering keeps character strings, figures, and symbols visible while zooming.
- You can insert up to 32 layers without zooming by selecting the menu items in the layer menu. (The layer menu is located above the left portion of the toolbar of the graphical display.)
- Individual planes can be enabled or disabled temporarily for better visibility.



[rguios20-031209-01.tif, 2, en_US]

Figure 4-4 1 Single-line Diagram - 2 Views (Different Zooming Levels)

4.2.4 Selecting Schemes

The goal of this task is to visualize network energization, voltage levels, and network grouping in the single-line diagram by using schemes.

Selecting Schemes



To select the scheme for **Energization Coloring**, click this button on the toolbar of the single-line diagram.



To select the scheme for **Netgroup Coloring**, click this button on the toolbar of the single-line diagram.



To select the scheme for **Voltage Level Coloring**, click this button on the toolbar of the single-line diagram.

Select **Schemes** on the toolbar of the single-line diagram, to select **Energization**, **Voltage**, **Netgroup**, **Loop Parallel**, **Multiphase**, **Phase A**, **Phase B**, or **Phase C**.

4.2.5 Performing Operations on Field Resources

The goal of this task is to perform operations on field resources displayed in a single-line diagram. For example, a context menu (configured) in the IMM provides the possible operations:

Manually Updating Values (Manual Update/Manual Update Toggle)

For additional information on how to update values, refer to [11.2.1 Manually Updating Values](#).

Performing Interactive Tracing (Trace)

For additional information on how to perform interactive tracing, refer to [4.2.11 Performing Interactive Tracing](#).

Performing Supervisory Control (Control State)

For additional information on how to perform supervisory control, refer to [9.2.2 Controlling the State of a Switch](#).

Performing Manual Override

For additional information on how to perform manual override, refer to [11.2.2 Manually Overriding Values](#).

Acknowledging Alarms

For additional information on how to acknowledge an alarm using the affected field resource, refer to [7.2.1 Handling an Incoming Alarm](#).

Applying a Marker

For additional information on how to apply a marker, refer to [10.2.1 Setting a Marker with Reminder Function](#).

Calling the Property Window

For additional information on how to call the **Property Window**, refer to [4.2.8 Calling the Property Window](#).

4.2.6 Performing Operations on Measurements

The goal of this task is to perform operations on measurements displayed in a single-line diagram. For example, a context menu (configured) in the IMM provides the possible operations:

Manually Updating Values (Manual Update/Manual Update Toggle)

For additional information on how to update values, refer to [11.2.1 Manually Updating Values](#).

Performing Supervisory Control (Control State)

For additional information on how to perform supervisory control, refer to [9.2.2 Controlling the State of a Switch](#).

Performing Manual Override

For additional information on how to perform manual override, refer to [11.2.2 Manually Overriding Values](#).

Applying a Marker

For additional information on how to apply a marker, refer to [10.2.1 Setting a Marker with Reminder Function](#).

Acknowledging Alarms

For additional information on how to acknowledge an alarm using the affected field resource, refer to [7.2.1 Handling an Incoming Alarm](#).

Calling the Property Window

For additional information on how to call the **Property Window**, refer to [4.2.8 Calling the Property Window](#).

Changing the Priority of a Value (Manual Value Selection Adjustment)

For additional information on how to change the priority of a value, refer to [5.2.4 Manually Adjusting the Priority of a Value](#).

Calling the Display Search Results Dialog (Display Callup)

For additional information on how to call a display, refer to [4.2.10 Using the Display Search Result Dialog](#).

4.2.7 Performing Operations for Multi Selections

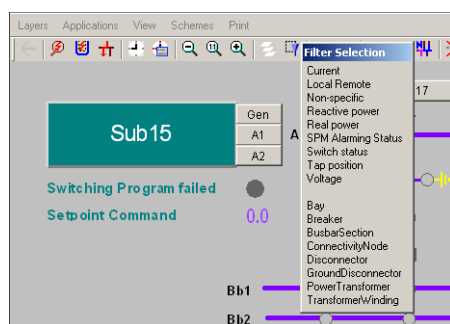
The operator has the possibility to acknowledge alarms directly in the single-lines in the following ways:

- Single selection:
All alarms of one object using context menu.
- Multi selection:
All alarms of all selected objects using context menu.
- Button acknowledge all visible alarms:
All alarms of visible objects are acknowledged.

The operator has the possibility of filter option:

- Click  on the tool bar.

The **Filter Selection** dialog opens.



[rguios15-021209-01.tif, 1, en_US]

Figure 4-5 Filter Selection Dialog

- Select an entry from the filter list.
For example, **Current**.
- The selected elements are highlighted.
For example, all available **Current** measurements of the current single-line diagram.

4.2.8 Calling the Property Window

The goal of the task is to call the **Property Window**.

For example, the **Property Window** provides an overview of limits, quality codes, and the raw data archive.

Calling the Property Window

- On a single-line diagram, right-click the desired measurement or value.
- Alternatively, on the Runtime Explorer, right-click an item in the content area.
- Select the menu item **Property Window** from the context menu.

The **Property Window** opens.



NOTE

The **Property Window** is a generic tool and the number of displayed tabs depends on the selected measurement or value.

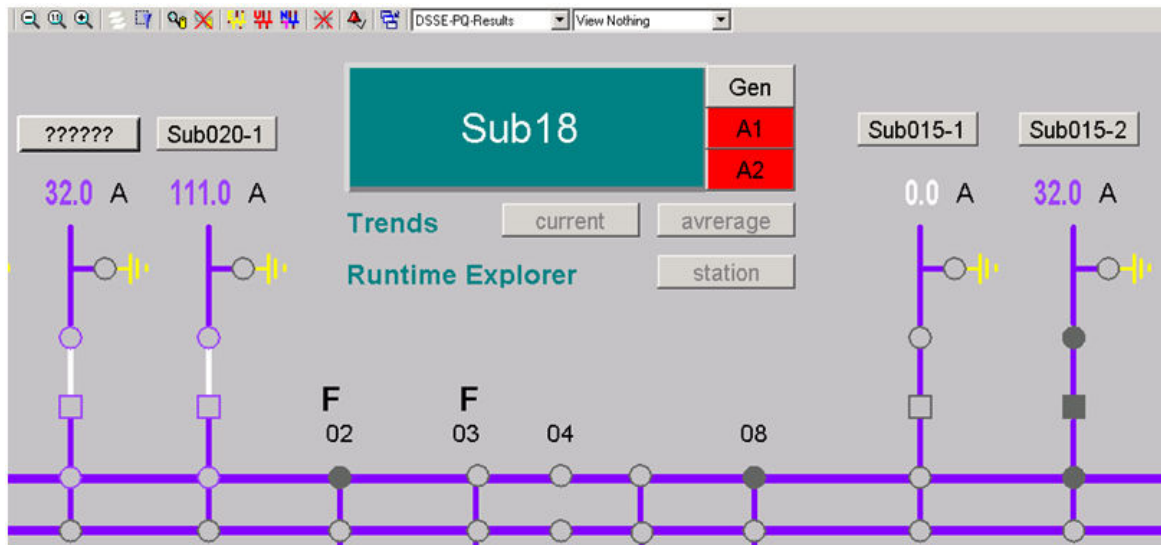
4.2.9 Calling Application Buttons

The goal of the topic is to call an application.

For example, Runtime explorer, single-line diagram, and word document.

Calling an Application Button

- Click an application button on the PAT or a single-line diagram
For example, **Sub19**, **current**, or **station**.
The pre-configured application opens.



[rguios01-050110-01.tif, 2, en_US]

Figure 4-6 Single-line Diagram with Application Buttons



NOTE

For example, you can invoke pre-configured MS Word or Excel documents, Runtime Explorers, **Trends and Charts** or single-line diagrams with application buttons.

4.2.10 Using the Display Search Result Dialog

The goal of the task is to select an object and to open the corresponding displays with the **Display Search Result** dialog.

Calling a Display or Calling the Display Search Result Dialog

- On the single-line diagram, the Runtime Explorer or a summary, select the desired object and right-click.
- Select **Display Callup** from the context menu.
The display opens immediately, if only 1 display is attached with the object or The **Display Search Result** dialog opens, if more than 1 display is attached with the object.
- Alternatively you can click the appropriate button on the PAT.
The **Display Search Result** dialog opens.

Using the Display Search Result Dialog

- Navigate through the (result) tree of the **Display Search Result** dialog, using the **+** and **-** characters to expand or to collapse the tree.
- Select the desired view.
- Double-click, or click the button **Show**, or use drag and drop.

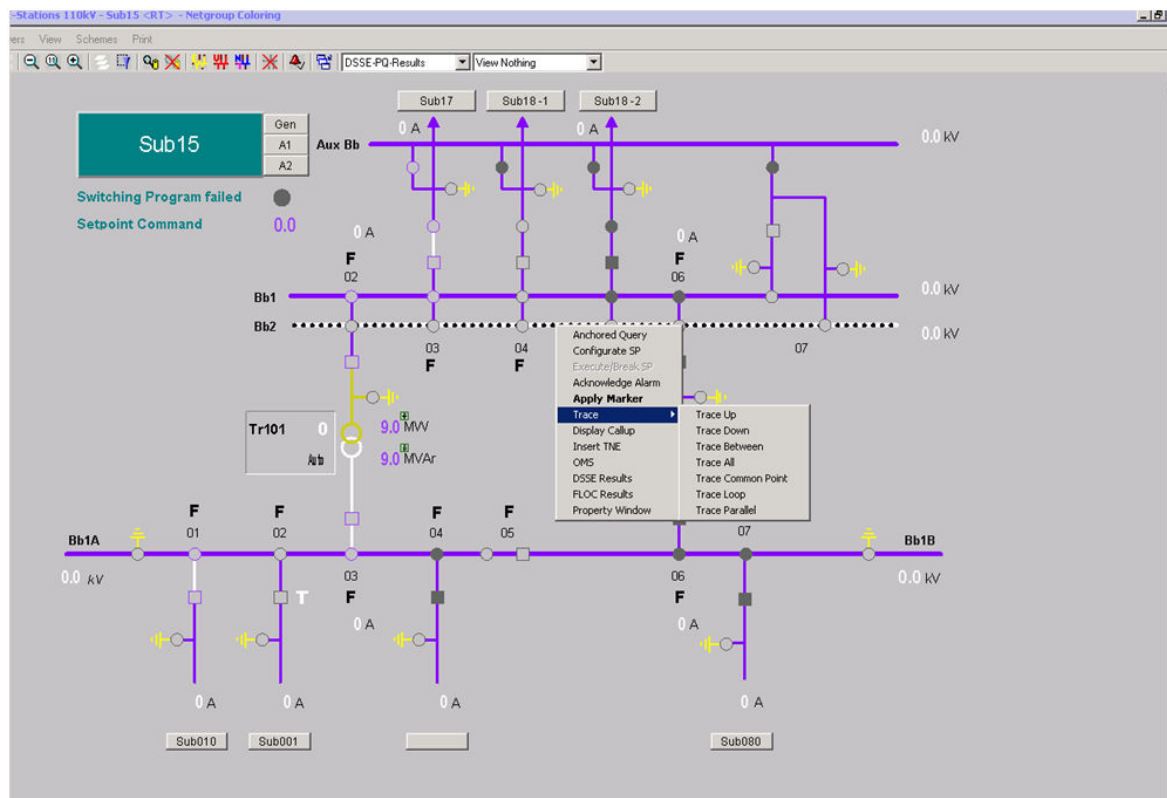
The selected view is displayed.

4.2.11 Performing Interactive Tracing

The goal of this task is to highlight special network situations with the tracing function.

Performing Interactive Tracing

- In the single-line diagram, right-click the object that serves as the initial point of the tracing function.



[rguios23-031209-01.tif, 2, en_US]

Figure 4-7 Invoking Interactive Tracing in the Single-line Diagram

- From the context menu, select **Trace**.
The **Trace** menu opens, displaying the following functions:
 - **Trace Up**
This function highlights the connections from the selected object upstream.
This trace is possible for objects in a DMS (Distribution Management System) network only.
 - **Trace Down**
This function highlights the connections from selected object downstream.
This trace is only possible for objects in a DMS network.
 - **Trace Between**
This function highlights the connections between 2 selected objects.
This trace can be invoked for objects in DMS networks as well as EMS (Energy Management System) networks.
 - **Trace All**
This function highlights all connections from/to selected object.
This trace can be invoked for objects in DMS networks as well as EMS networks.
 - **Trace Common Point**
This function highlights the connections from multiple selected objects towards the common infeed.
This trace is only possible for objects in a DMS network.
 - **Trace Loop**
This function highlights the connections from 1 side of selected object to the opposite side of the same object.
This trace can be invoked for objects in DMS networks as well as EMS networks.
 - **Trace Parallel**
This function highlights the connections from selected object towards multiple infeeds.
This trace can be invoked for objects in DMS networks as well as EMS networks.
- From the **Trace** menu, select the desired tracing function.

4.2.12 Using the Display Recall

The goal of this task is to display the previously opened single-line diagram.

Using the Display Recall

- Click the **Display Recall** button on the single-line diagram.

The previously opened single-line diagram opens in the same window.

4.2.13 Displaying Values of Network Applications

The goal of this task is to switch the display of the single-line diagram from SCADA values to values of the desired Transmission Network Application (TNA) function.



NOTE

If Transmission Network Applications (TNA) are available in the system, values of TNA functions can be displayed in a single-line diagram.
For example, **State Estimator** or **Power Flow**.

Displaying TNA Values in the Single-line Diagram

- To display a list of TNA functions, click the **TNA** button on the toolbar of the single-line diagram.



[rguios24-031209-01.tif, 2, en_US]

Figure 4-8 Singe -line Diagram with TNA Button

- Select the desired TNA function.
For example, **State Estimator**, **Security Dispatch** or **Voltage Scheduler**.

All values displayed in the single-line diagram are switched from the current source to the selected TNA function, except for those values that are defined as static values. The currently selected source function is displayed in the window title. In addition, a special symbol beside each value indicates the selected source function.

**NOTE**

The symbols indicating source functions can be determined with the Graphics Editor.

4.2.14 Handling the Embedded Tabular Display

The goal of the task is to handle an embedded tabular display of a single-line diagram.

**NOTE**

The embedded tabular display is used as additional visualizing display in single-line diagrams. It is configured in the **Graphics Editor** in the IMM. By default, the configured values are displayed. If a specific column is configured as a drag and drop target within the **IMM Graphics Editor**, the default object for this column can be temporarily replaced. The default value is used again when the worldmap containing the embedded tabular display is updated by the **IMM Graphics Editor**.

Displaying a Value in the Embedded Tabular Display

- In a single-line diagram with an embedded tabular display, select the desired measurement.
- Use drag and drop to display the measurement in the embedded tabular display (The embedded tabular display is configured to allow replacement of measurements).

Updating a Value in an Embedded Tabular Display Manually

- Right-click in a single cell of an embedded tabular display in a single-line diagram.
- Select **Edit Mode** from the context menu.
- Enter the desired value into the cell.
- Right-click again in the cell.
- Select **Save & Exit edit Mode**.

The new value is saved to the tabular display.

Exporting the Tabular Display to MS Excel

- Right-click in a tabular display of a single-line diagram.
- Select **Export to MS Excel** from the context menu.
MS Excel opens immediately and the whole table is exported to MS Excel.

- Save the file.

4.2.15 Handling the Embedded Trends And Charts Display

The goal of the task is to handle an embedded **Trends And Charts** display of a single-line diagram.



NOTE

The embedded **Trends And Charts** display is used as additional visualizing display in single-line diagrams. It is configured in the Graphics Editor in the IMM.

Displaying a Value in the Embedded Trends And Charts Display

- In a single-line diagram with an embedded **Trends And Charts** display, select the desired measurement.
- Use drag and drop to display the measurement in the embedded **Trends And Charts** display.

Using Additional Trends And Charts Functions

The normal **Trends And Charts** functions are also available for embedded **Trends And Charts** displays. For additional information on **Trends And Charts**, refer to Trends And Charts.

4.2.16 Printing the Single-line Diagram

The goal of this task is to print the visible part of a single-line diagram.



NOTE

This operation exemplifies the printing of the visible part of a single-line diagram. It is also possible to print the complete diagram.



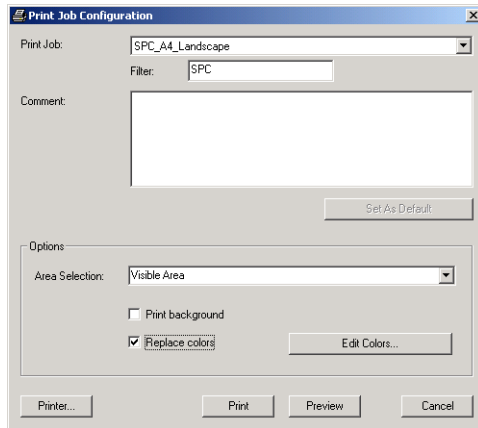
NOTE

This operation exemplifies the printing of a single-line diagram based on particular print settings. To print the diagram based on the default settings, click the **Print** icon on the toolbar. The diagram is printed immediately.

Initiating the Printing Process

- On the menu bar of the single-line diagram, select **Print...** from the **Print** menu.

The **Print Job Configuration** dialog opens.



[scawm007-170209-01.tif, 1, en_US]

Figure 4-9 Print Job Configuration Display

Configuring the Print Job

- On the dialog, select a **Print Job** from the list box.



NOTE

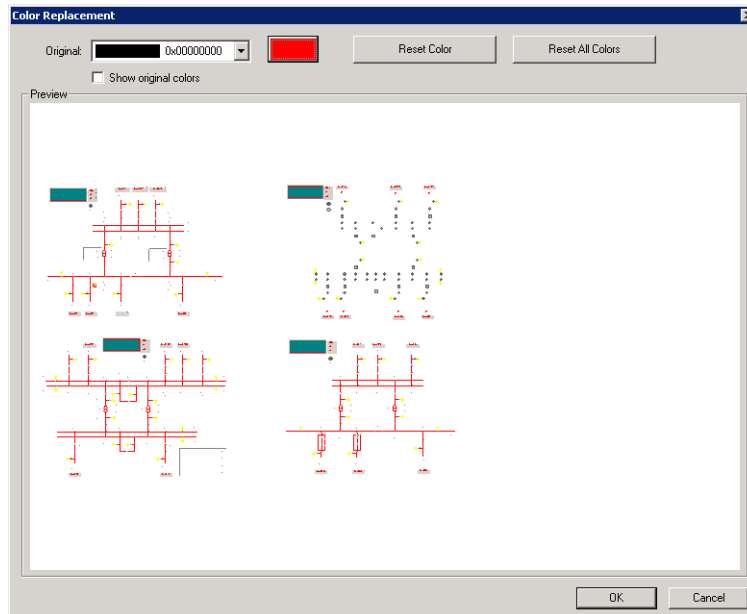
If no filter is defined, all available print jobs are displayed in the list. To filter the entries of the list, enter a string in the **Filter** text box. As a result, the list comprises only print jobs with names containing the entered filter string.

-
- From the list box **Area Selection**, select **Visible Area**.
 - To print the background of the single-line diagram, mark the **Print background** check box.

Editing Colors for the Printout

- Mark the **Replace Colors** check box.
- Click **Edit Colors....**

The **Color Replacement** display opens.



[scawm008-031207-01.tif, 1, en_US]

Figure 4-10 Color Replacement Display

- From the **Original** list box, select the color you want to replace.
- Click the color button to the right of the list box.
The **Color** dialog opens.
- Select the new color and click **OK**.
- To replace further colors, repeat the previous steps.
- To save the color replacement settings, click **OK**.

Selecting the Printer

- In the **Print Job Configuration** dialog, click **Printer**.
The **Print Setup** dialog opens.
- Select the printer from the **Name** list box.
- To specify the printer settings, click **Properties ...**.
- In the printer configuration dialog, change the printer settings as desired and click **OK**.
- In the **Print Setup** dialog, click **OK**.

Previewing and Printing

- In the **Print Job Configuration** dialog, click **Preview**.
The preview display opens.
- Click **Print**.



NOTE

Printing of Worldmaps is not available on TS Web Session.

4.2.17 Procedural Search

The goal of this task is to identify the exact faulted equipment from the list of possible faulted equipments that caused a fault in the topological network.



NOTE

Procedural search functionality can be started only after the occurrence of an earth fault and detection of faulted equipments by DNA-Fault Location (FLOC). This implies that procedural search can be used only if DNA-FLOC is running and configured correctly for earth fault detection.

Starting and Stopping Procedural Search

On occurrence of an earth fault, DNA FLOC populates the list of faulted equipments in Topological Processor Application Suite Repository (TP ASR).

- Click **Start Search** on the Power 5 Application Toolbar (PAT).
-



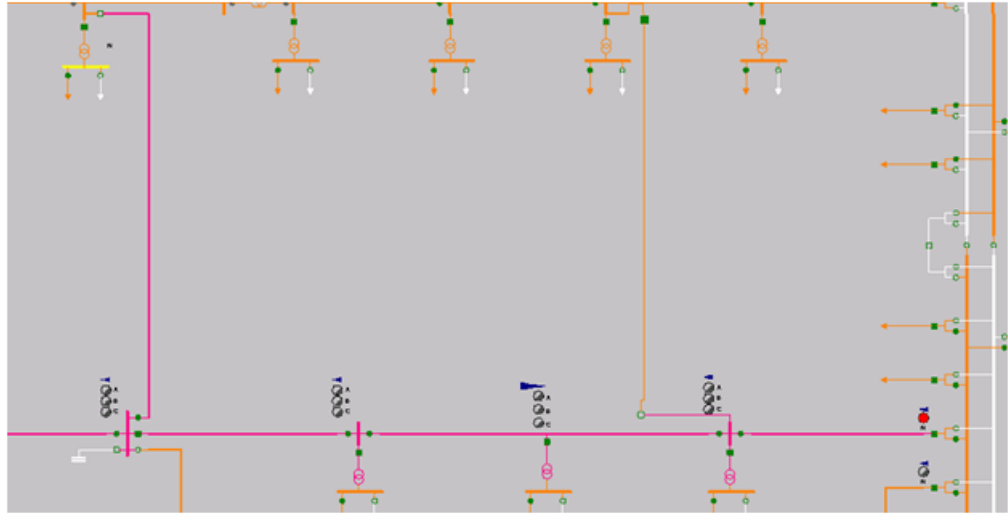
NOTE

The search is relevant only for RT and it is not for study or test mode.

- Perform switching actions on the single-line diagram to narrow down the fault and identify the exact faulted equipment.

The procedural search cases can be highlighted using the color changes in the network display. You can configure the colors in IMM under **/Presentation/Color Definitions/Color Palettes/Colors - Network State**

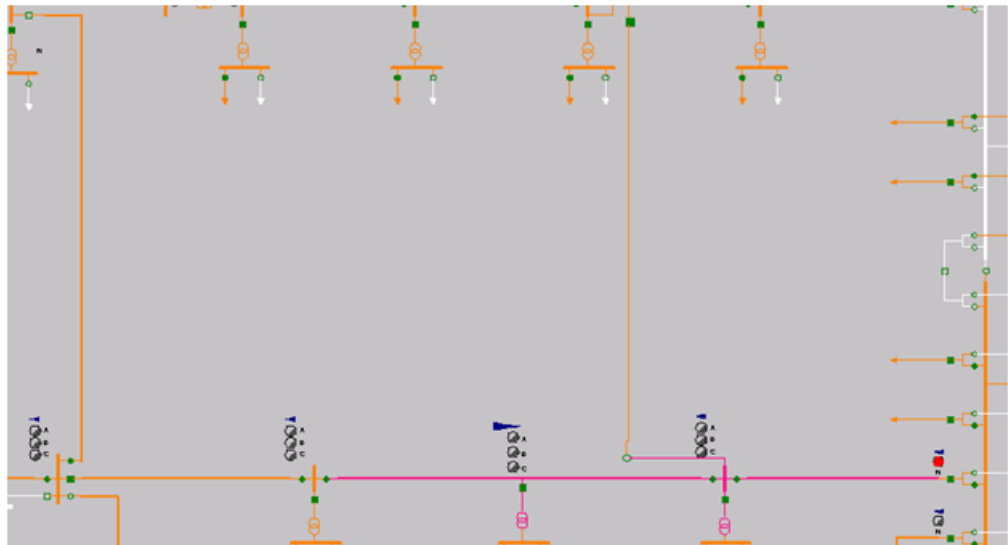
- Energized and fault candidate (normal FLOC color)



[sc_Energizedandfaulted, 1, en_US]

Figure 4-11 Energized and Fault Candidate

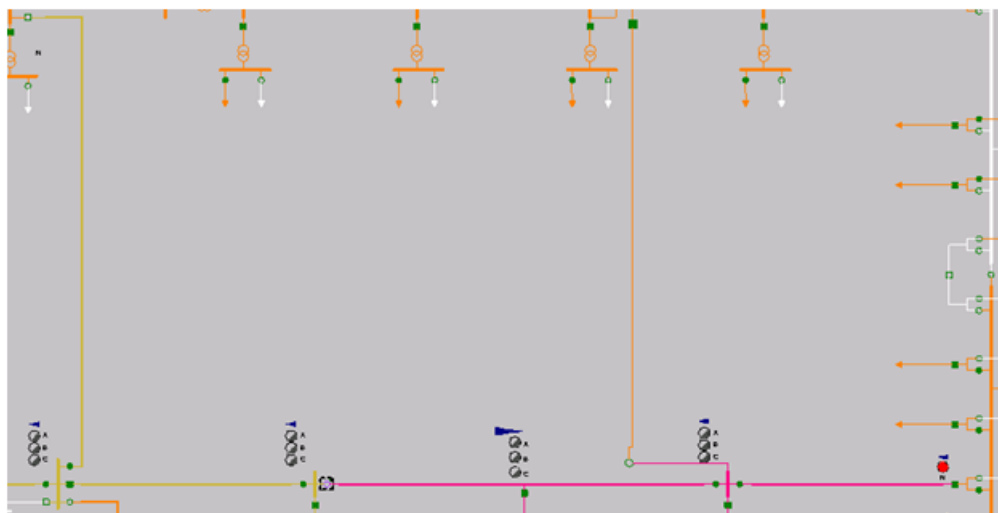
- Energized and not a fault candidate (normal energization color)



[sc_Energized and no fault, 1, en_US]

Figure 4-12 Energized and not a fault candidate

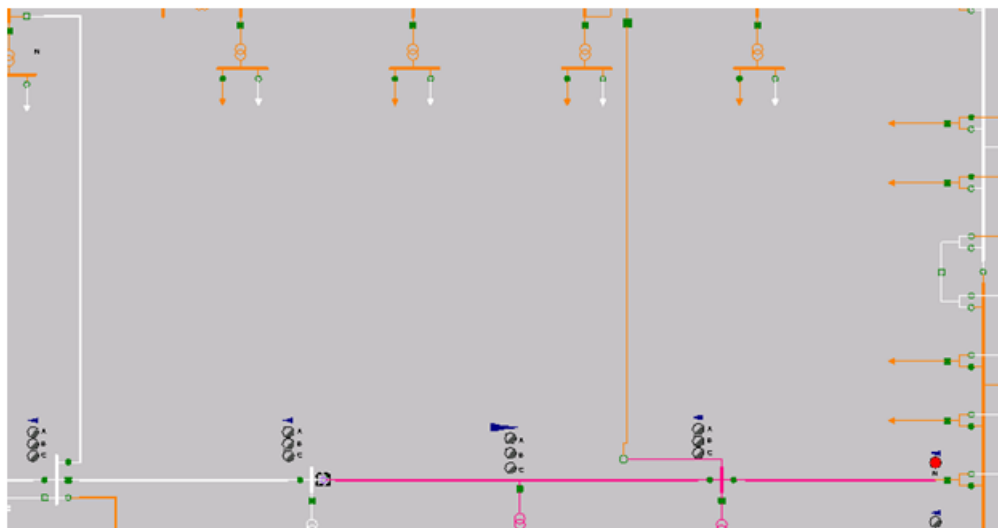
- De-energized and fault candidate (as configured in IMM)
User can configure the colors in IMM under **/Presentation/Color Definitions/Color Palettes/Colors - Network State/Deenergized_ProceduralSearch**.



[sc_De-energized and fault, 1, en_US]

Figure 4-13 De-energized and fault candidate

- De-energized and not a fault candidate (normal de-energization color)

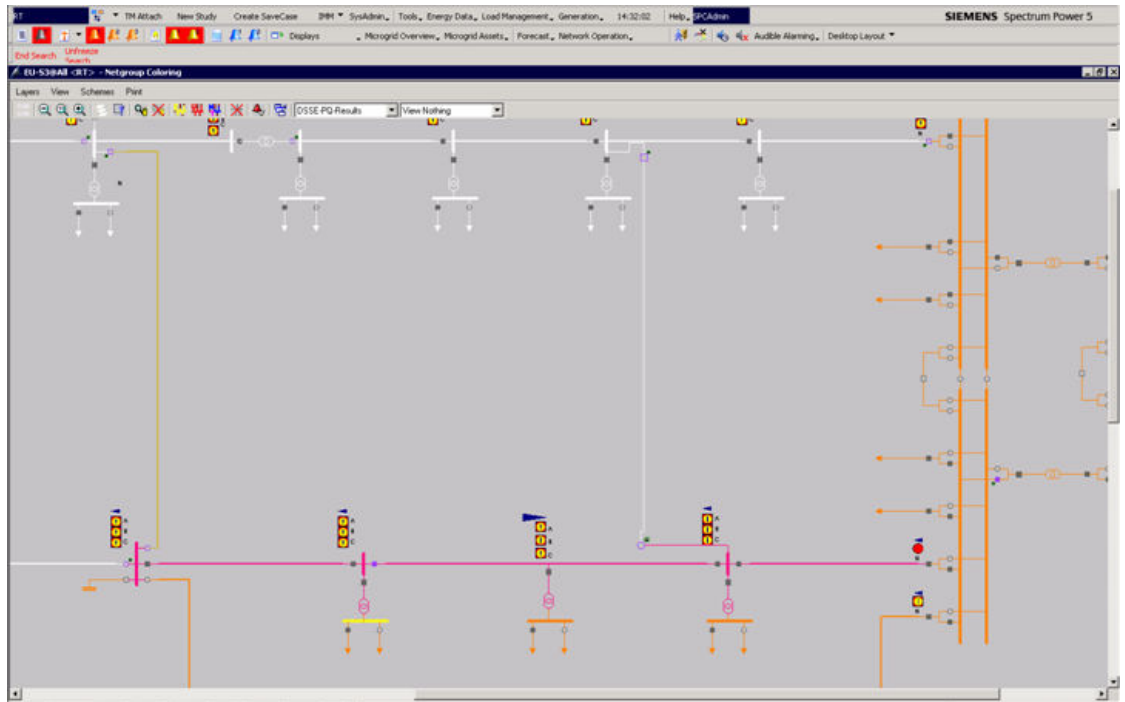


[sc_De-energized and no fault, 1, en_US]

Figure 4-14 De-energized and not a fault candidate

Procedural search either removes equipments (indicating that the equipments are either energized or not faulted) from the procedural search case or leaves the search case unchanged.

- Click **Freeze**.
You can freeze the search and candidate list is frozen.

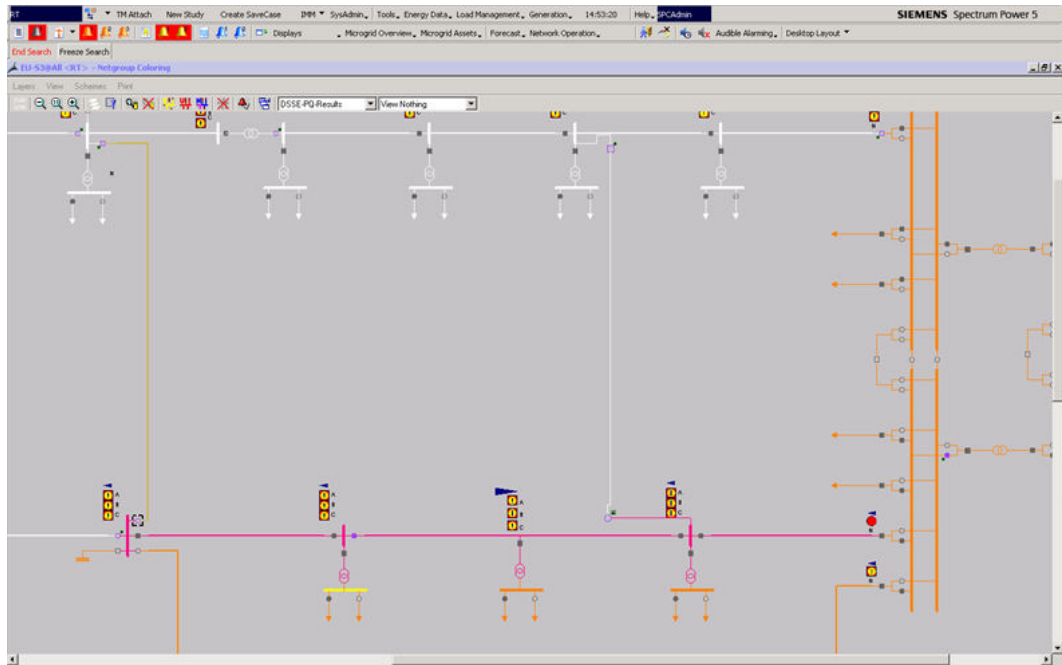


[sc_Freeze screen, 1, en_US]

Figure 4-15 Freeze Screen

You can either select Unfreeze or End Search option.

- Click **Unfreeze**.
 - If there is no earth fault, the equipments that are energized and non-faulted are removed from the candidate list.
 - If there is still an earth fault, then equipments that are de-energized are removed from the candidate list.



[sc_UnFreeze screen, 1, en_US]

Figure 4-16 Unfreeze Screen



NOTE

During shutdown or restart of Power 5 on RTS system, the frozen state is not persistent.



NOTE

During RTS switchover or failover if search is not frozen, search is automatically closed only if number of candidate list has reached zero. Remaining candidate items are recorded in general summary only during end procedural search operation

- Click **End Procedural Search** to stop the search.

Procedural search coloring is affected on all displays on all UIs. The information updated in the UI by a procedural search case remains the same even if the UI is restarted or display is opened from a different machine. The Procedural Search toggle button status also remains the same in all systems. A new search case can be created only after closing an existing one using End Procedural Search button. The procedural search fault items are recorded in the General Summary.

Path 3	Path 4	Path 5	Status	Value	Ind	Limit	Unit	User	Console	Comment
WindingH			App		Spont					Procedural Search Fault ter
WindingL			App		Spont					Procedural Search Fault ter
Procedural Earth Fault Search St	_PS Status		App	End	Spont					Procedural Search Fault ter
Procedural Earth Fault Search St	_PS Status		App	End	Alarm					Procedural Search Fault ter
Procedural Earth Fault Search St	_PS Status		App	Start	Spont					Procedural Search Fault ter
Procedural Earth Fault Search St	_PS Status		App	Start	Alarm					Procedural Search Fault ter
WindingH			App		Spont					Procedural Search Fault ter
WindingH			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
WindingL			App		Spont					Procedural Search Fault ter
WindingL			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
WindingH			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
WindingL			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
Procedural Earth Fault Search St	_PS Status		App	Frozen	Spont					Procedural Search Fault ter
Procedural Earth Fault Search St	_PS Status		App	Frozen	Alarm					Procedural Search Fault ter
WindingH			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
WindingL			App		Spont					Procedural Search Fault ter
WindingL			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter
			App		Spont					Procedural Search Fault ter

[sc_General summary, 1, en_US]

Figure 4-17 General Summary



NOTE

The start, end, freeze and unfreeze procedural search actions are entered as an event into the General Summary.

4.3 User Interface Description

4.3.1 Toolbar of the Single-line Diagram

Each single-line diagram provides a menu bar and a toolbar. The toolbar provides quick access to frequently used functions.

Icons of the Toolbar of a Single-line Diagram

	Button Display Recall: Clicking this button displays the previously opened single-line diagram.
	Application button Fault Location: Clicking this button opens the respective application.
	Application button Fault Isolation and Service Restoration: Clicking this button opens the respective application.
	Application button Distribution System Power Flow: Clicking this button opens the respective application.
	Button Energization Coloring: This coloring scheme displays the equipment according to its supply state.
	Button Netgroup Coloring: The netgroups (geographical, organized or operational area) are displayed in different colors.
	Button Voltage Level Coloring: Basically the same as the Energization Coloring scheme, but energized equipment is represented in a color indicating the respective voltage level.
	Zoom button: Clicking this button zooms the single-line diagram to the default zoom level.
	Zoom button: Zoom out
	Zoom button: Zoom In
	Button Navigator open: This buttons opens the navigator window.
	Button Navigator close: This buttons closes the navigator window.
	Button Reset Tracing: The function resets the highlighting of elements, for example, tracing or alarm coloring.
	Button Arrange/close window: The function opens a dialog with a list of all single-line diagrams that are currently open. Selecting an item of the list brings the respective single-line diagram to the front. If many single-line diagrams are open, you can click the icon, select all diagrams and open a context menu. Select the Close, Cascade, Tile vertically and Tile horizontally function to arrange the diagrams.
	Button Open Layer Dialog: The function opens a window with list of layers available. The layers can be selected and deselected.
	Button Filter Selection: The function opens a dialog to select a filter criteria. For example, Breaker, to select the breakers of the current single-line diagram.
	Button Acknowledge The function acknowledges all visible alarms of the current single-line diagram.

**NOTE**

The availability of the application buttons depends on the installed applications.

**NOTE**

In-Service status coloring and Merged/Split bus coloring are additional features, which must be configured with the start of the implementation of the system.

The In-Service status coloring displays all paths between energizers and loads.

The Topology Processor (TP) calculates the statuses of network equipment.

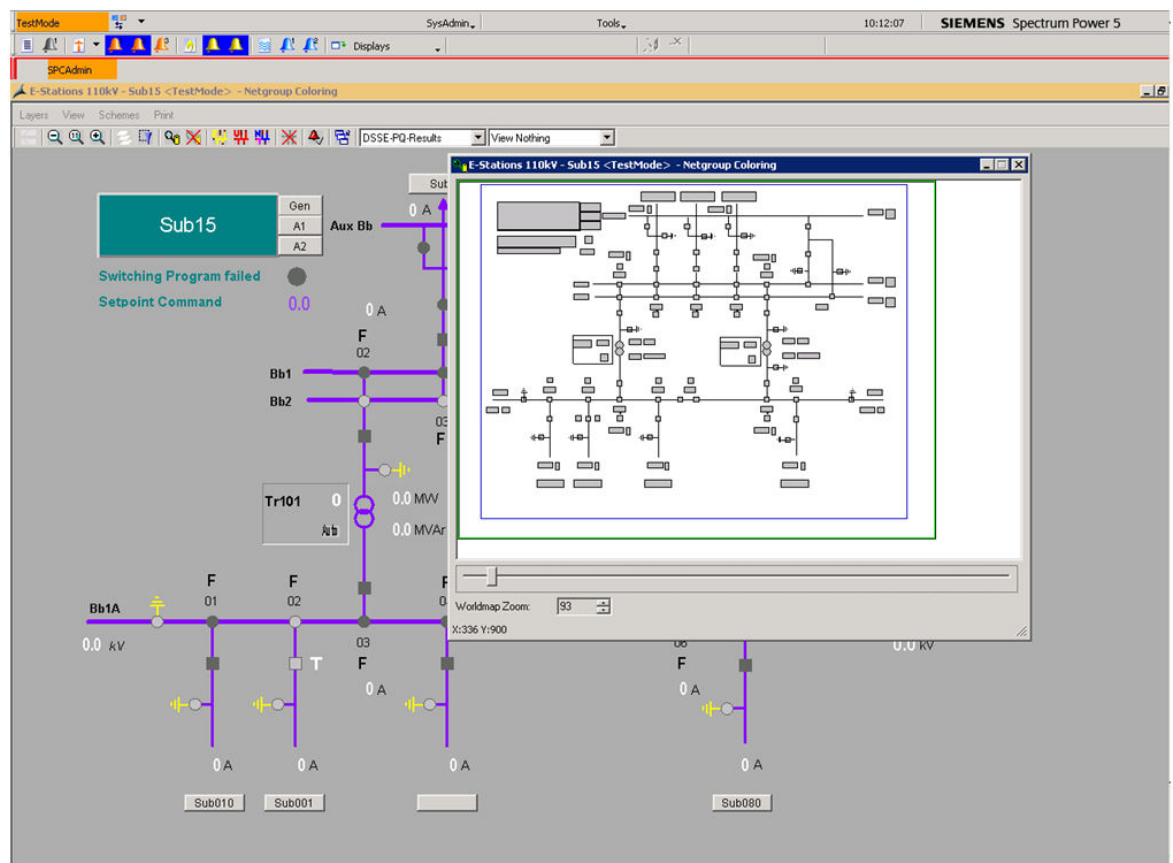
For example, open-end branch and split-bus.

The single-line diagrams are updated accordingly with predefined colors.

Both features are only operating with the high-voltage network and the **Energization Coloring** scheme.

4.3.2 Navigator Window

An optional navigation window can be used for panning and zooming within a large single-line diagram. The green rectangle of the navigation window represents the segment, displayed in the associated main window.



[rguios19-031209-01.tif, 2, en_US]

Figure 4-18 Single-line Diagram with Navigator Window in the Top Right Corner

4.3.3 Application Values in Single-line Diagrams

Application values can be shown next to the electrical equipment in single line diagrams, either in a graphical way or as numbers.

The following 5 types of application value displays are provided:

- Value table
- Pie
- Gauge
- Flow direction
- Anchored Query

Selection of Application Value Displays

You have the possibility to select the application values that are shown in the single line diagram via an application value profile and a visibility matrix, that can be selected from the toolbar.

Application Value Profile

You can select from one of several profiles in this drop-down box. An application value profile defines for each type of electrical equipment, which types of application values are shown in which way.

For example a **simple profile** might specify that the flow direction is shown on AC line segments.

Visibility Matrix

In addition you can select a visibility profile. This visibility profile defined which types of application value displays are visible.

For example the profile **value tables** indicates that only value tables are shown, but no pies, gauges, flows, anchored queries.

Application Value Displays

Value Tables

Value tables are used to show selected application values.

Pie

Pie chart is a way to represent data graphically. It is a circular chart in which the area shaded depends on the quantity of measurement. In a pie chart, quantity of measurement is displayed graphically by filling respective area of the chart and also by displaying percentage text.

Gauge

It is used to show various quantities, to indicate a value's position within a range. This control is created and placed automatically near to the Bus. All values in the Gauge Symbol are specified in percentage (%) values. For graphical representation, the respective percentage value is filled with a specified color.

Anchored Query

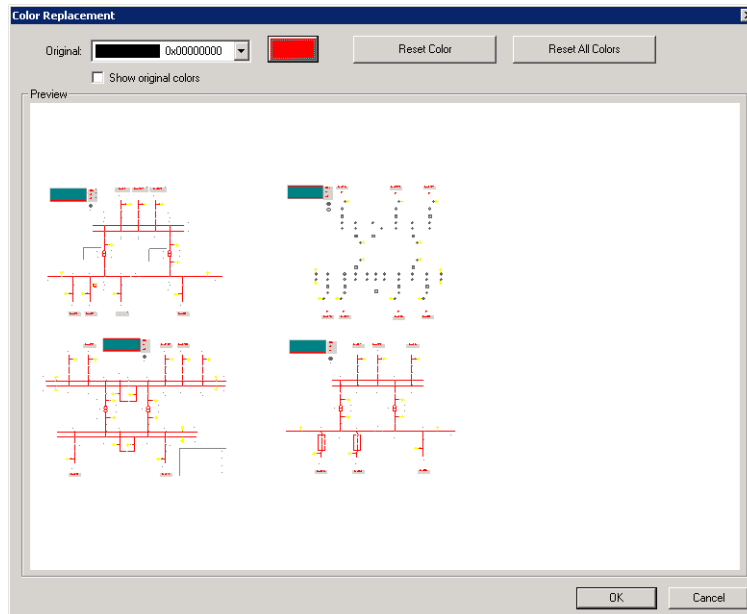
You can invoke **Anchored Query** from the context menu. An **Anchored Query** of equipment is like a property dialog that will show certain application values of the equipment.

Flow Arrows

Direction of power flow is shown in a network diagram or worldmap. Arrow symbols on lines are used to indicate power flow direction.

4.3.4 Color Replacement Dialog

The **Color Replacement** dialog enables you to change the color settings of the single-line diagram for the printout.



[scawm008-031207-01.tif, 1, en_US]

Figure 4-19 Color Replacement Dialog

List Box: Original

This list box displays all colors used in the single-line diagram.

Button for New Color Selection

The button to the right of the **Original** list box opens the **Color** dialog for selecting a new color.

Button: Reset Color

This function resets the definition of the currently selected color.

Button: Reset All Colors

This function resets the definition of all colors.

Check Box: Show Original Colors

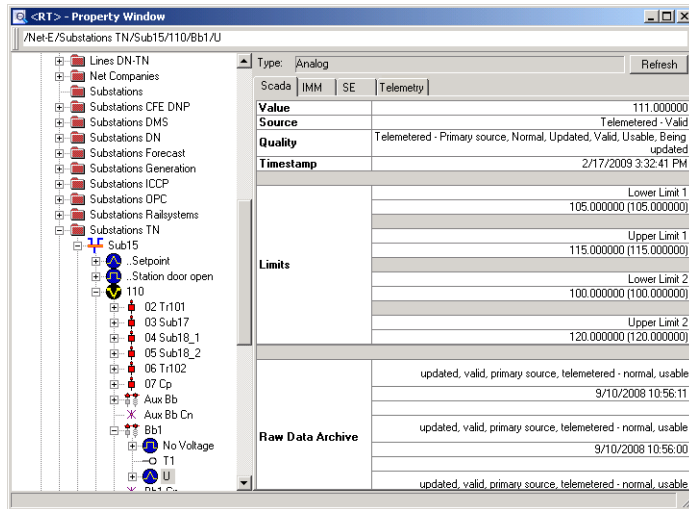
If this check box is marked, the **Preview** area of the **Color Replacement** dialog shows the single-line diagram in its original colors and not according to the chosen color settings.

Button: OK

This function saves the chosen color settings.

4.3.5 Property Window

The **Property Window** allows you an overview of the selected measurement or value, displaying dynamic and static information. The **Property Window** is a generic tool and the displayed data depends on the selected measurement or value.



[slidr001-200209-01.tif, 1, en_US]

Figure 4-20 Property Window Showing an Overview of the Selected Value

Text Box Right on the Top

The path of the selected measurement or value is displayed.

Text Box: Type

The type of the selected measurement or value is displayed.

Button: Refresh

Clicking this button refreshes the **Property Window**.

Tabs

- **SCADA:** This tab displays the Value, Quality, Timestamp of the selected value, Raw Data Archive etc.
- **Telemetry:** This tab displays the front-end configuration.
- **IMM:** This tab displays the IMM configuration.
- **SE:** This tab displays the value of the State Estimator.
- **Calculation:** This tab displays details of the used calculations.

**NOTE**

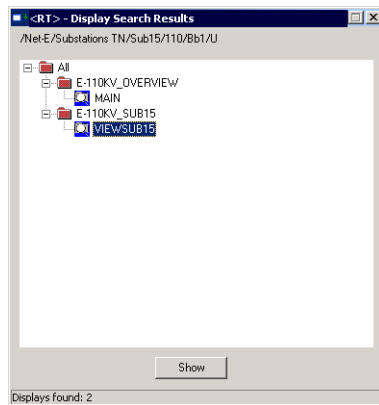
The **Raw Data Archive** shows the last 10 values of a blocked measurement.

**NOTE**

IMM data are only displayed when PSOS is running.

4.3.6 Display Search Results Dialog

The **Display Search Result** dialog provides all displays which are attached to a selected object in a single-line diagram, the Runtime Explorer or a summary.



[scafe021-200209-01.tif, 1, en_US]

Figure 4-21 Display Search Result Dialog

Tree View

Use the + and - characters to expand or collapse the tree and to navigate through the results.

Button: Show

After selecting a view, click this button to display the view.



NOTE

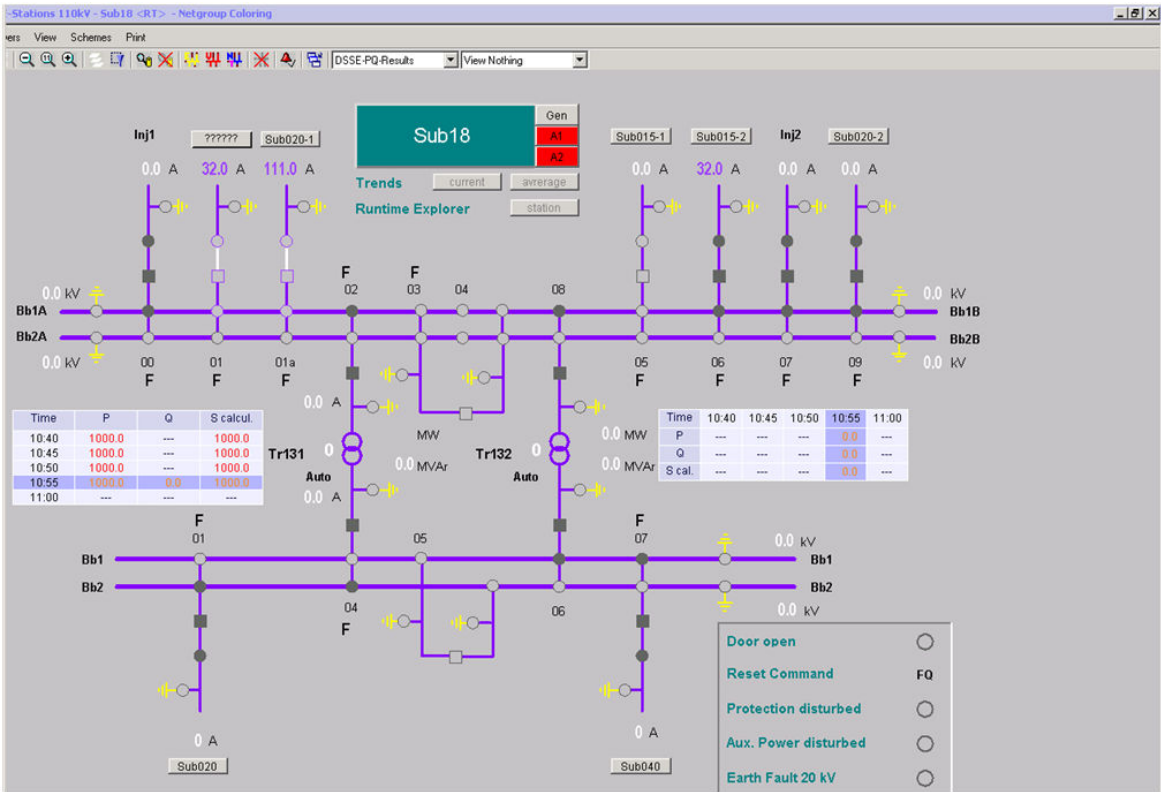
The **Display Callup** dialog provides all available displays, the **Display Search Result** dialog provides only the displays attached to the selected object.

4.3.7 Embedded Tabular and Embedded Trends and Charts Display

In the IMM Graphics Editor you can configure embedded tabular and embedded **Trends And Charts** as additional visualizing displays in single-line diagrams.

Embedded Tabular

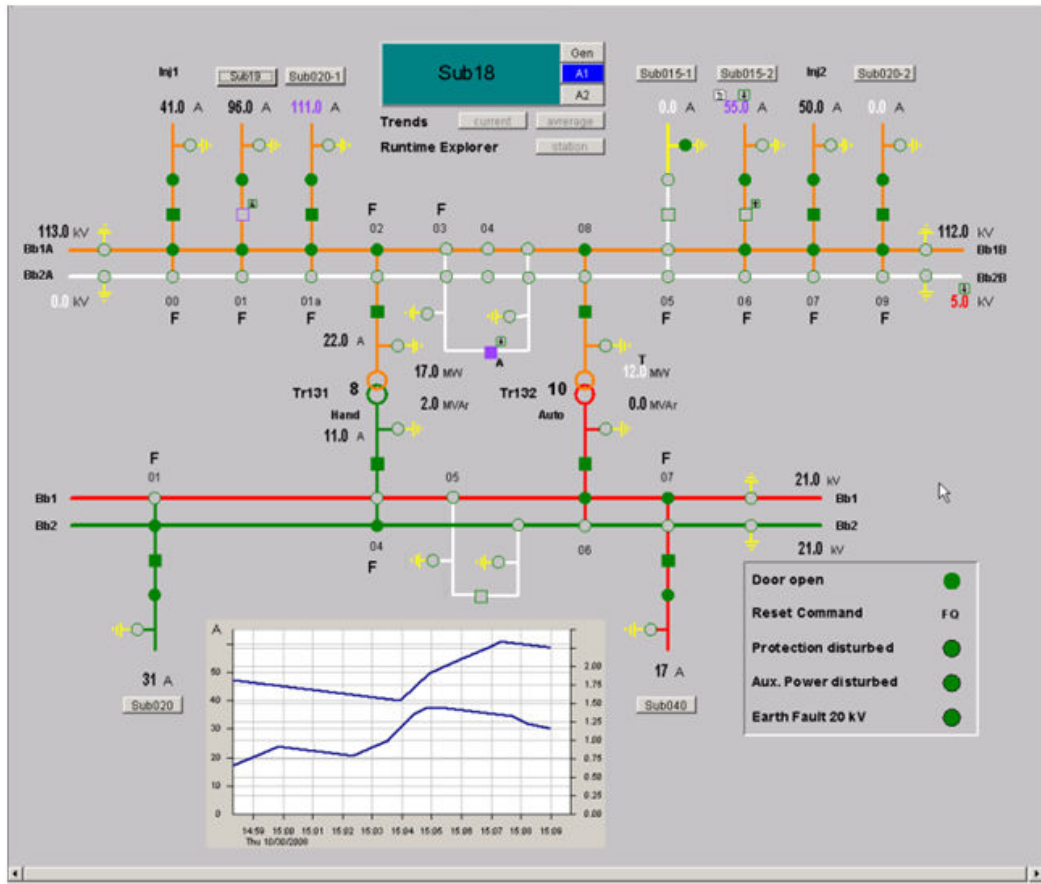
The tabular display is embedded in the single-line diagram.



[rguios25-031209-01.tif, 2, en_US]
Figure 4-22 Embedded Tabular Display

Embedded Trends And Charts

The **Curves and Trends** display is embedded in the single-line diagram.

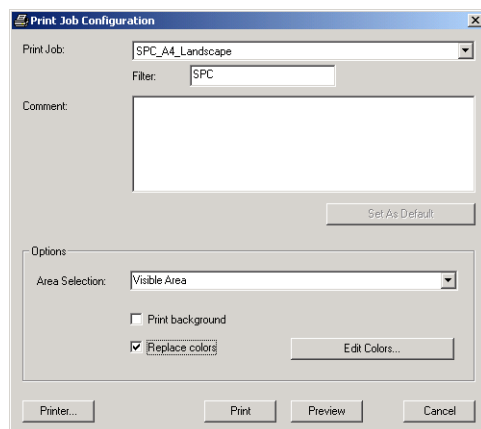


[rguios26-031209-01.tif, 2, en_US]

Figure 4-23 Embedded Trends And Charts Display

4.3.8 Printing Job Configuration Dialog

On the **Print Job Configuration** dialog, you can select a print job and modify the print settings.



[scawm007-170209-01.tif, 1, en_US]

Figure 4-24 Print Job Configuration Dialog

List Box: Print Job

This list box displays the currently selected print job. Clicking the arrow button at the right side opens a list of available print jobs.

Text Box: Filter

This text box allows you to filter for print jobs by using filter strings.

Button: Set as Default

Clicking this button stores the selected print job as the default job for further printouts.

Text Box: Comment

In this text box, you can enter a comment.

Options

Several options can be selected to determine the layout of the print output.

- **Area Selection**
The list box contains options for printing the visible area or the whole single-line diagram.
- **Print background**
Mark the check box to print with background.
- **Replace colors**
Mark the check box to replace printout colors of the single-line diagram.
The function **Edit Colors** opens the **Color Replacement** dialog (refer to [4.3.4 Color Replacement Dialog](#)).

Buttons

- **Printer**
This function opens a display for selecting the printer and printer options.
- **Preview**
This function opens the print preview display.
- **Print**
This function prints the single-line diagram according to the chosen print job and options.

5 Runtime Explorer

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5.1 Introduction

The Runtime Explorer is a generic tool showing all real-time data as configured in the Application Suite Repository (ASR). The data fields of the Runtime Explorer are updated during runtime, thus providing the same information as for the single-line diagrams.

Any alarm condition assigned to a network component, is shown in the Runtime Explorer in addition to the **Alarm Summary** entry. You can acknowledge an alarm condition either in the Runtime Explorer, in the single-line diagram, or in the corresponding **Alarm Summary**.

The qualities shown in the runtime explorer views are a subset of the qualities used by Spectrum Power 5. The most important data are shown to the user. While using the filter for qualities of Runtime Explorer, you can select atomic qualities. The filtering will take into account the atomic qualities for the filtering, but still will show the subset of readable qualities.

For example, telemetered updated is shown in the runtime explorer view. Valid is selected as atomic quality in the filter, the entry is still shown, because the information is valid, even the text for valid is not shown.

Main Features

- All open tables are automatically updated with current network component and alarm status.
- Predefined and ad hoc filters focus the view of the user on the current working area. Thus, a table of all manually updated values or currently set markers can be easily obtained.
- No configuration work of the Runtime Explorer is necessary. It is a fully generic tool, automatically defined by the data model.

5.2 Operational Steps

5.2.1 Calling the Runtime Explorer

The goal of this task is to start the Runtime Explorer.

Calling the Runtime Explorer

The following methods are used to start the Runtime Explorer:

- Select **Tools** and use drag and drop operation on the PAT.
- Select **Tools > Runtime Explorer** on PAT.
- Click the appropriate application button on a single-line diagram or the PAT.

For additional information, refer to [4.2.9 Calling Application Buttons](#).

The Runtime Explorer opens.



NOTE

To change the view of the Runtime Explorer, click the appropriate button on the toolbar.
For example, **Limit View** and **Update View**.

5.2.2 Performing Operations on Measurements

The goal of this task is to perform operations on measurements displayed in the Runtime Explorer.

For example, a context menu (configured in the IMM) provides the possible operations:

Manually Updating Values (Manual Update/Manual Update Toggle)

For additional information on how to update values, refer to [11.2.1 Manually Updating Values](#).

Performing Supervisory Control (Control State)

For additional information on how to perform supervisory control, refer to [9.2.2 Controlling the State of a Switch](#)

Performing Manual Override

For additional information on how to perform manual override, refer to [11.2.2 Manually Overriding Values](#).

Applying a Marker

For additional information on how to apply a marker, refer to [10.2.1 Setting a Marker with Reminder Function](#).

Acknowledging Alarms

For additional information on how to acknowledge an alarm, refer to [7.2.1 Handling an Incoming Alarm](#).

Changing the Priority of a Value (Manual Value Selection Adjustment)

For additional information on how to change the priority of a value, refer to [5.2.4 Manually Adjusting the Priority of a Value](#).

Calling the Property Window

For additional information on how to call the **Property Window**, refer to [4.2.8 Calling the Property Window](#).

Calling the Display Search Results Dialog (Display Callup)

For additional information on how to call a display, refer to [4.2.10 Using the Display Search Result Dialog](#).

5.2.3 Performing Operations on Field Resources

The goal of this task is to perform operations on a field resource displayed in the tree view of the Runtime Explorer.

For example, a context menu (configured in the IMM) provides the following operations:

Manually Updating Values (Manual Update/Manual Update Toggle)

For additional information on how to update values, refer to [11.2.1 Manually Updating Values](#)

Performing Supervisory Control (Control State)

For additional information on how to perform supervisory control, refer to [9.2.2 Controlling the State of a Switch](#).

Acknowledging Alarms

For additional information on how to acknowledge an alarm using the affected field resource, refer to [7.2.1 Handling an Incoming Alarm](#).

Applying a Marker

For additional information on how to apply a marker on a field resource, refer to [10.2.1 Setting a Marker with Reminder Function](#).

Performing Interactive Tracing (Trace)

For additional information on how to perform interactive tracing, refer to [4.2.11 Performing Interactive Tracing](#).

Performing Manual Override

For additional information on how to perform manual override, refer to [11.2.2 Manually Overriding Values](#)

Calling the Property Window

For additional information on how to call the **Property Window**, refer to [4.2.8 Calling the Property Window](#)

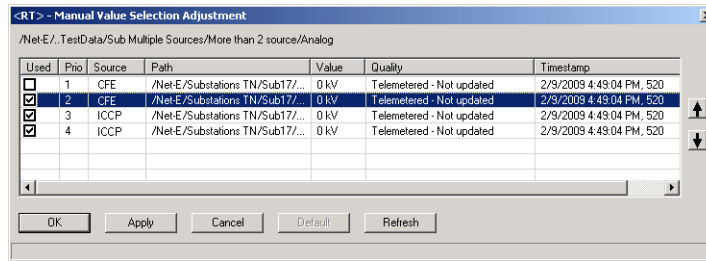
5.2.4 Manually Adjusting the Priority of a Value

The goal of the task is to adjust the priority of a value manually, to influence the value selection process of a measurement.

Manually Adjusting the Priority of a Value

- Select the desired measurement on the Runtime Explorer or the Single-line diagram.
- Right-click the desired measurement.
- Select **Multiple Source Selection** from the context menu.

The **Manual Value Selection Adjustment** dialog opens.



[scare022-200209-01.tif, 1, en_US]

Figure 5-1 Manual Value Selection Adjustment Dialog

- Click the desired value.
The row turns blue and the arrow buttons on the right turn bold.
- Click the arrow buttons to the right to increase or decrease the priority of the value.
- Click **OK**.



NOTE

If the check box of a value in the column **Used** is disabled, the value is not considered in the hierarchy.

For additional information on source value selection, refer to Technical Specification SCADA.

5.2.5 Working with Filters

5.2.5.1 Creating a Filter

The goal of this task is to define filters to restrict the information displayed in the **Runtime Explorer**. The 5 different categories of filter criteria are available:

- **Name**
- **Type**
- **Value**
- **Source/Quality**
- **Timestamp**

When defining the filter, you can specify only 1 filter criteria or combine several criteria.



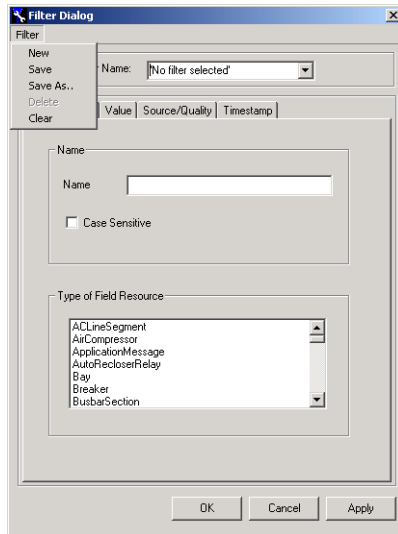
NOTE

The filter is applied on the **Explorer View**, the **Substation View**, the **Update View**, the **Limit View**, and the **Marker View**.

Creating the Filter

- Click the **Configure Filter** icon on the toolbar of the Runtime Explorer.

The **Filter** dialog opens.



[sca023-170209-01.tif, 1, en_US]

Figure 5-2 Filter Dialog

- Select **Filter > New** from the **Filter** dialog.

The dialog displays **New Filter** in the list box.

Defining a Name and Field Resource Filter

- Enter a string in the **Name** text box in the **Name** tab.
- Click **Apply**.

The string is used as a **Name** filter.



NOTE

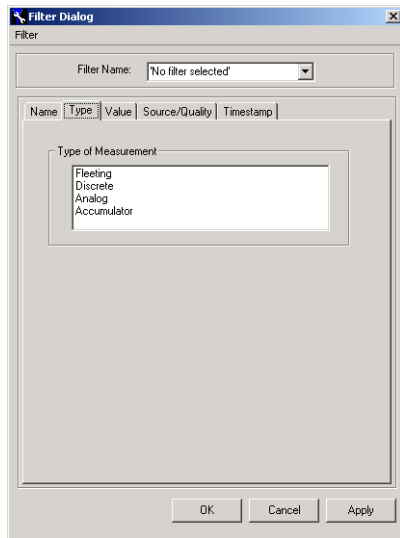
Enable the check box **Case Sensitive** to handle the string in the **Name** text box as case sensitive.

-
- Select the filter criteria from the **Type of Field Resource** list box.
 - Click **Apply**.

The filter criteria is used as a **Field Resource** filter.

Defining a Measurement Filter

- Click the **Type** tab.



[scares024-170209-01.tif, 1, en_US]

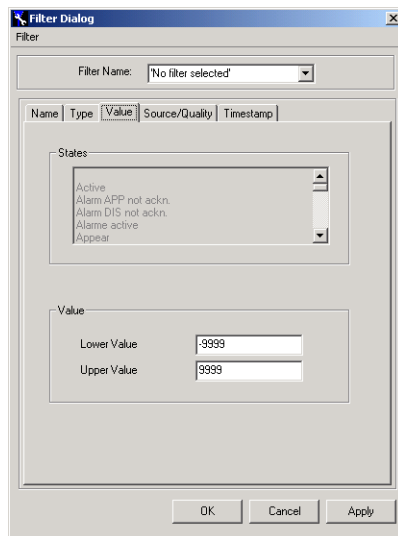
Figure 5-3 Measurement Filter

- Select 1 or more criteria from the **Type of Measurement** list box.
- Click **Apply**.

The filter criteria is used as a **Measurement** filter.

Defining a Value Filter

- Click the **Value** tab.



[scares025-170209-01.tif, 1, en_US]

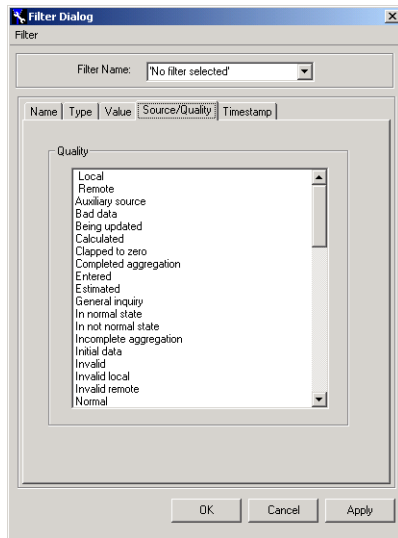
Figure 5-4 Value Filter

- Enter the **Lower Value** and **Upper Value** limits.
- Select a state from the **States** list box.
- Click **Apply**.

The measurements are filtered according to the set filter criteria.

Defining a Source/Quality Filter

- Click the **Source/Quality** tab.



[scare026-170209-01.tif, 1, en_US]

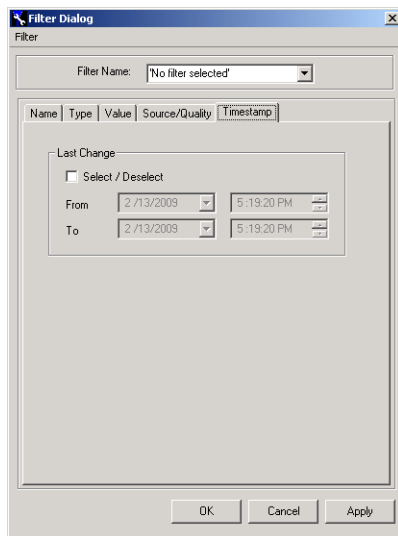
Figure 5-5 Source/Quality Filter

- Select **Source** or **Quality** from the corresponding list box.
- Click **Apply**.

The measurements are filtered according to the set filter criteria.

Defining a Timestamp Filter

- Click the **Timestamp** tab.



[scare027-170209-01.tif, 1, en_US]

Figure 5-6 Timestamp Filter

- Enable or disable the **Select/Deselect** check box.
- Define the time stamp of the last change with the corresponding list boxes **From** and **To**.
- Click **Apply**.

The measurements are filtered according to the set filter criteria.

Saving the New Filter

- Select **Filter > Save As ..** from the **Filter** dialog.
The **Name** dialog opens.
- Enter the filter name in the **Name** text box.
- Click **OK**.

The new filter is saved.

Deleting a Filter

- Select a filter in the filter list box.
- Select **Filter > Delete** from the **Filter** dialog.
- Confirm the dialog.

The selected filter is deleted.

5.2.5.2 Editing a Filter

The goal of this task is to edit the criteria of an existing filter.

Calling a Filter

- Click the **Configure Filter** icon on the toolbar of the Runtime Explorer.
The **Filter** dialog opens.
- Select the filter you want to edit from the list box.

Changing the Name and Field Resource Filter

- Change the string in the **Name** text box of the **Name** tab.
- Mark the check box **Case Sensitive** to handle the string in the **Name** text box as case sensitive.
- Change the field resource in the **Type of Field Resource** list box.
- Click **Apply**.

Changing the Value Filter

- Click **Value** tab.
- Change the selection in the **States** list box.
- Change the **Upper Value** and the **Lower Value**.
- Click **Apply**.

Changing Source/Quality Filter

- Click **Source/Quality** tab.
- Change the selected filter criteria.
- Click **Apply**.

Changing the Timestamp Filter

- Click **Timestamp** tab.
- Change the time stamp filter criteria.
- Click **Apply**.

Saving the Changes

- Select **Filter > Save As ..** from the menu bar of the **Filter** dialog.
The **Name** dialog opens.
- If desired, change the filter name in the **Name** text box.
- Click **OK**.

The new filter is saved.

Overwrite the Current Filter

- Select **Filter > Save** from the menu bar of the **Filter** dialog.
The currently selected filter is overwritten with the new selected values.



NOTE

The default filter cannot be overwritten.

Setting the Filter Values to Default

- Select **Filter > Clear** from the menu bar of the **Filter** dialog.
The values of the currently selected filter are set to default.

5.2.5.3 Switching a Filter On/Off

The goal of the task is to activate or to deactivate the currently selected filter.



NOTE

The tooltip of the **Filter ON/OFF** button indicates the current switch position.
If the filter is switched on, the current filter is displayed in the status bar of the Runtime Explorer.

Switching a Filter On/Off

- Click the appropriate toolbar button on the Runtime Explorer to switch the currently selected filter on or off.

The filter criteria are applied on the displayed data.

5.2.5.4 Deleting a Filter

The goal of this task is to delete an existing filter.

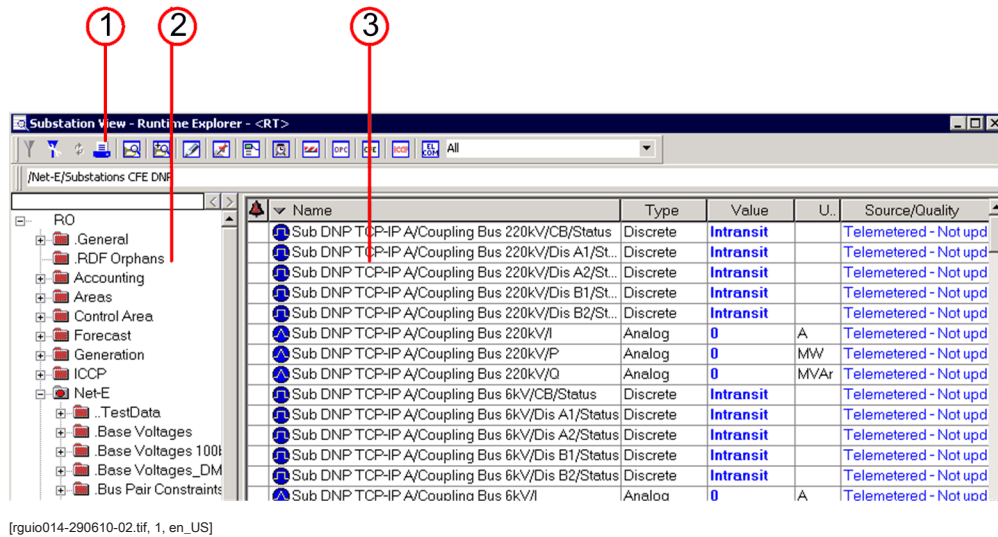
Deleting a Filter

- Select a filter in the list box of the **Filter** dialog.
- Click **Filter > Delete** on the **Filter** dialog.
- Confirm with **OK**.

The selected filter is deleted.

5.3 User Interface Description

5.3.1 Runtime Explorer



[rguio014-290610-02.tif, 1, en_US]

Figure 5-7 Substation View of the Runtime Explorer

- (1) Toolbar
- (2) Tree view
- (3) Content area

Predefined and ad hoc data shown in the tree view and the content area depend on the selected view. For additional information about the available views, refer to the subsequent sections.



NOTE

An additional tooltip occurs, if the mouse pointer is next to a column in the content area and the column width is not completely enlarged. The yellow frame indicates the use of a filter.

Toolbar

The toolbar provides the following buttons:

- Ad hoc filter on/off
- Filter configuration dialog
- Explorer View
- Substation View (default view)
- Update View
- CFE View
- ICCP View
- Marker View
- Multiple Source View



Limit View



Refresh Button (only active after IMM activation)



Print Button

- Limit Dialog button to open the **Limit Set** dialog (only visible when the **Limit View** has been selected).

The toolbar provides the following list boxes (used as predefined filters):

- Marker selection
Filters the marker view content (only visible when the **Marker View** is selected).
- Limit selection
Filters for certain limits (only visible when the **Limit View** is selected).
- Multiple source selection
Filters for certain sources (only visible when **Multiple Source View** is selected).

The toolbar provides the following text boxes:

- Find Box
On entering the search string (with wildcards) the complete tree view is retrieved. You can navigate through the results in the tree view with the arrow buttons right to the Find Box.
- The text box beneath the toolbar buttons, shows the actual path in the tree view. It is possible to change the path manually.



NOTE

The content of the toolbar (available views) and the view, displayed when starting the Runtime Explorer, can be configured in the IMM.

Tree View

The left part of the Runtime Explorer shows the tree view. The tree view comprises all objects (instances). To expand or collapse the tree view (or parts thereof), click the **+** or **-** characters in the hierarchy.

Content Area

The right part of the Runtime Explorer shows the measurements and process data of the selected objects in a tabular form with multiple columns.

To sort the table, click in the header of the column that functions as the sort key. After the first click, an arrow is displayed in the header of the column, showing the sort direction. The status bar beneath the content area shows the selected filter, the number of displayed stations and date and time.

5.3.2 Explorer View

The **Explorer View** of the Runtime Explorer lists the domain objects and corresponding measurement objects exactly 1 level below the selected tree object.

The benefit of the **Explorer View** is the validation of the correct data model structure, for example, after a data error is logged.

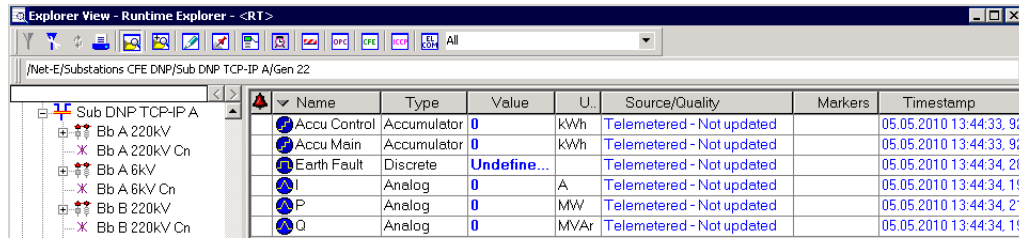


Figure 5-8 shows the Explorer View of the Runtime Explorer. The left pane displays a tree structure with 'Sub DNP TCP-IP A' selected. The right pane shows a table with the following data:

Name	Type	Value	U.	Source/Quality	Markers	Timestamp
Accu Control	Accumulator	0	kWh	Telemetered - Not updated		05.05.2010 13:44:33, 92
Accu Main	Accumulator	0	kWh	Telemetered - Not updated		05.05.2010 13:44:33, 92
Earth Fault	Discrete	Undefined...		Telemetered - Not updated		05.05.2010 13:44:34, 28
I	Analog	0	A	Telemetered - Not updated		05.05.2010 13:44:34, 19
P	Analog	0	MW	Telemetered - Not updated		05.05.2010 13:44:34, 21
Q	Analog	0	MVA	Telemetered - Not updated		05.05.2010 13:44:34, 19

[rguio008-290610-02.tif, 1, en_US]

Figure 5-8 Explorer View of the Runtime Explorer

Content Area

The content area displays all underlying objects that are assigned to the selected object in the object tree.

5.3.3 Substation View

The **Substation View** allows you to find all measurements that are below a selected item in the object tree. It recursively searches all objects and displays the measurements associated with a selected object. This view provides you all measurements of a specific substation, displayed in 1 table. Although this view displays only measurement objects, they are from any level in the object tree hierarchy.

The benefit of the **Substation View** is to get the same process information (without topology) as usually provided in single-line diagrams, however with no construction effort.

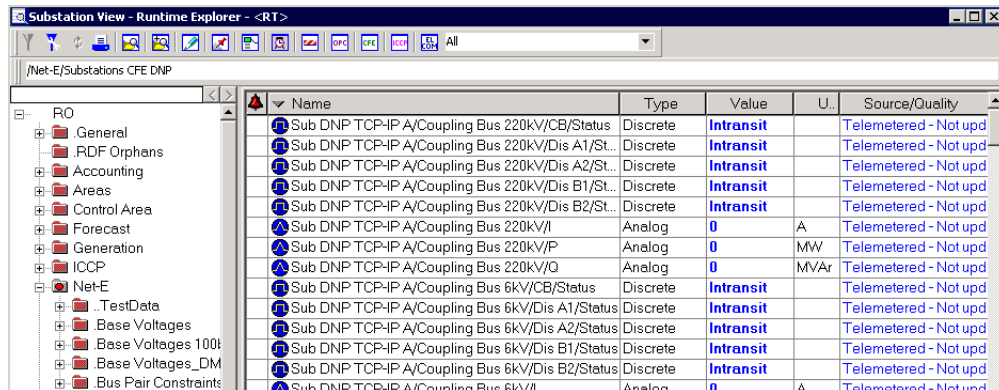


Figure 5-9 shows the Substation View of the Runtime Explorer. The left pane displays a tree structure with 'RO' selected. The right pane shows a table with the following data:

Name	Type	Value	U.	Source/Quality
Sub DNP TCP-IP A/Coupling Bus 220kV/CB/Status	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/Dis A1/St...	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/Dis A2/St...	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/Dis B1/St...	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/Dis B2/St...	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/I	Analog	0	A	Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/P	Analog	0	MW	Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 220kV/Q	Analog	0	MVA	Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 6kV/CB/Status	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 6kV/Dis A1/Status	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 6kV/Dis A2/Status	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 6kV/Dis B1/Status	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 6kV/Dis B2/Status	Discrete	Intransit		Telemetered - Not upd
Sub DNP TCP-IP A/Coupling Bus 6kV/I	Analog	0	A	Telemetered - Not upd

[rguio007-290610-03.tif, 1, en_US]

Figure 5-9 Substation View of the Runtime Explorer

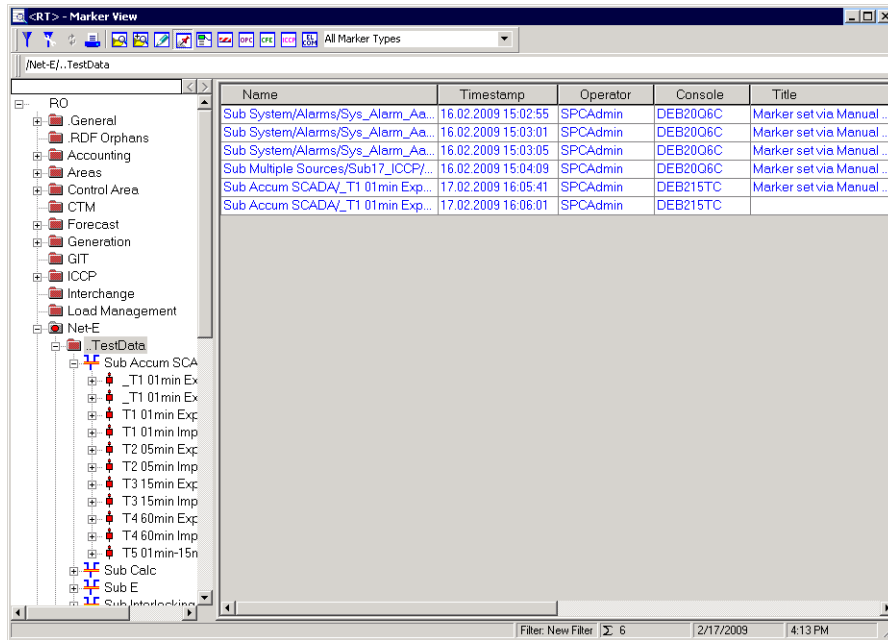
Content Area

The content area shows current states and values of all underlying objects that are assigned to the selected object in the object tree.

5.3.4 Marker View

Markers and tags are used to identify field devices (switches, lines, data points, protection devices) for special processing. Tags are markers used to identify points with control restrictions.

The benefit of the **Marker View** is to get a fast and complete overview in 1 table of the markers that are currently set in the whole network.



[scare017-200209-01.tif, 1, en_US]

Figure 5-10 Marker View of the Runtime Explorer

Content Area

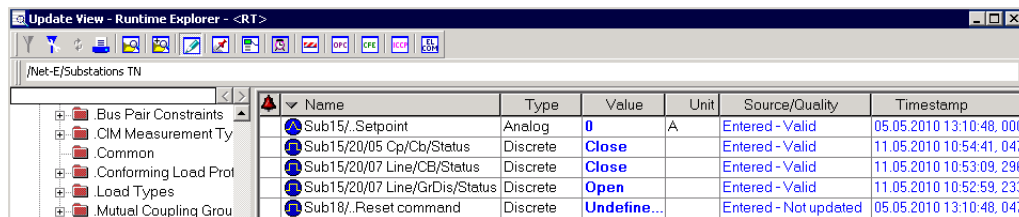
The content area shows all currently set markers of all objects that are assigned to the selected object in the object tree.

List Box for Filter Selection

The list box on the toolbar enables you to filter the displayed markers. Only markers of the selected marker type are displayed.

5.3.5 Update View

The **Update View** provides an overview of all manual updates in the system in a single display. The quality code of the manually updated values is entered and not updated as displayed in the source quality column.



[rguio002-280610-02.tif, 1, en_US]

Figure 5-11 Update View of the Runtime Explorer

Content Area

The content area shows all current manual updates of all objects that are assigned to the selected object in the object tree.

For example, a substation.

5.3.6 CFE View

The **CFE View** shows CFE status information for CFE interfaces and CFE links.

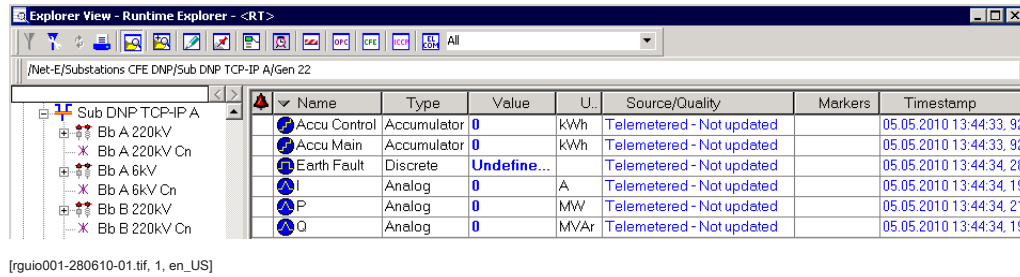


Figure 5-12 CFE View of the Runtime Explorer

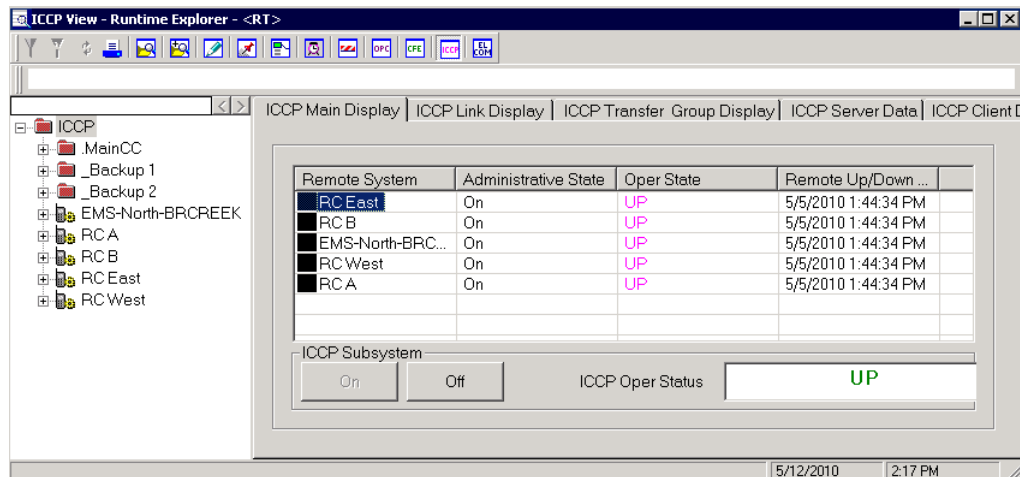
Content Area

The content area comprises the following tabs:

- **Single**
Shows the CFE interfaces.
- **Summary**
Shows the CFE interfaces and all interface links.
- **Failures**
Shows the CFE interfaces and interface links with the **Info** status **Failure**.

5.3.7 ICCP View

The **ICCP View** provides an overview of the administrative and operating state of the ICCP subsystem, ICCP links, transfer groups, server and client data.



[rguio003-280610-02.tif, 1, en_US]

Figure 5-13 ICCP View of the Runtime Explorer

Content Area

The content area comprises the following tabs:

- **ICCP Main Display**
Shows the overall administrative and operating state of the ICCP subsystem and of each ICCP remote. It provides you with the capability to enable/disable ICCP and to connect/disconnect with the listed remote systems.

- **ICCP Link Display**
Shows the operating state of each ICCP link with the remote system.
- **ICCP Transfer Group Display**
Shows the administrative and operating state of transfer groups for the selected remote control center.
- **ICCP Server Data**
Shows the current server data which is transmitted to the selected remote control center.
- **ICCP Client Data**
Shows the configuration of the selected transfer group and the dynamic changes of the received data.

For additional information on ICCP, refer to Reference Guide ICCP Protocol (ICCP).

5.3.8 Limit View

The **Limit View** provides a comprehensive overview of the currently set limits and their state.

Name	Type	Meas Value	Unit	Default	Entered	Limit Set	Limit Set Name
P/Default LimitSet/OG1	High Warning	0	MW	10			
P/Default LimitSet/OG2	High Emergency	0	MW	12			
Q/Default LimitSet/OG1	High Warning	0	MVA	10			
Q/Default LimitSet/OG2	High Emergency	0	MVA	12			

[rguio004-280610-02.tif, 1, en_US]

Figure 5-14 Limit View of the Runtime Explorer

Content Area

The content area shows the limits of all objects that are assigned to the object selected in the object tree.

List Box for Filter Selection

The list box enables you to filter according to certain criteria, like **All limits**, for example.

Button: Limit Dialog

Clicking this button opens the **Limit Set dialog** for changing the active limit set. For additional information on limit sets, refer to [13.3.2 Limit Set Dialog](#).

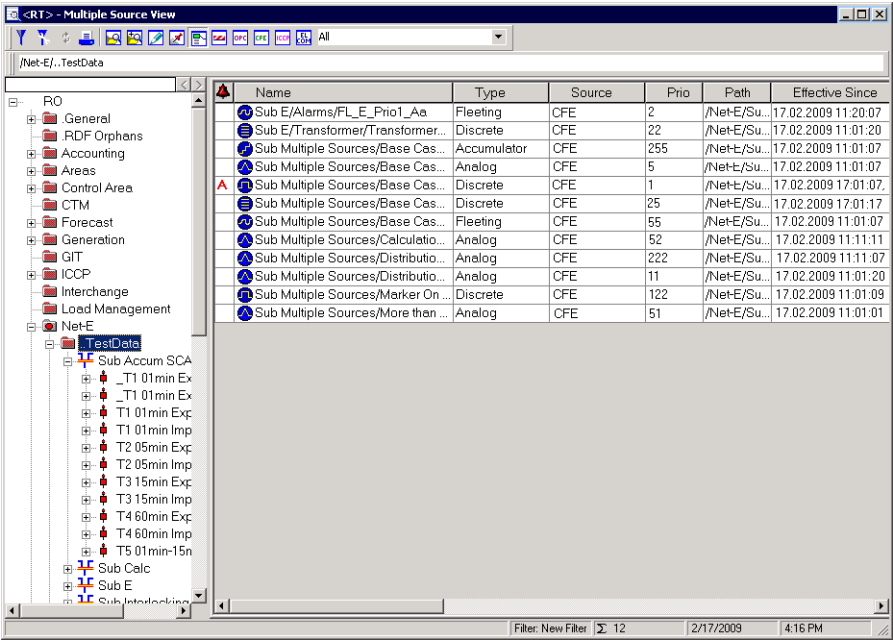


NOTE

If a default limit is violated the value in the concerning column changes the color.

5.3.9 Multiple Source View

The **Multiple Source View** provides an overview of the multiple values (received from different sources) of a measurement and the result of the selection process.



[scare021-200209-01.tif, 1, en_US]
Figure 5-15 Multiple Source View of the Runtime Explorer

Content Area

The content area shows information of each single source like value source, current value, current quality of the value, for example.

List Box for Filter Selection

The list box enables you to filter according to certain criteria like **all source** or **no primary set sources**.

6 Trends And Charts

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6.1 Introduction

Trends And Charts enables you to view and create 2-dimensional plots (charts or curves) of existing data values. The graphical tool processes both, real-time, historical, and COP data. The **Trends And Charts** user interface displays up to 16 curves in 1 chart. To design the charts and curves, you can set a wide range of parameters.

Other tasks of **Trends And Charts** are:

- You can configure the curve styles interactively. The flood fill feature fills the area between a curve and the x-axis or between 2 curves with a configurable color and pattern.
- The curves are updated with newly arriving values.
- A selected curve point can be displayed as a digital value together with its archive time by moving the rule.
- Indicators show invalid measurements.
- Curve displays can be printed or saved to a file.
- Curve data can be exported to Microsoft Excel.

6.2 Operational Steps

6.2.1 Starting Trends And Charts

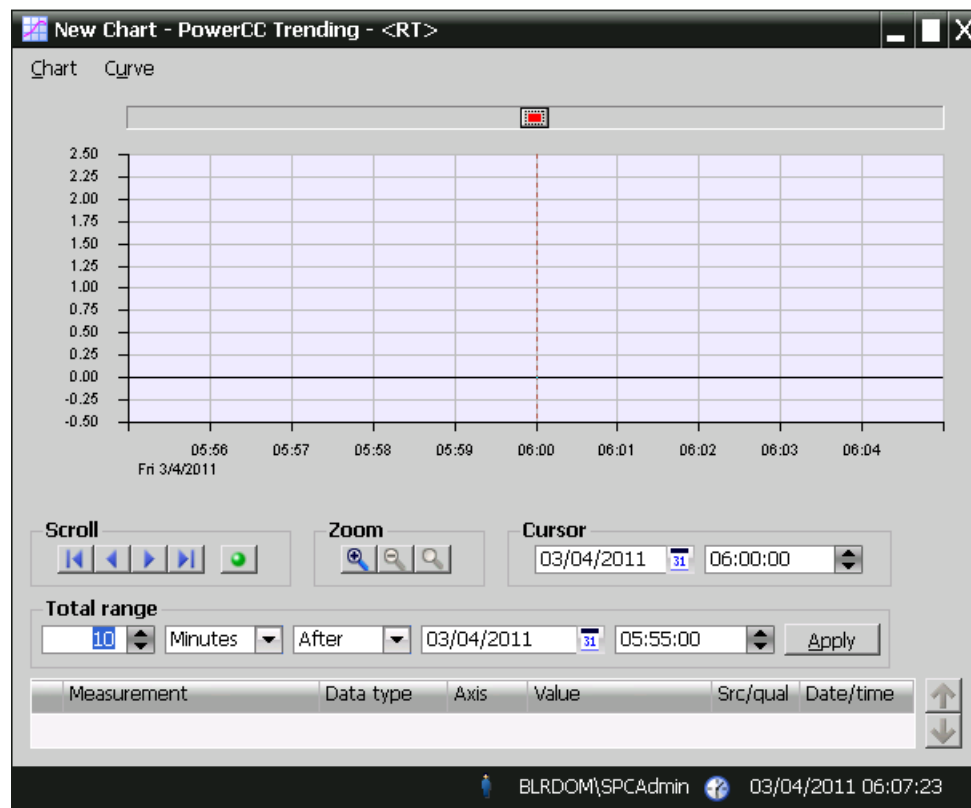
The goal of this task is to start the **Trends And Charts**.

Starting the Trends And Charts

The following methods are used open **Trends And Charts**:

- On the PAT, select **Tools > Trends And Charts**.
- Alternatively, select **Tools** and use drag and drop operation.

The **Power 5 Trending** window opens.



[uiost001-090311-01.tif, 1, en_US]

Figure 6-1 Power 5 Trending Window

6.2.2 Displaying a Chart

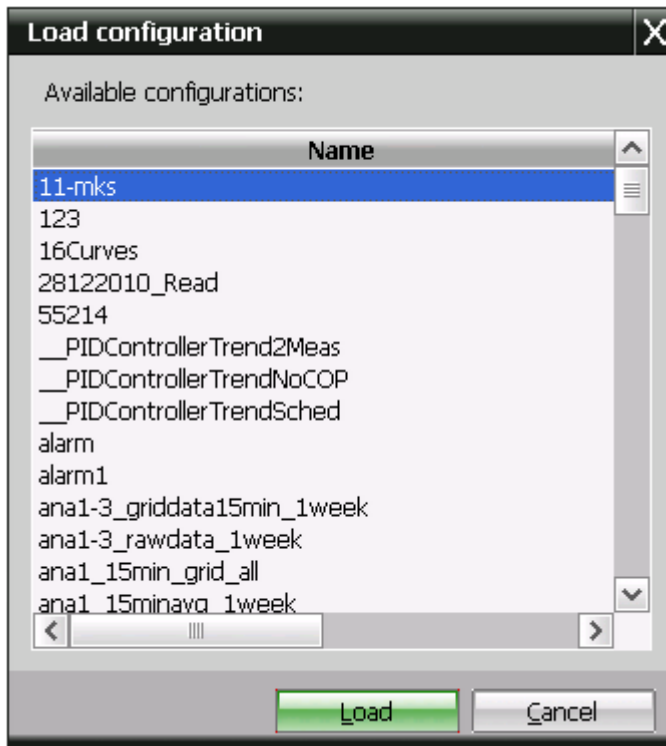
The goal of the task is to open a pre-configured or user-defined chart in the **Power 5 Trending** window.

Displaying a Chart

- On the **Power 5 Trending** window, select **Chart > New**.
All signals are disconnected and deleted, all curves and axes are cleared from the chart.

- Select **Chart > Open**.

The **Load Configuration** dialog with a list of the available chart configurations opens.



[uiost017-160311-01.tif, 1, en_US]

Figure 6-2 Load Configuration Dialog

- Select a chart configuration from the available configuration list.
- Click **Load**.

The chart configuration is displayed in the **Power 5 Trending** window.

Displaying a Chart Using Application Button

- On the PAT or a single-line diagram, click the appropriate application button.
For additional information, refer to [4.2.9 Calling Application Buttons](#).

The pre-configured chart is displayed.

6.2.3 Displaying a Chart Using Runtime Explorer

The goal of the task is to display a chart using the Runtime Explorer.

Displaying a Chart Using Runtime Explorer

- Right-click an instance in the tree view of the Runtime Explorer and select **Trending**.

The **Power 5 Trending** window opens, displaying the selected instance.



NOTE

The **Trending** option will be enabled for certain type of measurements only.
For example, Analog, Digital, and Accumulator.

6.2.4 Displaying a Chart Using Single-line Diagram

The goal of the task is to display a chart using a single-line diagram.

Displaying a Chart Using Single-line Diagram

- Right-click a value on a single-line diagram and select the menu item **Trending**.

The **Power 5 Trending** window opens, displaying the selected value.

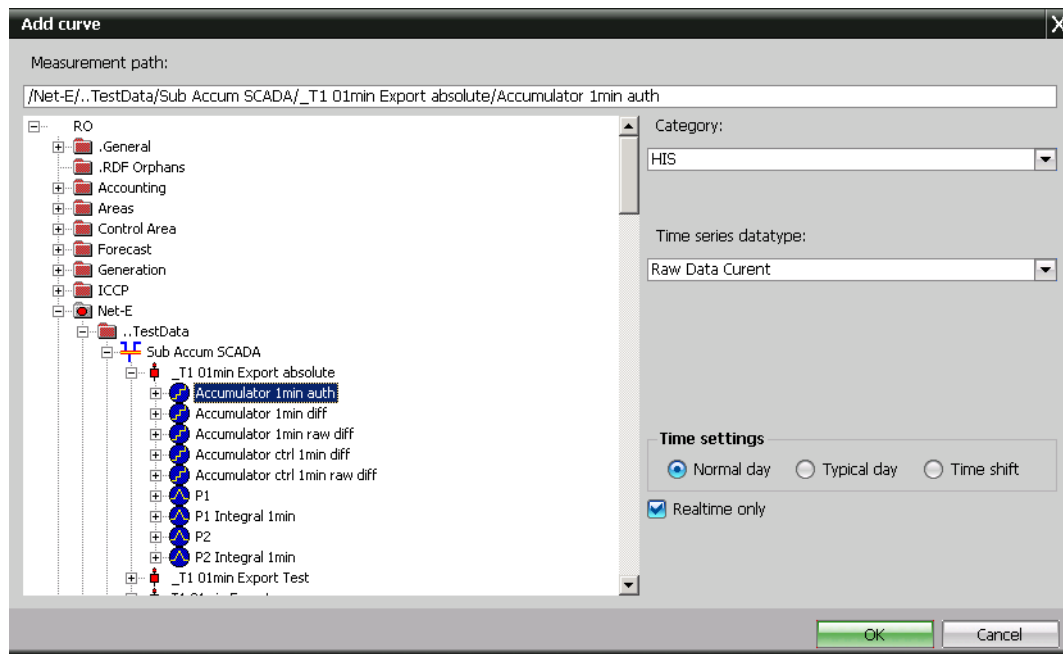
6.2.5 Adding a Curve

The goal of this task is to add a curve to a chart configuration.

Adding a Curve Using Signal Settings Dialog

- On the **Power 5 Trending** window, select **Curve > Add Curve > Signal**.

The **Add Curve** dialog opens.



[uiost003-090311-01.tif, 1, en_US]

Figure 6-3 Add Curve Dialog

- Select an object from the instance tree.
The selected instance is displayed in the text box above the instance tree.
- Select a **Category** and a **Time Series Data Type** from the corresponding list boxes.
- Enter the value in the **Query Cycle** (or use the up/down arrow buttons) and select the unit from the corresponding list box.

- Select the required **Time settings** option:
 - Normal day - The data is shown according to the reference date selected in the **Power 5 Trending** window.
 - Typical day (floating) - Always data for the selected date is shown.
 - Time shift - Data for curve minus particular number of hours, days, weeks, months or years is shown.
- Click **OK**.

The curve is displayed in the chart.

Adding a Curve Using Runtime Explorer

- On the Runtime Explorer select the desired view.
- In the tree view, select a value and use drag and drop into the measurement table and Chart area of to display the curve in the **Power 5 Trending** window.

Adding a Curve Using Single-line Diagram

- On a single-line diagram, select the desired measurement.
- Use drag and drop to display the curve in the **Power 5 Trending** window.

6.2.6 Handling Charts and Curves

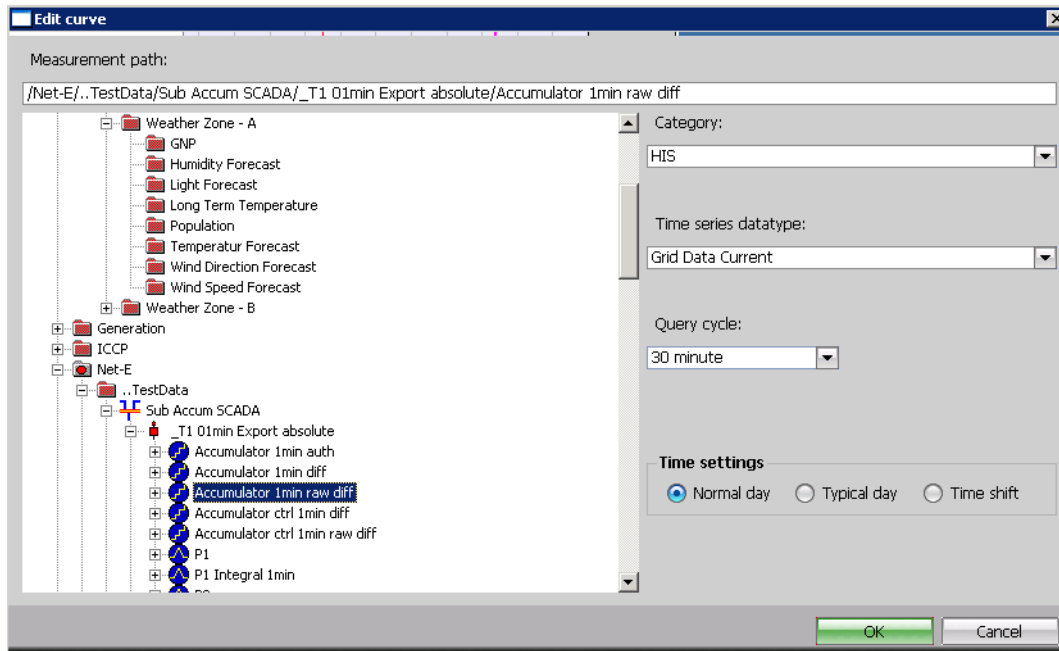
6.2.6.1 Editing a Curve

The goal of this task is to edit the currently displayed curve.

Editing a Curve

- On the **Power 5 Trending** window, select **Curve > Edit Curve**

The **Edit Curve** dialog opens.



[uiost022-160311-01.tif, 1, en_US]

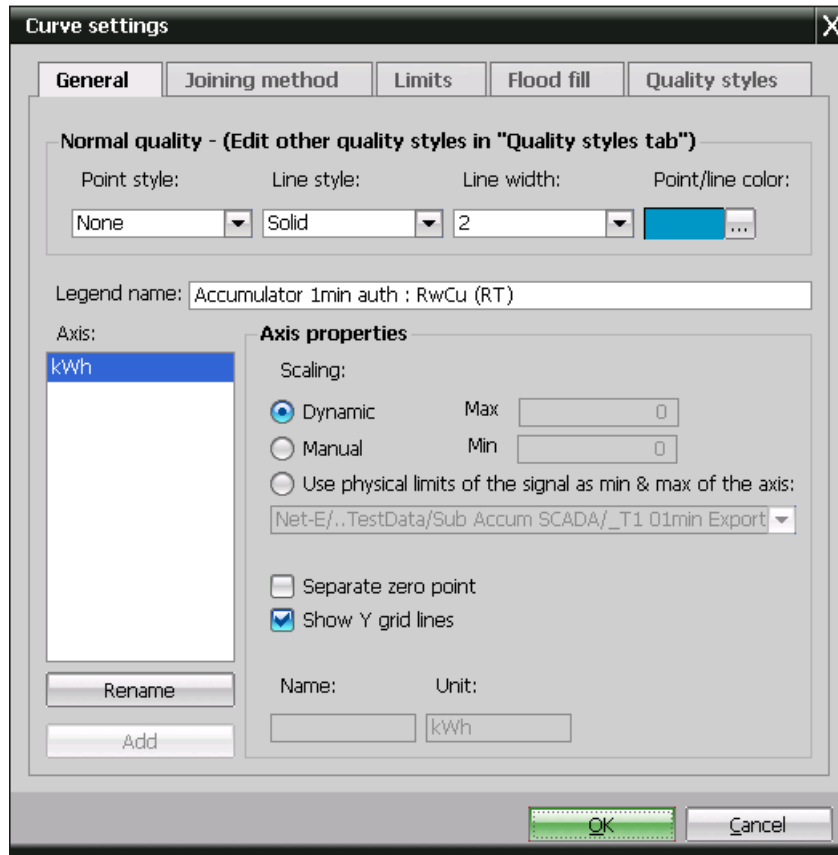
Figure 6-4 Edit Curve Dialog

6.2.6.2 Editing the Layout of a Curve

The goal of this task is to edit the layout of the selected curve.

Editing the Curve Layout

- On the **Power 5 Trending** window, select **Curve > Edit Display...**
The **Curve Settings** dialog opens.



[uiost005-090311-01.tif, 1, en_US]

Figure 6-5 Curve Settings Dialog

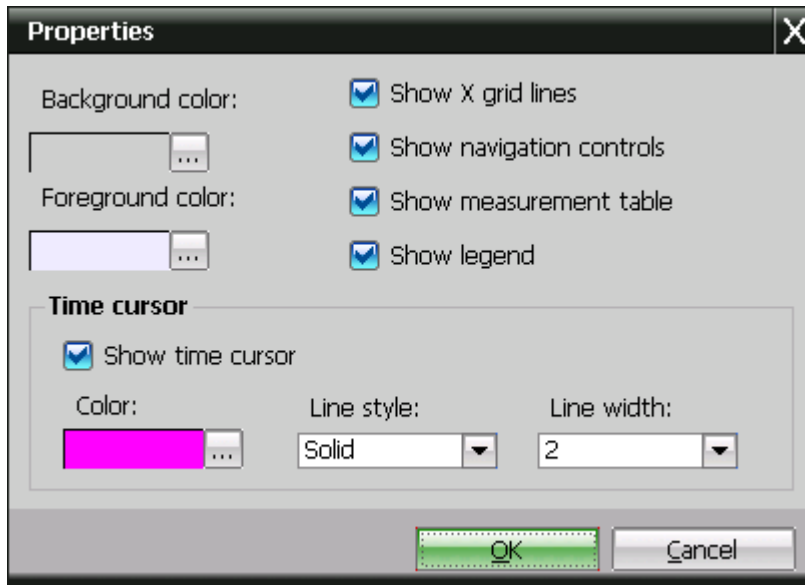
- Define the various curve settings grouped by tabs, within this dialog.
- Click **OK** to confirm the entries.

6.2.6.3 Editing Chart Colors and Other Properties

The goal of this task is to edit the colors and other properties of the chart.

On the **Power 5 Trending** window, select **Chart > Properties..**

The **Properties** dialog for editing chart colors and other chart-specific properties opens.



[uiost006-090311-01.tif, 1, en_US]

Figure 6-6 Properties Dialog

Editing Colors

- To change the color of the background, click the button to the right of the **Background color** field. The **Color** dialog opens.
- Select a color and click **OK**.
- To change the chart foreground color, click the button to the right of the **Foreground color** field. The **Color** dialog opens.
- Select a color and click **OK**.
- On the **Properties** dialog, click **OK**.
The foreground and background colors change according to the selected colors

Editing Other Properties

- To show grid lines of the x-axis, enable the check box **Show X grid lines**.
- To show the controls (**Scroll** or **Zoom**, for example) in the lower part of the **Power 5 Trending** window, enable the **Show Navigation Controls** check box.
- To show the measurement table, enable the **Show Measurement Table** check box.
- To show the legend, enable the **Show Legend** check box.
- Select the presentation (Color, line style, line width) of the time cursor if desired.
- To save the changes, click **OK** in the **Properties** dialog.
The changes are displayed in the chart.

6.2.6.4 Browsing through the Chart

The goal of the task is to browse through the chart.

Browsing Through the Chart

- Use the **Scroll** or **Zoom** buttons to browse through the charts and to zoom in/out the curves.
- Specify the date or time field **Cursor** to position the vertical chart ruler.

- Specify the time intervals of the chart in the text boxes of the **Total Range** area.
- Click the **Apply** button.

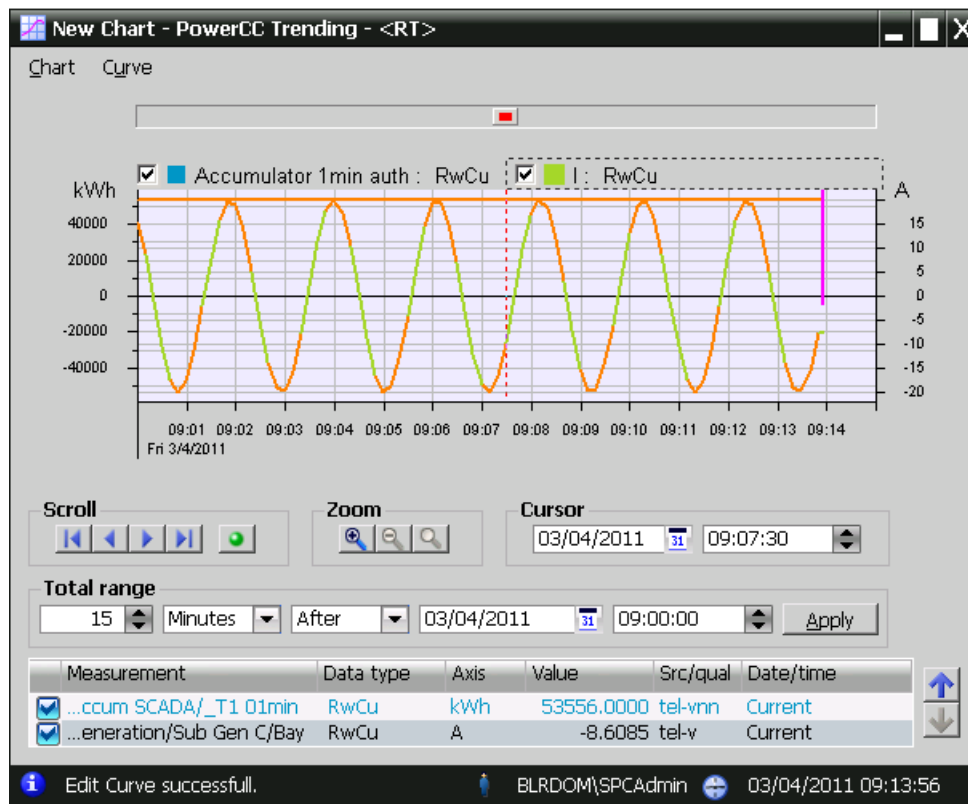
The specified time intervals are displayed in the chart.

You can move the vertical chart ruler by using the ruler slider above the chart area.

The vertical chart ruler specifies the instantaneous values of the curves. The appropriate instantaneous values are displayed in the measurement list.

6.2.7 Performing Operations on the Power 5 Trending Window

The goal of the task is to perform operations on the **Power 5 Trending** window using context menu and the measurement list.



[uiost008-090311-01.tif, 1, en_US]

Figure 6-7 Curves And Trends Dialog

Performing Operations Using Context Menu

- Right-click in the chart area.
Select one of the following items from the context menu.
 - Select **Zoom In**:
The chart area is zoomed in.
 - Select **Zoom Out**:
The chart area is zoomed out.
 - Select **Total Range**:
This operation brings the chart area back to original size after zooming in.
 - Select **Zoom to Selection**:
A magnifier is displayed to enlarge parts of the chart area.
 - Select **Refresh Chart**:
The chart area is refreshed.
 - Select **Export to MS Excel**:
Chart data are exported to MS Excel.

Handling the Measurement List

- Enable the check box of a curve in the measurement list.
The curve is displayed in the chart area.
- Alternatively, disable the check box of a curve in the measurement list.
The curve is not displayed in the chart area.
- Select a curve in the measurement list, by clicking on the item.
The item turns blue and the arrow buttons on the right turn bold.
- Click the arrow button up/down.
The arrows changes the measurement position in the table. It also changes the Z-order of the curves in the Chart.
The enabling and disabling of the arrows depend on the number of curves and their position in the table.

The curve is increased or decreased in the measurement list.

The actions on the check box on measurement table is also in synchronous with the check box on legends. Also, on selection of a curve in measurement table or legend, causes the curve to be highlighted in the chart area.

Performing Operations on the Measurement List Using Context Menu

- Right-click an item in the measurement list.
Select one of the following items from the context menu:
 - Select
The **Signal Setting** dialog opens.
 - Select **Paste Curve**.
A copied curve is pasted to the measurement list.
 - Select **Edit Curve**
The **Signal Setting** dialog opens.
 - Select **Edit Display**
The **Curve Settings** dialog opens.
 - Select **Remove Curve**.
The selected curve is removed from the chart.

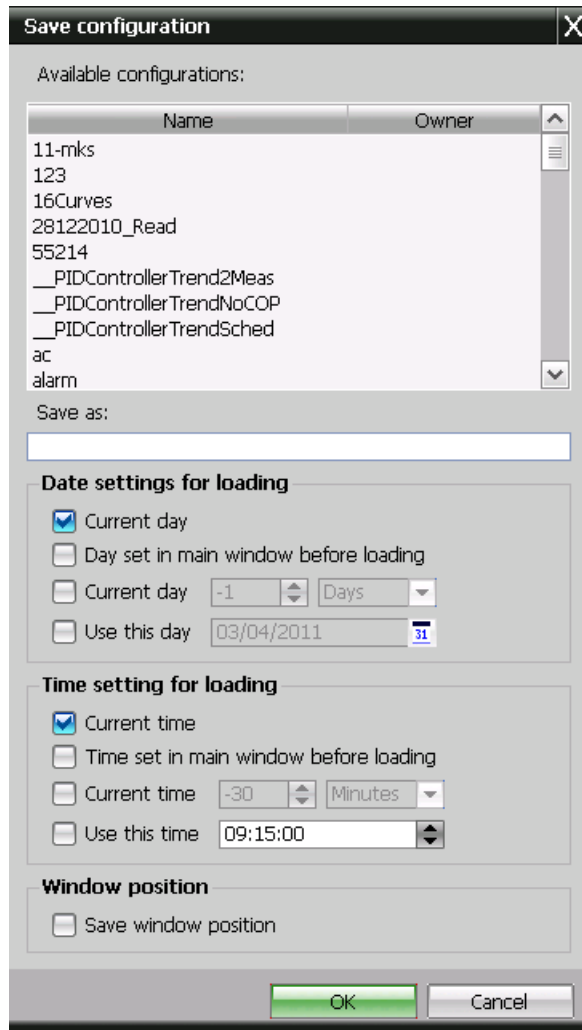
6.2.8 Saving a Chart

The goal of the task is to save a new or modified chart to the available chart configurations in the database.

Saving a New Chart

- On the **Power 5 Trending** window, select **Chart > Save ...** to save the current chart configuration.

The **Save Configuration** dialog opens. The upper part of the dialog shows the list of stored chart configurations.

The image shows a 'Save configuration' dialog box. At the top, it has a title bar with a close button. Below the title bar, it says 'Available configurations:' followed by a list box containing the following items: '11-mks', '123', '16Curves', '28122010_Read', '55214', '__PIDControllerTrend2Meas', '__PIDControllerTrendNoCOP', '__PIDControllerTrendSched', 'ac', and 'alarm'. Below the list box is a 'Save as:' text field. Underneath that are three sections: 'Date settings for loading' with options for 'Current day', 'Day set in main window before loading', 'Current day' (with a spinner set to -1 and a dropdown for 'Days'), and 'Use this day' (with a date field '03/04/2011' and a calendar icon); 'Time setting for loading' with options for 'Current time', 'Time set in main window before loading', 'Current time' (with a spinner set to -30 and a dropdown for 'Minutes'), and 'Use this time' (with a time field '09:15:00' and a time picker icon); and 'Window position' with a 'Save window position' checkbox. At the bottom are 'OK' and 'Cancel' buttons.

[uiost007-090311-01.tif, 1, en_US]

Figure 6-8 Save Configuration Dialog

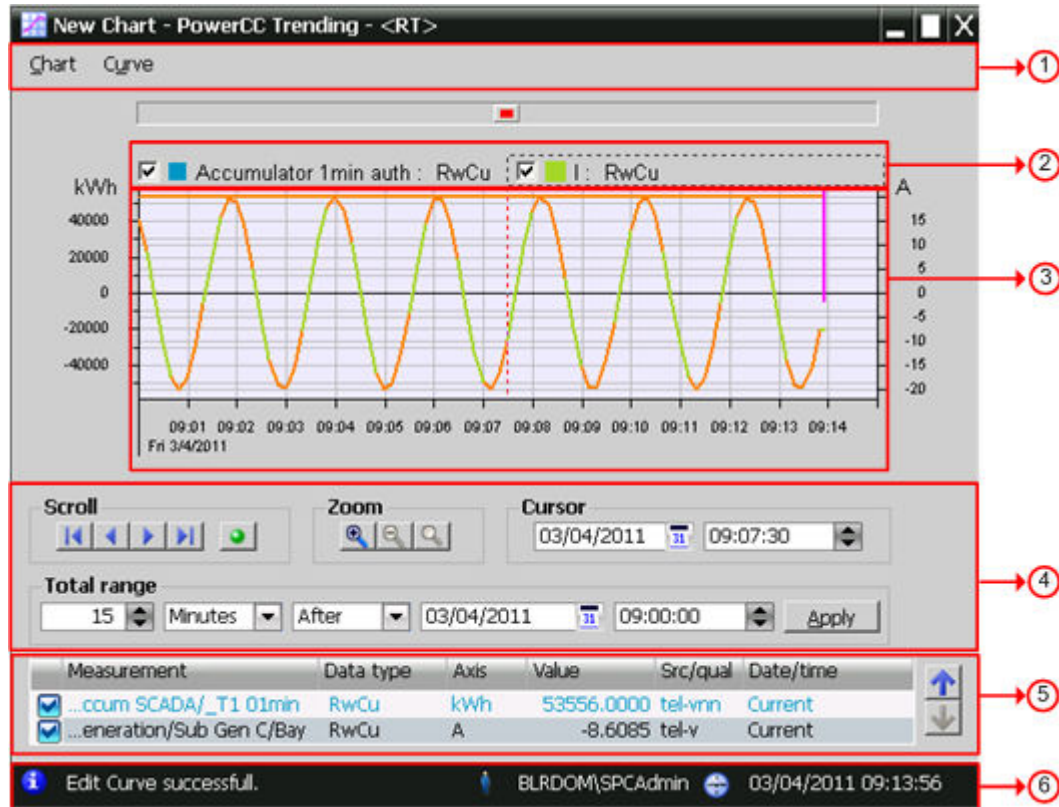
- Enter a new name for the chart configuration in the **Save as** text box.
- Enter the values for **Date settings for loading** and **Time settings for loading** in the lower part of the dialog.
- Confirm with **OK**.

The chart is saved to the database.

6.3 User Interface Description

6.3.1 Trends And Charts

Trends and Charts is the display of historical data in an easy to understand method. **Trends And Charts** displays forecast data and schedules as well. This can be a real-time or historical trend.



[uiost014-160311-01.tif, 1, en_US]

Figure 6-9 Trends And Charts

- (1) Menu bar
- (2) Legend for Curves
- (3) Chart area
- (4) Graphic presentation control
- (5) Measurement list
- (6) Status bar

Menu Bar

The menu bar provides the following menu items:

- **Chart**

This function opens the **Chart** menu, displaying the following menu items:

- **New**
Clears previous curves and axes from the chart.
- **Open...**
Shows a display with the list of available chart configurations stored in the database. You can select a chart configuration from this list.
- **Save...**
Saves the current chart configuration.
- **Save As...**
Opens the **Save Configuration** dialog without a configuration name in the **Save as** field.
- **Delete...**
Opens a display which contains a list of all stored chart configurations. Select one of these configurations to delete it.
- **Export to MS Excel**
Curve data are exported to MS Excel.
- **Print**
Prints the current chart on the default printer.
- **Print to File...**
Opens a display for entering a file name. The current chart configuration is stored in this file (in bitmap format).
- **Properties**
Opens a display for editing chart colors and other chart-specific properties.
- **Refresh Chart**
Clears all curve data and reloads it from the archive.
- **Exit**
Closes the **Curves and Trends** display.

- **Curve**

This function opens the **Curve** menu, displaying the following menu items:

- **Add Curve > Signal**
Opens the **Add Curve** dialog.
- **Add Curve > Calculation**
Opens the **Add Calculation Curve** dialog.
- **Paste Curve**
Adds the measurement that has been selected from the single-the diagram to the measurement list and adds the corresponding curve to the chart area.
- **Edit Curve...**
Opens the **Edit Curve** dialog.
- **Edit Display...**
Opens the **Curve Settings** dialog with the **General**, **Limits**, **Joining Method**, **Flood Fill** and the **Quality Styles Tab** for editing the layout of the currently selected curve.
- **Remove Curve**
Deletes the current signal and the curve from the chart. If the axis of the curve is unused, it is also deleted.

Legend for Curves

The legends are provided for the curves shown in the Trend Chart. By default these legends will not be shown. You can enable it using the **Show Legends** check box in the **Properties** dialog.

The legend shows the following:

- Check box
 - Toggling the check box will toggle the visibility of curve in the Trend Chart. This option is in synchronous with the check box in the Measurement table.
- A color box with the normal quality color of the curve.
- Legend name
 - By default the legend name consists of the Measurement name, Data type and Timeshift/Typical day/Typical day floating details (if applicable). You can set a custom legend name using the **Curve Display Properties** dialog.

Selection of legend leads to highlighting of curve in the Chart. Legends also have tooltip.



NOTE

The legends are especially helpful in case of embedded Trend controls in worldmaps. Using legends instead of Measurement table allows more space for the chart in the limited space within the Trend control.

Chart Area

In this area the selected curves are displayed.

- You can change the graphic representation of the curves using the interactive elements below the chart area.
- The appropriate instantaneous values are displayed in the measurement list.
- The vertical chart ruler specifies the instantaneous values of the curves.
- The vertical chart ruler can be moved by using the ruler slider above the chart area.
- You can open a context menu to perform operations in the chart area. For example you can export the whole chart area to MS Excel.

For additional information on the context menu, refer to [6.2.7 Performing Operations on the Power 5 Trending Window](#).

Graphic Presentation Control

- You can browse through the chart by using the **Scroll** buttons. The button enables or disables the update of the chart.
- With the **Zoom** buttons, you can zoom in/out the curves.
- You can specify the date/time field **Cursor** to position the vertical chart ruler.
- The list boxes **Total Range** specifies the time intervals of the charts.
- After clicking the button **Apply**, the specified time intervals are displayed in the chart.

Measurement List

The measurement list shows all appropriate pathnames of the measurement (signal) instances of the selected curves. Additionally, the list includes the corresponding y-axis label, the value, and quality code as well as the typical day (if it is defined) of the curve.

- Check box of a measurement:
Enabling/disabling the check box displays/fades out the selected curve.
- Arrow buttons on the right:
Only active if a curve is selected (blue). With the arrow buttons you can increase/decrease the selected curve in the measurement list.

- Context menu of the measurement list:
You can open a context menu to perform operations on the measurement list.
For additional information on the context menu, refer to [6.2.7 Performing Operations on the Power 5 Trending Window](#).
- Typical day and Time shift
You can define the curve with relative time shift.

A curve in the chart will be highlighted by clicking on the corresponding entry in the Measurement table. When a different row is selected in the Measurement table, the previous curve will be de-highlighted and the current curve will be highlighted.

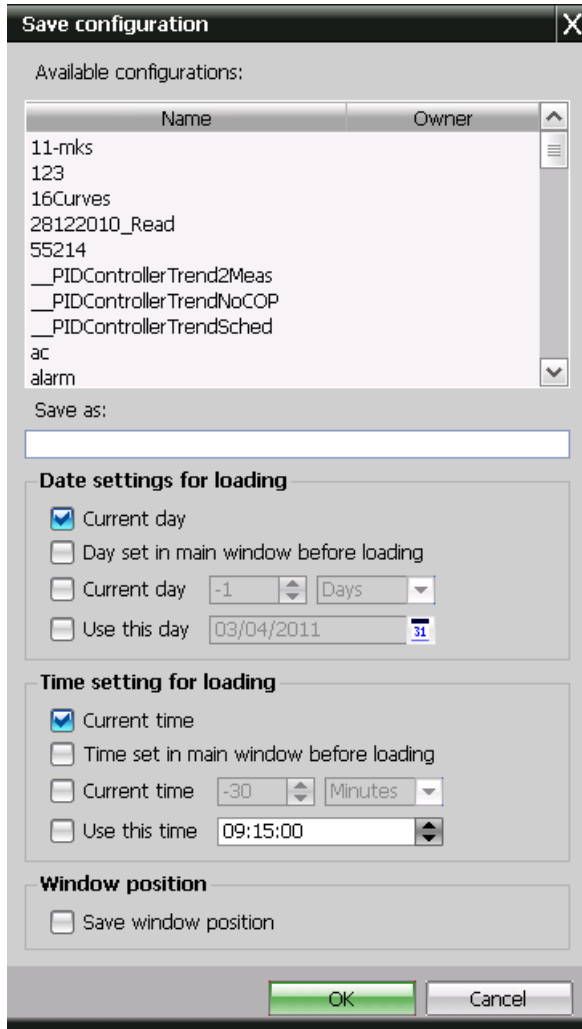
Status Bar

The status bar shows the chart data processing status. It also shows the current user and current time. For example, retrieving data.

6.3.2 Save Configuration Dialog

The upper part of this display shows a list of already stored configurations and a text box for entering a new name for the configuration.

In the lower part of the display, you can determine the date and time settings of the chart.



[uiost007-090311-01.tif, 1, en_US]

Figure 6-10 Save Configuration Dialog

Text Box: Available Configurations

Shows all available configurations.

Text Box: Save As

Here you can enter the name of the configuration.

Date Settings for Loading

- **Current day**
Displays the curve starting at the day of loading.
- **Day set in main window before loading**
Displays the curve starting at the day set in the trend control display.
- **Current day definition**
The starting day of the curve can be defined relative to the current day.
For example, **-2 days** means starting 2 days before the current day.
- **Use this day**
The starting day of the curve can be selected in the calendar field.

Time Settings for Loading

- **Current time**
Displays the curve starting at the time of loading.
- **Time set in main window before loading**
Displays the curve starting at the time set in the trend control display.
- **Current time definition**
The start time of the curve can be defined relative to the current time.
For example, **-30 minutes** means starting 30 min before the current time.
- **Use this time**
The start time of the curve can be entered in the text box.

Window Position

- **Save window position**
Saves the position of the trend control display on the screen and opens the display at the same position when the chart is called up.

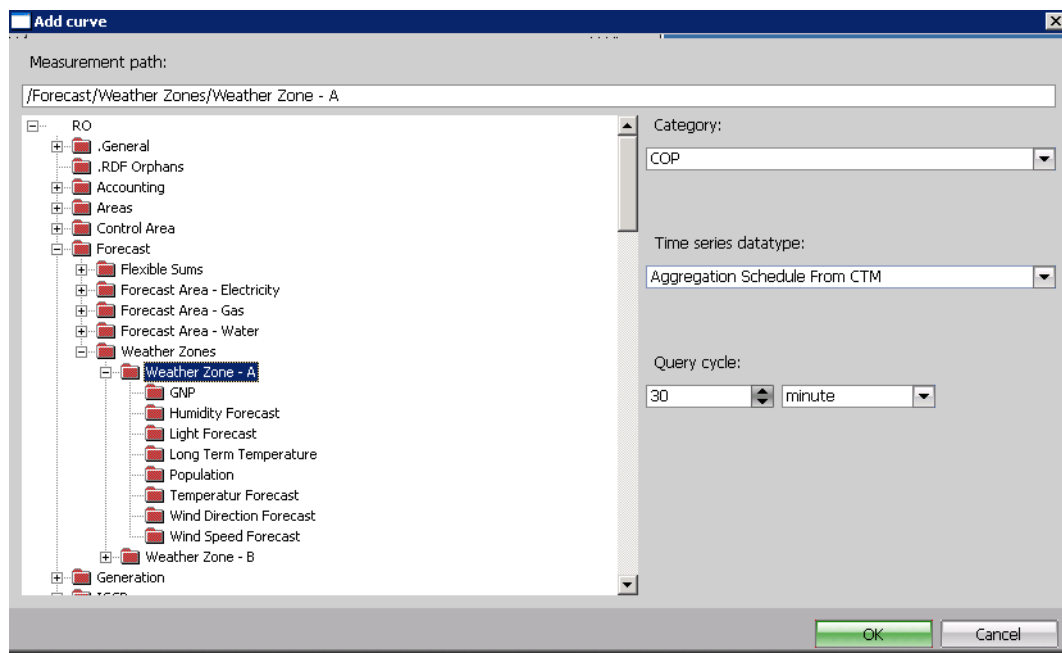
6.3.3 Add Curve Dialog

The **Add Curve** dialog is used to define parameters for the signals and the corresponding curves within the **Spectrum Power 5 Trend Control**.



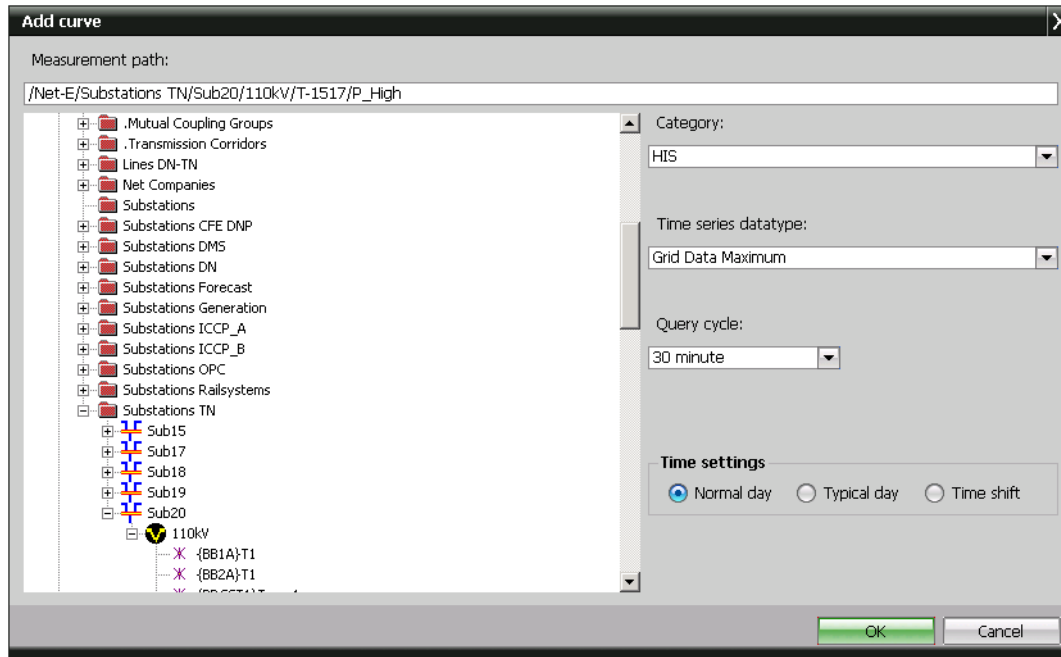
NOTE

The layout of the right section of the **Add Curve** dialog depends on the selected data category.



[uiost023-160311-01.tif, 1, en_US]

Figure 6-11 Add Curve Dialog for Chosen Category COP



[uiost024-160311-01.tif, 1, en_US]

Figure 6-12 Add Curve Dialog for Chosen Category HIS



NOTE

The layout of the right section of the **Add Curve** dialog of the chosen category HIS depends on the selected time series data type and time settings.

Category

Defines the signal source category:

- **COP** displays schedule data for the curve.
- **HIS** displays historical and current data for the curve.

Time Series Datatype

Defines the time series datatype for the curve. The available selections are configured for the project.

Query Cycle (for Category COP)

Defines the number of cycles and the unit of the query for the curve. Available units are second, minute, hour, day, and week.

Query Cycle (for Category HIS)

Defines the cycle of the query for the curve. Possible query cycles are dependent from the selected time series datatype.

The **Query** cycle is not available for time series datatype **Raw Data Current**.

Time Settings (for Category HIS)

Defines the time settings for the curve trended.

- **Normal**
No special time settings, the current time is used.

- **Typical day**
Defined by the absolute date in the past with a drop-down calendar field after the selection. The typical day is repeated for every day, except **Floating** is checked. Then the typical days are changed according to time history when scrolling to another day.
- **Time Shift**
Define the time offset within the 2 entry fields after the selection. The curve drawn according to the specified time offset to current time at runtime.

Realtime only (for Category HIS)

The **Realtime only** selection box is only shown if time series datatype **Raw Data Current** and **Time Settings Normal** has been selected. If realtime only is selected, the curve is a pure realtime trend and does not display values stored in the **Historical Information System (HIS)**.

6.3.4 Add Calculation Curve Dialog

The **Add Calculation Curve** dialog is used to define parameters of a calculation curve within the Spectrum Power 5 **Trend Control**.

Add calculation curve

Name:
Sum Bay 02 03

Formula:
A+B

Unit / Axis:
A

Operands:

#	Name	Type	Unit
A	Net-E/Substations TN/Sub15/110/02 Tr101/I	1 min GrCu	A
B	Net-E/Substations TN/Sub15/110/03 Sub17/I	1 min GrCu	A

Ok Cancel

[scuios01-250412-01.tif, 1, en_US]

Figure 6-13 Add Calculation Curve Dialog

Name

The name of the calculation curve as it is showed up in the legend.

Table Operands

The list of available operands are:

- #
Identifier of the operand that is used in the formula
- Name
Name of the operand
- Type
Data type of the operand (grid)
- Unit
Unit of measure of the operand

Formula

This formula is for the calculation. The operands have to be designated by their identifier as indicated in the table **Operands**. Only operands, which are provided on a grid (for example, Grid Data Current) can be used in formulas. The resolution of the grid must be identical (for example, 1 minute) for all operands.

Unit / Axis

This is a unit of measure of the formula.

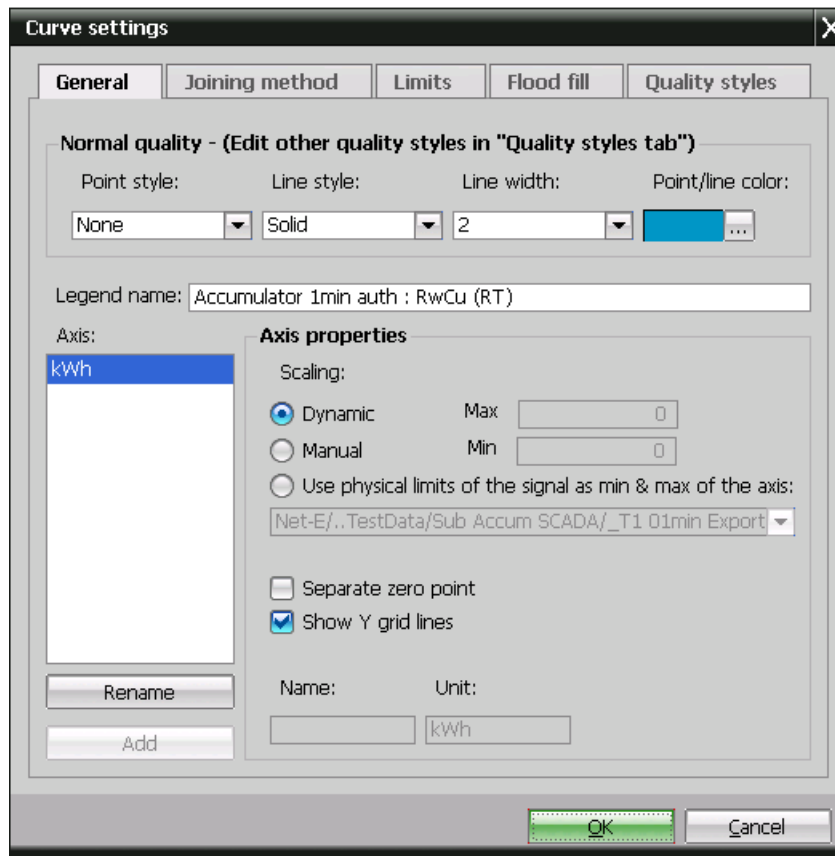


NOTE

Make sure that the joining method of the calculation is the same as the one used for the operands. Otherwise you might get unexpected results.

6.3.5 Curve Settings Dialog

The **Curve Settings** dialog is used to define the layout for the configured curves within the **Spectrum Power 5 Trend Control**.



[uiost009-090311-01.tif, 1, en_US]

Figure 6-14 Curve Settings Dialog – General Tab

Point Style, Line Style, Line Width, and Point/Line Color

Defines the point style, line style, and line width (specified in points) of the curve with the list boxes. Select the color from the color palette.

Legend Name

The legend name consists of the Measurement name, Data type and Timeshift/Typical day/Typical day floating details (if applicable). You can set a custom legend name using the **Curve Display Properties** dialog.

Axis

The corresponding y-axis of the curve is selected out of the available y-axis within the chart configuration.

- **Rename**
Enables renaming of the available y-axis within the chart. The entry fields Name and Unit is enabled to allow the changes.
- **Add**
Adds an additional y-axis and the unit of the y-axis within the chart and assigns this axis to the selected curve. The entry fields **Name** and **Unit** is enabled to allow the changes.

Axis Properties

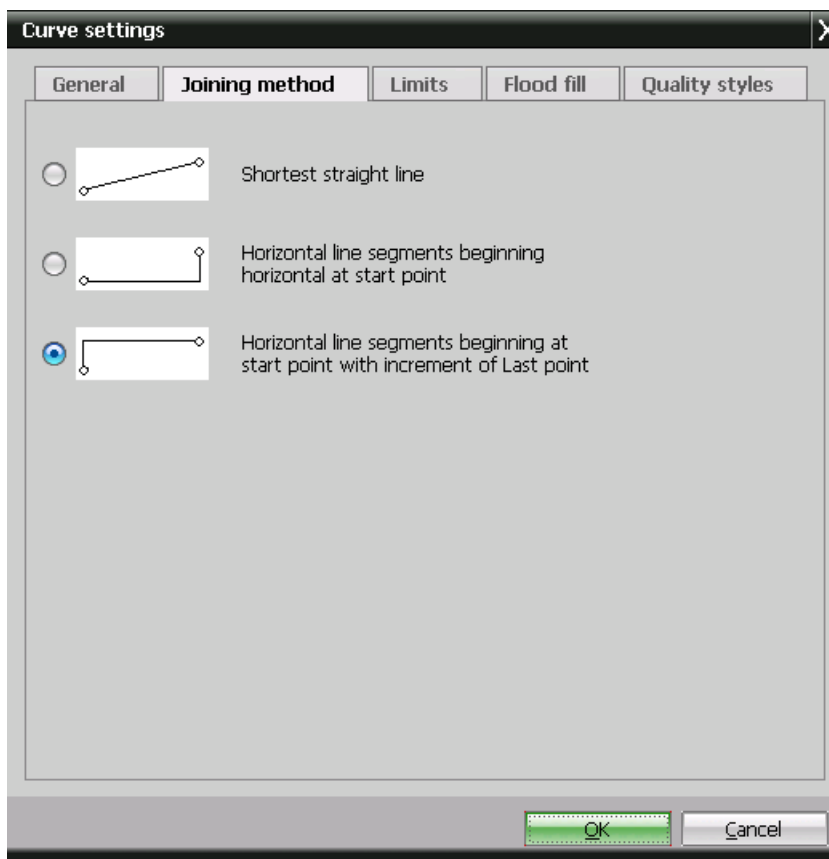
The corresponding y-axis properties within the chart configuration is selected for the curve.

- **Dynamic**
Allows dynamic scaling of the y-axis of the corresponding curve according the current values at runtime.

- **Manual**
Forces a fix scale of the y-axis of the corresponding curve at runtime. Within the Min and Max fields the values of the manually scaled y-axis is configured. Both fields are enabled if Manual has been selected.
- **Use physical limits of the signal as Min & Max of the axis**
Sets the **Min** and **Max** values to the physical limits of the related signal source of the corresponding curve. The list box below is enabled if more than 1 curves share the y-axis. You can then choose the signal source for the physical limits
- **Separate zero point**
If **Separate zero point** is unchecked, multiple y-axis use a common zero point. Otherwise the zero points of the different y-axis are separated.
- **Show Y grid lines**
Y-grid lines or no y-grid lines for the selected y-axis are shown at runtime.
- **Name and Unit**
Refer to Axis.

**NOTE**

Multiple curves may use the same y-axis. The changes you are configuring for **Axis** and **Axis Properties** apply to all curves using the same y-axis.



[uiost010-090311-01.tif, 1, en_US]

Figure 6-15 Curve Settings Dialog – Joining Method Tab

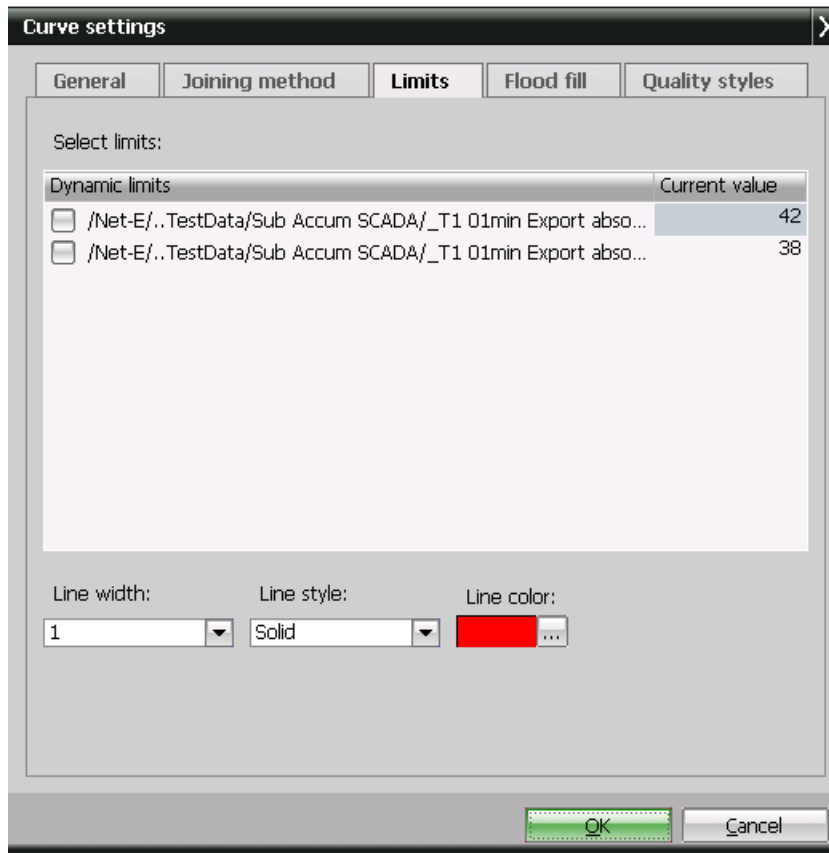
Option Buttons

Defines the type of connection between 2 curve points for the selected curve.



NOTE

By nature, trending current values always use the display type **Horizontal Line Segment beginning horizontal at Start Point**.



[uiost011-090311-01.tif, 1, en_US]

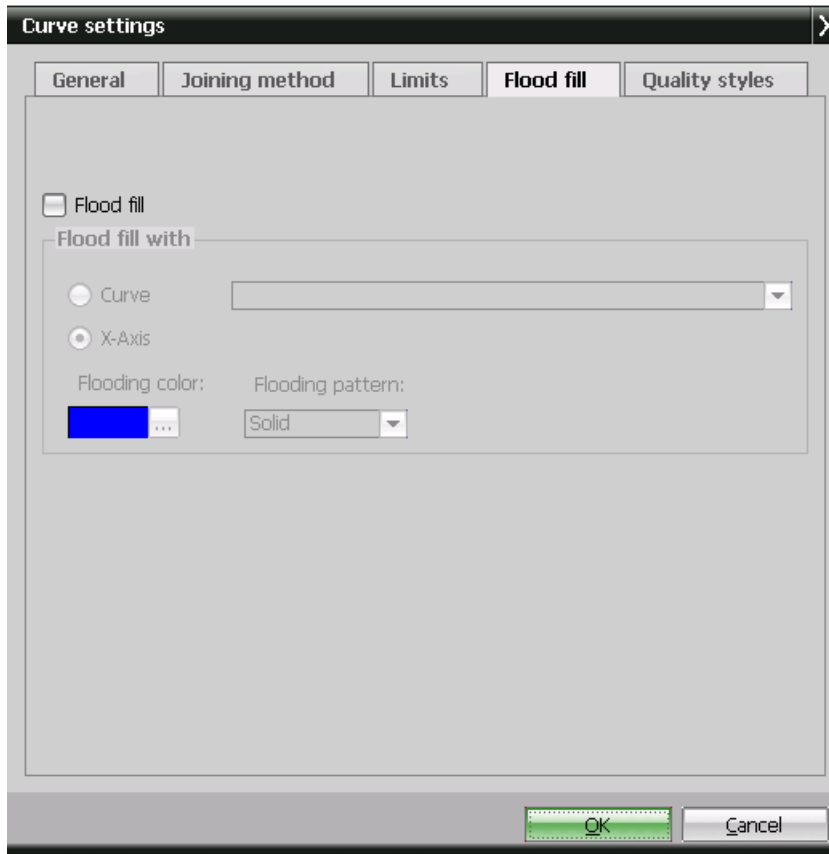
Figure 6-16 Curve Settings Dialog – Limits Tab

Select Limits

Shows the defined limits for the selected measurements to be trended, together with the current limit value. Select the limits to display as a line within the chart with the check boxes on the left side.

Line Width, Line Style, and Line Color

Defines the limit line style and limit line width (specified in points) of the selected dynamic limit with the list boxes. Choose the color from the color palette. Unchecked limits are not shown within the chart.



[uiost012-090311-01.tif, 1, en_US]

Figure 6-17 Curve Settings Dialog – Flood Fill Tab

Flood Fill

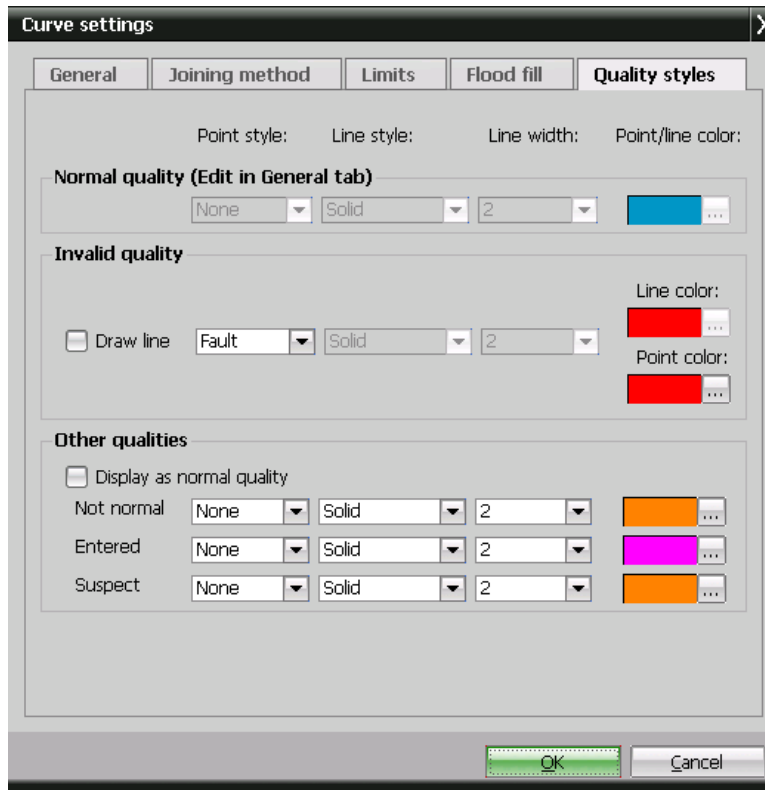
The check box enables filling the space between the selected curve and another one or the x-axis with a selectable color and pattern. The settings are used if **Flood Fill** is enabled.

Flood Fill with

Defines the corresponding flooding properties of the selected curve.

- **Curve and X-Axis**
Defines flooding of the selected curve with the x-axis or with another curve configured for the chart (from the list box on the right)
- **Flooding color**
Choose the color from the color palette
- **Flooding pattern**
Choose the pattern for flooding the area between the selection.

Quality Styles



[uiost013-090311-01.tif, 1, en_US]

Figure 6-18 Quality Styles

Point Style, Line Style, Line Width, and Point/Line Color

Defines the point style, line style, and line width (specified in points) of the curve with the list boxes. Select the color from the color palette for each of the 5 categories of the quality codes.

The 5 categories of the quality codes are:

- Normal Quality
- Invalid Quality
- Not Normal
- Entered
- Suspect



NOTE

If the “Display as normal quality” check box is checked, the quality colors are not applied to the curve. All qualities will be shown as normal color.

7 Alarm Processing

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7.1 Introduction

In an abnormal situation, an alarm indicator alerts you and guides to the message summary or the graphical display, displaying the device causing the alarm condition.

Generic cross associations from message summaries to graphical displays and from graphical displays to message summaries render fast failure tracking down to the alarm source.

The following indications indicate an alarm in the control center:

- A flashing symbol on the PAT
- A flashing symbol in the single-line diagram
- An alarm indicator in the **Alarm Summary** and in the substation view of the Runtime Explorer (the letter **A** in the first column indicates the condition **object in alarm**)
- An audible alarm through the loudspeaker

Additionally, external acoustic devices can also raise an alarm (central audible alarming).

Central Audible Alarming

Central audible alarming allows activation of alarm devices on different locations in case of incoming alarms. Alarming hardware connected to an RTU is supported.

Central audible alarming can be enabled or disabled on the PAT. Alarm devices can be reset either locally or through an adjusting command initiated by a Spectrum Power 5 operator.

Central audible alarming monitors the status of the alarm devices. If a device is already active, central audible alarming does not send adjusting commands. If a device is reset, central audible alarming again sends adjusting commands when new alarms come in.



NOTE

A new logged in user has always an active central audible alarming device.

Wireless Alarming

Alarms can be sent as text messages to mobiles, US style pagers, and to a telephone as a voice message stating the priority of the alarm, the area of responsibility, affected equipment path, and the current value of the equipment. Hence, the user responsible is informed of the area where the problem occurred through the voice message. The operator can acknowledge the alarm by entering the corresponding PIN in the same call.



NOTE

Wireless Alarming and Central Audible Alarming forwards only appearing alarms. Disappearing alarms are not considered.

- In the **General Summary**, the following events are reported:
 - Activation or deactivation of Wireless Alarming
 - Failure while sending the alarm using the specified channel because of technical problems
 - The shift plan becomes invalid
 - Message is not acknowledged in time
- Text messages (SMS) contain the complete alarm message and voice messages consist of the general text including the priority of the alarm, the area of responsibility, affected equipment path, and the current value of the equipment.
- Alarms can be filtered according to the area of responsibility, alarm priority, and alarm category.



NOTE

If an acknowledgment does not arrive in time, the Spectrum Power 5 escalation strategy looks up the next operator on duty in a prepared standby shift list and repeats the alarm forwarding. Acknowledgment of the alarm using phone or mobile does not acknowledge the alarm in the corresponding **Alarm Summary**. It only prevents forwarding an alarm to the next operator on duty.

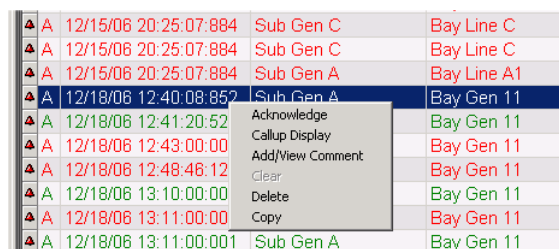
7.2 Operational Steps

7.2.1 Handling an Incoming Alarm

The goal of this task is to acknowledge an incoming alarm.

Handling an Incoming Alarm

- Click  on the PAT to switch off the audible alarming (for permanent tones only).
The color of the icon changes to yellow.
- Follow the alarm indication and navigation provided by the PAT.
- Right-click in the flashing presentation symbol of the respective single-line diagram.
- Alternatively, right-click the alarm entry in the **Runtime Explorer**.
- Alternatively, right-click the alarm entry in the **Alarm Summary**
For additional information, refer to [8.3.4 Alarm Summary](#).
- Select the menu item **Acknowledge Alarm** (**Acknowledge** in an **Alarm Summary**) from the context menu.



A	12/15/06 20:25:07:884	Sub Gen C	Bay Line C
A	12/15/06 20:25:07:884	Sub Gen C	Bay Line C
A	12/15/06 20:25:07:884	Sub Gen A	Bay Line A1
A	12/18/06 12:40:08:852	Sub Gen A	Bay Gen 11
A	12/18/06 12:41:20:52		Bay Gen 11
A	12/18/06 12:43:00:00		Bay Gen 11
A	12/18/06 12:48:48:12		Bay Gen 11
A	12/18/06 13:10:00:00		Bay Gen 11
A	12/18/06 13:11:00:00		Bay Gen 11
A	12/18/06 13:11:00:001	Sub Gen A	Bay Gen 11

[scaui002-031207-01.tif, 1, en_US]

Figure 7-1 Acknowledging an Alarm in the Alarm Summary

The alarm is acknowledged and the alarming indications are stopped. An alarm must only be acknowledged in one alarming instance to be acknowledged in all alarming instances. You are allowed to continue with corresponding control operations. Also the PAT and audible alarming are updated.

7.2.2 Inhibiting Audible Alarming

The goal of the task is to inhibit audible alarming.

Inhibiting Audible Alarming



NOTE

If the font color of the button **Audio Inhibit** is red, audible alarming is inhibited.

- Click the **Audio Inhibit** button on the PAT
The function cuts off the loudspeaker for all future alarms until you click **Audio Inhibit** again.

7.2.3 Inhibiting Central Audible Alarming

The goal of the task is to inhibit central audible alarming.

Inhibiting Central Audible Alarming



NOTE

The color red indicates that the type of audible alarming is switched off.

- If audible alarming is inhibited, click the button **Audio Inhibit**.
- Click **Audible Alarming** menu on the PAT.
- Select the type of central audible alarming from the **Audible Alarming** menu.
The following alarming types are provided in the **Audible Alarming** menu:
 - **GAA On/Off**
This command switches on/off central audible alarming.
 - **GAA Off - Prio 1**
This command switches on/off central audible alarming for alarms with priority 1.
 - **GAA Off - Prio 2**
This command switches on/off central audible alarming for alarms with priority 2.
 - **GAA Off - Prio 3**
This command switches on/off central audible alarming for alarms with priority 3.

7.2.4 Wireless Alarming

7.2.4.1 Activating Wireless Alarming

The goal of the task is to activate Wireless Alarming.

Activating Wireless Alarming



NOTE

If Wireless Alarming is already switched off, the font color of **WA On/Off** is red.

- Select **Audible Alarming > WA On/Off** on the PAT.

7.2.4.2 Configuring Wireless Alarming

The goal of the task is to configure Wireless Alarming.

Pre-requisite

The ISDN Karte Fritz!card / AVM and CAPI Treiber' are installed in the Power 5 system.
For additional information on installation, refer to IG-INST-EN.

Procedure

In IMM, configure **Alarm Filters** and **Communication Devices** under `/ .General/Wireless Alarming`.
Link **Communication Devices** to **Personnel** (under `/ .General/Personnel`) against **WA Channel**, **First Spare WA Channel**, or **Second Spare WA Channel** attribute in **Links** tab. Wireless Alarming contacts the person through these configured channels.

Define the shift plan of the personnel responsible for each area in the Shift Plan excel template and import this template to Shiftplan Editor.

For additional information on importing Shiftplan Editor, refer [7.2.4.4 Importing a Shift Plan](#).



NOTE

Set the value of the wireless alarming attribute **Code Page Type** to **CP8859-1 (West European)**, if WALA is intended for Western European languages.

7.2.4.3 Activating the Shift Plan

The goal of this task is to call the **Shift Plan Editor** and to open a shift plan.

Calling the Shift Plan Editor

The following methods are used to open the **Shift Plan Editor**:

- On the PAT select **Tools > Shift Plan Editor**.
- On the PAT select **Tools** and use drag and drop operation.

The **Shiftplan Editor** dialog opens.

Opening a Shift Plan

- Select **File > Open...** from the **Shiftplan Editor**.
The **Select a shiftplan ...** dialog opens.
- Select a menu item from the **Please select a shiftplan** list box.
- Confirm with **OK**.

The shift plan opens in the **Shiftplan Editor** dialog.

7.2.4.4 Importing a Shift Plan

The goal of this task is to import a shift plan

Importing a Shift Plan

- On the **Shiftplan Editor**, select **File > Import...**
The **Import shift plan from ...** dialog opens.
- Select the path and file name of the shift plan you want to import.
- Click **Open** to import the shift plan from the specified Excel file.

The selected shift plan is imported from the Excel file.

7.2.4.5 Exporting a Shift Plan

The goal of this task is to store a shift plan.

Exporting a Shift Plan

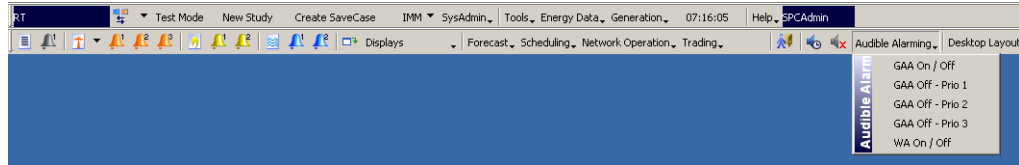
- On the **Shift Plan Editor**, select **File > Export...**
The **Export shift plan to ...** dialog opens.
- Select a path and enter a file name.
- Click **Save**.

The shift plan is saved as an MS Excel file.

7.3 User Interface Description

7.3.1 Alarm Processing

Alarm processing in SCADA uses different ways to inform you about incoming alarms.



[rguios27-031209-01.tif, 1, en_US]

Figure 7-2 PAT with Open Audible Alarming Menu

Warning Indication: Flashing Icons

Flashing icons on the left side of the PAT indicate alarms of different priorities. Up to 3 alarm priorities are available for the electricity, gas, and water network.

Clicking an icon opens the general **Alarm Summary** or the **Alarm Summary** of the corresponding priority.

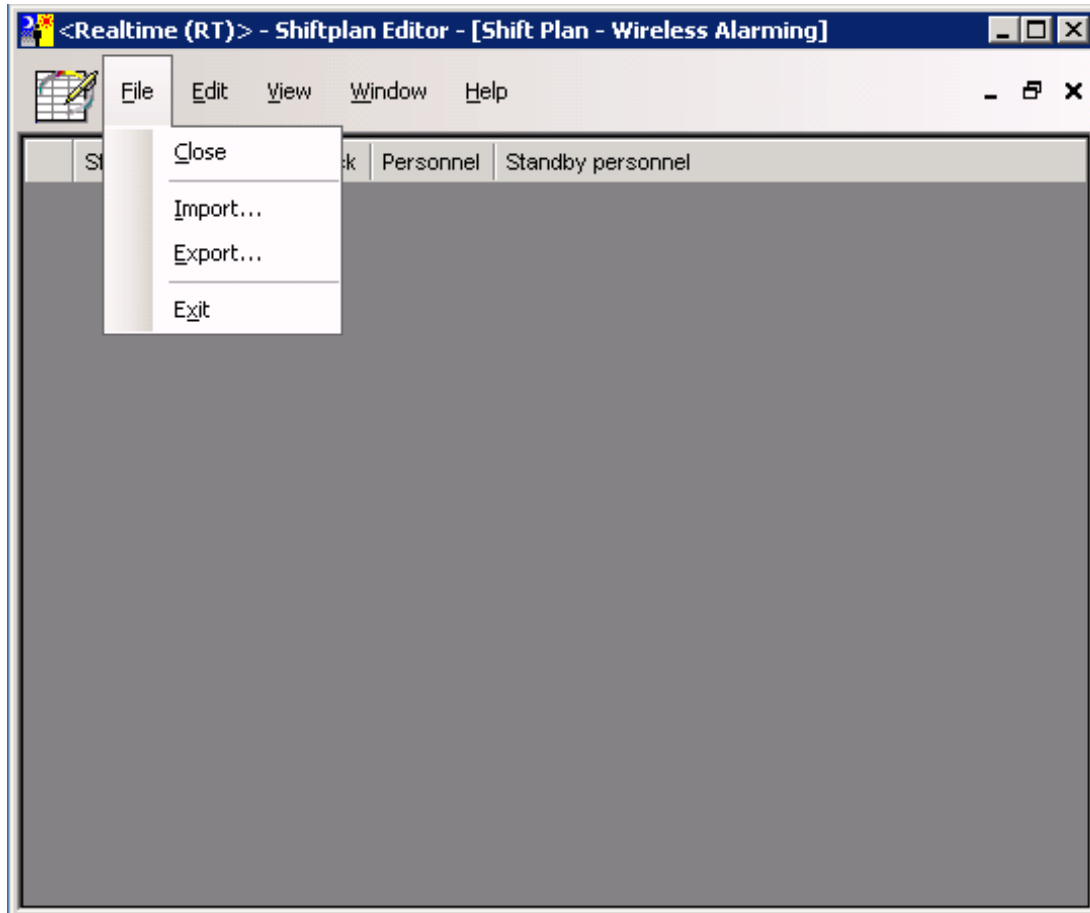
For example, the **Alarm Summary**, Prio 1.

Alarm Processing Buttons

- **Silence**
Clicking this icon switches off the audible alarming.
- **Audio Inhibit**
Clicking this icon cuts off the loudspeaker.
- **Audible Alarming**
Clicking this button opens the **Audible Alarming** menu. You can select **Audio Inhibit** and **Silence** to reactivate the alarm processing conditions. Additionally, you can select **Wireless Alarming** and **Central Audible Alarming** (Prio 1-3) to configure the alarm processing individually.

7.3.2 Shift Plan Editor

The **Shift Plan Editor** allows you to update the shift plan, and an overview of the shift teams, and the appropriate team members. Shift plans can be imported from or exported to Excel files.



[scaui003-031207-01.tif, 2, en_US]

Figure 7-3 Shift Plan Editor

Menu Bar

- **File**
 - **Close**
Closes the currently selected shift plan.
 - **Import**
Opens the **Import shift plan to...** dialog. Enter the path to import a shift plan.
 - **Export**
Opens the **Export shift plan to...** dialog. Enter the path to export a shift plan.
 - **Exit**
Closes the Shift Plan Editor.
- **Edit**
 - **Delete**
Deletes the currently chosen shift plan

- **View**
 - **Tasks**
Shows the dialog **Task** with the name and description of the task.
 - **Persons**
Shows the dialog **Persons** with the names and descriptions of the persons on duty.
 - **Refresh**
Refreshes the selected shift plan.
- **Window**
Shows window features like **Tile horizontally**, **Arrange windows**, for example.
- **Help**
 - **About..**
Opens the dialog **About Shiftplan editor**.

8 Alarm Lists and Summaries

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8.1 Introduction

The following summaries are provided for event/alarm monitoring:

- **General Summary**
- **Alarm Summary**
- **Abnormal Status Summary**
- **Test Summary**
- **Marker Summary**

A new event is entered into all alarm summaries that are configured for the respective message type and in case it differs from the normal state into the **Abnormal Status Summary**. If configured, a new event is entered into the message archive (in addition to the summary) and sorted according to the time stamp of the event.

Alarm summary includes Gas, Water, and Electricity. It is possible to configure upto 3 priorities.

8.2 Operational Steps

8.2.1 Displaying a Summary

The goal of the task is to display a message summary.
For example, the **General Summary**.

Displaying a Summary

- Click the appropriate button on the PAT to invoke the summary.
For example, **Electricity** summary.

The summary opens.

8.2.2 Calling a Display Using Summary Message

The goal of the task is to open the display that refers to the summary entry.
For example, the single-line diagram.

Calling a Display Using Summary Message

- Right-click the message in the summary.
- Select **Callup Display** from the context menu.

The corresponding display opens.



NOTE

If more than 1 display is attached to the message the **Display Search Results** dialog opens.
For additional information, refer to [4.3.6 Display Search Results Dialog](#).

8.2.3 Adding a Comment to the General Summary

The goal of this task is to add a comment to the **General Summary**.

Adding a Comment

- Right-click a status bar in the **General Summary**.
- Select **Add/View Comment** from the context menu.
The **Comment Dialog** opens.
- Enter the comment into the **New Comment** text box.
- Click **Add**.

The comment is added to the status bar in the **General Summary**.



NOTE

The maximum length of the comment is 500 characters. The current number of characters is displayed in the status bar of the dialog.

8.2.4 Handling Alarm Lists

The goal of this task is to acknowledge an alarm in the **Alarm Summary**.

Acknowledging an Alarm

- Right-click an alarm message entry.
- Select **Acknowledge** from the context menu.

Each acknowledgment is entered as an event into the **General Summary**.

For additional information on alarm acknowledgment, refer to [7.2.1 Handling an Incoming Alarm](#).

Multi Selection

In addition, you can acknowledge multiple alarms in the alarm summary in the following ways:

- Using multi selection
All selected alarms are acknowledged independent of whether they are visible or not.
- Using button acknowledge all alarms
All alarms in the list are acknowledged.
- Using button acknowledge all visible alarms
All visible alarms are acknowledged.

Deleting an Entry in the Alarm Summary

- Right-click an alarm message entry.
- Select **Clear** from the context menu.
- Alternatively, select **Delete** from the context menu.



NOTE

This menu item is only enabled, if you have modify access right for **System Administration**.
For additional information, refer to Reference Guide System Software (SYSW).

8.2.5 Handling Markers in the Marker Summary

The goal of this task is to handle markers in the **Marker Summary**.

Opening the Marker Summary

- To invoke the **Marker Summary**, click the arrow button next to the **Electricity - Summary** button and select **Markers**.

The **Marker Summary** opens.

Adding a Reminder to a Marker

- Select the marker by clicking the respective entry in the **Marker Summary**.
- Click  on the toolbar of the **Marker Summary**.
Alternatively, you can right-click the marker entry in the summary.
- Select **Edit Note** from the context menu.
The **Edit Note** dialog opens.
- Enter a **Title** for the note.
- Enter the note in the **Add Description** text box.
- Activate the **Reminder** check box.
- Select date, time, and **Periodicity** from the respective list boxes.
- Click **OK**.

Acknowledging an Expired Reminder

- Right-click the marker entry with the expired reminder in the summary.
- Select **Acknowledge Reminder** from the context menu.

The reminder disappears from the summary.

Adding a General Note to the Marker Summary

- click the  on the toolbar of the **Marker Summary**.
The **New Note Dialog** opens.
- Enter a **Title** for the note.
- Enter the note in the **Add Description** text box.
- If necessary, activate the **Reminder** check box.
- Select date, time, and **Periodicity** from the respective list boxes.
- Click **OK**.

The general note is added to the **Marker Summary**.

Adding a Comment to a Marker

- Click the respective entry in the **Marker Summary** to select the marker.
- Click the **Add/View comment** on the toolbar of the **Marker Summary** button.
Alternatively, right-click the marker entry in the summary.
- Select **Add/View Comment** from the context menu.
The **Comment Dialog** opens.
- Enter the comment in the **New Comment** text box.
- Click **Add**.

Deleting a Marker

- Click the respective entry in the **Marker Summary** to select the marker you want to delete.
- Click the  button on the toolbar of the **Marker Summary**.
Alternatively, right-click the marker in the summary.
- Select **Delete Marker** from the context menu.

The entry disappears from the **Marker Summary**.

8.2.6 Filtering Specific Messages in a Summary

The goal of this task is to define filter criteria for displaying specific messages in the corresponding summary.

Calling the Filter Dialog

- Click the  button in the toolbar of the corresponding message summary.
The **Filter Dialog** opens.
- Select **Filter > New** on the menu bar.

Defining Filter Criteria: Date and Source

- Select the data source (**Alarm System** or **Alarm Archive**) from the corresponding list box on the **Date and Source** tab,
- Define the **Range** by specifying a start date and time (**From**) and end date and time (**To**).
Alternatively, you can specify a time range regarding the last **x** days/hours/minutes.



NOTE

No date range restrictions are checked by default.

Defining Filter Criteria: Path

- On the **Path** tab, define 2 filter criteria for each path column.
- Select a logic (**And** or **Or**) to link the filter criteria for each path column.

Defining Filter Criteria: Advanced

- Open the **Advanced** tab for further filter configuration.
- Define criteria for category, priority, and message class from the corresponding list boxes.
Alternatively, you can enter the corresponding criteria into the text boxes **Value**, **Ind** (Indicator), **Comment**, **Status**, **User**, **Console**, **Limit**, **Unit**.
- Check **Yes** to search for acknowledged alarms or check **No** to search for unacknowledged alarms.

Saving and Activating the Filter

- Select **Filter > Save** to save the filter.
If no filter name is specified, the **Save Filter As** dialog opens.
- Enter the **Filter Name**.
- Click **Activate** to switch on the filter.



NOTE

The **Activate** button is only enabled, if the filter is switched off.

- Click **Apply**, to apply the new or changed filter to the summary.

The color of the table grid frame is yellow. The filter name is displayed in the status bar. If no filter name is specified, the text **Temporary filter** is displayed.



NOTE

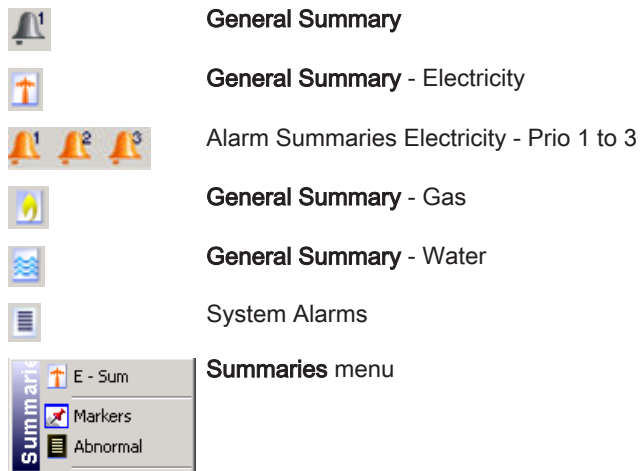
The **Apply** button is enabled as soon as a filter criterion field is changed.

8.3 User Interface Description

8.3.1 Summary Buttons on the PAT

The Power 5 Application Toolbar contains buttons for calling message summaries.
For example, the **General Summary**.

Buttons and the Summary Menu on the PAT



8.3.2 Toolbar of a Summary

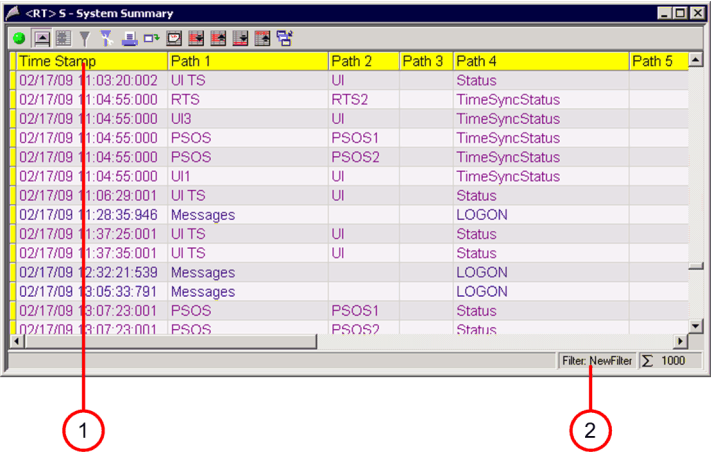
The toolbar of a summary provides quick access to frequently used functions.

Icons on an Alarm Summary Toolbar

-  Button to enable/disable summary updating:
Disabling allows to concentrate on a previous alarm condition without being distracted by more recent alarms.
-  Button to enable/disable auto scrolling:
Enabling of auto scroll pushes the summary to the begin/end position if a new alarm arrives.
-  Click this button after a sorting step has modified the layout, the initial layout can be established again.
-  Button **Filter is ON/OFF**:
If an ad-hoc filter is configured, the button is enabled. This allows you to switch on/off the configured ad-hoc filter.
-  Button **Configure filter**:
Clicking this button invokes a filter menu. You can enter a character string for the search. The use of wildcard characters is possible.
-  Button **Print summary**:
Print list or export list to MS Excel.
Printing of Alarm lists is supported on TS Web Session.
-  Button to display **Display Callup**:
The **Display Callup** dialog opens.
-  Button **Add/View comment**:
Clicking this button invokes a menu. You can view an existing comment or enter a new comment for the selected line. A new comment can be entered into the **Comment** column of the selected line.

-  Button to acknowledge selected alarms:
The acknowledgment indication (for example, **A**) in the second column is removed. The acknowledgment action is entered into the **General Summary**.
-  Button to acknowledge all visible alarms:
The acknowledgment indication of all visible lines is removed. The acknowledgment action is entered into the **General Summary** with all the lines that have been acknowledged.
-  Button to acknowledge all alarms:
The acknowledgment of all alarms in the alarm list. The acknowledgment action is entered into the **General Summary**.
-  Buttons to step up/down to the next not acknowledged line.
-  Button **Clear summary**:
Clears the content of the **Alarm Summary**. This item is assigned to the system administrator in case of alarm orphans.
-  Buttons for page down/up
-  Buttons to step to the end/top:
Go to the end or beginning of the summary.
-  Button **Arrange/close window**:
The function opens a dialog with a list of all summaries and single-line diagrams currently open. Selecting an item of the list brings the respective summary to the front. If many diagrams and summaries are open, you can click the icon, select all diagrams and open a context menu. Select the **Close**, **Cascade**, **Tile vertically** and **Tile horizontally** function to arrange the diagrams.

8.3.3 General Summary



[scams001-250209-01.tif, 1, en_US]

Figure 8-1 General Summary

- 1 Yellow filter indicates that a filter is active or that the archive is the data source
- 2 Filter name

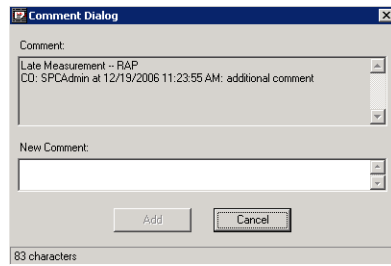
Right-clicking a status bar opens the **General Summary** context menu with the following items:

Display Callup

Opens the appropriate display, for additional information on the **Display Callup**, refer to [4.2.10 Using the Display Search Result Dialog](#).

Add/View Comment

Opens the **Comment** dialog:



[scams003-051207-01.tif, 1, en_US]

Figure 8-2 Comment Dialog

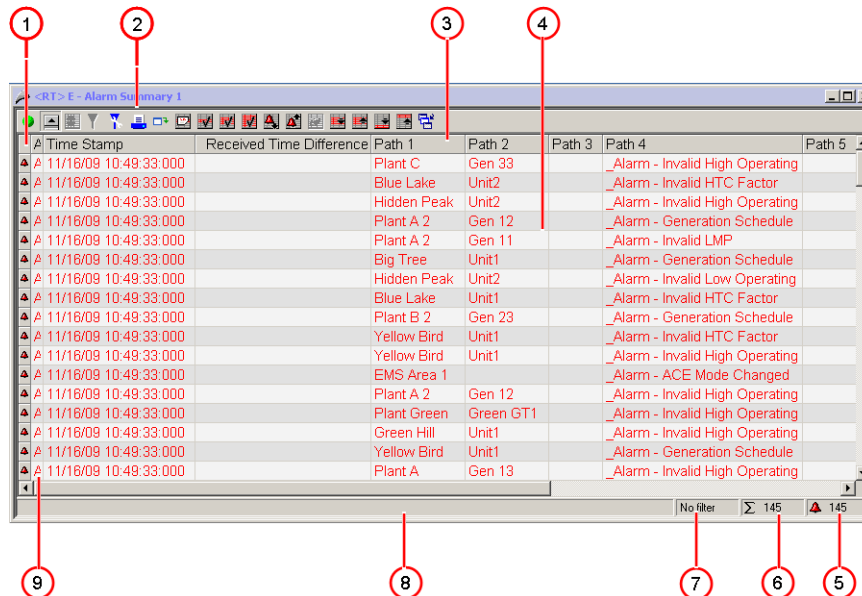
- **Text Box: Comment**
Displays all user comments.
- **Text Box: New Comment**
You can enter a new comment into this text box. The maximum length of the comment is 500 characters. The current number of characters is displayed in the status bar of the dialog.
- **Button: Add**
Adds the entered comment to the selected message. An indication string, for example, **CO**, the name of the user as well as the current time stamp is automatically added to the new comment.



NOTE

Comments can only be added, if you have modify access right for Network Management.

8.3.4 Alarm Summary



[rguios30-031209-01.tif, 1, en_US]

- (1) Icon for acknowledged alarms
- (2) Toolbar
- (3) Pathname

- (4) Grid
- (5) Number of acknowledged alarms
- (6) Total number of alarms
- (7) Filter
- (8) Status bar
- (9) String for acknowledged alarm

Right-clicking a status bar opens the **Alarm Summary** context menu with the following items:

Acknowledge

Acknowledges the corresponding message entry.

Display Callup

Opens the appropriate display, for additional information on the **Display Callup**, refer to [4.2.10 Using the Display Search Result Dialog](#).

Add/View Comment

For additional information on comments, refer to [8.3.3 General Summary](#).

Copy

Copies the corresponding alarm message entry to the clipboard.

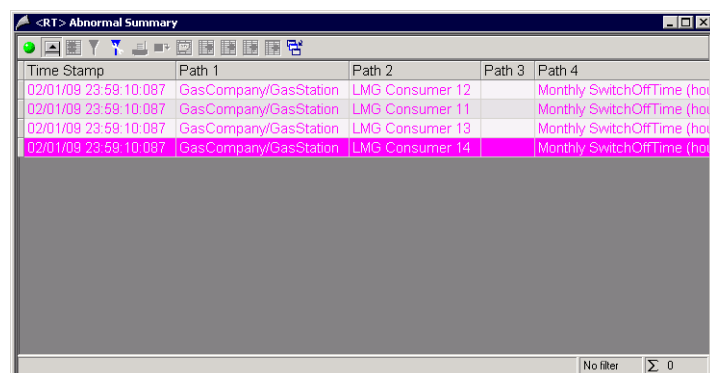
Delete

Removes the corresponding alarm message entry. This menu item is enabled, if you have modify access right for **System Administration**. For additional information on access rights, refer to Reference Guide System Software (SYSW).

8.3.5 Abnormal State Summary

The **Abnormal State Summary** lists the following abnormal states:

- For example, alarm state of general events, protection devices, housekeeping events
- Abnormal switch position
- Analog values out of normal range as defined by the operating limits



Time Stamp	Path 1	Path 2	Path 3	Path 4
02/01/09 23:59:10:087	GasCompany/GasStation	LMG Consumer 12		Monthly SwitchOffTime (ho
02/01/09 23:59:10:087	GasCompany/GasStation	LMG Consumer 11		Monthly SwitchOffTime (ho
02/01/09 23:59:10:087	GasCompany/GasStation	LMG Consumer 13		Monthly SwitchOffTime (ho
02/01/09 23:59:10:087	GasCompany/GasStation	LMG Consumer 14		Monthly SwitchOffTime (ho

[scad0008-190209-01.tif, 1, en_US]

Figure 8-3 Abnormal Status Summary

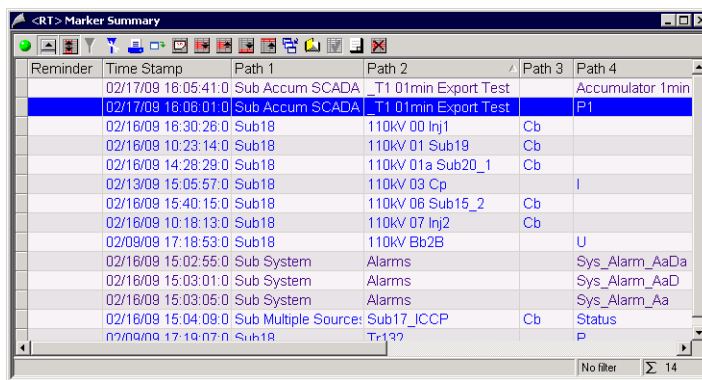
8.3.6 Marker Summary

The layout of the **Marker Summary** resembles the layout of an **Alarm Summary**. The **Marker Summary** has additional columns and toolbar buttons. The following additional columns are available:

- Description
- History
- Reminder
- Title

For the handling of markers, the following additional toolbar buttons are provided:

- Edit a note
- Acknowledge an expired reminder
- Create a general note
- Delete a marker



Reminder	Time Stamp	Path 1	Path 2	Path 3	Path 4
	02/17/09 16:05:41:0	Sub Accum SCADA	T1 01min Export Test		Accumulator 1min
	02/17/09 18:06:01:0	Sub Accum SCADA	T1 01min Export Test		P1
	02/16/09 18:30:26:0	Sub18	110kV 00 Inj1	Cb	
	02/16/09 10:23:14:0	Sub18	110kV 01 Sub19	Cb	
	02/16/09 14:28:29:0	Sub18	110kV 01a Sub20_1	Cb	
	02/13/09 15:05:57:0	Sub18	110kV 03 Cp	I	
	02/16/09 15:40:15:0	Sub18	110kV 06 Sub15_2	Cb	
	02/16/09 10:18:13:0	Sub18	110kV 07 Inj2	Cb	
	02/09/09 17:18:53:0	Sub18	110kV Bb2B		U
	02/16/09 15:02:55:0	Sub System	Alarms		Sys_Alarm_AaDa
	02/16/09 15:03:01:0	Sub System	Alarms		Sys_Alarm_AaD
	02/16/09 15:03:05:0	Sub System	Alarms		Sys_Alarm_Aa
	02/16/09 15:04:09:0	Sub Multiple Source:	Sub17_ICCP	Cb	Status
	02/16/09 17:19:07:0	Sub18	Tr1??		P

[scare014-260209-02.tif, 1, en_US]

Figure 8-4 Marker Summary

Right-clicking a line in the **Marker Summary** opens a context menu with the following items:

Display Callup

Opens the appropriate display, for additional information on the **Display Callup**, refer to [4.2.10 Using the Display Search Result Dialog](#).

Add/View Comment

For additional information on comments, refer to [8.3.3 General Summary](#).

Copy

Copies the selected line.

Edit Note

Opens the **New Note** dialog where the note of the selected marker can be edited. For additional information on notes, refer to [10.3.2 New Note Dialog](#).

Acknowledge Reminder

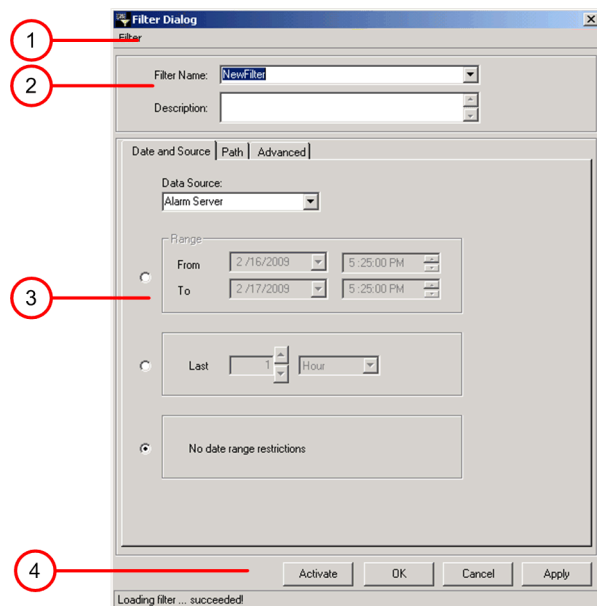
Acknowledges an expired reminder of a marker.

Delete Marker

Deletes the selected marker.

8.3.7 Filter Configuration Dialog

In the **Filter** configuration dialog, you can define filter criteria to display only specific messages in the corresponding summary.



[scams010-260209-02.tif, 1, en_US]

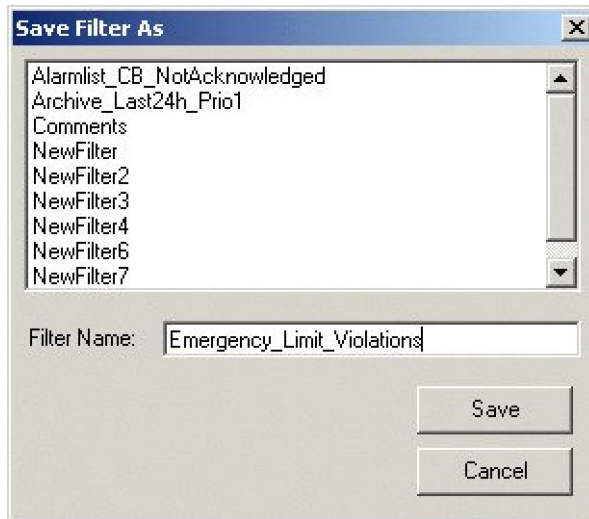
Figure 8-5 Filter Configuration Dialog

- (1) Menu bar
- (2) Header
- (3) Tabs
- (4) Buttons

Menu Bar

- **New**
Creates a filter
The selected filter is cleared and a default name **New Filter** is inserted into the list box **Filter Name**.

- **Save**
Saves the current filter
If no filter name is specified, the **Save Filter As** dialog opens.



[scams011-051207-01.tif, 1, en_US]

Figure 8-6 Save Filter As Dialog

You can enter a filter name into the text box **Filter Name**. After clicking the button **Save**, the filter is saved with the specified filter name.

- **Save As...**
Opens the **Save Filter As** dialog
- **Delete**
Clears the current filter and removes the corresponding alarm control filter instance
After the menu item **Delete** was selected, a confirmation dialog opens asking you to confirm the deletion process.
- **Clear**
Resets all filter criteria fields

Header

- **File Name**
Your can select a filter from this list box. When creating a filter, you can enter the filter name into this field.
- **Description**
You can enter a comment about the specified filter into this text box.

Tabs

- **Date and Source tab**

The screenshot shows a 'Filter Dialog' window with a 'Filter' section at the top containing 'Filter Name' (set to 'NewFilter') and 'Description'. Below this are three tabs: 'Date and Source' (selected), 'Path', and 'Advanced'. The 'Date and Source' tab contains a 'Data Source' dropdown menu set to 'Alarm Server'. Underneath is a 'Range' section with three radio button options: 'From' (with date and time pickers for 2/16/2009 and 5:25:00 PM), 'To' (with date and time pickers for 2/17/2009 and 5:25:00 PM), and 'Last' (with a numeric spinner set to 1 and a unit dropdown set to 'Hour'). A fourth radio button option, 'No date range restrictions', is also present. At the bottom of the dialog are 'Activate', 'OK', 'Cancel', and 'Apply' buttons. A status bar at the very bottom indicates 'Loading filter ... succeeded!'.

[scams012-250209-01.tif, 1, en_US]

Figure 8-7 Date and Source Tab

- **Data Source**
You can select the data source (alarm system or alarm archive) from this list box.
After selecting the data source, you can select one of the following time range restriction to display messages within a specified time range.
- **Range From/To**
You can specify the start date/time and the end date/time.
- **Last**
You can specify the last specified number of days/hours/minutes.
- **No date range restrictions**
There are no time restrictions. This is the default setting.

- **Path tab**

The screenshot shows a 'Filter Dialog' window with a 'Path' tab selected. At the top, there is a 'Filter Name' dropdown menu set to 'NewFilter' and a 'Description' text box. Below these are three tabs: 'Date and Source', 'Path', and 'Advanced'. The 'Path' tab contains five identical sections, each labeled 'Path 1' through 'Path 5'. Each section has two text input fields and two radio buttons labeled 'And' and 'Or'. The 'Or' radio button is selected in all sections. At the bottom of the dialog are four buttons: 'Activate', 'OK', 'Cancel', and 'Apply'.

[scams013-250209-01.tif, 1, en_US]

Figure 8-8 Path tab

For each path column, 2 filter criteria can be defined. The filter criteria can be linked by a logic **And** or **Or**.

In this example, you can search for all messages of the substation name **sub a** or **sub b**, where the bay name contains the strings **trafo** or **a2**.

- **Advanced tab**

[scams014-250209-01.tif, 1, en_US]

Figure 8-9 Advanced tab

- **Cat/Prio/MsgCl**
You can select corresponding criteria for category, priority, and message class from the list boxes. Multiple items can be selected from 1 list.
- **Value, Ind (Indicator), Comment, Status, User, Console, Limit, Unit**
You can enter corresponding criteria into these text boxes.
- **Acknowledged**
You can search for acknowledged alarms (**Yes** is checked) or not acknowledged alarms (**No** is checked).

Buttons

- **Activate**
Applies the filter configuration changes. After clicking the button, the **Filter configuration** dialog is closed and the appropriate **Filter On/Off** toolbar button is switched on.
The button **Activate** is only enabled, when the filter is switched off.
- **OK**
Applies the filter configuration changes and closes the **Filter configuration** dialog.
The button **OK** is enabled as soon as a filter criterion field has changed.
- **Apply**
Applies the filter configuration changes and applies the new filter criteria to the summary, if the filter is already switched on.
The button **Apply** is enabled as soon as a filter criterion field has changed.
The following effects in a message summary visualize an activated filter:
 - The **Filter On/Off** toolbar button is pressed.
 - The color of the table grid frame is yellow.
 - The filter name is displayed in the status bar. If no filter name is specified, the text **Temporary filter** is displayed.
 - If the message summary shows data from the alarm archive, the additional archive icon is displayed in the right corner of the status bar.

9 Supervisory Control

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9.1 Introduction

Supervisory control is done on a selected object within a graphical display (single-line diagram) or a tabular display (Runtime Explorer).

Depending on the selected network component, a certain dialog is used for supervisory control.

Each control action is monitored for time-out. If the target is not reached in time, the supervisory control application stops the control sequence with a hint.

Supervisory control considers the following special cases:

- The current switch position can be sent out as the desired control target.
- A set point can be sent.
- Step position commands can be sent to tap position devices.

For example, transformers.

9.2 Operational Steps

9.2.1 Calling the Control State Dialog

The goal of this task is to operate an equipment. There are 2 alternative ways to open the **Control State** dialog:



NOTE

The operations described in this document present only 1 way to call up the Control State dialog. You can select a different alternative.

Calling the Control State Dialog Using a Single-line Diagram

- Right-click the switch device that you intend to open or close and select **Control State** from the context menu in a single-line diagram.

Calling the Control State Dialog Using the Runtime Explorer

- Select **Tools > Runtime Explorer** from the PAT.
- Select the substation from the tree view.
- Right-click the device in the list.
- Select **Control State** from the context menu.

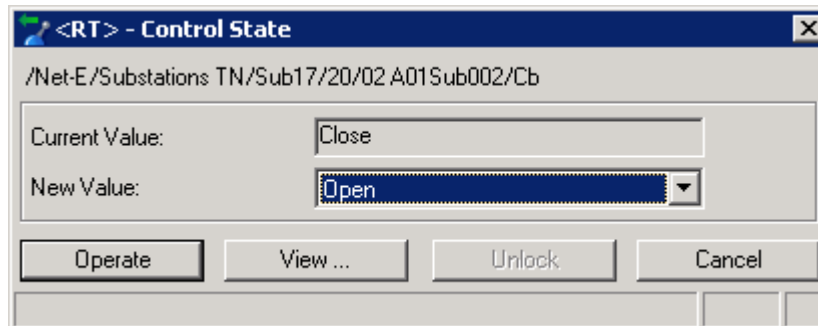
9.2.2 Controlling the State of a Switch

The goal of this task is to open or close a switch.
For example, a circuit-breaker.

Controlling the State of a Switch

- Right-click switch whose states are going to change in the single-line diagram.
- Select **Control State** from the context menu.

The **Control State** dialog opens.



[sc_Circuit-breaker with Assigned Control Dialog, 1, en_US]

Figure 9-1 Circuit-breaker with Assigned Control Dialog

- Select the state **Open** from the list box **Control to** to open a closed switch.
- Select the state **Closed** from the list box **Control to** to close an open switch.
- Click **Operate**.

The state of the selected switch device changes accordingly.



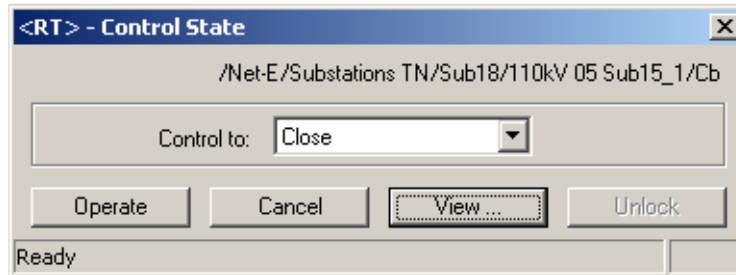
NOTE

Interlocking conditions can block the control of network components.
For additional information, refer to [9.3.5 Control State Dialog - Interlocking Conditions](#).

9.3 User Interface Description

9.3.1 Control State Dialog

The list box **Control to** shows the expected status, which usually is the opposite of the current switch position.



[scacd004-051207-01.tif, 1, en_US]

Figure 9-2 Switching Command Control Dialog

List Box: Control to

The following states are selectable:

- Close
- Open

Button: Operate

Clicking this button applies the selected state to the device in the single-line diagram.

Button: View

Clicking this button opens a menu with 3 items:

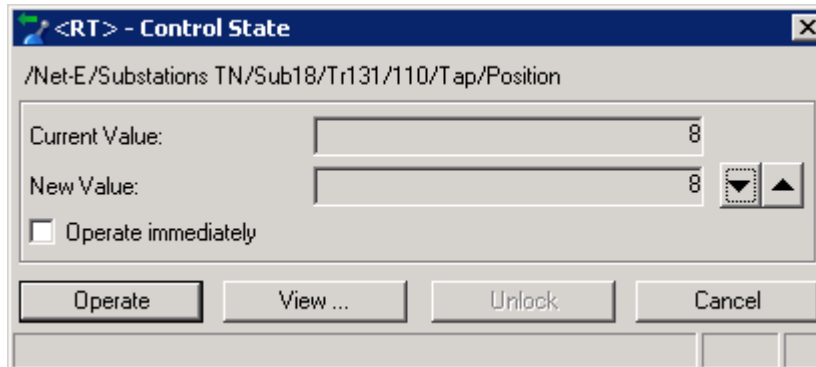
- **Interlock Violations**
Show all interlock violations
- **Interlock Rules**
Show all interlock rules
- **Hide Interlock/Violations**
Only active, if **Interlock Violations** or **Interlock Rules** is active
Selecting this item closes the display of **Interlock Violations** or **Interlock Rules**.

Button: Unlock

Only active if the check box of an interlocking violation or rule is marked. Clicking this button disables the interlock rules and violations. The violated condition has to be configured as a soft condition.

9.3.2 Control State Dialog - Step Position Control

An alternative design of the **Control State Dialog**, for example when changing the current value of a transformer.



[sc_Step updown Control Dialog, 1, en_US]

Figure 9-3 Step up/down Control Dialog

Text Box Current and Arrow Buttons to the Right

Shows the value of the device, the arrow buttons are used to increase/decrease the value. The button between the arrow button resets the changed value.

Check Box: Operate Immediately

Marking this check box enables you to increase/decrease the current value of the transformer stepwise with the arrow buttons, without clicking the button **Operate**.

Button: Operate

Clicking this button applies the selected state to the device in the single-line diagram.

Button: View ...

Clicking this button opens a menu with 3 items:

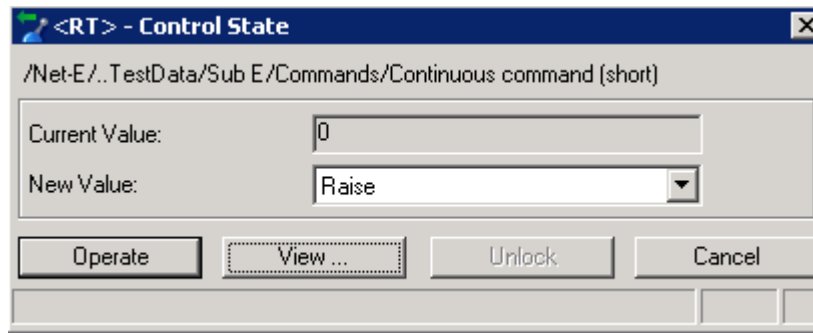
- **Interlock Violations**
Show all interlock violations
- **Interlock Rules**
Show all interlock rules
- **Hide Interlock/Violations**
Only active, if **Interlock Violations** or **Interlock Rules** is active
Selecting this item closes the display of **Interlock Violations** or **Interlock Rules**.

Button: Unlock

Only active if the check box of an interlocking violation or rule is marked. Clicking this button disables the interlock rules and violations. The violated condition has to be configured as a soft condition.

9.3.3 Continuous Command Dialog

This dialog changes the state of a device. For example, the current value of a sliding controller.



[sc_Continuous Command Dialog, 1, en_US]

Figure 9-4 Continuous Command Dialog

Button: Lower

Clicking this button lowers the current state of the device.

Button: Raise

Clicking this button raises the current state of the device.

Button: View...

Clicking this button opens a menu with 3 items:

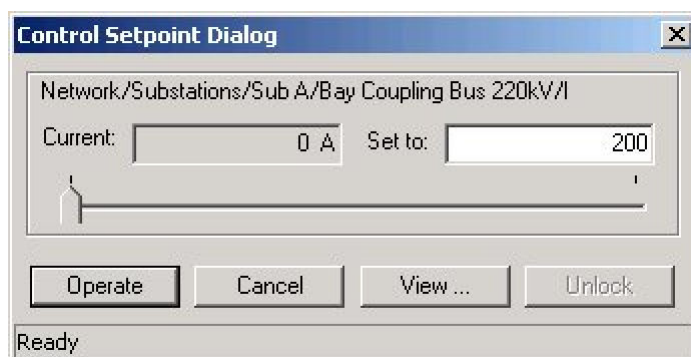
- **Interlock Violations**
Show all interlock violations
- **Interlock Rules**
Show all interlock rules
- **Hide Interlock/Violations**
Only active, if **Interlock Violations** or **Interlock Rules** is active
Selecting this item closes the display of **Interlock Violations** or **Interlock Rules**.

Button: Unlock

Only active if the check box of an interlocking violation or rule is marked. Clicking this button disables the interlock rules and violations. The violated condition has to be configured as a soft condition.

9.3.4 Control Setpoint Dialog

This dialog sets the set point value for the selected device.



[scacd007-051207-01.tif, 1, en_US]

Figure 9-5 Set Point Control Dialog

Text Box: Set to

You can enter the value of the set point.

Button: Operate

Clicking this button applies the selected value to the device in the single-line diagram.

Check box: Set Blocked Marker

Enabling this check box applies a blocked marker to prevent updating from the process.

Button: View...

Clicking this button opens a menu with 3 items:

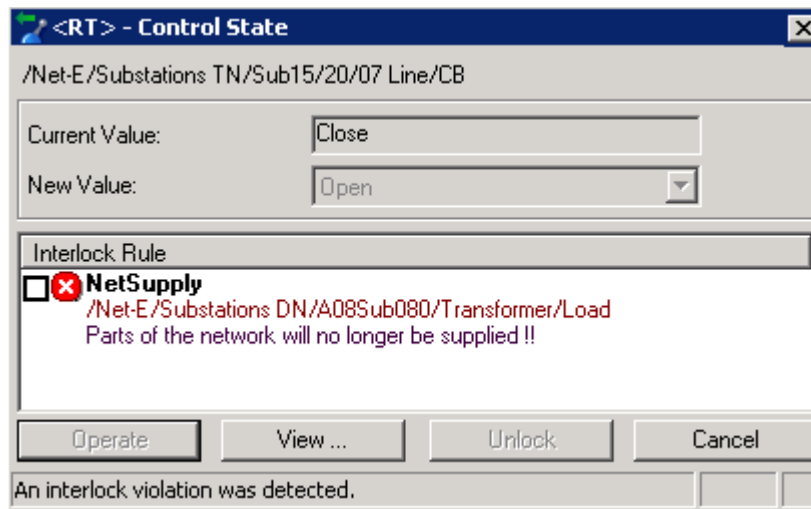
- **Interlock Violations**
Show all interlock violations
- **Interlock Rules**
Show all interlock rules
- **Hide Interlock/Violations**
Only active, if **Interlock Violations** or **Interlock Rules** is active
Selecting this item closes the display of **Interlock Violations** or **Interlock Rules**.

Button: Unlock

Only active if the check box of an interlocking violation or rule is marked. Clicking this button disables the interlock rules and violations. The violated condition has to be configured as a soft condition.

9.3.5 Control State Dialog - Interlocking Conditions

During a supervisory action, all general, topological, and special interlocking conditions are checked. If at least 1 condition is violated, the next step is rejected, and the **Interlocking Condition** dialog opens:



[sc_Interlocking Conditions Dialog, 1, en_US]

Figure 9-6 Interlocking Conditions Dialog

List Box: Control to

The following states are selectable:

- Close
- Open

Button: Operate

Applies the selected state to the device in the single-line diagram.

Button: View

Clicking this button opens a menu with 3 items:

- **Interlock Violations**
Show all interlock violations
- **Interlock Rules**
Show all interlock rules
- **Hide Interlock/Violations**
Only active, if **Interlock Violations** or **Interlock Rules** is active
Selecting this item closes the display of **Interlock Violations** or **Interlock Rules**.

Button: Unlock

Only active if the check box of an interlocking violation or rule is marked. Clicking this button disables the interlock rules and violations. The violated condition has to be configured as a soft condition.

10 Marking and Tagging

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10.1 Introduction

Markers and tags identify network components for special processing. For example, network components can be switches, lines, data points, protection devices. Any marker or tag is attached to a presentation object in the corresponding tabular or graphical displays.

Applying and removing of a marker is entered into the **General Summary** and the message archive. The message includes the acronym of the user.

Marker types named tags are considered as an interlocking condition for supervisory control.

10.2 Operational Steps

10.2.1 Setting a Marker with Reminder Function

The goal of this task is to set a marker with a note and reminder for a switch device.

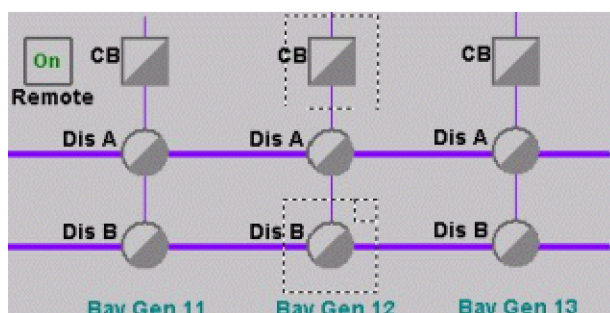
Setting a Marker

- Right-click the switch device to mark in the single-line diagram.



NOTE

Applying a marker is possible for more than 1 marker at a time. The marker refers to all selected objects.

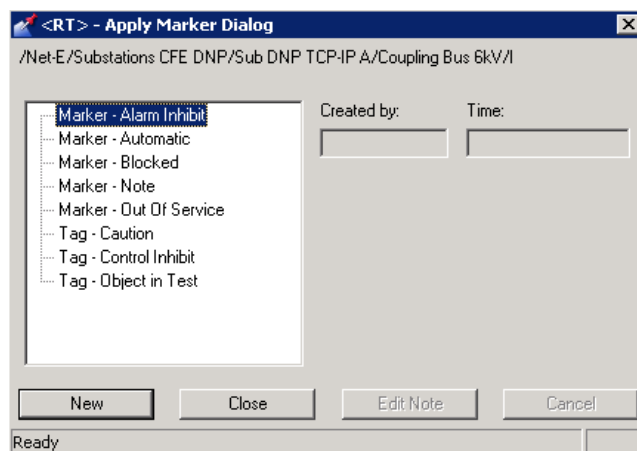


[scamd005-051207-01.tif, 1, en_US]

Figure 10-1 Multiple Object Selection

- Select **Apply Marker** from the context menu.

The **Apply Marker** dialog opens.



[rguio014-280610-02.tif, 1, en_US]

Figure 10-2 Apply Marker Dialog

- Select a marker.
- Click **New**.
- Click **Apply**.

The marker is applied to the selected object.



NOTE

You can set a marker with reminder function in the Runtime Explorer.



NOTE

The items **Marker - Automatic** and **Marker - Out of Service** are only used for service configurations.



NOTE

The markers **Alarm Inhibit**, **Object in Test**, and **Blocked** are applied on RTUs also.

Adding a Note and Setting a Reminder

- Click **Edit Note** on the **Apply Marker** dialog.

The **New Note** dialog opens.

The screenshot shows the 'New Note Dialog' window. It contains the following fields and controls:

- Path:** A text box containing the path `/Net-E/Substations TN/Sub15/110/07 Cp/GrDis 2`.
- Title:** An empty text box.
- Reminder:** A checkbox that is checked.
- Date:** A date picker showing 2/17/2009.
- Time:** A time picker showing 5:34:18 PM.
- Periodicity:** A numeric input field showing 0.
- Unit:** A dropdown menu currently set to 'Minute'.
- Attachment:** A text box with a browse button.
- Description:** A large text area with a 'Description' label and a 'Description' button.
- Add Description:** A text area with an 'Add Description' label and an 'Add Description' button.
- History:** A text area with a 'History' label and a 'History' button.
- Buttons:** 'Ok' and 'Cancel' buttons at the bottom.

[scamd002-250209-01.tif, 1, en_US]

Figure 10-3 New Note Dialog

- In the **New Note** dialog, enter a note title and a description.
 - If necessary, enable the check box **Reminder** and select the date and **Periodicity**.
 - Click **OK**.
-



NOTE

The **Marker Summary** displays all currently set markers.

10.2.2 Deleting a Marker

The goal of this task is to delete a marker of a switch device.

Deleting a Marker Using the Single-line Diagram

- Right-click the switch device whose marker you want to delete in the single-line diagram.
- Select **Apply Marker** from the context menu.
The **Apply Marker Dialog** opens.
- On the **Apply Marker** dialog, click **Delete**.
- Click **Apply**.

Deleting a Marker Using the Marker Summary

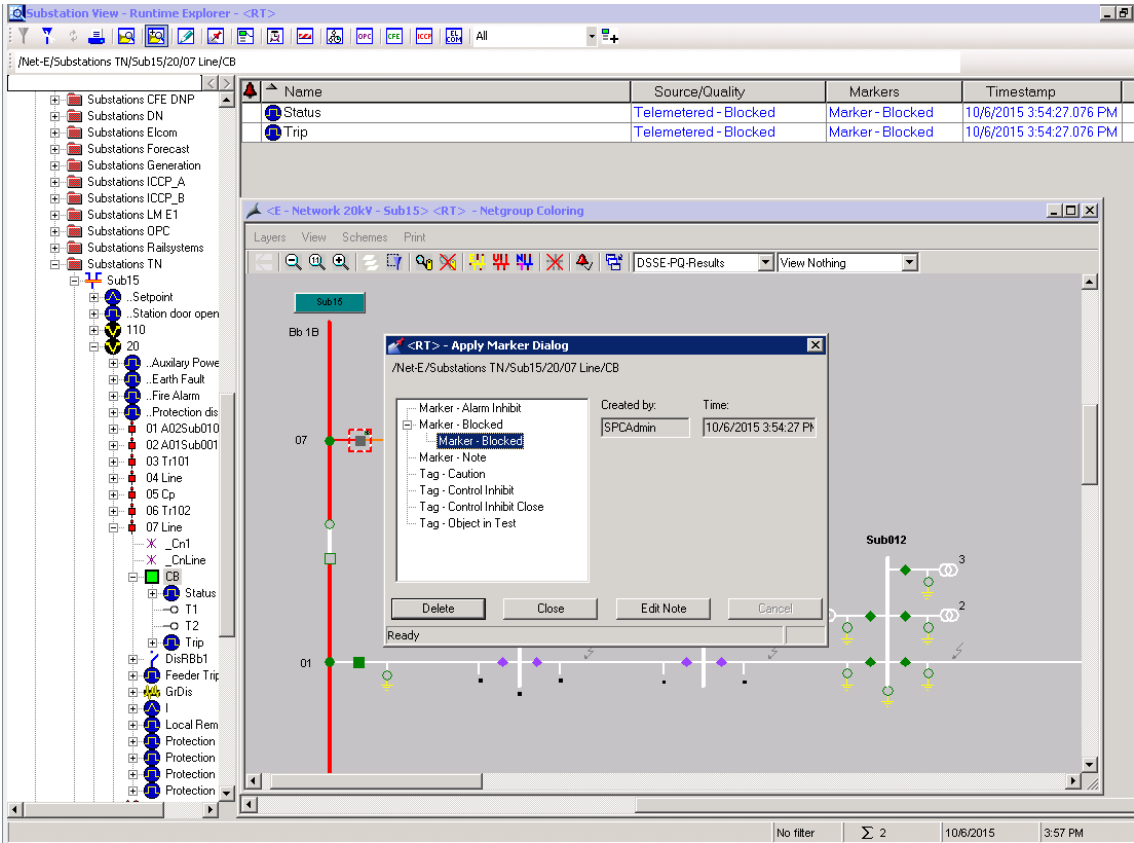
- Click the arrow to the right of the **General Summary** button on the PAT.
- Select **Markers** from the **Summaries** menu.
- Right-click the marker entry you want to delete in the **Marker Summary**.
- Select **Delete Marker** from the context menu.

The corresponding **Marker Summary** entry is deleted.

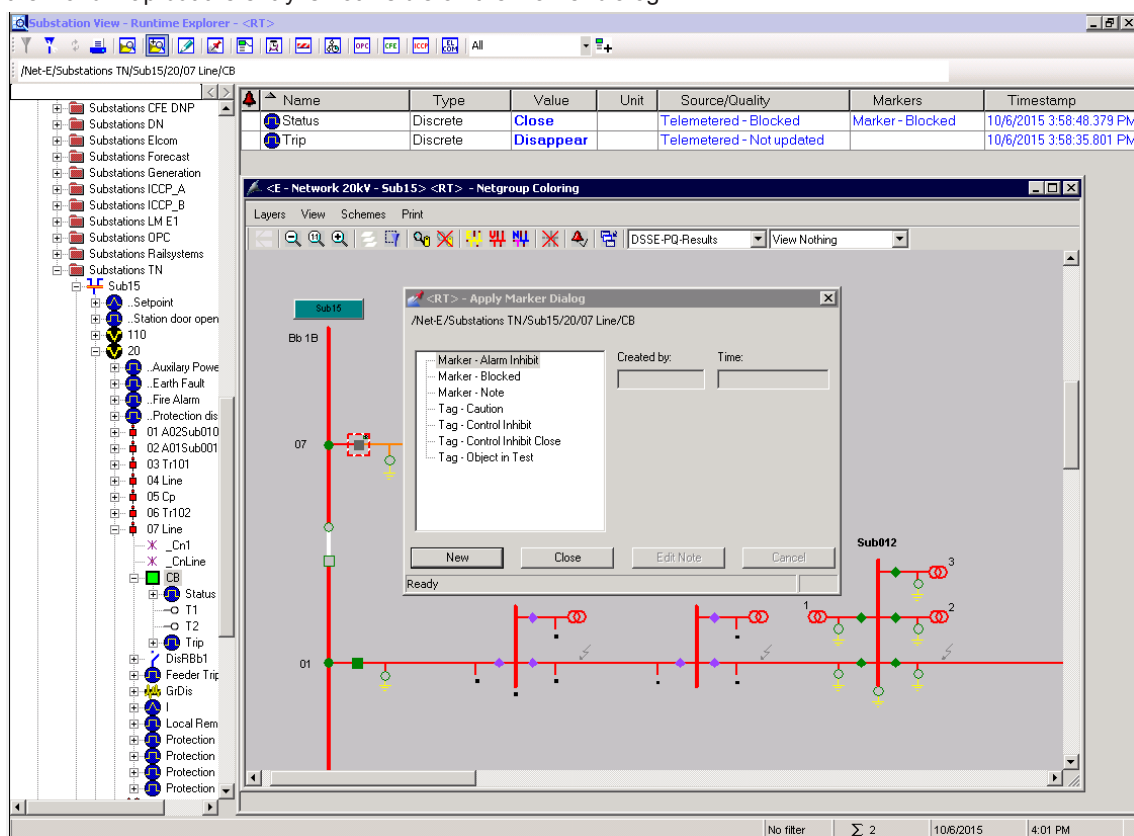


NOTE

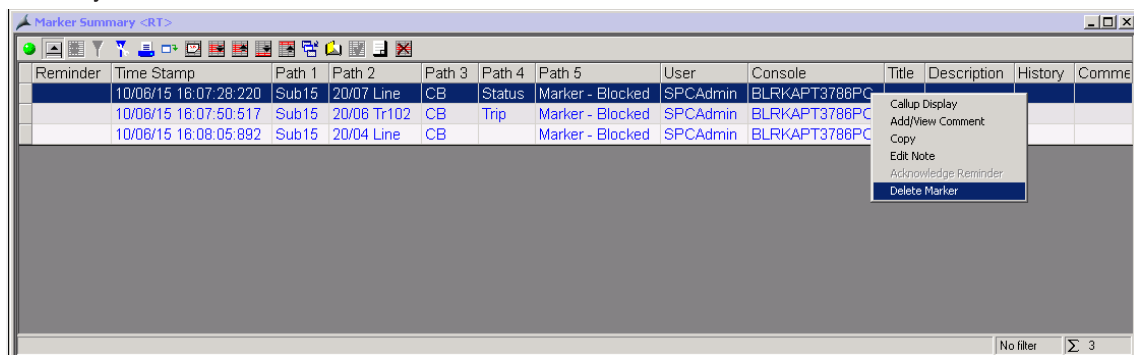
Markers are allowed to be deleted from the same hierarchy at which they were inserted. For example, if a marker is inserted at the Field Resource (for example, Circuit Breaker) level, it can be viewed for the inherited status measurements in Runtime explorer, but cannot be deleted.



If it is inserted at the measurement level (for example, Status) in Runtime explorer, an indication is seen in the world map but the entry is not visible on the Marker dialog.



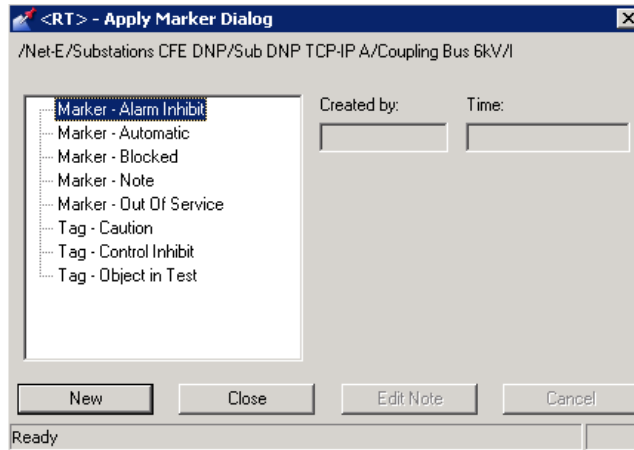
However, all the markers entered can be seen in the **Marker Summary** and can be deleted from this summary view.



10.3 User Interface Description

10.3.1 Apply Marker Dialog

The **Apply Marker Dialog** enables you to select an existing marker or to create a new one. Additional information can be entered for each marker.



[rguio015-280610-02.tif, 1, en_US]

Figure 10-4 Apply Marker Dialog

Button: Edit Note

Clicking this button opens the **New Note Dialog**. Here, you can enter a note.

Button: Delete

Clicking this button deletes the selected marker.

Button: Close

Clicking this button closes the dialog.

List Box with Tags and Markers

- **Alarm Inhibit** (Marker)
The data point still updates all summaries, network diagrams, databases, and archive, however does not raise an alarm condition and is not entered into any **Alarm Summary**.
- **Blocked** (removed from operation - Marker)
The data point does not update any network diagram, database, or summary.
- **Note** (Marker)
Note attached to a field device (technological note) or general note not attached to a field device.
- **Caution** (Tag)
This device seems to be suspicious. This tag serves as an interlocking condition for supervisory control.
- **Control Inhibit** (Tag)
This tag serves as an interlocking condition for supervisory control (all directions).
- **Object in Test** (Tag)
The data point is bypassed around the main system, does not raise an alarm condition, and updates a **Test Summary**.
- **Control Inhibit Close** (Tag)
This tag serves as an interlocking condition for supervisory control (only the direction close).

10.3.2 New Note Dialog

The **New Note** dialog shows the history of the marker (when it was created and modified) as well as the previously entered descriptions.

[scamd002-250209-01.tif, 1, en_US]

Figure 10-5 New Note Dialog

Text Box: Path

The path of the selected device is displayed.

Text Box: Title

You can enter the title of the note.

Text Box: Attachment

Click the button to the right of the text box, to select a file and to attach it to the new note.

Text Box: Description

The text box displays current and previous descriptions for this marker.

Check Box: Reminder

You can set a reminder using the corresponding list boxes.

Text Box: Add Description

Use the text box for an additional description.

Text Box: History

The text box displays the history of changes. It specifies when the marker was created and modified.

11 Manual Updating

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11.1 Introduction

A manual update is only allowed for invalid values (in opposition to manual override). A manually updated value is overwritten by the real field value as soon as data acquisition for this data point starts and supplies a valid value. A manually updated value is not removed from operation automatically. Manual update is possible for more than 1 object at a time, the update dialogs are applied on all selected objects.

The manual override function updates the analog or digital value, and implicitly applies the marker **Blocked** (removed from operation) to the object. If the real-time data acquisition still works for this object, the real field value does not automatically update this object. The manual value is kept until you remove the marker.

11.2 Operational Steps

11.2.1 Manually Updating Values

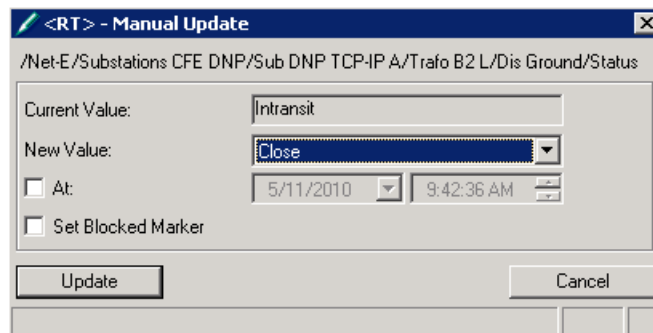
The goal of the task is to update measured values of a field resource manually.

Manually Updating a Value

The following 2 methods are used to update measured values manually:

- In the single-line diagram, right-click the object to update.
Alternatively, in the Runtime Explorer, right-click the measurement you want to update manually.
- Select **Manual Update** from the context menu.

The **Manual Update** dialog opens.



[rguio009-290610-01.tif, 1, en_US]

Figure 11-1 Manual Update Dialog



NOTE

There is a different dialog for an analog and digital values (refer to [11.3.1 Manual Update Dialog](#)).

- Select a value from the list box **New Value**.
- Select the **At** check box.
- Select the required date and time.
- Select **Set Blocked Marker**
- Click **Update**.

The switch device is updated with the new values. The manual updating activity is entered into the **General Summary** and the message archive.

Manually Updating a Value - Toggle Option

- Right-click the device you want to update manually in the single-line diagram.
Right-click the measurement you want to update manually alternatively, on the Runtime Explorer.
- Select **Manual Update Toggle**.
The switch device changes the state immediately to the opposite one.

The switch device is updated with the new values.



NOTE

If the current state of the selected element is **Disturbed** or **Intransit** or manual update is not allowed, the corresponding manual update display is opened as described in the preceding sections.

11.2.2 Manually Overriding Values

The goal of this task is to override the measured value of a field resource manually.

Manually Overriding Switch Device Values

- Right-click the object you want to override manually in the single-line diagram.
Right-click the measurement you want to override manually alternatively, in the Runtime Explorer.
- Select the menu item **Manual Override** from the context menu.
The **Manual Override** dialog opens.
- Select an item from the list box **Update to**.
- Enter the update time in the text box **Update Time**.
- Click **Update**.

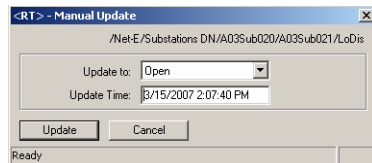
The switch device is updated with the new values.

11.3 User Interface Description

11.3.1 Manual Update Dialog

The **Manual Update** dialog enables you to update a value manually.

Manual Update of Digital Values

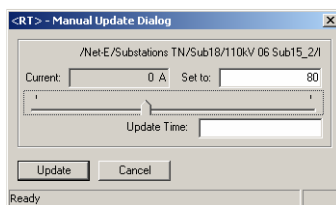


[scamu001-051207-01.tif, 1, en_US]

Figure 11-2 Digital Value Manual Update Dialog

- List box: **Update to**
The following states are selectable:
 - **Open**
 - **Close**
 - **Disturbed**
 - **Intransit**
- Text box: **Update Time**
Here you can enter the update time.
- Button: **Update**
Clicking this button updates the value.

Manual Update of Analog Values



[scamu002-051207-01.tif, 1, en_US]

Figure 11-3 Analog Value Manual Update Dialog

- Text box: **Current**
Showing the current value.
- Text box: **Set to**
You can enter the desired value.
- Text box: **Update Time**
Here you can enter the update time.
- Check box: **Set Blocked Marker**
Enabling this check box applies a blocked marker to prevent updating from the process.
- Button: **Update**
Clicking this button updates the value.

The current system time is assumed as the update time of the status or value, otherwise enter the value to the text box **Update Time**.

12 Archive Replay

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12.1 Introduction

In a study, you can create a replay case by defining a time interval in the past. All digital values that are stored in the archive for that time interval are automatically added to this case. You can also add selected analog values to the replay case.

Those replay cases can be stored as specific files in the file system for later analysis. A file for a replay case contains all necessary information for the replay of the data that means the historical data for all digitals in the selected time interval as well as the historical raw data of the selected analog measurements.

Based on a replay case you can replay the data in a study. You can analyze the data at a certain point in time by single-line diagrams and curve charts. You can replay the data in real time (slower or faster than real time), stop and resume the execution and jump to a certain point in time. The time selection can be both explicit using date and time or using the selection of an event from a special summary.



NOTE

During archive replay, it is possible to change the status of digitals or insert/remove Temporary Network Elements (TNEs). This can be done by manual manipulation or by initializing the Study from RT. By this changes of Study state, it is possible that the current loaded replay case does not match to the current Study state. When you continue the Study, an unexpected system reaction is possible. You have to consider this association.

Restrictions

Only those digital values and analog values that are archived are taken into account for the replay.

For example, the topological coloring in the replay study is only correct, when all the switching states are archived.

Save Case Taker

The Save Case Taker is an application in Spectrum Power 5 that creates one save case per day. If a maximum number of save cases is reached, the oldest one will be deleted. The activation of Save Case Taker, the creation time of save cases, and maximum number of save cases is configured using IMM.



NOTE

As the archive replay requires save cases, activate the Save Case Taker.

12.2 Operational Steps

12.2.1 Overview

There are the following 2 main use cases for Archive replay:

- Creation of a new replay case.
- Evaluation of an existing replay case.

Those use cases are described in the following sections in more detail.

12.2.2 Creating a New Replay Case

The goal of this task is to create a new replay case for a certain period in the past. The creation of the replay case is only possible for a few days in the past. The number of days depends on the configuration and is 10 by default.

The following steps are performed for this task:

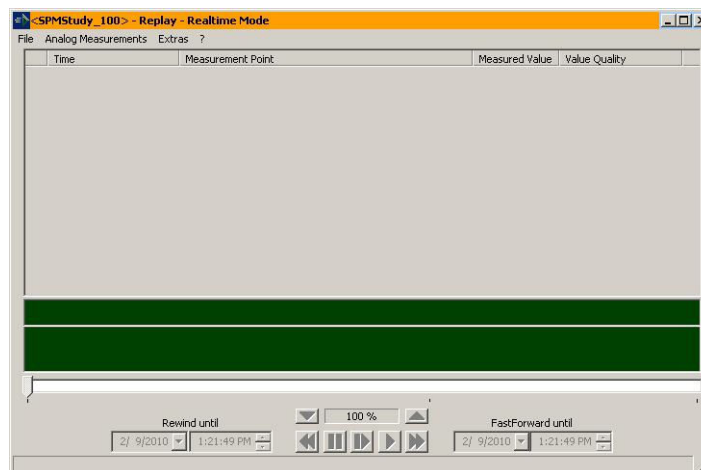
- Switch to a study.
- Open the replay user interface using PAT.



[uioar001-120111-01.tif, 1, en_US]

Figure 12-1 Open the Replay User Interface Using the PAT

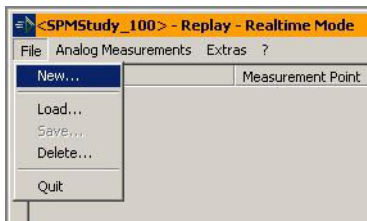
The base screen of the replay user interface is shown to the user.



[uioar002-120111-01.tif, 1, en_US]

Figure 12-2 Basic User Interface of the Replay User Interface

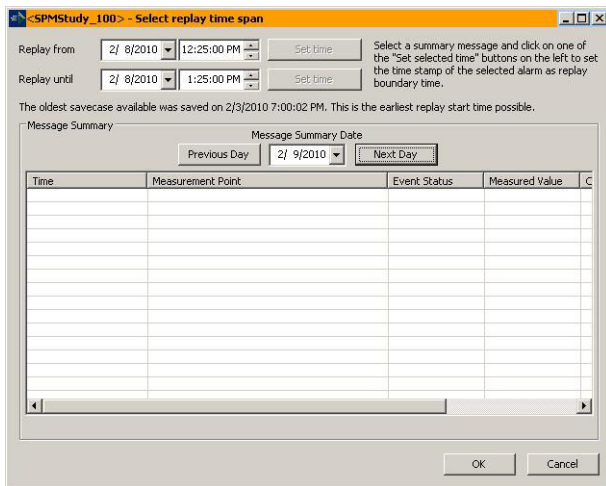
- Select **File > New** from the menu bar of the replay user interface to create a new replay case.



[uioar003-120111-01.tif, 1, en_US]

Figure 12-3 Create a New Replay Case Using the Replay Interface

The screen for the definition of the replay case is displayed.

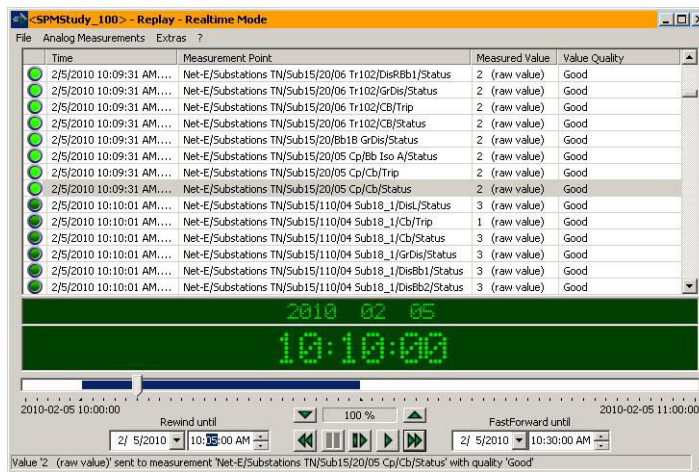


[uioar004-120111-01.tif, 1, en_US]

Figure 12-4 User Interface for the Definition of the Replay Case

- Enter the date and time in the fields **Replay from** and **Replay until** to define the replay case.
Another possibility to define the date and time is using the replay summary:
Selecting a date in the **Message Summary Date** field, the entries of that date that were recorded in the replay summary are shown in the table. You can now select an entry of the summary and press one of the **Set Time** buttons in order to take over the time stamp of that event as either start or end time of replay case.
Confirm the creation of the replay case by pressing the **OK** button.

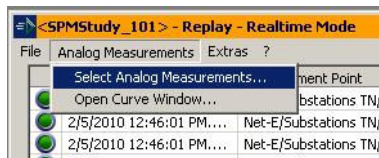
- Once the dialog closes, the base window of the replay user interface shows all changes of digital values that were recorded in the defined time frame.



[uioar005-120111-01.tif, 1, en_US]

Figure 12-5 Base User Interface of Replay User Interface After a Replay Case has been Defined

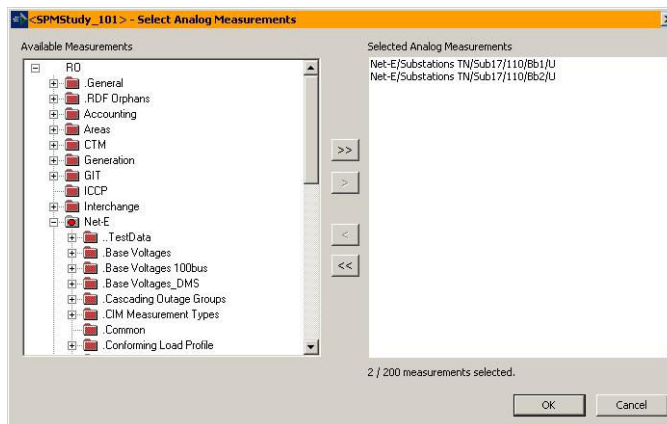
By default a replay case only contains the archived digital values and TNEs in the defined time span. That means, when evaluating the replay case by default, no analog values are available. If analog values are also needed for the evaluation of the replay case, those analog values have to be added explicitly. This is done by selecting **Analog Measurements > Select Analog Measurements ...**



[uioar006-120111-01.tif, 1, en_US]

Figure 12-6 Select Analog Measurements for the Replay Case

The dialog **Select Analog Measurements** is displayed.



[uioar007-120111-01.tif, 1, en_US]

Figure 12-7 Dialog for the Selection of the Analog Values

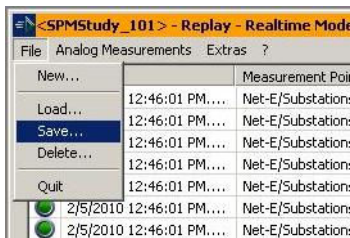
- You can add and remove individual analog measurements using the buttons > and <.



NOTE

Only archived analog values can be added to a replay case. In the example above, 2 analog measurements have been selected. The maximum number of analog measurements that can be selected per replay case is defined in the configuration, by default the maximum is set to 200.

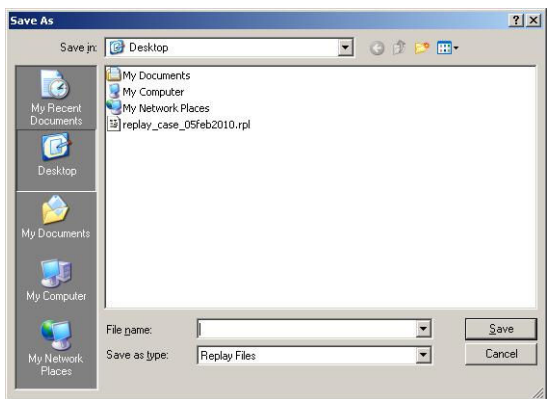
- Confirm the selection using the **OK** button.
The corresponding analog values are read from the archive and added to the replay case.
- Save the replay case using **File > Save** from the base user interface.



[ui0ar008-120111-01.tif, 1, en_US]

Figure 12-8 Saving a Replay Case

- Specify the file name.



[ui0ar009-120111-01.tif, 1, en_US]

Figure 12-9 Dialog for Saving of Replay Case Files

Replay case files are identified by the file extension **rpl**.

The default directory that is used is defined during configuration. The replay case file contains all necessary information (digital and analog values) that is needed for later evaluation.

12.2.3 Evaluating an Existing Replay Case

The goal of this task is to evaluate an existing replay case, that was created as described in the previous section.

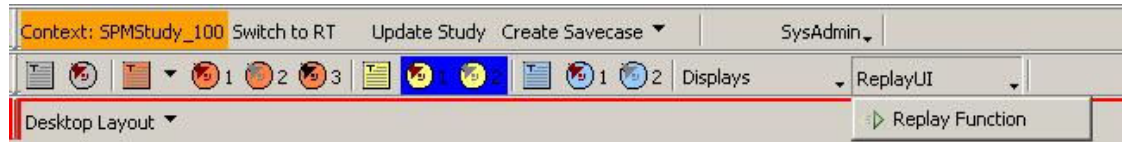
The following steps are performed for this purpose:



NOTE

The first 3 steps are the same as in the last section. When the replay user interface is already open those steps can be omitted.

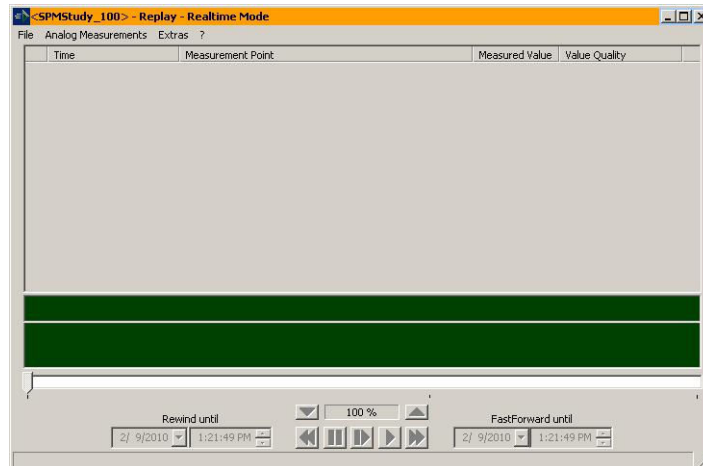
- Switch to study.
- Open the replay user interface using PAT.



[uioar001-120111-01.tif, 1, en_US]

Figure 12-10 Open the Replay User Interface Using the PAT

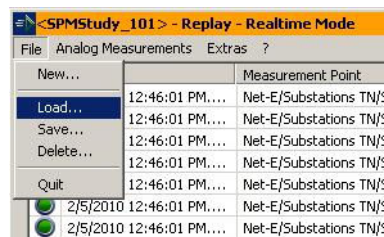
The base screen of the replay user interface is displayed.



[uioar002-120111-01.tif, 1, en_US]

Figure 12-11 Basic User Interface of the Replay User Interface

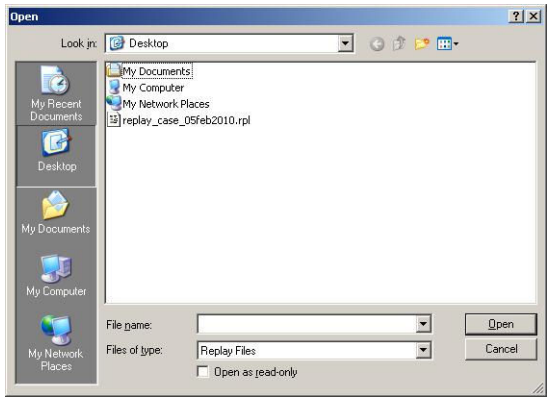
- Select **File > Load** to load an existing replay case.



[uioar010-120111-01.tif, 1, en_US]

Figure 12-12 Loading an Existing Replay Case

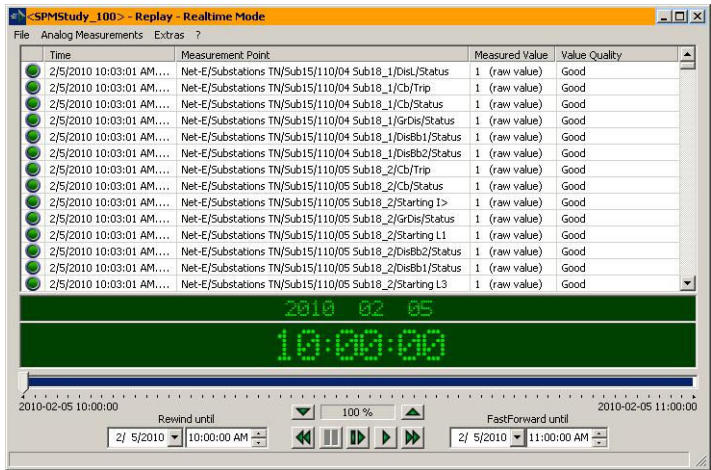
- In the next dialog, select the replay case file you want to analyze and confirm using the **OK** button.



[uioar011-120111-01.tif, 1, en_US]

Figure 12-13 Dialog for Opening an Existing Replay Case

The basic information of the replay case is shown in the base user interface.



[uioar012-120111-01.tif, 1, en_US]

Figure 12-14 Base Replay User Interface after Replay Case has been Loaded

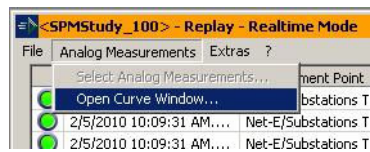
- You can analyze the situation at the simulation time by opening single-line diagrams. In those diagrams all digital values and all analog values that are part of the replay case have the values they had at the simulation time.



NOTE

If the digital measurements for all relevant switching states are part of the replay case, also the correct topological coloring at the simulation time is shown in the single-line diagram.

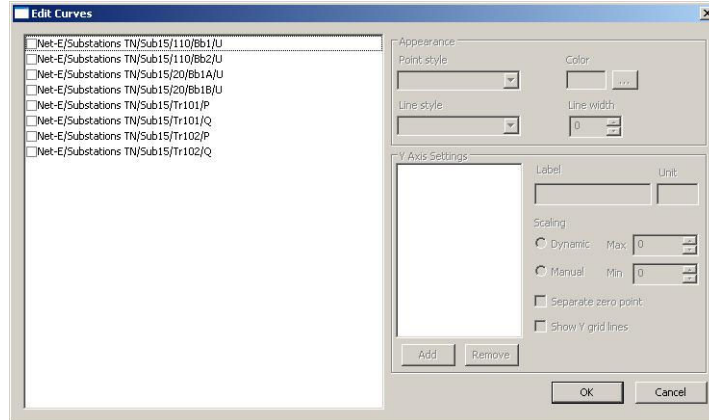
- In addition, you can analyze the analog measurements that are part of the replay case in curve displays. For this purpose, select the menu item **Analog Measurements > Open Curve Window ...**



[uioar013-120111-01.tif, 1, en_US]

Figure 12-15 Opening the Curve Window

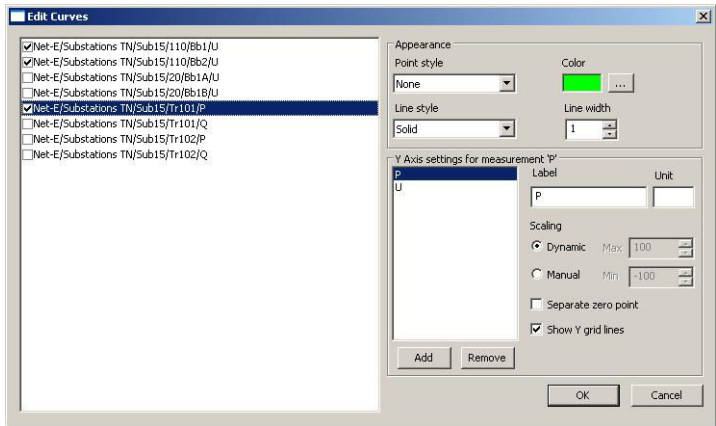
- In the curve window, you can select the measurements you want to see in one curve display.



[uioar014-120111-01.tif, 1, en_US]

Figure 12-16 Dialog for Editing Curves

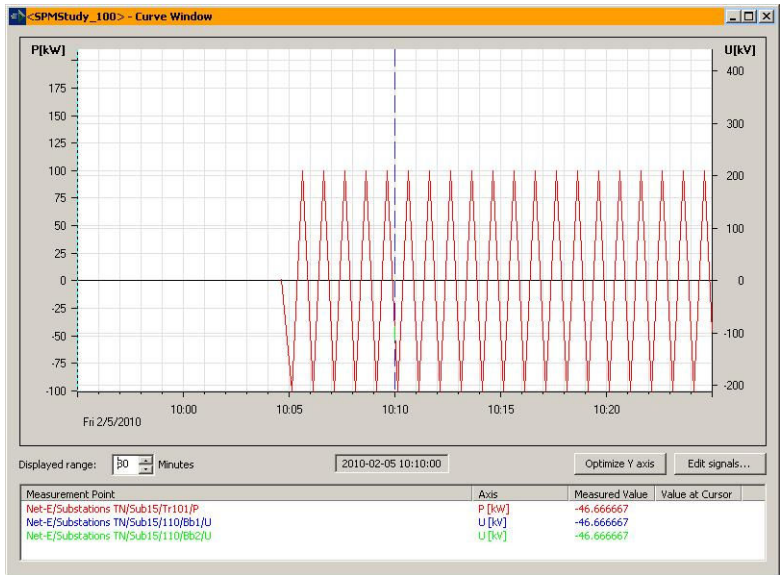
- For each selected measurement, you can specify the appearance of the curve in terms of point style, color, line style, and line width. In addition, select the y-axis for the curve.



[uioar015-120111-01.tif, 1, en_US]

Figure 12-17 Dialog for Editing Curve with Measurement Selected

After clicking the **OK** button the corresponding curve window is shown.



[uioar016-120111-01.tif, 1, en_US]

Figure 12-18 Sample Curve Window

- Within the curve window, the last time stamp shown is always the current simulation time. Using the text boxes displayed range you can specify, how far in the past the values are shown from this point in time. You can open several curve windows at a time.

12.3 User Interface Description

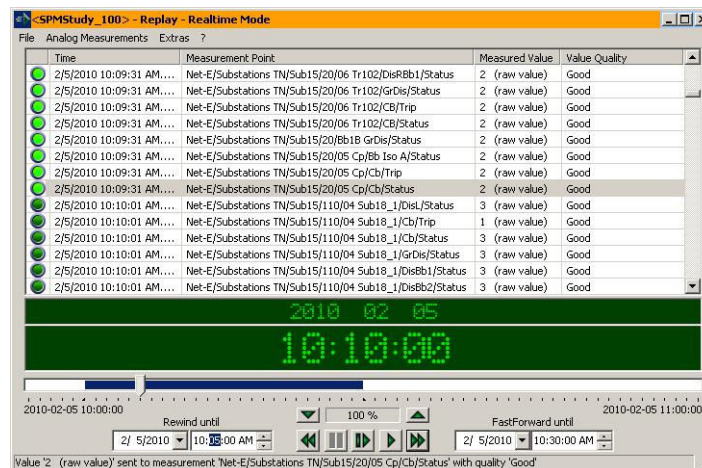
The user interface provides the following information:

- Start time of the simulation indicated in large green letters In the example above: **February 5th, 2010, 10:00:00**.
- Time span of the replay case shown underneath the slider In the example above: **February 5th, 2010, 10:00:00 until 11:00:00**.
- All changes of digital values that have occurred during the time span of the replay case, shown in the table.

The following possibilities are available to change the simulation time:

- Replay the data by hitting the **play** button.
The speed at which the data is replayed can be influenced using the **slower** and **faster** buttons next to the percentage field. By default the data is replayed in real time (100%) that means each minute one minute of simulated data is replayed. While the data is replayed you can stop the replay by clicking the **stop** button.
- Alternatively, replay the data step by step.
This is done by clicking the **stepwise execution** button. In this case only the next digital event is replayed.
- In addition, you can go directly to a certain point in time.
There are 2 alternatives to do that.
 - Enter the time in **rewind until** (or **fast forward until**) field.
 - Press the **rewind** (**fast forward**) button.
 - Alternatively, select an event from the table and select **rewind** or **fast forward** from the context menu.

The current situation is always presented as shown in the following screen shot.



[uioar005-120111-01.tif, 1, en_US]

Figure 12-19 Replay User Interface During Replay



NOTE

The following information can be obtained from the user:

- Current simulation time, in this case **February 5th, 2010, 10:10:00**.
 - Events that have been simulated are marked in the table with a light green circle. The last simulated event is selected in the table.
 - Slider Indicates the simulation time in relation to the time span of the replay case.
 - Blue area Within the slider highlights, the **Rewind** until and **Fastforward** until dates.
-

13 Limit Monitoring

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13.1 Introduction

Default limits are configured in the Information Model Management (IMM). Each limit can be defined either as upper or lower limit. A dead band per limit prohibits the creation of unnecessary limit violation alarms. The operator can update or overwrite default limiting values manually.

The priority of limit sets is:

- Priority 1: Dynamic limit (the limit is manually overwritten by an operator)
- Priority 2: Limit set or seasonal limit
- Priority 3: Default limit (set in the IMM)

13.2 Operational Steps

13.2.1 Defining Dynamic Limits

The goal of the task is to define analog limiting values in network diagrams and to disable default limits, if necessary.

Updating Values in the Dynamic Limits Dialog

- Right-click an analog value in the single-line diagram.
- Select **Dynamic Limit** from the context menu.
The **Dynamic Limits** dialog opens.
- Define the analog limiting values.
- Click **Apply**.

The new limiting values are activated.

Disabling of Default Limits

- Right-click an analog value in the single-line diagram.
- Select **Dynamic Limit** from the context menu.
The **Dynamic Limits** dialog opens.
- Click the button **Disable Limit**.

The default limits are disabled.

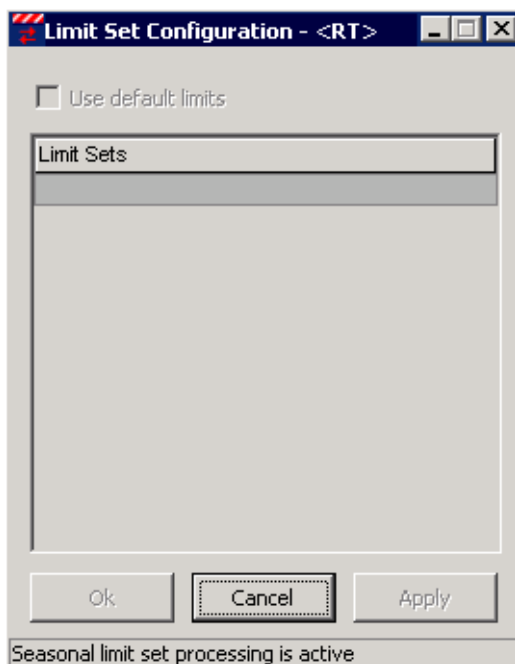
13.2.2 Applying Limit Sets Manually

The goal of the task is to apply limit sets manually.

Applying Manual Limit Sets

- Select **Tools > Limit Set Dialog** on the PAT.
Alternatively, click the **Limit Dialog** button on the toolbar of the Runtime Explorer (in **Limit View** mode).

- The **Limit Set Configuration** dialog opens.



[rguio010-280610-01.tif, 1, en_US]

Figure 13-1 Limit Set Configuration

- Select the new limit set from the list box **Change to Limit Set**.
- Click **Change** to change the limit set.

The new limit set is applied.

13.3 User Interface Description

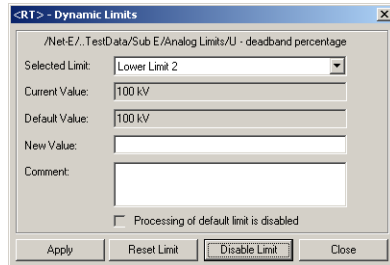
13.3.1 Dynamic Limit Dialog

The **Dynamic Limits** dialog enables you to define analog limiting values directly in a single-line diagram. You can create a new dynamic limit or reset a limiting value to the default value.



NOTE

Default limits are configured in the Information Model Management (IMM).



[scald004-030908-01.tif, 1, en_US]

Figure 13-2 Dynamic Limits Dialog

List Box: Selected Limit

You can select the corresponding limit from the **Selected Limit** list box.

Text Box: Current Value

The current value is displayed.

Text Box: Default Value

The default value is displayed.

Text Box: New Value

You can enter the new limiting value into the **New Value** text box.

Text Box: Comment

You can enter a comment here.

Check Box: Processing of Default Limit Is Disabled

Clicking this check box enables/disables the processing of default limits.

Button: Apply

The function activates the new limiting value.

Button: Reset Limit

The **Reset Limit** function resets the new limiting value to the default value.

Button: Disable Limit

The function disables/enables the default limits.

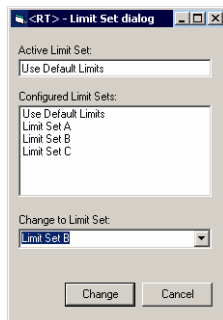
13.3.2 Limit Set Dialog

The **Limit Set** dialog shows the currently active limit set as well as a list of all defined limit sets.



NOTE

The configuration in the Information Model Management (IMM) determines whether seasonal or manual limit sets are applied. If seasonal limit sets are configured, the dialog items are disabled. Otherwise, limit sets can be activated manually.



[scald003-051207-01.tif, 1, en_US]

Figure 13-3 Limit Set Dialog

Text Box: Active Limit Set

The text box displays the active limit set.

Text Box: Configured Limit Sets

The text box displays the limit sets, which are configured in the IMM.

List Box: Change to Limit Set

You can select a new limit set from the list box.

Button: Change

Clicking this button changes the selected limit set to the active limit set.

14 Switching Procedure Management

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14.1 Introduction

Switching Procedure Management (SPM) allows you to prepare and perform switching procedures on the power system. Following are some examples of SPM:

- Prepare and perform supervisory control operations on the power system or the network.
For example, change the connections from a main busbar to a reserve busbar in a primary station.
- Isolate an area in fault conditions or for clearance operations.
- Restore power following an unplanned or planned outage.

Types of Switching Procedures

A switching procedure consists of a sequence of operational steps and organizational steps.

There are 2 different types of switching procedures:

- Switching orders
- Switching programs

Although similar in structure, switching orders and switching programs provide different functionality.

Switching Order

A switching order is a work order that contains a specific sequence of operational steps and organizational steps. It is a formal document used for planned work.

A switching order can be created by:

- A user in response to a request for work on the network.
- A system application through the provided interface.
For example, isolation and restoration procedures created by Fault Isolation and Service Restoration (FISR).

A switching order provides written instructions related to isolation, de-energization, energization, or procedural modification of a portion of the utility network. It is also used to provide clearance records.

SPM manages the complete lifecycle of switching orders from creation through archiving or deletion. It maintains and records the state of the switching orders through the various phases involved. Each switching order has its own unique identification number.

Switching order is approved before execution and is executed only once.

Switching Program

A switching program is a collection of operational steps and organizational steps. It is used to perform a specific action on the power system.

A switching program is created to:

- Reduce the workload of the operation personnel for repetitive activities.
- Accommodate frequently used sequences of remote control operations.
For example, change the connections from a main busbar to a reserve busbar in a primary station.

Since the switching program automatically performs the necessary checks, switching programs are implemented to ascertain safe switching. For example, interlocking validation.

A switching program can be executed several times. Execution can be started manually or according to a defined schedule.

14.2 Workflow

14.2.1 Executing a New Switching Procedure

The goal of this task is to perform a new switching procedure. To achieve this goal perform the following:

- Create the procedure
- Test the procedure
- Approve the procedure
- Run the procedure

The following operation explains the creation of a switching procedure that automatically disconnects and reconnects a line. A wait step between the switching steps is added.

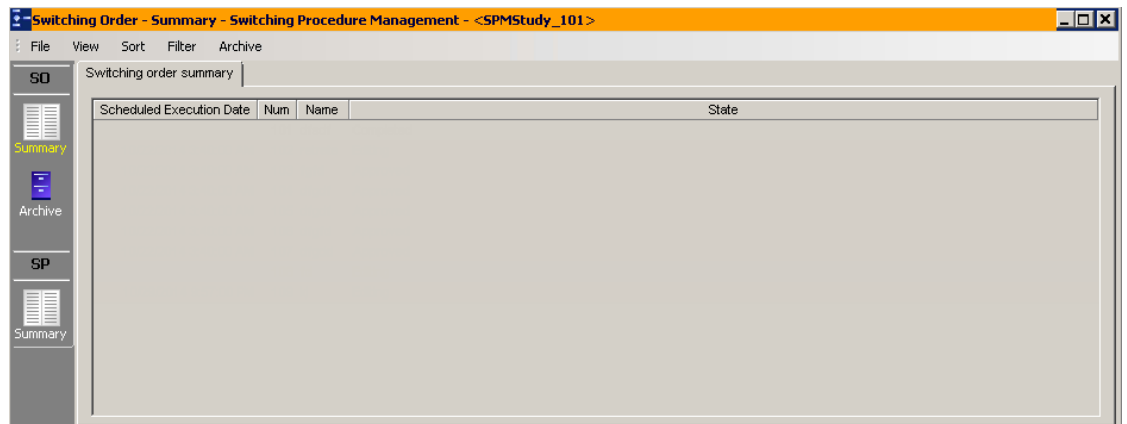


NOTE

A switching procedure can be created, tested, and approved in study context only.
The approved procedure can be executed in real-time (RT) context only.

Creating a Switching Procedure

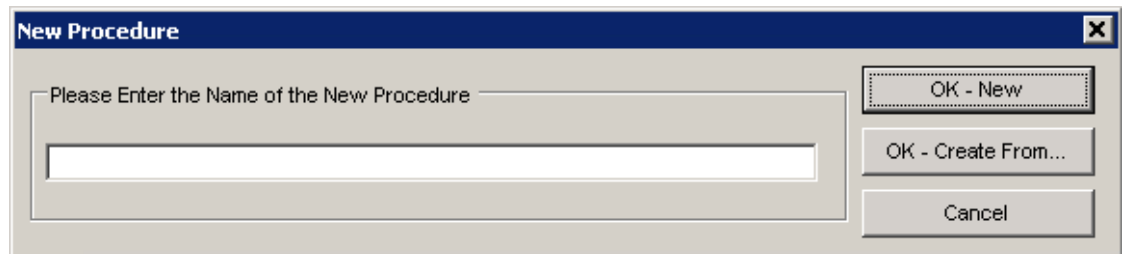
- To switch to study context, click **Switch to Study** on the Power 5 Application Toolbar (PAT).
- On the PAT, select **Network Operation > Switching Procedure Management**.



[sc_fig14-1, 1, en_US]

Figure 14-1 SPM Dialog in Study Context

- Select **File > New Procedure** from the menu bar of the **SPM** dialog. The New Procedure dialog opens.

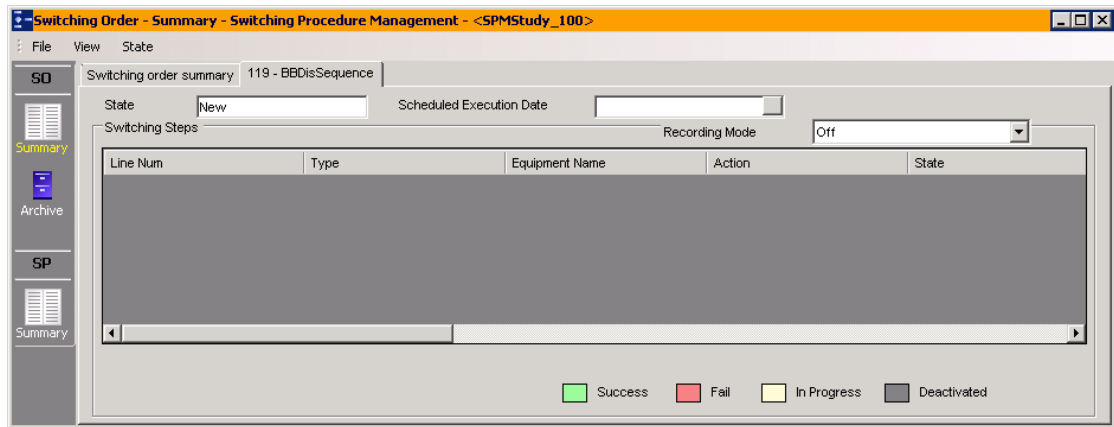


[sc_fig14-2, 1, en_US]

Figure 14-2 Creating a Switching Procedure

- In the **New Procedure** dialog, enter the name of the procedure.

- Click **OK-New** to create an empty switching procedure.
SPM opens the new switching order.

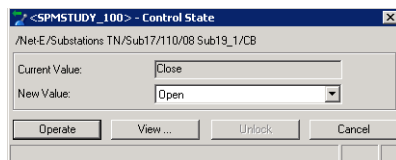


[sc_14-3, 2, en_US]

Figure 14-3 New Switching Procedure

Adding Switching Steps to Disconnect the Line

- Set the **Recording mode** to **With Execution**.
- Right-click the circuit-breaker and select **Control State** from the context menu.
The **Control State** dialog opens.



[rgscad02-210110-01.tif, 1, en_US]

Figure 14-4 Control State Dialog

- Set **Control** to **Open** and click **Operate**.
- Execute the same steps for both disconnectors next to the circuit-breaker.

Inserting a Wait Step

- In the **Switching Steps** area of the **SPM** dialog, right-click the switching step before which you want to insert a wait step.
- Select **Insert Wait Step** from the context menu.

Adding Switching Steps to Reconnect the Line



NOTE

To add the necessary steps for reconnecting the line, you can copy the steps that disconnect the line and paste them into the list in reversed order and function.

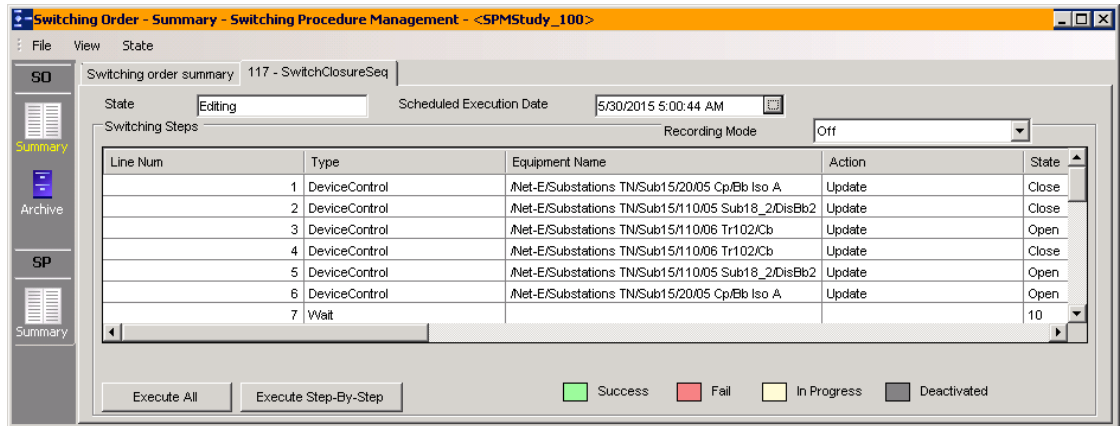
- In the **Switching Steps** area of the **SPM** dialog, mark all the switching steps except for the wait step.



NOTE

To mark more than 1 entry in the **Switching Steps** list, press and hold the **Ctrl** key and click the respective switching steps.

- Right-click the marked switching steps and select **Copy** from the context menu.
- Right-click the marked switching steps again and select **Invert Paste** from the context menu.



[sc_14-5, 2, en_US]

Figure 14-5 Complete Switching Procedure

Testing and Approving the New Procedure

- In the **SPM** dialog, click **Execute all**.
The switching procedure is executed.
- Check in the **Execution Result** column of the **Switching Steps** list if the execution of each step was successful.
- Open the **General Summary** and check whether the execution was successful.

Time Stamp	Path 1	Path 2	Path 3	Path 4	Stat	Value	Ind	User	Console	Comment
06/02/08 10:42:37:354	SPM Configuration			SPM Message	App	StartExecuteAll	Control	SPCAdmin	DEB20NVC	Execute All of procedure 101 is started.
06/02/08 10:42:41:839	Sub17	110/08 Sub19_1	CB	Status	App	Open	Control	SPCAdmin	DEB20NVC	
06/02/08 10:42:42:214	Sub17	110/08 Sub19_1	CB	Status	App	Open	Ctrl Resp			
06/02/08 10:42:43:745	Sub17	110/08 Sub19_1	DisL	Status	App	Open	Control	SPCAdmin	DEB20NVC	
06/02/08 10:42:44:183	Sub17	110/08 Sub19_1	DisL	Status	App	Open	Ctrl Resp			
06/02/08 10:42:45:808	Sub17	110/08 Sub19_1	DisBb1	Status	App	Open	Control	SPCAdmin	DEB20NVC	
06/02/08 10:42:46:183	Sub17	110/08 Sub19_1	DisBb1	Status	App	Open	Ctrl Resp			
06/02/08 10:42:57:386	Sub17	110/08 Sub19_1	DisBb1	Status	App	Close	Control	SPCAdmin	DEB20NVC	
06/02/08 10:42:58:183	Sub17	110/08 Sub19_1	DisBb1	Status	App	Close	Ctrl Resp			
06/02/08 10:42:59:386	Sub17	110/08 Sub19_1	DisL	Status	App	Close	Control	SPCAdmin	DEB20NVC	
06/02/08 10:43:00:183	Sub17	110/08 Sub19_1	DisL	Status	App	Close	Ctrl Resp			
06/02/08 10:43:01:996	Sub17	110/08 Sub19_1	CB	Status	App	Close	Control	SPCAdmin	DEB20NVC	
06/02/08 10:43:02:152	Sub17	110/08 Sub19_1	CB	Status	App	Close	Ctrl Resp			
06/02/08 10:43:02:433	SPM Configuration			SPM Message	App	CloseExecuteAll	Control	SPCAdmin	DEB20NVC	Execute All of procedure 101 is completed.

[eaas0001-100608-07.tif, 1, en_US]

Figure 14-6 General Summary



NOTE

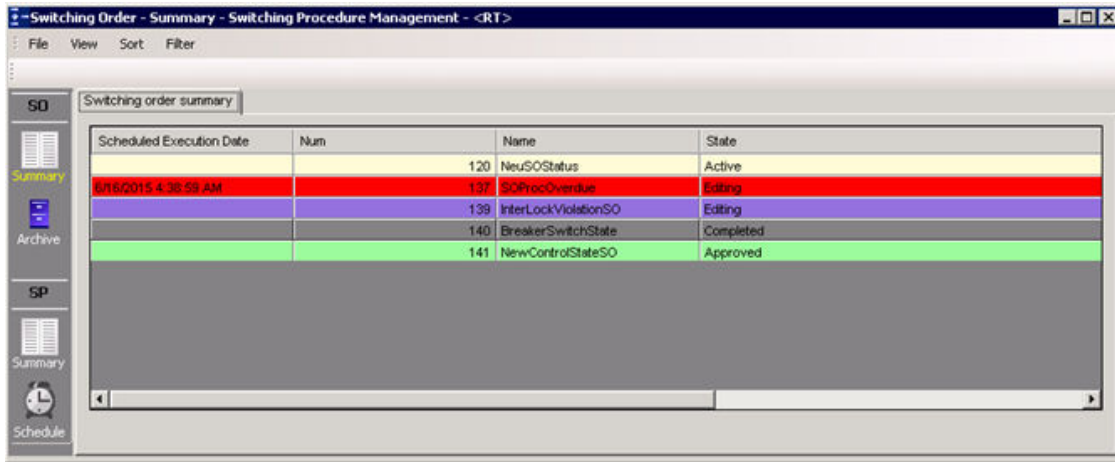
Checking the successful execution is also possible using single-line diagram.

Open the corresponding single-line diagram and check whether the created switching order represents the correct switching state.

- If the execution was successful, select **State > Approved** from the menu bar of the **SPM** dialog.

Executing the Approved Switching Procedure

- To switch to real-time context, click **Switch to RT** on the PAT.
- On the PAT, select **Network Operation > Switching Procedure Management**.



Scheduled Execution Date	Num	Name	State
	120	NeuSOStatus	Active
6/16/2015 4:38:59 AM	137	SOProcOverdue	Editing
	139	InterLockViolationSO	Editing
	140	BreakerSwitchState	Completed
	141	NewControlStateSO	Approved

[sc_SPMdialog_nolegend, 1, en_US]

Figure 14-7 SPM Dialog in Real-time Context

- To open the switching procedure, double-click the respective entry in the **SO Summary** of the **SPM** dialog.
A tab with the details of the new switching procedure opens.
- Click **Execute all**.
The new switching procedure is run in real-time context.

14.3 Switching Orders

14.3.1 Overview

14.3.1.1 Setup of Switching Order Dialog

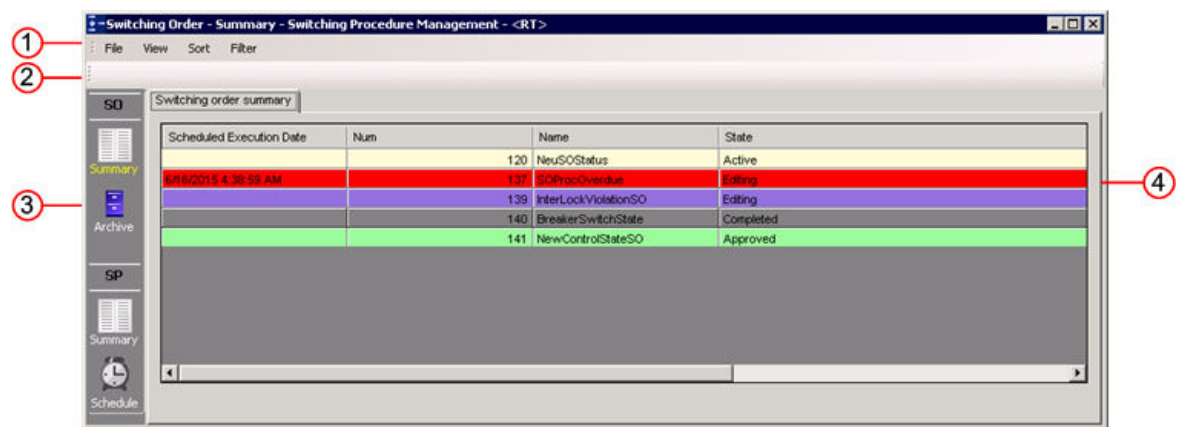
The **Switching Order** dialog consists of several displays to provide an overview of all active and archived switching orders as well as detailed information about the selected switching order.

The same displays are used in real-time context and in study context. The color of the title bar of a display indicates whether it is in real-time context or in study context.



NOTE

The colors for distinguishing between real-time context and study context are selected by the user.



[sc_14-8, 2, en_US]

Figure 14-8 Switching Order Dialog Overview

1 Menu Bar

Provides menu items to select the switching order functions depending on the selected display. For example, creating new switching orders.

The menu items displayed vary according to the selected context (study or real time context).

For example, the creation of a switching order is only possible in study context.

2 Toolbar

Provides shortcut icons to save the selected switching order and to generate a report, depending on the selected display.

3 View Bar

Allows you to select the different switching order displays, **SO Summary** display and **Archive** display.

You can also use the view bar to toggle between the **Switching Order** dialog and the **Switching Program** dialog.

4 Work Section

Displays the necessary information for all switching orders functions, depending on the selected display.

14.3.1.2 Life Cycle Management

The life cycle of a switching order consists of the following states:

States	Description
New	When you create a switching order, its status is set to New .

States	Description
Editing	When you turn on the recording mode, the status automatically changes from New to Editing .
Checked	After the switching order is tested, set the status to Checked .
Approved	After the review of a checked switching order, set the status to Approved .
Active	After a switching order is invoked, its status is set to Active .
Suspended	This state is set when a switching order stops during execution by a break step or a break command or if the execution failed.
Completed	If the switching order cannot be implemented anymore, set the status of a switching order to Completed . The state is also set to Completed automatically when the last step of the switching order is executed.
Archived	After a switching order is archived, its status is set to Archived .

Before each status change, SPM checks whether you are authorized to perform the status change.

SPM saves the associated information of each status change (date or time, status, user name). This history is displayed in the SPM summary display.

14.3.1.3 Switching Order Structure

A switching order consists of the following parts:

- **General Information**
General information is stored in the header section and contains name, number, type, status, scheduled start date, and comments of the switching order.
- **Operational Steps**
Contains information for supervisory control like equipment name, action, status, and operator comment.
- **Organizational Steps:**
 - **Wait**
Stops the execution for a defined period of time.
 - **Break**
Stops the execution until you continue it.
 - **Comment**
Allows you to add comments to the switching program.
 - **Conditional steps**
 - If else
 - While

14.3.2 Operational Steps

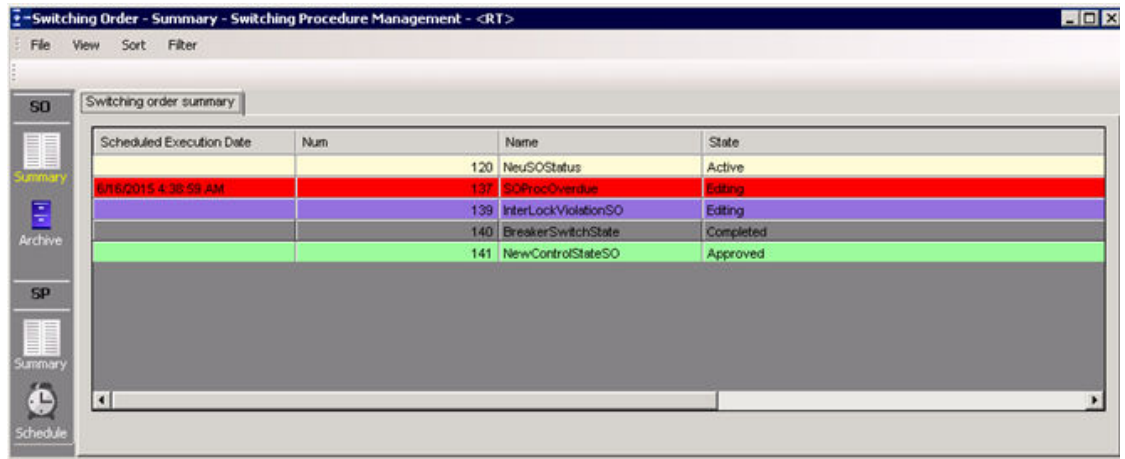
14.3.2.1 Starting Switching Order Dialog

- From the Power 5 Application Toolbar (PAT), choose **Network Operation > Switching Procedure Management**.

For additional information, refer to [14.3.3.1 Power Application Toolbar \(PAT\)](#).

The **Switching Order** dialog opens.

For additional information, refer to [14.3.1.1 Setup of Switching Order Dialog](#).



[sc_SPMdialog_nolegend, 1, en_US]

Figure 14-9 SO Summary Display

- Select the **SO Summary** display from the view bar of the dialog.

14.3.2.2 Creating Switching Orders

The goal of this task is to create a new switching order. Switching orders can be created from scratch or by using a template.

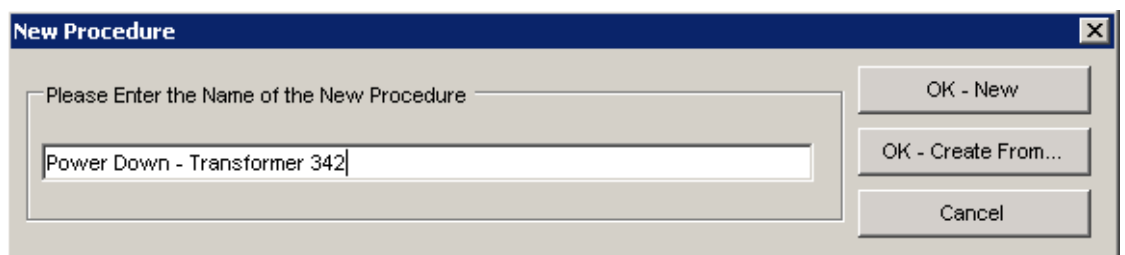


NOTE

Creation and modification of switching orders is possible only in study context.

Creating Empty Switching Orders

- Open the **Switching Order** dialog and select the **SO Summary** display.
 - Select **File > New Procedure** from the menu bar.
- The **New Procedure** dialog opens and prompts to enter the name for the new switching order.



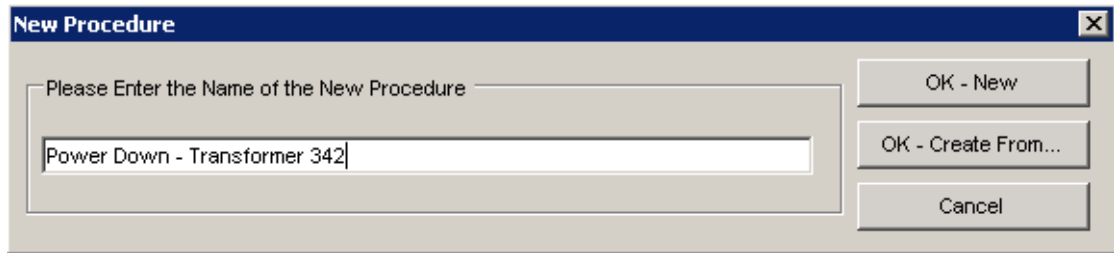
[sc_14-10, 1, en_US]

Figure 14-10 New Procedure Dialog

- Click **OK-New** to create an empty switching order.
SPM creates a new empty switching order and opens it in a separate detail tab. The state of the new switching order is set to **New**. The type of the switching order is set to **SO**.
SPM creates an entry for the modification history containing the creation date and the user name automatically.

Creating Switching Orders Based On Existing Switching Orders

- Open the **Switching Order** dialog and select the **SO Summary** display.
- Select **File > New Procedure** from the menu bar.
The **New Procedure** dialog opens and prompts to enter the name for the new switching order.

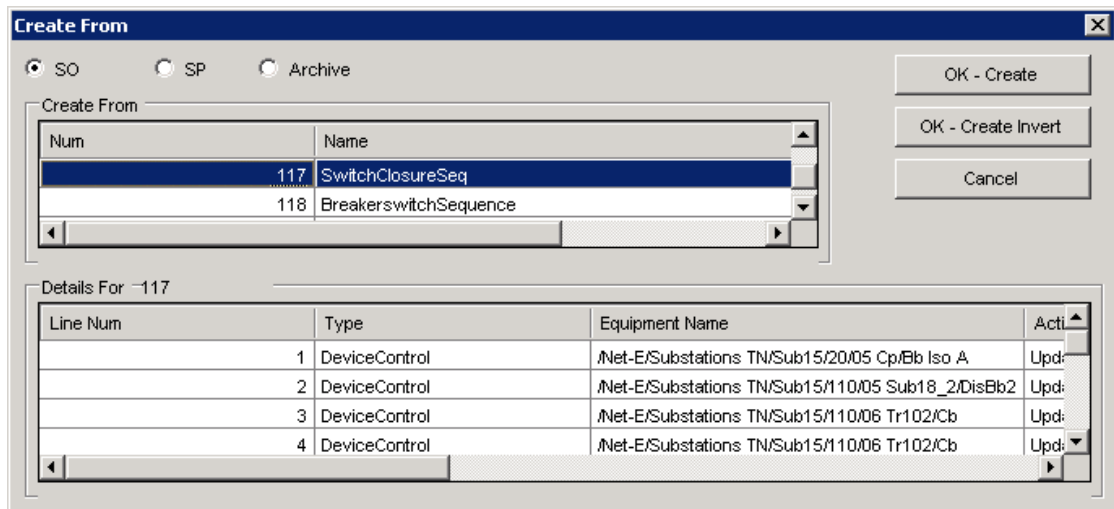


The **New Procedure** dialog box has a title bar with a close button. It contains a text input field with the placeholder text "Please Enter the Name of the New Procedure". The input field contains the text "Power Down - Transformer 342". To the right of the input field are three buttons: "OK - New", "OK - Create From...", and "Cancel".

[sc_14-10, 1, en_US]

Figure 14-11 New Procedure Dialog

- Click **OK-Create From** to select an existing switching order for the creation of the new switching order.
The **Create From** dialog opens.



The **Create From** dialog box has a title bar with a close button. It contains three radio buttons: **SO** (selected), **SP**, and **Archive**. Below the radio buttons is a section titled "Create From" containing a list box with two items: "117 SwitchClosureSeq" (selected) and "118 BreakerswitchSequence". To the right of the list box are three buttons: "OK - Create", "OK - Create Invert", and "Cancel". Below the list box is a section titled "Details For 117" containing a table with four columns: "Line Num", "Type", "Equipment Name", and "Acti".

Line Num	Type	Equipment Name	Acti
1	DeviceControl	/Net-E/Substations TN/Sub15/20/05 Cp/Bb Iso A	Upd
2	DeviceControl	/Net-E/Substations TN/Sub15/110/05 Sub18_2/DisBb2	Upd
3	DeviceControl	/Net-E/Substations TN/Sub15/110/06 Tr102/Cb	Upd
4	DeviceControl	/Net-E/Substations TN/Sub15/110/06 Tr102/Cb	Upd

[sc_14-12, 2, en_US]

Figure 14-12 Create From Dialog

- Select the desired switching order from the **Create From** list.
The **Details** list shows all steps of the selected switching order.
- Click **OK-Create** to create a new switching order containing the steps in original order.
Alternatively, click **OK-Create Invert** to create a new switching order containing in reverse order.
SPM creates a new empty switching order and opens it in a separate detail tab. The state of the new switching order is set to **New**. The type of the switching order is set to **SO**.
SPM automatically creates an entry for the modification history containing the creation date and the user name.

14.3.2.3 Modifying Existing Switching Orders

The goal of this task is to open, change, save, and delete an existing switching order.

**NOTE**

The modification of switching orders is possible only in study context.

Selecting Switching Orders

- Open the **Switching Order** dialog and select the **SO Summary** display.
- Double-click a switching order in the list of existing switching orders.
SPM opens the selected switching order in a separate tab.

Line Num	Type	Equipment Name	Action	State
1	DeviceControl	/Net-E/Substations TN/Sub15/20/05 Cp/Bb Iso A	Update	Close
2	DeviceControl	/Net-E/Substations TN/Sub15/110/05 Sub18_2/DisBb2	Update	Close
3	DeviceControl	/Net-E/Substations TN/Sub15/110/06 Tr102/Cb	Update	Open
4	DeviceControl	/Net-E/Substations TN/Sub15/110/06 Tr102/Cb	Update	Close

Date	User Name	State
05/29/2015 03:38:56 PM	SPCAdmin	Editing
05/29/2015 03:29:36 PM	SPCAdmin	Editing
05/29/2015 12:36:23 PM	SPCAdmin	Editing
05/29/2015 10:44:21 AM	SPCAdmin	Completed
05/29/2015 10:39:16 AM	SPCAdmin	Editing
05/29/2015 10:38:52 AM	SPCAdmin	New

[SO_Details, 2, en_US]

Figure 14-13 SO Summary Display - Detail Tab

Modifying Switching Order Information

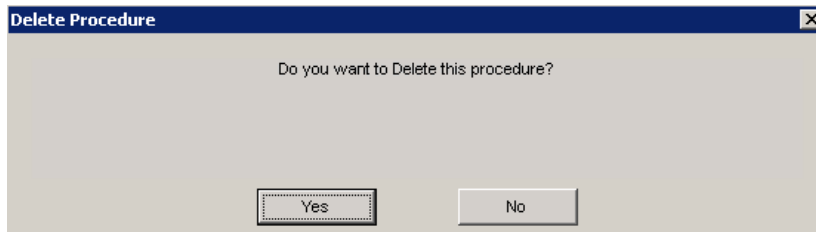
- Ensure that the switching order details are toggled on using the **View > Details** menu.
- Enter **Name** (up to 80 characters) and **Comments** (up to 255 characters) for the selected switching order in the corresponding text boxes.
The comment field is typically used to make special notes about the switching order.
- Select the **Reason** of the outage from the listbox.
For additional information on configuring the **Reason** in IMM, refer TS-SCADA-EN.
- Enter the details for the Switching Order in the Additional field section provided at the bottom of the SO Summary display.
For additional information on configuring the **Additional Field** in IMM, refer TS-SCADA-EN.
- Change the state for the selected switching order by selecting the corresponding menu item in the **State** menu.
- Click the button right of the text box to enter the **Scheduled Execution Date**.
The **Set Time** dialog opens and prompts to enter date and time directly or to select it from the calendar.

Saving Modified Switching Orders

- Select **File > Save** to save the new or changed switching order to the database.

Deleting Switching Orders

- Open the **Switching Order** dialog and select the **SO Summary** display.
- Right-click the switching order to be deleted and select **Delete Procedure** from the context menu. The Confirmation dialog opens.



[sc_14-14, 1, en_US]

Figure 14-14 Confirmation Dialog

- Click **No** to retain the procedure.
- Click **Yes** to delete the procedure.
SPM deletes the selected switching order after confirmation.

14.3.2.4 Inserting New Steps

The goal of this task is to add operational steps or organizational steps to a switching order.

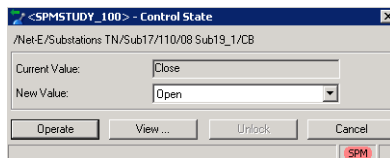


NOTE

The recording mode always adds new steps at the end of the switching order. You must cut and paste the steps to the correct position manually.

Inserting Operational Steps in Recording Mode

- Select and open the desired switching order.
- Select **With Execution** from the **Recording Mode** list box.
SPM changes the state of the switching order to **Editing** automatically.
- Invoke the network diagram that displays the device to be controlled.
- Select the desired element and open the **Control State** dialog.



[rgscad03-210110-01.tif, 1, en_US]

Figure 14-15 Control State Dialog

The text "SPM" in the right bottom area indicates that the SPM Recording mode is switched on. The red background color indicates that recording mode with execution is selected.

- Perform the normal supervisory control action.
After selecting the button **Operate**, the supervisory control function is executed in the power system and is recorded as operational step of the switching order.
A message in the status bar of the **Control State** dialog indicates if the step is added successfully.
- Perform the other desired supervisory control actions in the same or other network diagrams.

- After you have performed all desired supervisory control actions, return to the **SPM Summary Display**. The currently selected switching order contains all steps that are performed in the network displays.
- Select **Off** from the **Recording Mode** list box to finish the recording mode.

Inserting Organizational Steps

To insert an organizational step (Break Step, Wait Step, or Comment Step), the following steps are necessary:

- Select the operational step after which the organizational step shall be inserted.
- Right-click the operational step and select **Insert Break Step**, **Insert Wait Step** or **Insert Comment Step** from the context menu.
The organizational step is inserted to the switching order.
- Enter an appropriate comment if desired.

Inserting Value Simulation Step

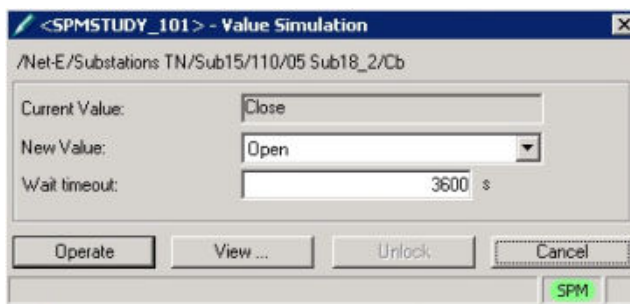
Value Simulation function is provided in study mode for switches that are telemetered, but not telecontrolled. **Value Simulation** allows the analysis of scenarios where the switch is controlled locally or its value changes spontaneously. For this purpose, switch to study mode, right-click the corresponding switch and select **Value Simulation**. After entering the target state and confirming the operation, the new value of the switch is simulated.

This operation can also be recorded in the switching sequence. In this case, the function is translated into a wait step with a user defined wait time out.

When the switching procedure is executed in real time context and arrives at the wait step, it waits until the switch arrives at the defined target position. If the target position is not achieved within the defined time out period, the switching sequence fails and eventually stops.

Recording the Switching Sequence in Study Mode

- Switch to the Study Context
- Select and open the desired switching order.
- Select **With Execution** or **Without Execution** from the Recording Mode list box.
SPM changes the state of the switching order to **Editing** automatically.
- Invoke the network diagram that displays the device to be controlled.
- Select the desired element and open the **Value Simulation** dialog.

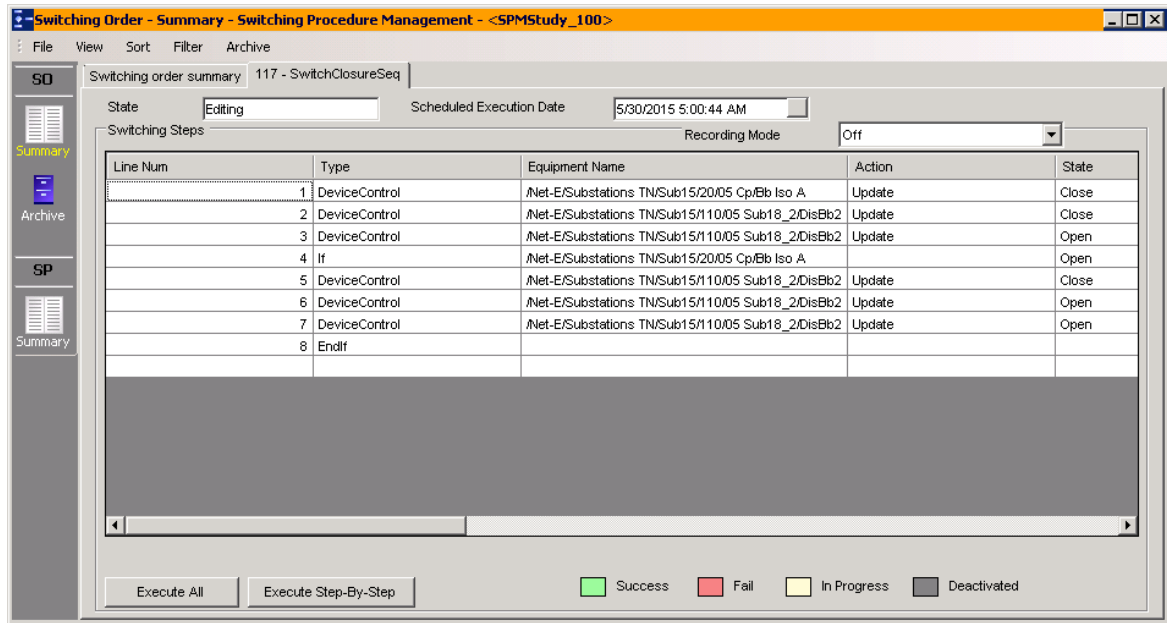


[sc_ValueSimulationDialog, 1, en_US]

Figure 14-16 Value Simulation Dialog

The text **SPM** in the right bottom area indicates that the SPM Recording mode is switched on. The red background color indicates that recording mode with execution is selected and the green background color indicates that recording mode without execution is selected.

- Enter the wait timeout value in the **Wait timeout** text field.
- Click **Operate**.
The corresponding switching step is recorded for that particular switching order.



[sc_14-17, 2, en_US]

Figure 14-17 Switching Order - Steps

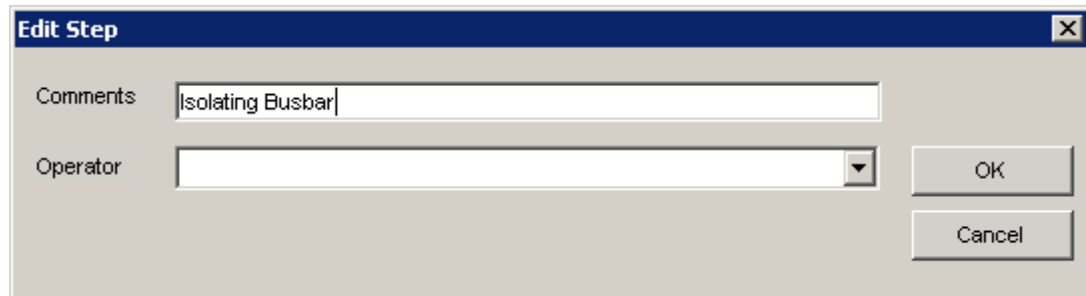
14.3.2.5 Modifying Existing Steps

The goal of this task is to change, copy, activate, deactivate, or delete existing steps of a switching order.

Modifying Comments and Operator

- Select and open the switching order that contains the desired step.
- Right-click the step and select **Edit Step...** from the context menu.

The **Edit Step** dialog opens and prompts to enter a comment from the selected step.



[sc_14-18, 1, en_US]

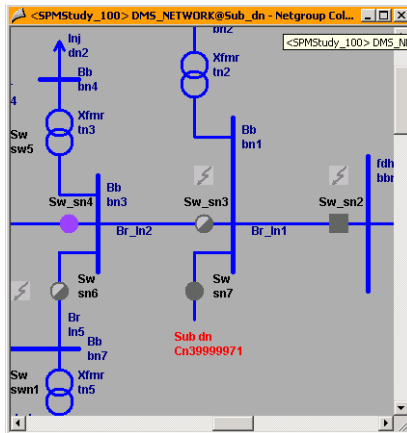
Figure 14-18 Edit Step Dialog

- Enter your comments in **Comments** text box.
- Select the **Operator** from the list box.
For additional information on configuring the operator in IMM, refer TS-SCADA-EN.
- Click **OK** button.

The comment and the operator is now displayed in the respective line of the switching order.

Calling Up a Network Display

- Select and open the switching order that contains the desired step.
- Right-click the step and select **Callup Network Display** from the context menu.
The network display showing the corresponding element is opened.



[spmso006-150508-01.tif, 1, en_US]

Figure 14-19 Network Diagram

Copying/Cutting/Pasting Steps

- Select and open the switching order that contains the desired step.
- Right-click the step and select **Copy** from the context menu to copy the selected step into the clipboard.
Alternatively, right-click the step and select **Cut** from the context menu to copy the selected step into the clipboard and delete it from the current position.
- Select the step before which you want to paste the selected steps.
- Right-click and select **Paste** from the context menu to paste the selected steps in original order.
Alternatively, right-click and select **Invert Paste** from the context menu to paste the selected steps in reverse order.

Deactivating or Activating Steps

- Select and open the switching order that contains the desired step.
- Right-click the step and select **Deactivate Step** from the context menu.
Alternatively, right-click the step and select **Undo Deactivate Step** from the context menu.
SPM automatically changes the color of the step to indicate that the step is inactive or active.

Deleting Steps

- Select and open the switching order that contains the desired step.
- Right-click the step and select **Delete** from the context menu.
SPM deletes the selected step without further acknowledgment.

14.3.2.6 Setting Archive Preferences

The goal of this task is to configure the preferences for the automatic archive procedure.

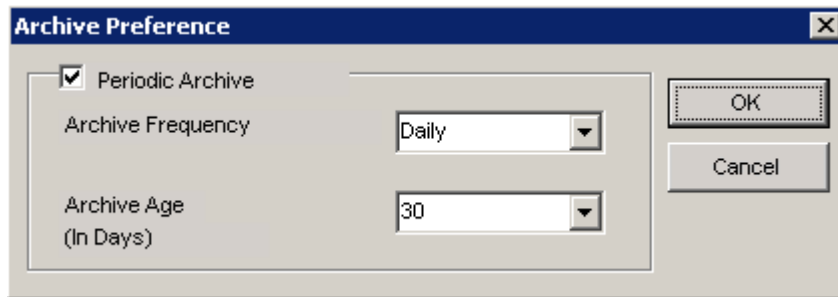


NOTE

At a specified frequency, **Archive Frequency**, all switching orders are processed by the archive procedure:

- Switching orders with state **completed**, that are older than a specified number of days (**Archive Age**), are automatically archived.
- Switching orders, that are still in state **new**, **editing**, **checked**, or **approved**, are not included in the archiving procedure.

- Select **Archive > Preference** from the menu bar of the switching order dialog. The **Archive Preference** dialog opens.



[sc_14-20, 1, en_US]

Figure 14-20 Archive Preference Dialog

- Select **Periodic Archive** check box to activate the automatic archive procedure.
- Select **Archive Frequency** (Daily, Weekly, Monthly) and the **Archive Age** (in days).
- Click **OK** to save the changes made in the archive preferences.

14.3.2.7 Simulating Switching Orders

The goal of this task is to simulate a switching order without execution.

This is done to test the switching order and to simulate the effects of the operations.

Therefore you have to switch to the study context.

Simulating Switching Programs

- Switch to the Study Context
- Select and open the desired switching order.
- Click **Execute all** in the **Switching Order** dialog.
The switching order is executed in the study context.

See the effects of the simulation

- Select **Study > Simulate** from the menu bar.
The corresponding worldmap shows the effect of the switching order like energization, deenergization by coloring the concerned network area.
- Alternatively, select **Study > Simulate PF** from the menu bar.
The Distribution System Power Flow display is opened showing the power flow data as if the switching order is executed.

14.3.2.8 Executing Switching Orders from Switching Order Dialog

The goal of this task is to execute a switching order from the **Switching Order** dialog.

Switching orders are invoked in study context or in real-time context. Invoking a switching order in study context means testing the switching order without effect on the real-time system. Invoking a switching order in real-time context means to execute it in the real-time system.

**NOTE**

The following example describes the execution in real-time context. Hence the term “execute” is used. The execution in study context is similar.

Switching orders can be executed step by step or they can be started and allowed to run to completion. Organizational steps can pause and wait for user response or delay for a specified amount of time.

**NOTE**

Step-by-step execution is restricted to the execution of the next not-executed step. It is not affected by the currently selected line.

Executing Complete Switching Orders

- Select and open the desired switching order.
- Click the **Execute All** button to begin the execution of the switching order.
It is mandatory in SPM to have the switching order approved before executing it.
The steps of the switching order change to the “In Progress” color. As the steps are executed, the color changes depending on the state of the step.
Normal interlocking conditions are checked during execution and any violations cause the procedure to stop. The aborted step changes to the “Fail” color.

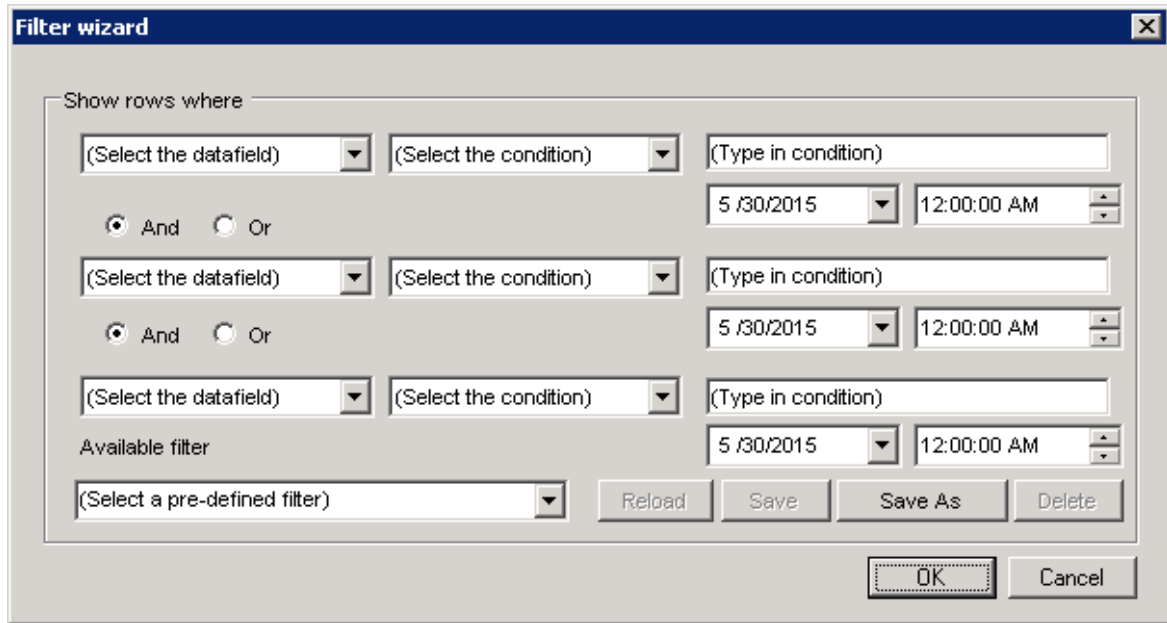
Executing Step By Step

- Select and open the desired switching order.
- Click the **Execute Step-by-step** button to begin the execution of the next, non-executed step.
It is mandatory in SPM to have the switching order approved before executing it.
The step of the switching order change to the “In Progress” color. As the step is executed, the color changes depending on the state of the step.
Normal interlocking conditions are checked during execution and any violations cause the procedure to stop. The aborted step changes to the “Fail” color.

14.3.2.9 Filtering Lists

The **Filter Wizard** allows custom filtering of the lists of the following displays:

- **Switching Order Summary** display
- **Switching Order Archive** display
- **Switching Program Summary** display
- **Switching Program Schedule** display

The image shows a 'Filter wizard' dialog box with a title bar containing a close button. The main area is titled 'Show rows where' and contains three identical filter criteria sections. Each section has three dropdown menus: '(Select the datafield)', '(Select the condition)', and '(Type in condition)'. Below each section are radio buttons for 'And' (selected) and 'Or'. The third section is followed by an 'Available filter' section with a dropdown menu labeled '(Select a pre-defined filter)'. At the bottom right of the main area are buttons for 'Reload', 'Save', 'Save As', and 'Delete'. At the very bottom of the dialog are 'OK' and 'Cancel' buttons. The 'Type in condition' dropdowns are currently set to '5 /30/2015' and '12:00:00 AM'.

[spmco001-150508-01.tif, 2, en_US]

Figure 14-21 Filter Wizard

Filtering Lists

- Right-click on an entry in one of the supported lists and select **Filter Wizard** from the context menu. The **Filter Wizard** dialog opens.
- Select up to 3 filter criteria (data field, operator, and condition).
- Click **OK** to execute the filtering criteria. SPM displays the filtered list according to the selected filter criteria.



NOTE

The **Filter Wizard** allows you to save the preferred filtering criteria in order to reuse them. Defined filtering settings can be selected, saved, or deleted from the **Available Filter** list box.

14.3.2.10 Generating Reports

The goal of this task is to create a report. Report are available for both switching orders and switching programs.

Generating Reports

- Select and open the desired switching order or switching program.
- Click **Report** icon in the toolbar or select **File > Generate Report** from the menu.
SPM automatically creates an Excel file containing the requested report.
This Excel file can now be printed, or saved as a file.

Switching Procedure Management Report							
				11/13/14 10:04			Additional Field Name
Procedure Number:	3						
Name:	Pre-Defined SP						
State:	Completed						
Scheduled Date:							
OR Number:							
Type:	SP						
Comments:							
Reason:	UnPlanned Activity						
Line Num:	1	Type:	DeviceControl				
Action:	Marker	State:	Set				
Equipment Name:	/Net-E/Substations TN/Sub15/110/03 Sub17/Cb			Marker:	Marker - Alarm Inhibit		
Execution Date:	11/10/2014 9:19						
Line Num:	2	Type:	DeviceControl				
Action:	Marker	State:	Set				
Equipment Name:	/Net-E/Substations TN/Sub15/110/03 Sub17/DisBb1			Marker:	Marker - Alarm Inhibit		
Execution Date:	11/10/2014 9:19						
Line Num:	3	Type:	DeviceControl				
Action:	Marker	State:	Set				
Equipment Name:	/Net-E/Substations TN/Sub15/110/04 Sub18_1/DisBb1			Marker:	Marker - Alarm Inhibit		
Execution Date:	11/10/2014 9:19						

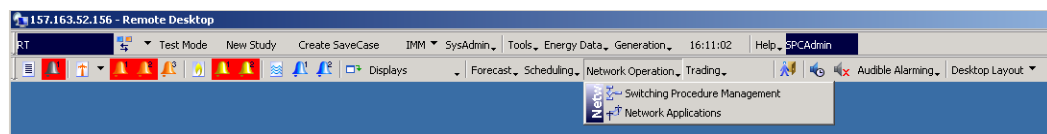
[sc_14-23_latest, 1, en_US]

Figure 14-22 Example Report Output

14.3.3 User Interface Description

14.3.3.1 Power Application Toolbar (PAT)

The **Switching Order** dialog can be started using the Power 5 Application Toolbar (PAT).

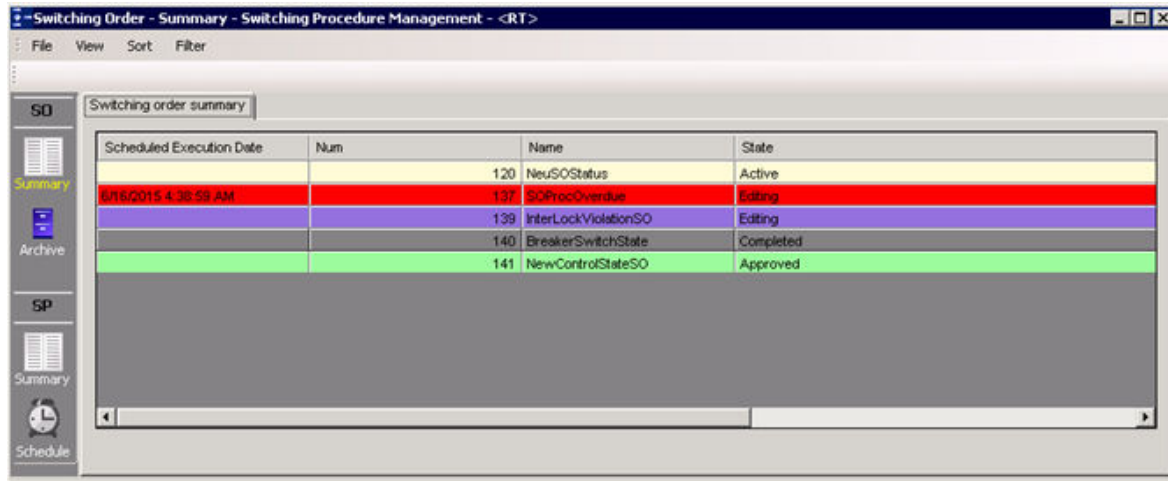


[rgscad06-210110-01.tif, 1, en_US]

Figure 14-23 PAT Network Operation Menu

14.3.3.2 Summary Display - Summary Tab

The summary tab of the **SO Summary** display provides an overview of all switching orders.



[sc_SPMdialog_nolegend, 1, en_US]

Figure 14-24 SO Summary Display - Summary Tab

Menu Bar

Allows you to create a new switching order, to customize the display, to sort and filter the list of switching orders, and to set the archive preferences.

Switching Orders List

Provides a list of all switching orders containing scheduled execution time and date, unique switching order number, name and state of all switching orders.

The background color of a line indicates the state of the respective switching order:

State	Color
New	White
Editing	Blue
Checked	Light Blue
Approved	Green
Active	Yellow
Completed	Gray
Suspended	Red
Selected	Dark Blue

Switching Order Detail

Contains detailed information about the selected switching order as well as a history of all modifications.

The **Switching Order Detail** section is toggled on and off by selecting the **View > Details** menu item.

14.3.3.3 Summary Display - Detail Tab

If you create a new switching order or open an existing switching order, the **SO Summary** display opens a new tab for the switching order.

This tab provides detailed information about the selected switching order. The name of the new tab consists of the switching order number and name.

Switching Order Summary - Switching Procedure Management - <SPMStudy_100>

File View Sort Filter Archive

SO Switching order summary 117 - SwitchClosureSeq

State Scheduled Execution Date

Switching Steps Recording Mode

Line Num	Type	Equipment Name	Action	State
1	DeviceControl	/Net-E/Substations TN/Sub15/20/05 Cp/Blo Iso A	Update	Close
2	DeviceControl	/Net-E/Substations TN/Sub15/110/05 Sub18_2/DisBb2	Update	Close
3	DeviceControl	/Net-E/Substations TN/Sub15/110/06 Tr102/Cb	Update	Open
4	DeviceControl	/Net-E/Substations TN/Sub15/110/06 Tr102/Cb	Update	Close

Execute All Execute Step-By-Step

Success Fail In Progress Deactivated

Type OR Number Name Comments Reason

Date	User Name	State
05/29/2015 03:38:56 PM	SPCAdmin	Editing
05/29/2015 03:29:36 PM	SPCAdmin	Editing
05/29/2015 12:36:23 PM	SPCAdmin	Editing
05/29/2015 10:44:21 AM	SPCAdmin	Completed
05/29/2015 10:39:16 AM	SPCAdmin	Editing
05/29/2015 10:38:52 AM	SPCAdmin	New

[SO_Details, 2, en_US]

Figure 14-25 SO Summary Display - Detail Tab

Menu Bar

Allows you to create a new switching order, to generate a report, to customize the display, and to change the state of the selected switching order.

Toolbar

Provides shortcut icons to save the selected switching order and to generate a report.

Switching Order Steps

- The **Switching Steps** table provides a list of all defined steps of the selected switching order, including detailed information.
- The **Schedule Execution Date** text box allows to select the scheduled execution date for the selected switching order.
- The **Recording Mode** list box allows to toggle the recording mode to add steps to the switching order.
- The **Execute All** button allows the execution of the whole switching order, the **Execute Step-By-Step** button allows the step-by-step execution of the switching order.

Switching Order Detail

Contains detailed information about the selected switching order and history of all modifications.

The **Switching Order Detail** section can be toggled on and off by selecting the **View > Details** menu item.

14.3.3.4 Archive Display

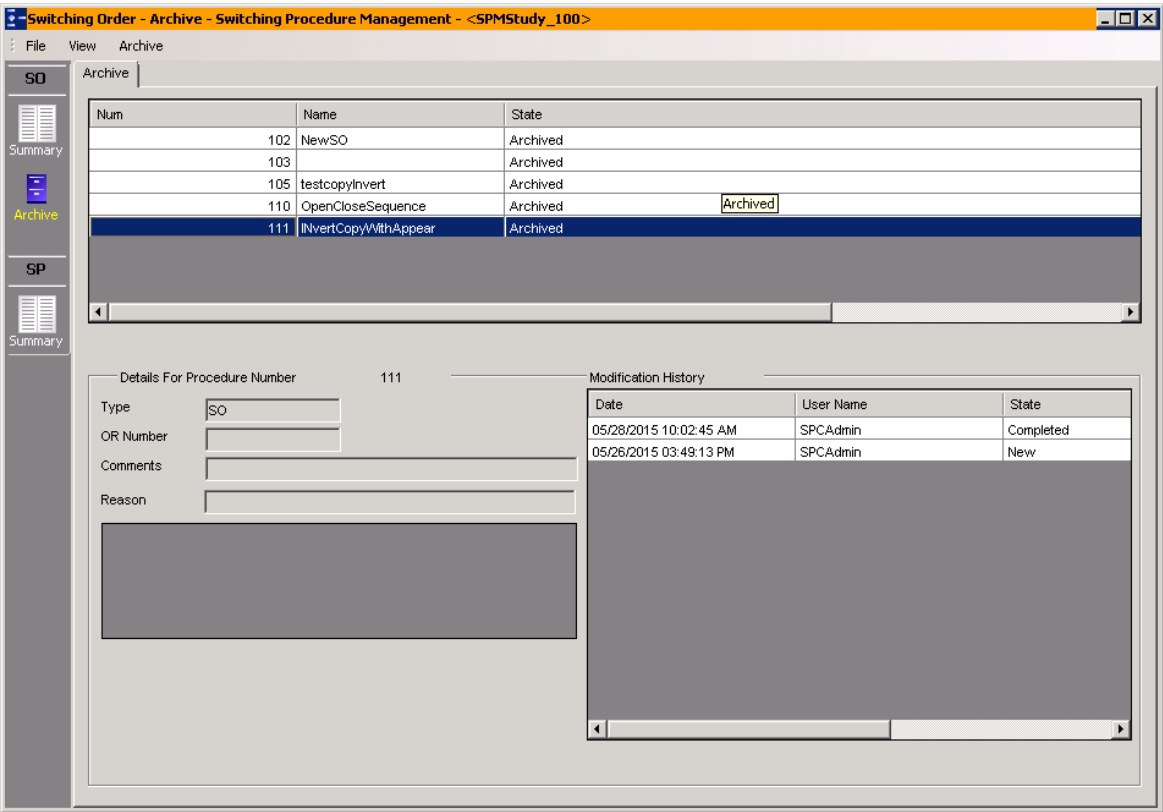
Switching orders with state **Completed** are moved to the archive for historical storage and analysis on a periodic base (Daily, Weekly, Monthly).

The archive is maintained manually by the administrator. Offline functions can access the archive for additional analysis as desired.

Summary Tab

The summary tab of the **Archive** display provides an overview of all archived switching orders.

For additional information about the provided information, refer to [14.3.3.2 Summary Display - Summary Tab](#). The **Summary** tab is read-only, no modifications is made to an archived switching order.



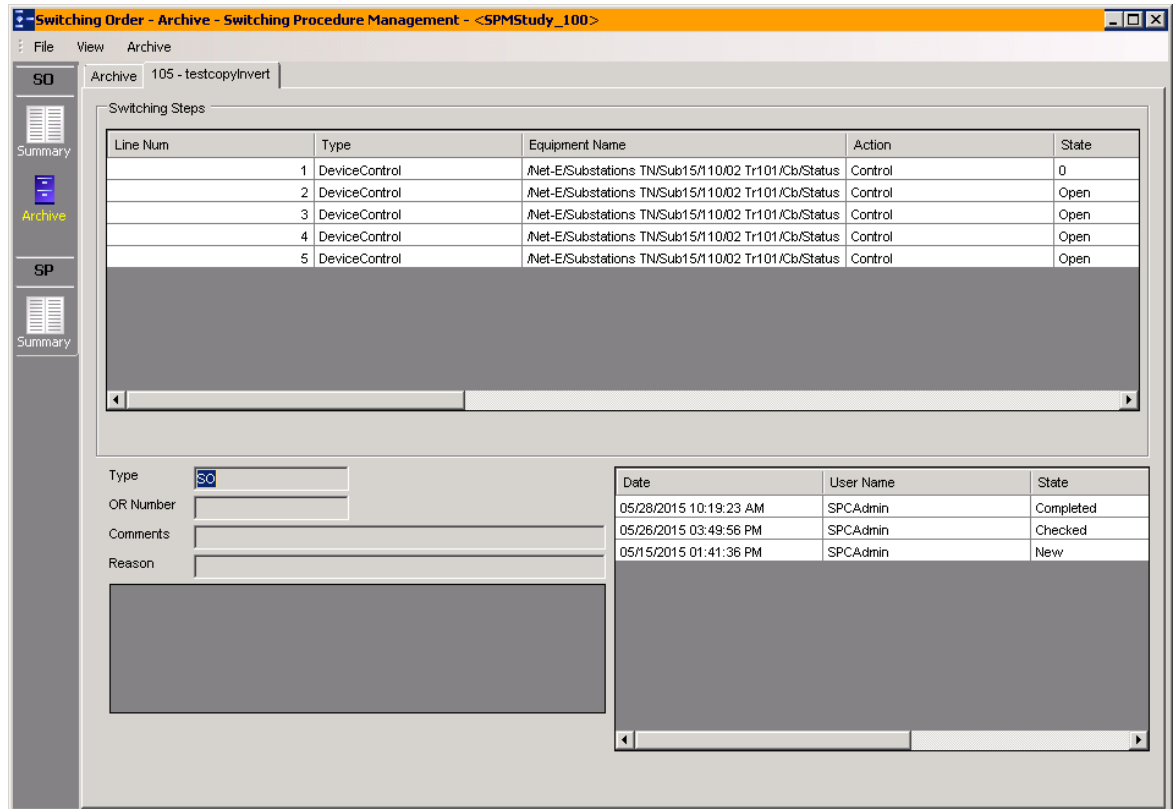
[sc_14-27, 2, en_US]

Figure 14-26 Archive Display - Summary Tab

Detail Tab

If you double-click an archived switching order, the **Archive** display opens a new tab for the switching order. This tab provides detailed information about the archived switching order. The name of the new tab consists of the switching order number and name.

For additional information about the provided information, refer to [14.3.3.3 Summary Display - Detail Tab](#). The **Detail** tab is read-only, no modifications are made to an archived switching order.



[sc_14-28, 2, en_US]

Figure 14-27 Archive Display - Detail Tab

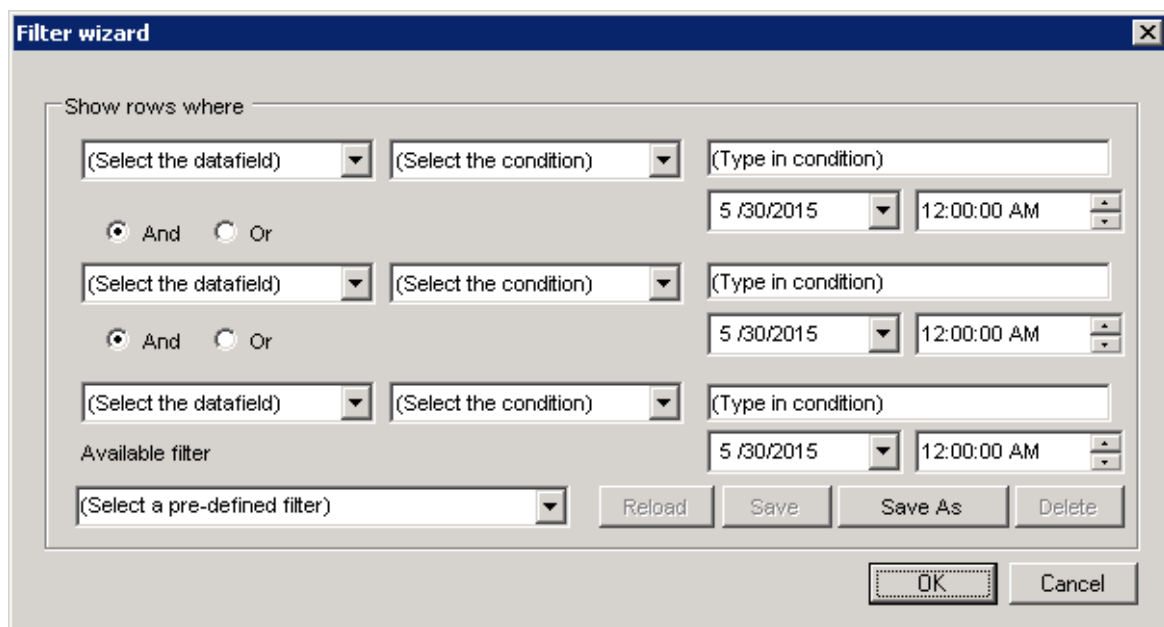
14.3.3.5 Filter Wizard

The **Filter Wizard** allows custom filtering of the following lists:

- Switching orders summary list
- Switching programs summary list

Filter wizard can be invoked by right-clicking on an entry in one of the supported lists and selecting the **Filter Wizard** from the context menu.

You can define filter criteria and store them under a name, which allows later call-up, use and modification.



The "Filter wizard" dialog box is used to create filters. It features a "Show rows where" section with three rows of filter criteria. Each row contains a dropdown for "(Select the datafield)", a dropdown for "(Select the condition)", and a text input for "(Type in condition)". Below each row are radio buttons for "And" and "Or". The first two rows have additional dropdowns for "5 /30/2015" and "12:00:00 AM". At the bottom, there is an "Available filter" section with a dropdown for "(Select a pre-defined filter)", and buttons for "Reload", "Save", "Save As", and "Delete". The dialog also includes "OK" and "Cancel" buttons at the bottom right.

Filter wizard

Show rows where

(Select the datafield) (Select the condition) (Type in condition)

☒ And ☐ Or

(Select the datafield) (Select the condition) (Type in condition)

☒ And ☐ Or

(Select the datafield) (Select the condition) (Type in condition)

Available filter

(Select a pre-defined filter)

Reload Save Save As Delete

OK Cancel

[spmco001-150508-01.tif, 2, en_US]

Figure 14-28 Filter Wizard Dialog

14.4 Switching Programs

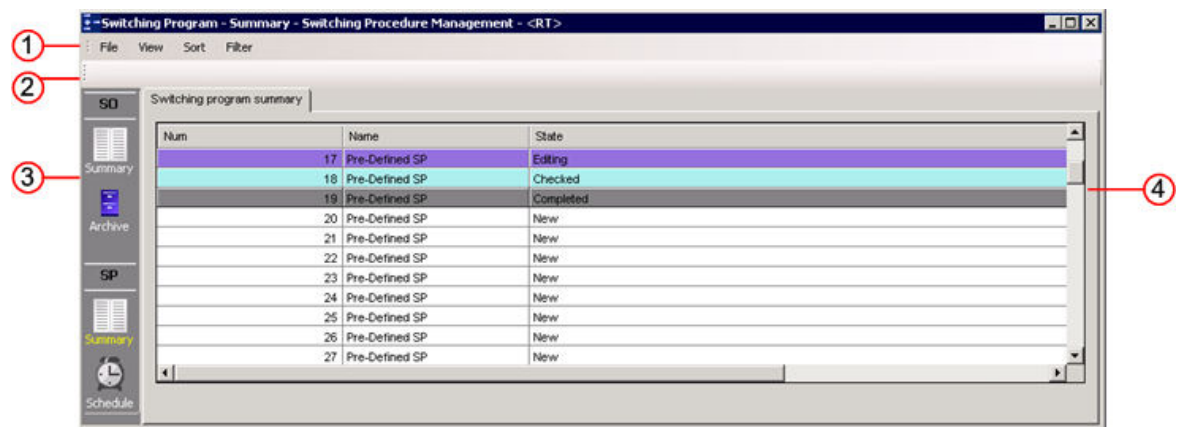
14.4.1 Introduction

14.4.1.1 Setup of Switching Program Dialog

The switching program dialog consists of several displays to provide an overview of all switching programs as well as detailed information about the selected switching program.

The dialog allows to define schedules for the scheduled execution of switching programs and to manages templates for the creation of new switching programs.

The same displays are used in real-time context and in study context. The color of the title bar of a display indicates whether it is in real-time context or in study context.



[sc_14-31, 2, en_US]

Figure 14-29 Switching Program Dialog Overview

1 Menu Bar

Provides menu items to select the switching program functions (for example, creating new switching programs), depending on the selected display.

Depending on the selected context (study context or real-time context) not all menus/menu items are available.

For example, the creation of a switching program is only possible in study context.

2 Toolbar

Provides shortcut icons for the user to save the selected switching program and to generate a report, depending on the selected display.

3 View Bar

Allows the user to select the different switching program displays (**SP Summary** display and **Schedule** display)

The user can also use the view bar to toggle between the switching order dialog and the switching program dialog.

4 Work Section

Shows the necessary information for all switching program functions, depending on the selected display.

14.4.1.2 Life Cycle Management

The life cycle of a switching program consists of the following states:

States	Description
New	When you create a switching program, its status is set to New .
Editing	When you turn on the recording mode, the status automatically changes from New to Editing .

States	Description
Checked	After the switching program is tested, set the status to Checked .
Active	After a switching program is invoked, its status is set to Active .
Suspended	This state is set when a switching program stops during execution by a break step or a break command or if the execution failed.
Completed	If the switching program cannot be implemented anymore, set the status of a switching program to Completed . The state is also set to Completed automatically when the last step of the switching program is executed.

Before each status change, SPM checks whether you are authorized to perform the status change. SPM saves the associated information of each status change (date or time, status, user name). This history is displayed in the SPM summary display.

14.4.1.3 Switching Program Structure

A switching program consists of the following parts:

- **General Information**
General information is stored in the header section and contains name, number, type, status, and comments of the switching program.
- **Operational Steps**
Contains information for supervisory control like equipment name, action, status, and operator comment.
- **Organizational Steps**
The following organizational steps are defined:
 - **Wait**
Stops the execution for a defined period of time.
 - **Conditional** (If Else, End If, While, End While)
Enables to check switch positions and values. The maximum possible depth for If and While statements is 3.
 - **Comment**
Allows you to add comments to the switching program.

14.4.2 Operational Steps

14.4.2.1 Starting Switching Program Dialog

- From the Power 5 Application Toolbar (PAT), select **Network Operation > Switching Procedure Management**.
For additional information, refer to [14.3.3.1 Power Application Toolbar \(PAT\)](#).

- Select the **Switching Program** dialog from the view bar.
For additional information, refer to [14.4.1.1 Setup of Switching Program Dialog](#).

The screenshot shows the 'Switching Program - Summary' dialog. It has a menu bar (File, View, Sort, Filter) and a toolbar (SD, Summary, Archive, SP, Summary, Schedule). The main area is divided into two sections. The top section is a table with columns 'Num', 'Name', and 'State'. The bottom section is titled 'Details For Procedure Number 4' and contains fields for 'Type', 'Name', 'Comments', 'Reason', 'Value', and 'Time'. To the right of these fields is a 'Modification History' table.

Num	Name	State
1	Pre-Defined SP	Suspended
2	Pre-Defined SP	Active
4	Pre-Defined SP	Completed
5	Pre-Defined SP	Active
6	Pre-Defined SP	Active

Modification History		
Date	User Name	State
05/21/2015 10:26:05 AM	SPCAdmin	Completed
05/06/2015 03:40:05 PM	SPM	New

[SP_Summary, 2, en_US]

Figure 14-30 SP Summary Display

14.4.2.2 Creating Switching Programs

The goal of this task is to create a new switching program. Switching programs is created from scratch or by using a template.



NOTE

The creation of switching orders is possible only in study context.

Creating Empty Switching Programs

- Open the **Switching Program** dialog and select the **SP Summary** display.
- Select **File > New Procedure** from the menu bar.
The **New Procedure** dialog opens and allows you to enter the name for the new switching program.

The screenshot shows the 'New Procedure' dialog box. It has a title bar with a close button (X). The main area contains a text input field with the placeholder text 'Please Enter the Name of the New Procedure'. Below this is a text input field containing the text 'Power Down - Transformer 324'. To the right of the input fields are three buttons: 'OK - New', 'OK - Create From...', and 'Cancel'.

[sc_14-33, 1, en_US]

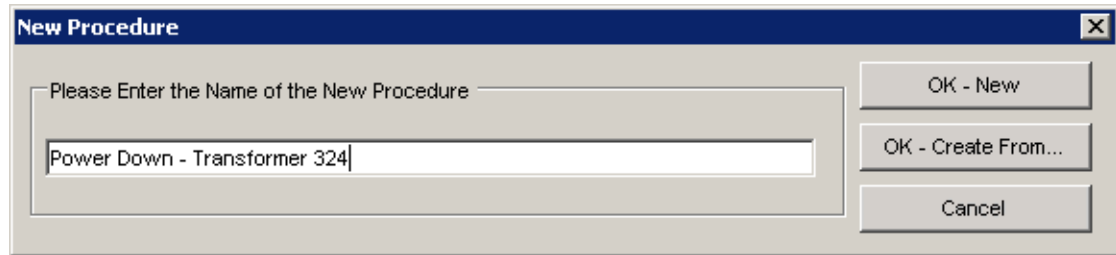
Figure 14-31 New Procedure Dialog

- Click **OK-New** to create an empty switching program.
SPM creates a new empty switching program and opens it in a separate detail tab. The state of the new switching program is set to **New**. The type of the switching program is set to **SP**.
SPM automatically creates an entry for the modification history containing the creation date and the user name.

Creating Switching Programs Based On Template

- Open the **Switching Program** dialog and select the **SP Summary** display.
- Select **File > New Procedure** from the menu bar.

The **New Procedure** dialog opens and allows you to enter the name for the new switching program.

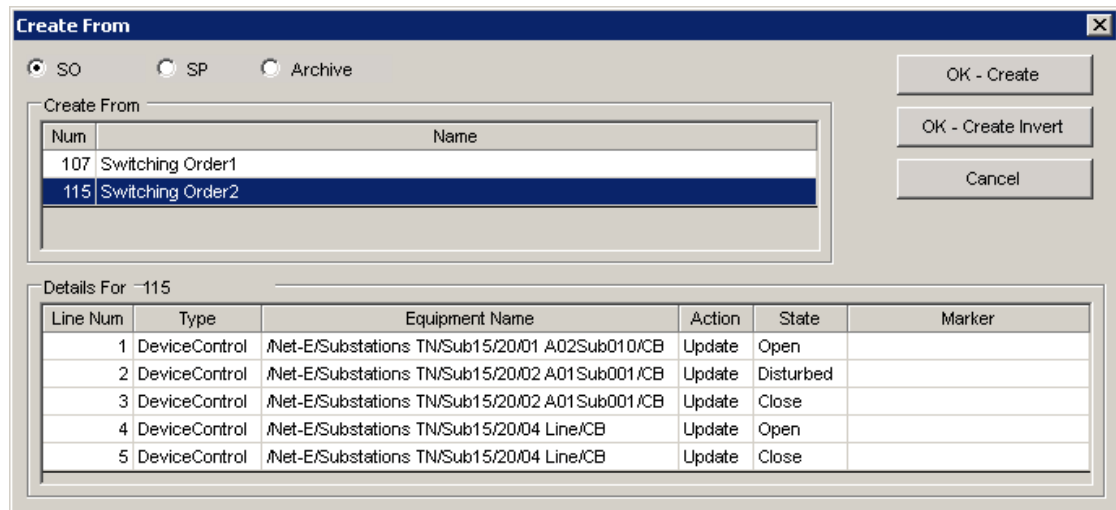


The **New Procedure** dialog box has a title bar with a close button. It contains a text input field with the placeholder text "Please Enter the Name of the New Procedure". Below this, the text "Power Down - Transformer 324" is entered. To the right of the input field are three buttons: "OK - New", "OK - Create From...", and "Cancel".

[sc_14-33, 1, en_US]

Figure 14-32 New Procedure Dialog

- Click the **OK-Create From** button to select a template for the creation of the new switching program. The **Create From** dialog opens.



The **Create From** dialog box has a title bar with a close button. It features three radio buttons: **SO** (selected), **SP**, and **Archive**. Below these is a section titled "Create From" containing a table with two columns: "Num" and "Name". The table lists two items: "107 Switching Order1" and "115 Switching Order2", with the second item selected. To the right of the table are three buttons: "OK - Create", "OK - Create Invert", and "Cancel". Below the table is a section titled "Details For 115" containing a table with columns: "Line Num", "Type", "Equipment Name", "Action", "State", and "Marker".

Line Num	Type	Equipment Name	Action	State	Marker
1	DeviceControl	/Net-E/Substations TN/Sub15/20/01 A02Sub010/CB	Update	Open	
2	DeviceControl	/Net-E/Substations TN/Sub15/20/02 A01Sub001/CB	Update	Disturbed	
3	DeviceControl	/Net-E/Substations TN/Sub15/20/02 A01Sub001/CB	Update	Close	
4	DeviceControl	/Net-E/Substations TN/Sub15/20/04 Line/CB	Update	Open	
5	DeviceControl	/Net-E/Substations TN/Sub15/20/04 Line/CB	Update	Close	

[sc_14-35, 1, en_US]

Figure 14-33 Create From Dialog

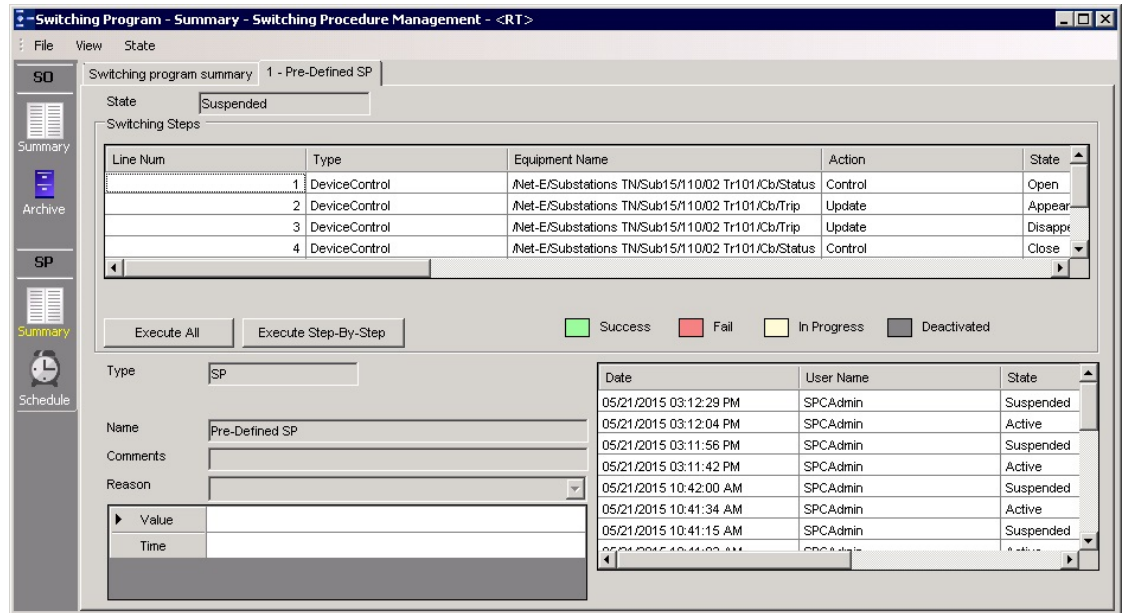
- Select the desired switching program from the **Create From** list. The **Details** list shows all steps of the selected switching program.
- Click **OK-Create** to create a new switching program containing the steps in original order. Alternatively, click **OK-Create Invert** to create a new switching program containing in reverse order. SPM creates a new empty switching program and opens it in a separate detail tab. The state of the new switching program is set to **New**. The type of the switching program is set to **SP**. SPM creates an entry for the modification history containing the creation date and the user name automatically.

14.4.2.3 Modifying Existing Switching Programs

The goal of this task is to open, change, save, and delete an existing switching program.

Selecting Switching Programs

- Open the **Switching Program** dialog and select the **SP Summary** display.
- Double-click a switching program in the list of existing switching programs.
SPM opens the selected switching program in a separate tab.



[SP_Details, 2, en_US]

Figure 14-34 SP Summary Display - Detail Tab

Modifying Switching Program Information

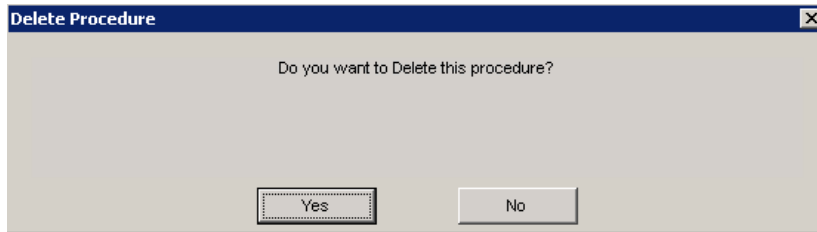
- Ensure that the switching program details are toggled on using the **View > Details** menu.
- Enter **Title** (up to 80 characters) and **Comment** (up to 255 characters) for the selected switching program in the corresponding text boxes.
The comment field is typically used to make special notes about the switching program.
- Select the **Reason** of the outage from the listbox.
For additional information on configuring the **Reason** in IMM, refer TS-SCADA-EN.
- Enter the details for the Switching Order in the Additional field section provided at the bottom of the SO Summary display.
For additional information on configuring the **Additional Field** in IMM, refer TS-SCADA-EN.
- Change the state for the selected switching program by selecting the corresponding menu item in the **State** menu.

Saving Modified Switching Programs

- Select **File > Save** to save the new or modified switching program to the database.

Deleting Switching Programs

- Open the **Switching Program** dialog and select the **SP Summary** display.
- Right-click the switching program to be deleted and select **Delete Procedure** from the context menu. The Delete Procedure Confirmation dialog opens.



[sc_14-14, 1, en_US]

Figure 14-35 Confirmation Dialog

- Click **No** to retain the procedure.
- Click **Yes** to delete the procedure.

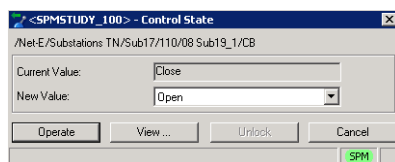
SPM deletes the selected switching program after confirmation.

14.4.2.4 Inserting New Steps

The goal of this task is to add operational steps or organizational steps to a switching program.

Inserting Operational Steps in Recording Mode

- Select and open the desired switching program.
- Select **Without Execution** from the **Recording Mode** list box. SPM automatically changes the state of the switching program to **Editing**.
- Invoke the network diagram that displays the device to be controlled.
- Select the desired element and open the **Control State** dialog.



[rgscad09-210110-01.tif, 1, en_US]

Figure 14-36 Control State Dialog

The string "SPM" in the right bottom area indicates that the SPM Recording mode is switched on. The green background color indicates that recording mode without execution is selected.

- Perform the normal supervisory control action. After selecting the button **Operate**, the supervisory control function is executed in the power system and is recorded as operational step of the switching program. A message in the status bar of the **Control State** dialog indicates if the step is added successfully.
- Perform the other desired supervisory control actions in the same or other network diagrams.
- After you have performed all desired supervisory control actions, return to the SPM Summary Display. The currently selected switching program contains all steps performed in the network displays.
- Select **Off** from the **Recording Mode** list box to finish the recording mode.

Inserting Wait Steps or Comment Steps

- Select the operational step after which the organizational step (wait step or comment step) can be inserted.
- Right-click the operational step and select **Insert Wait Step**, or **Insert Comment Step** from the context menu.
The organizational step is inserted to the switching program.
- Enter an appropriate comment if desired.

Inserting Conditional Steps

You can insert conditional steps that are executed only if certain network condition applies. For example, to insert an **If** condition, proceed as follows:

- Right-click the desired step and select **Insert Conditional Step > If** from the context menu.
- Insert the appropriate condition operations.
- If an **Else** condition is required, select **Insert Conditional Step > Else** from the context menu.
Follow the **Else** condition with all steps required for the else clause.
- Terminate the conditional execution by selecting **Insert Conditional Step > EndIf** from the context menu.

If-Not-Interlock and **While** conditions can be inserted in the same way.



NOTE

- Conditional checks can reference 1 device only.
- Conditional checks can be nested. The maximum depth is 3.
- When you are using an **If** block followed by an **Else** block, always one of them is executed depending on the network condition. It is not possible that both blocks are executed.
However, an **Else** block is not required.

14.4.2.5 Modifying Existing Steps

The goal of this task is to change, copy, activate, deactivate, or delete existing steps of a switching program.

Modifying Comments

- Select and open the switching program that contains the desired step.
- Right-click the step and select **Edit Step...** from the context menu.
The **Edit Step** dialog opens and allows you to enter a comment from the selected step.

The screenshot shows a dialog box titled "Edit Step". It has a "Comments" text box containing "Isolating Busbar" and an "Operator" text box which is empty. To the right of the "Operator" text box is a small downward-pointing arrow. At the bottom right of the dialog are "OK" and "Cancel" buttons.

[sc_14-18, 1, en_US]

Figure 14-37 Edit Step Dialog

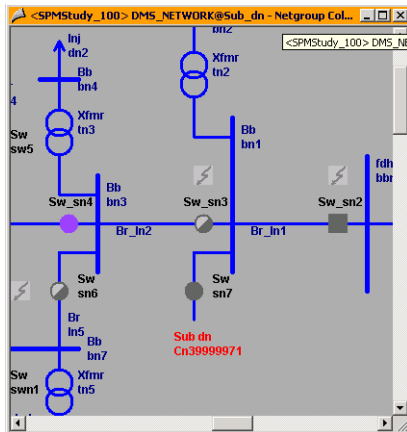
- Enter your comments in **Comments** text box.

- Select the **Operator** from the list box.
For additional information on configuring the operator in IMM, refer TS-SCADA-EN.
- Click **OK** button.

The comment and the operator is now displayed in the respective line of the switching procedure.

Calling Up a Network Display

- Select and open the switching program that contains the desired step.
- Right-click the step and select **Callup Network Display** from the context menu.
The network display showing the corresponding element is opened.



[spmso006-150508-01.tif, 1, en_US]

Figure 14-38 Network Diagram

Copying, Cutting, and Pasting Steps

- Select and open the switching program that contains the desired step.
- Right-click the step and select **Copy** from the context menu to copy the selected step into the clipboard.
Alternatively, right-click the step and select **Cut** from the context menu to copy the selected step into the clipboard and delete it from the current position.
- Select the step before which you want to paste the selected steps.
- Right-click and select **Paste** from the context menu to paste the selected steps in original order.
Alternatively, right-click and select **Invert Paste** from the context menu to paste the selected steps in reverse order.

Deactivating or Activating Steps

- Select and open the switching program that contains the desired step.
- Right-click the step and select **Deactivate Step** from the context menu.
Alternatively, right-click the step and select **Undo Deactivate Step** from the context menu.
SPM automatically changes the color of the step to indicate that the step is inactive or active.

Deleting Steps

- Select and open the switching program that contains the desired step.
- Right-click the step and select **Delete** from the context menu.
SPM deletes the selected step without further acknowledgment.

14.4.2.6 Handling Schedules

The goal of this task is to create, edit, or delete a schedule for the execution of an existing switching program.



NOTE

The scheduling functionality is only available in real-time context.

Creating Schedules

- Select the **Schedule** display from the view bar of the **Switching Procedure** dialog.
- Select **File > New Schedule** from the menu.
The **Enter Schedule** dialog opens.

Enter Schedule

Schedule Details

Schedule ID: 8

Type: [Dropdown Menu]

Type 1: Execute at Scheduled Date/Time
Type 2: Execute at Scheduled Date/Time, repeat after defined minutes
Type 3: Execute at Date/Time match

OK
Cancel

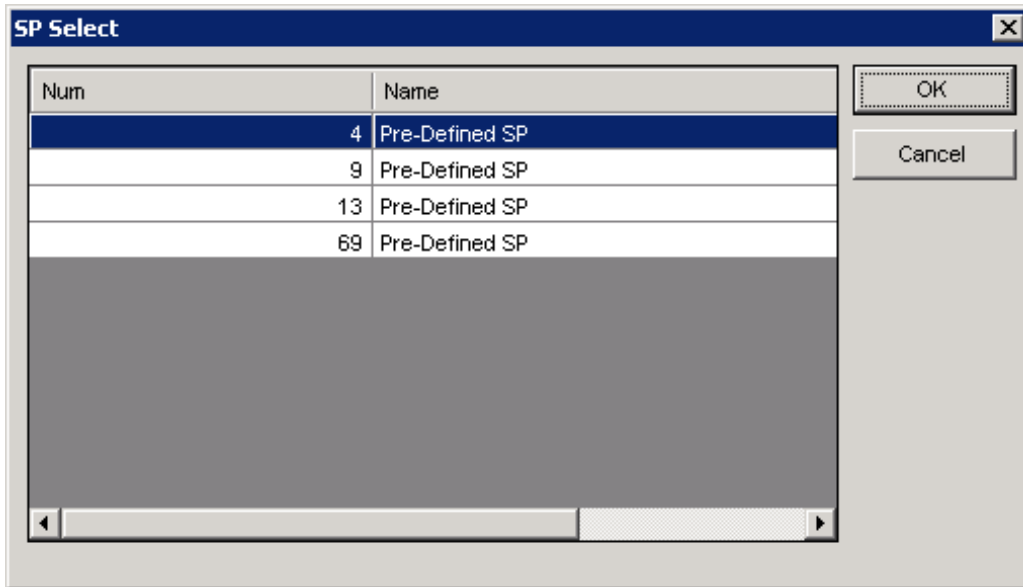
[sc_14-42, 1, en_US]

Figure 14-39 Enter Schedule Dialog

- Select the schedule type:
 - Type 1: The switching program is executed at the specified date and time.
 - Type 2: The switching program is executed first at the specified date and time. Execution is repeated with the specified frequency.
 - Type 3: The switching program is executed repeatedly at the specified day, hour, and minute.

- Click the button right of the **Num** text box to select the switching program with which the schedule can be defined.

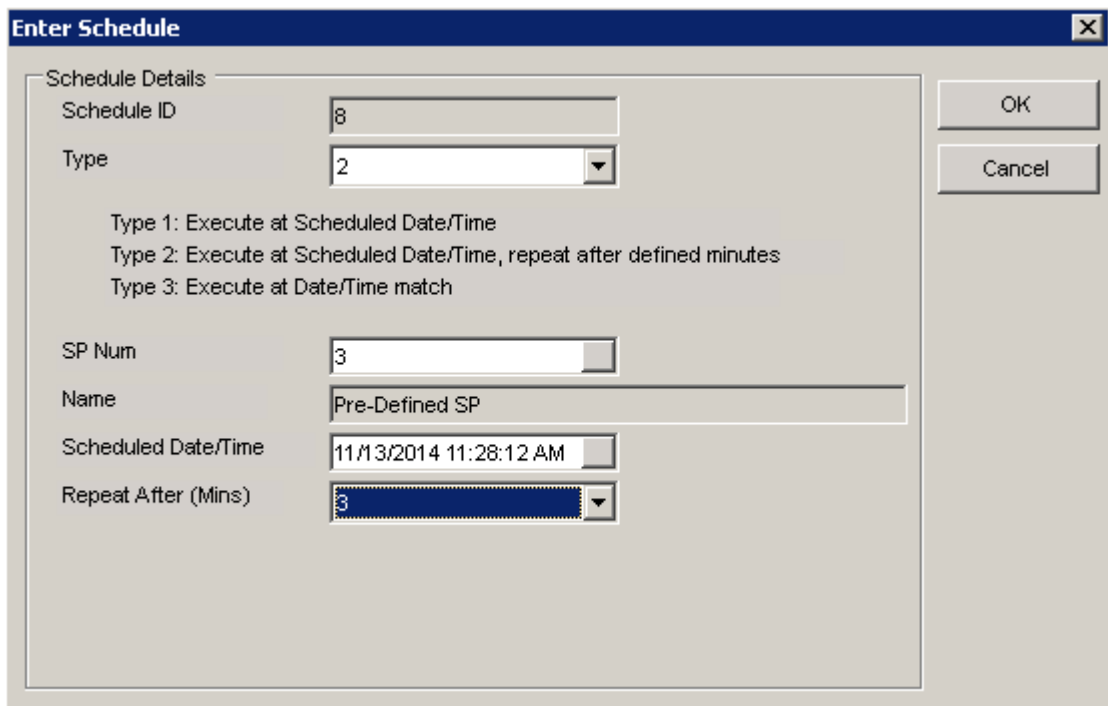
The **SP Select** dialog opens.



[sc_14-43, 2, en_US]

Figure 14-40 SP Select Dialog

- Select the desired switching program and click **OK**.
- Depending on the selected schedule type, enter **Scheduled Date/Time**, the **Repeat After** time interval or **Day**, **Hour**, and **Minute**.



[sc_14-44, 1, en_US]

Figure 14-41 Example of Completed Schedule Definition

- Click **OK** to save the schedule.
- The schedule is displayed in the list of schedules

Viewing Schedules

- Select the **Schedule** display from the view bar of the **Switching Procedure** dialog. All available schedules are displayed.

Schedule ID	Type	Num	Name	Scheduled Date/Time
9	2		11 Pre-Defined SP	5/28/2015 3:51:00 AM
10	2		4 Pre-Defined SP	5/30/2015 5:09:24 AM
8	1		109 New SP	5/29/2015 3:50:38 AM

[sc_14-45, 2, en_US]

Figure 14-42 Schedules Display

Editing Schedules

- Select the **Schedule** display from the view bar of the **Switching Procedure** dialog.
- Right-click the schedule you want to edit and select **Edit Schedule** from the context menu. The **Enter Schedule** dialog opens.
- Change **Type**, **Num**, and **Date/Time** for the schedule.
- Click **OK** to save the changes.

Deleting Schedules

- Select the **Schedule** display from the view bar of the **Switching Procedure** dialog.
- Right-click the schedule you want to delete and select **Delete Schedule** from the context menu. SPM deletes the selected schedule after confirmation of the user.

14.4.2.7 Simulating Switching Programs

The goal of this task is to simulate a switching program without execution. This is done to test the switching order and to simulate the effects of the operations. Therefore you have to switch to the study context.

Simulating Switching Programs

- Switch to the study context
- Select and open the desired switching program.
- In the **Switching Program** dialog click **Execute all**
The switching program is executed in the study context.

See the effects of the simulation

- Select **Study > Simulate** from the menu bar.
The corresponding worldmap shows the effect of the switching program like energization, deenergization by coloring the concerned network area.
- Alternatively, select **Study > Simulate PF** from the menu bar.
The Distribution System Power Flow display is opened showing the power flow data as if the switching program is executed.

14.4.2.8 Executing Switching Programs from Switching Program Dialog

The goal of this task is to execute a switching program from the **Switching Program** dialog.

Switching programs can be invoked in study context and in real-time context. Invoking a switching program in study context means testing the switching program without effect on the real-time system. Invoking a switching program in real-time context means to execute it in the real-time system.

Switching programs can be executed as one step at a time, step-by-step, or they can be started and allowed to run to completion. Organizational steps can pause and wait for user response or delay for a specified amount of time. By the use of conditional steps, the execution of steps is depending on certain network conditions.



NOTE

The following example describes the execution in real-time context, therefore the term “execute” is used. The execution in study context is similar.

Step-by-step execution is restricted to the execution of the next not-executed step. It is not affected by the currently selected line.

Executing Complete Switching Programs

- Select and open the desired switching program.
- Click **Execute All** to begin the execution of the switching program.
The steps of the switching program change to the “In Progress” color. As the steps are executed, the color changes depending on the state of the step.
Interlocking conditions are checked during execution and any violations cause the procedure to stop. The aborted step changes to the “Fail” color.

Executing Individual Steps

- Select and open the desired switching program.
- Right-click the step to be executed and select **Execute** from the context menu.
The step changes to the “In Progress” color.
Normal interlocking conditions are checked during execution and any violations reported.

Executing Step By Step

- Select and open the desired switching program.
- Click **Execute Step-by-step** to begin the execution of the switching program.
The steps of the switching program change to the “In Progress” color. As the steps are executed, the color changes depending on the state of the step.
Normal interlocking conditions are checked during execution and any violations cause the procedure to stop. The aborted step changes to the “Fail” color.

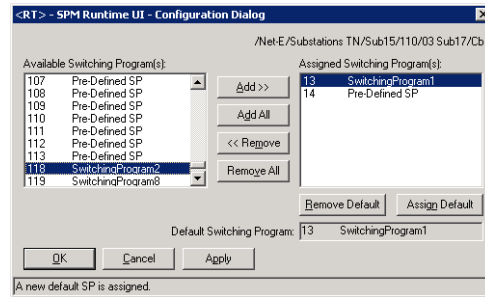
14.4.2.9 Executing Switching Programs from Network Diagram

The goal of this task is to execute a switching program by selecting a device in a network diagram.

Assigning Switching Programs

- Right-click the desired element in the network diagram and select **Configure SP** from the context menu.

The **SPM Runtime UI - Configuration** dialog opens.



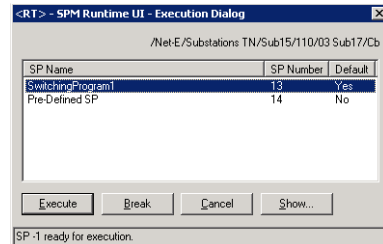
[spm010-150508-01.tif, 1, en_US]

Figure 14-43 SPM Runtime UI - Configuration Dialog

- Select the desired switching programs.
- Move the selected switching programs from the **Available Switching Program(s)** list to the Assigned Switching Programs list by clicking **Add >>**.
- To define the default switching program for the selected element, select the switching programs and click **Assign Default**.

Executing Switching Programs

- Right-click the desired element in the network diagram and select **Execute/Break SP** from the context.
- The **SPM Runtime UI - Execution** dialog opens. The default switching program is preselected.



[spm011-150508-01.tif, 1, en_US]

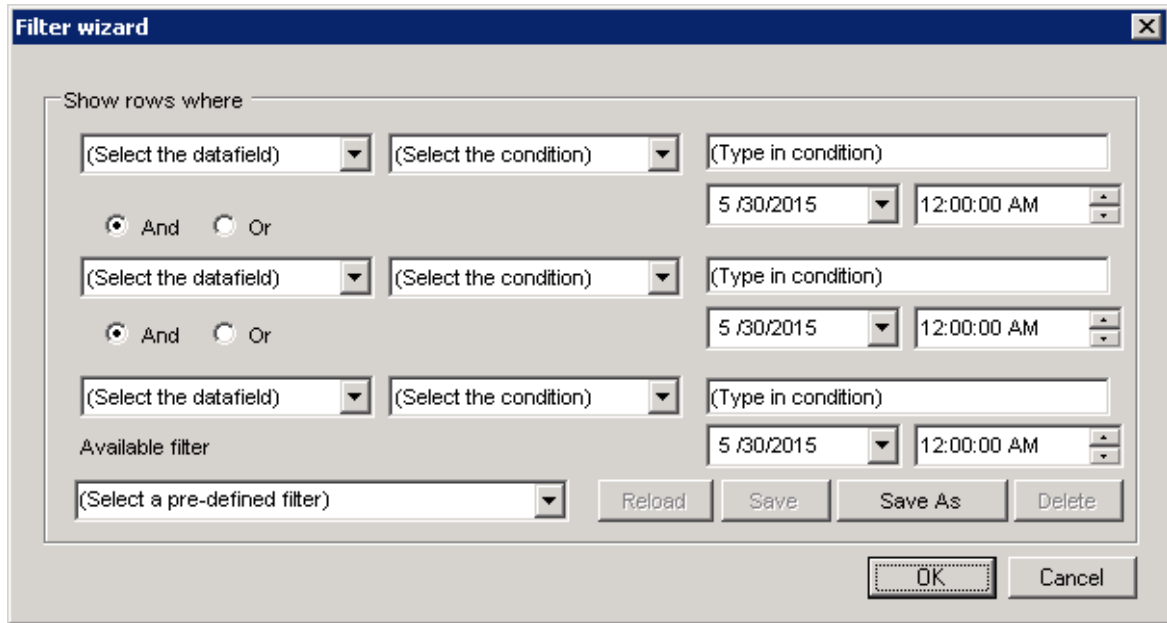
Figure 14-44 SPM Runtime UI - Execution Dialog

- Select the desired switching program from the list of the assigned switching programs.
- Click **Execute** to start the execution of the selected switching program.

14.4.2.10 Filtering Lists

The **Filter Wizard** allows custom filtering of the lists of the following displays:

- Switching Order Summary** display
- Switching Order Archive** display
- Switching Program Summary** display
- Switching Program Schedule** display

The image shows a 'Filter wizard' dialog box with a title bar containing a close button. The main area is titled 'Show rows where' and contains three identical filter criteria sections. Each section has three dropdown menus: '(Select the datafield)', '(Select the condition)', and '(Type in condition)'. Below each section are radio buttons for 'And' and 'Or'. The first two sections have '5 /30/2015' and '12:00:00 AM' selected in their respective dropdowns. Below these sections is an 'Available filter' section with a dropdown menu labeled '(Select a pre-defined filter)'. At the bottom right of the main area are four buttons: 'Reload', 'Save', 'Save As', and 'Delete'. At the very bottom of the dialog are 'OK' and 'Cancel' buttons.

[spmco001-150508-01.tif, 2, en_US]

Figure 14-45 Filter Wizard

Filtering Lists

- Right-click on an entry in one of the supported lists and select **Filter Wizard** from the context menu. The **Filter Wizard** dialog opens.
- Select up to 3 filter criteria (data field, operator, and condition).
- Click **OK** to execute the filtering criteria. SPM displays the filtered list according to the selected filter criteria.



NOTE

The **Filter Wizard** allows you to save the preferred filtering criteria in order to reuse them. Defined filtering settings can be selected, saved, or deleted from the **Available Filter** list box.

14.4.2.11 Generating Reports

The goal of this task is to create a report. Report are available for both switching orders and switching programs.

Generating Reports

- Select and open the desired switching order or switching program.
- Click **Report** icon in the toolbar or select **File > Generate Report** from the menu.
SPM automatically creates an Excel file containing the requested report.
This Excel file can now be printed, or saved as a file.

Switching Procedure Management Report							
		11/13/14 10:04				Additional Field Name Additional	
Procedure Number:	3						
Name:	Pre-Defined SP						
State:	Completed						
Scheduled Date:							
OR Number:							
Type:	SP						
Comments:	UnPlanned Activity						
Reason:	UnPlanned Activity						
Line Num:	1	Type:	DeviceControl				
Action:	Marker	State:	Set				
Equipment Name:	/Net-E/Substations TN/Sub15/110/03 Sub17/Cb			Marker:	Marker - Alarm Inhibit		
Execution Date:	11/10/2014 9:19						
Line Num:	2	Type:	DeviceControl				
Action:	Marker	State:	Set				
Equipment Name:	/Net-E/Substations TN/Sub15/110/03 Sub17/DisBb1			Marker:	Marker - Alarm Inhibit		
Execution Date:	11/10/2014 9:19						
Line Num:	3	Type:	DeviceControl				
Action:	Marker	State:	Set				
Equipment Name:	/Net-E/Substations TN/Sub15/110/04 Sub18_1/DisBb1			Marker:	Marker - Alarm Inhibit		
Execution Date:	11/10/2014 9:19						

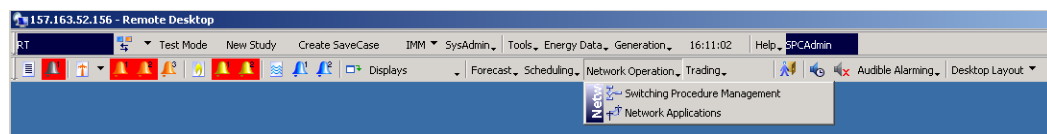
[sc_14-23_latest, 1, en_US]

Figure 14-46 Example Report Output

14.4.3 User Interface Description

14.4.3.1 Power Application Toolbar (PAT)

The switching program dialog can be started via the Power 5 Application Toolbar (PAT).

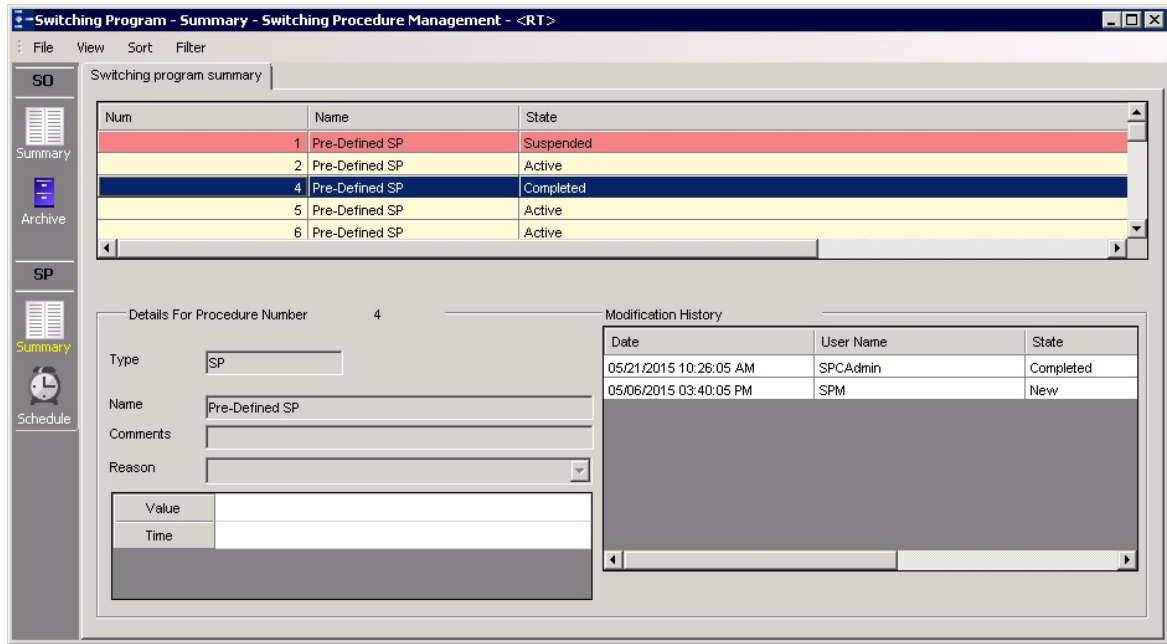


[rgscad10-210110-01.tif, 1, en_US]

Figure 14-47 PAT Network Operation Menu

14.4.3.2 Summary Display - Summary Tab

The summary tab of the **SP Summary** display provides an overview of all switching programs.



[SP_Summary, 2, en_US]

Figure 14-48 SP Summary Display - Summary Tab

Menu Bar

Allows the user to create a new switching program, to customize the display and to sort and filter the list of switching programs.

Switching Programs List

Provides a list of all switching programs containing unique switching program number, name, and state of all switching programs.

The background color of a line indicates the state of the respective switching program:

New:	White	Active:	Yellow
Editing:	Blue	Completed:	Gray
Checked:	Light Blue	Suspended:	Red
		Selected:	Dark Blue

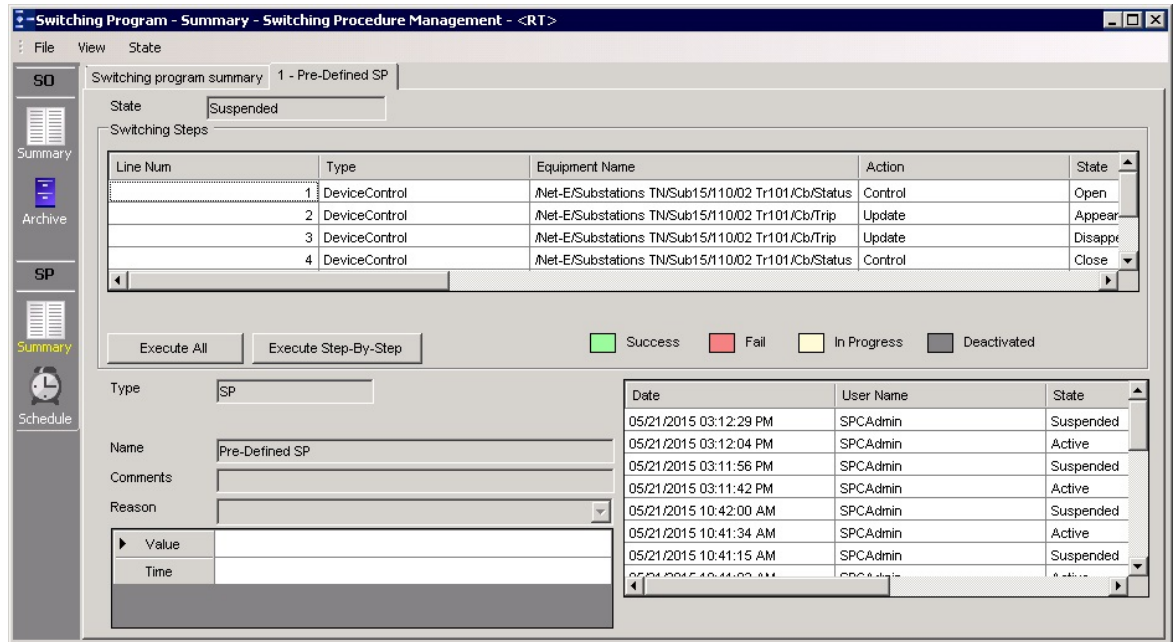
Switching Program Detail

Contains detailed information about the selected switching program as well as a history of all modifications. The **Switching Program Detail** section can be toggled on and off by selecting the **View > Details** menu item.

14.4.3.3 Summary Display - Detail Tab

If the user creates a new switching program or opens an existing switching program, the **SP Summary** display opens a new tab for the switching program.

This tab provides detailed information about the selected switching program. The name of the new tab consists of the switching program number and name.



[SP_Details, 2, en_US]

Figure 14-49 SP Summary Display - Detail Tab

Menu Bar

Allows the user to create a new switching program, to save a switching program as template, to generate a report, to customize the display, and to change the state of the selected switching program.

Toolbar

Provides shortcut icons for the user to save the selected switching program and to generate a report.

Switching Program Steps

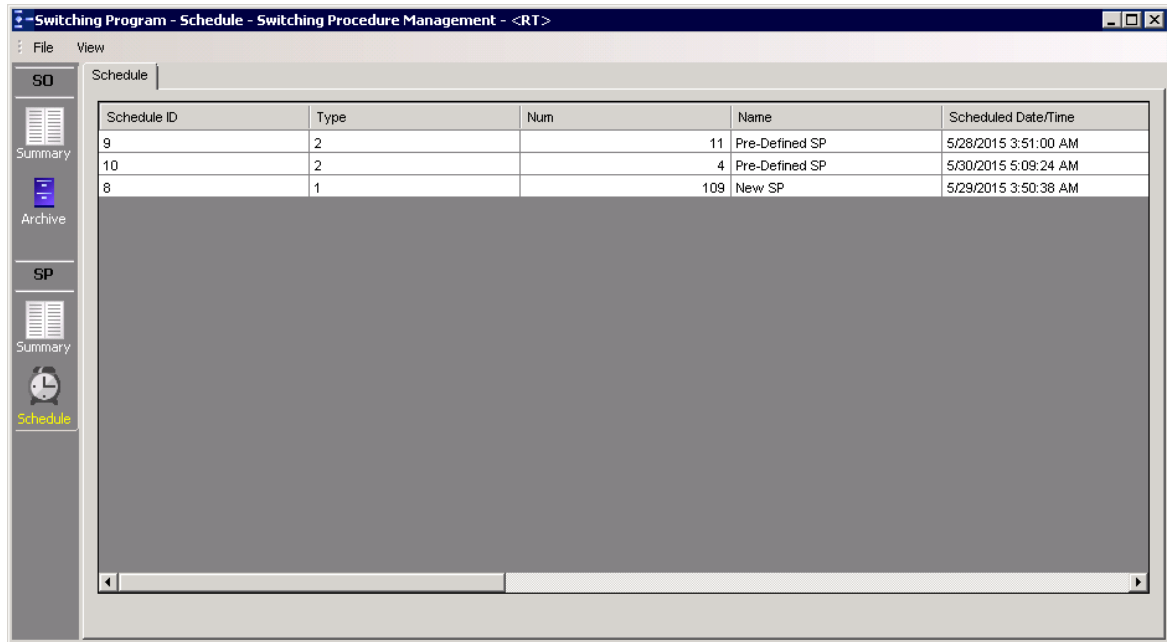
- The **Switching Steps** table provides a list of all defined steps of the selected switching program, including detailed information.
- The **Recording Mode** list box allows to toggle the recording mode to add steps to the switching program.
- The **Execute All** button allows the execution of the whole switching program, the **Execute Step-By-Step** button allows the step by step execution of the switching program.

Switching Program Detail

Contains detailed information about the selected switching program as well as a history of all modifications. The **Switching Program Detail** section can be toggled on and off by selecting the **View > Details** menu item.

14.4.3.4 Schedule Display

The **Schedule** display provides a list of all defined schedules for the scheduled execution of switching programs (once or periodically). The scheduled execution of switching programs is only available in real-time context.



The screenshot shows a software window titled "Switching Program - Schedule - Switching Procedure Management - <RT>". The window has a menu bar with "File" and "View". On the left is a sidebar with icons for "SD", "Summary", "Archive", "SP", "Summary", and "Schedule" (which is highlighted). The main area displays a table with the following data:

Schedule ID	Type	Num	Name	Scheduled Date/Time
9	2	11	Pre-Defined SP	5/28/2015 3:51:00 AM
10	2	4	Pre-Defined SP	5/30/2015 5:09:24 AM
8	1	109	New SP	5/29/2015 3:50:38 AM

[sc_14-45, 2, en_US]

Figure 14-50 Schedule Display

Menu Bar

Allows the user to create a new schedule and to edit or delete an existing schedule.

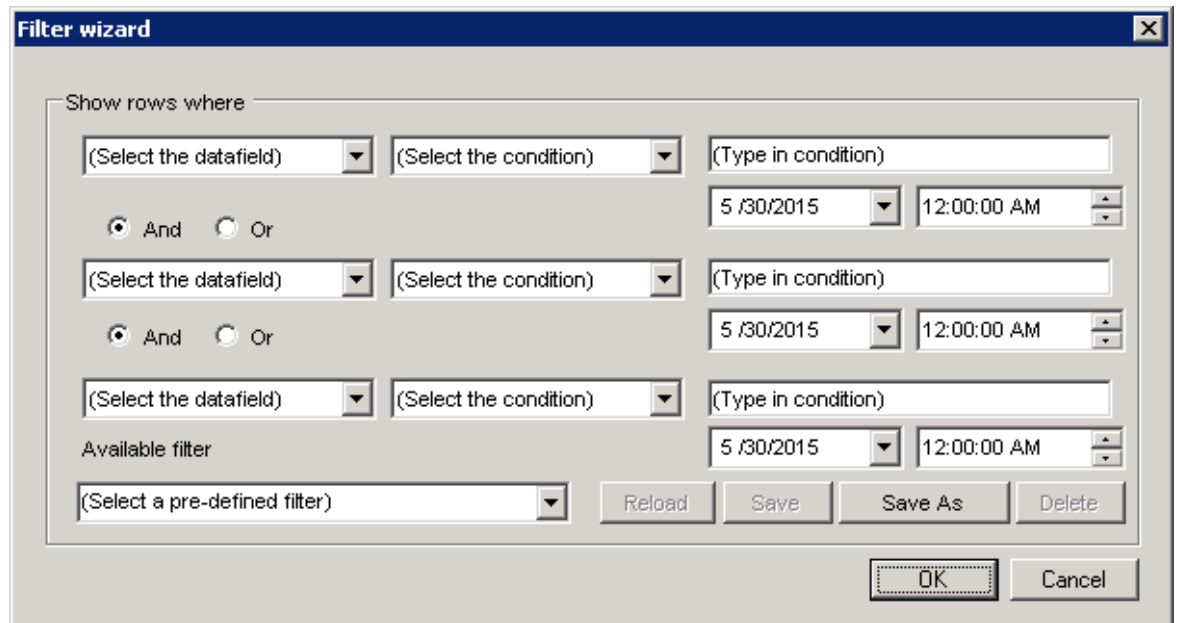
14.4.3.5 Filter Wizard

The **Filter Wizard** allows custom filtering of the following lists:

- Switching orders summary list
- Switching programs summary list

Filter wizard can be invoked by right-clicking on an entry in one of the supported lists and selecting the **Filter Wizard** from the context menu.

You can define filter criteria and store them under a name, which allows later call-up, use and modification.



The 'Filter wizard' dialog box is designed to help users create filters for data rows. It features a title bar with a close button. The main area is titled 'Show rows where' and contains three identical filter rows. Each row has three dropdown menus: '(Select the datafield)', '(Select the condition)', and '(Type in condition)'. Below the first two dropdowns are radio buttons for 'And' and 'Or'. The third dropdown is followed by two date/time pickers. The first row is pre-filled with '5 /30/2015' and '12:00:00 AM'. Below these rows is an 'Available filter' section with a dropdown for '(Select a pre-defined filter)'. At the bottom right are buttons for 'Reload', 'Save', 'Save As', 'Delete', 'OK', and 'Cancel'.

Filter wizard

Show rows where

(Select the datafield) (Select the condition) (Type in condition)

5 /30/2015 12:00:00 AM

And Or

(Select the datafield) (Select the condition) (Type in condition)

5 /30/2015 12:00:00 AM

And Or

(Select the datafield) (Select the condition) (Type in condition)

5 /30/2015 12:00:00 AM

Available filter

(Select a pre-defined filter)

Reload Save Save As Delete

OK Cancel

[spmco001-150508-01.tif, 2, en_US]

Figure 14-51 Filter Wizard Dialog

15 Temporary Network Elements

15.1	Introduction	256
15.2	Operational Steps	260
15.3	User Interface Description	307

15.1 Introduction

Temporary Network Elements (TNEs) are of special importance in case of network construction, during outages caused by severe weather or other disaster, or as emergency backup during maintenance of the network or power plants. In those scenarios, the power utilities are forced to still maintain the balance between generation and consumption as well as to resupply as fast as possible affected customers.

A utility typically owns a pool of specific Temporary Network Element types.

For example, jumpers, cuts, and grounds.

To put Temporary Network Elements into operation does always change the network topology of the network on a temporary base.

The Spectrum Power 5 function Temporary Network Elements (TNE), enables the operators and planners to make temporary changes to the single-line diagrams that reflect the changes to the network made in the field. These changes can be either planned, such as a lineman of a field crew doing maintenance work, or unplanned, such as a tree falling and breaking a power line. When the repair is completed, the change can be backed out and the network topology returns to its original state.

The Temporary Network Elements need to be visible within the Runtime UI displays and their impact on the network.

For example, network coloring in single-line diagrams and interlocking rules within SCADA.

Temporary Network Elements are considered within DNA and SCADA.

TNE Advanced Tooltip

The TNE affected objects show the configured tooltip for the object in the worldmap. The details which TNE is added can be seen with a display callup to the original worldmap where the TNEs were inserted, but are not shown additionally within the tooltip itself.



NOTE

The tooltip does not provide TNE information.

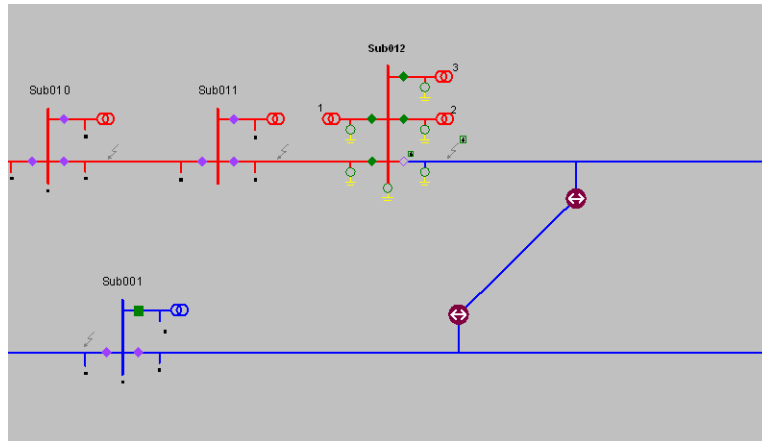
Jumpers

A temporary jumper is a short length of wire used temporarily to complete a circuit or to bypass a break in a circuit. A jumper connects 2 valid conductors topologically, similar to creating a power line (segment). The TNE Jumper is used to indicate that a lineman attached a temporary jumper wire to a power line (segment) or a busbar (segment) and then attached the other end to another power line (segment) or busbar (segment).



NOTE

Jumpers, which connect equipments that are part of different layers will also be displayed if only one of the connected equipment is shown.

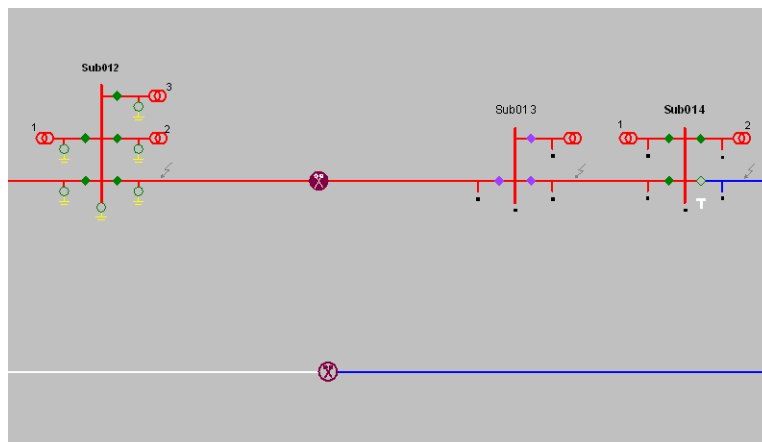


[rguios01-050811-01.tif, 1, en_US]

Figure 15-1 Jumpers

Cuts

A temporary cut is used to indicate a break in a power line (segment) which could occur for multiple reasons. A power line could break from natural causes, such as ice, fire, wind, or falling trees. A TNE Cut cuts a conductor topologically. It could be also purposefully cut or disconnected by a lineman performing maintenance.

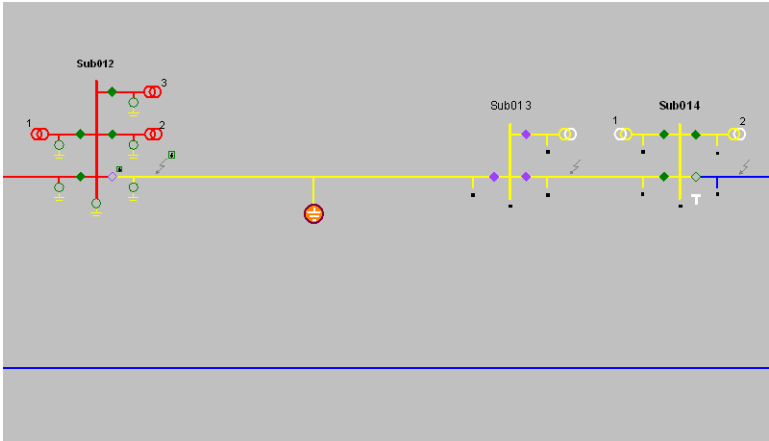


[rguios02-050811-01.tif, 1, en_US]

Figure 15-2 Cuts

Grounds

A temporary ground is a short length of wire attached to a power line (segment) or a busbar (segment) and then grounded at the other end. Either each phase is treated separately or combined by specific earthing devices. The result of a temporary ground can be understood, similar to closing a ground disconnect. The TNE Ground is used to indicate that a lineman has attached a temporary ground device at the specific conductor.



[rguios03-050811-01.tif, 1, en_US]

Figure 15-3 Ground

Temporary Network Elements in Spectrum Power 5 5.21

Spectrum Power 5 5.21 offers temporary network elements that model dynamic modifications to the network (jumpers, cuts, and grounds). All the temporary network elements can be inserted and removed at runtime, which shall be mirrored in an HIS Replay run also.

Symbols

The following table shows the standard presentation logic for the TNE disconnectors and ground disconnector.

Table 15-1 Standard Presentation Logic for the TNE Disconnectors

TNE	Object	Presentation in-active	Presentation active
Ground	Ground Disconnector		
Cut	Disconnector		
Jumper	Disconnector		

NOTE

The Presentation Logic for representation of TNEs in single-line diagrams is configurable within the configuration file, PDClientConfig.cfz, which is a part of Spectrum Power 5. The Presentation Logic IDs can be configured for Cuts, Grounds, and Jumper separately. After restart of the UI, the changed presentation will be used for TNEs.

NOTE

Ensure that PDClientConfig.cfz configuration file is edited using the Configuration Manager Editor. During the update, changes done in pdclient_wincc_config.xml file is automatically available in PDClientConfig.cfz file. Open PDClientConfig.cfz file to verify the updates.

For additional information on Configuration Manager Editor, refer RG-SYSW-EN.



NOTE

If a Presentation Logic GUID is inserted and is not a valid Presentation Logic GUID, no graphical representation will be shown for the symbol. The lines will be displayed. For example, the ground.

Maximum Number of TNEs on a Parent Object

A maximum of 10 TNEs can be placed on one parent object. If you try to insert the 11th TNE on the same parent object (for example, the same line) you will get an error message.

15.2 Operational Steps

15.2.1 Overview



NOTE

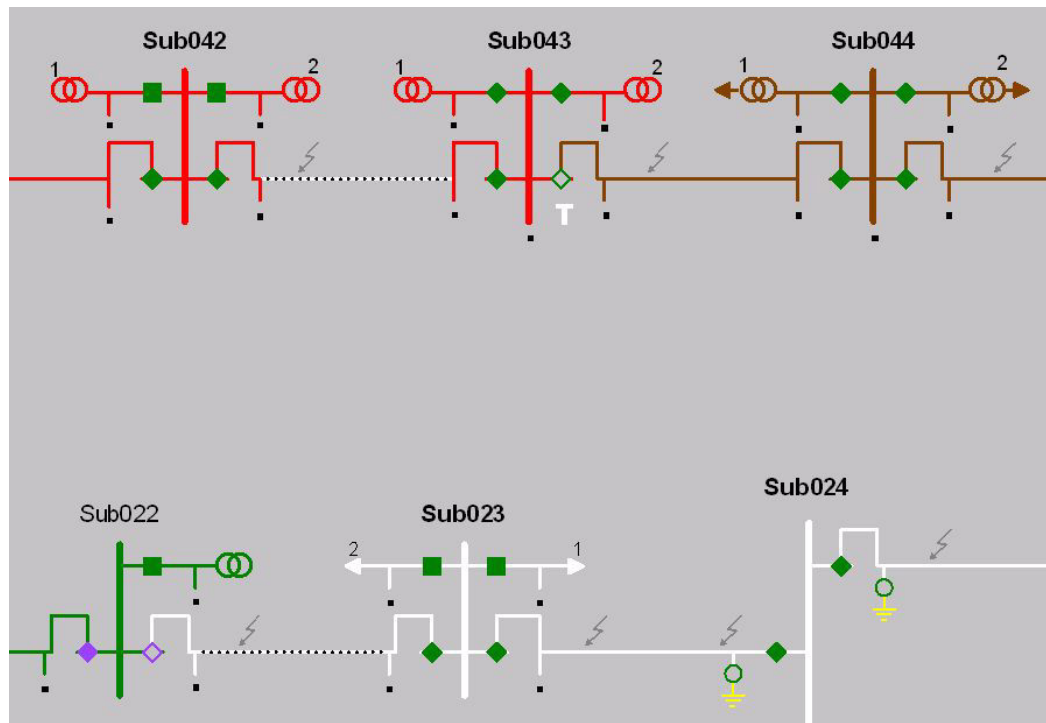
TNEs have to be preconfigured in the IMM and only the pool of preconfigured TNEs can be used at runtime.

15.2.2 Jumpers

15.2.2.1 Inserting a Jumper

The goal of this task is to insert a **Jumper** in the single-line diagram.

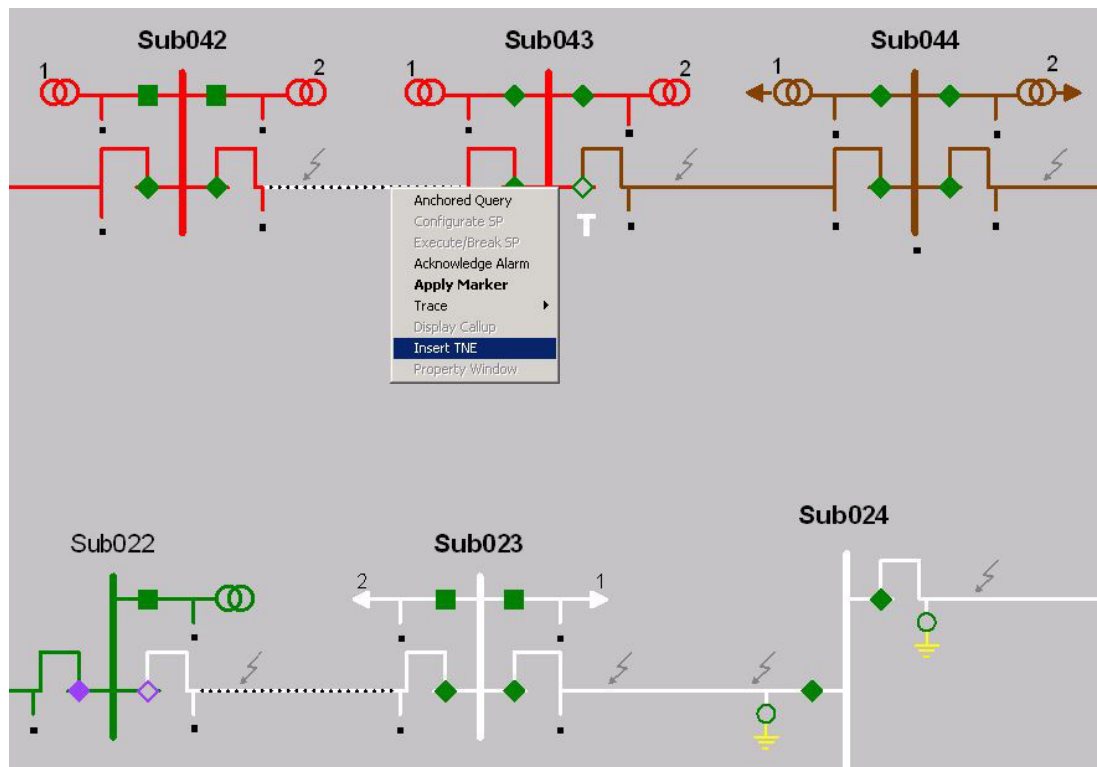
- From the worldmap, select the lines, the busbar segments, or the combination of both, with restriction of 2 objects to insert the **Jumper**.



[rguios01-020113-01.tif, 1, en_US]

Figure 15-4 Select Lines to Insert Jumper

- Right-click on the line.

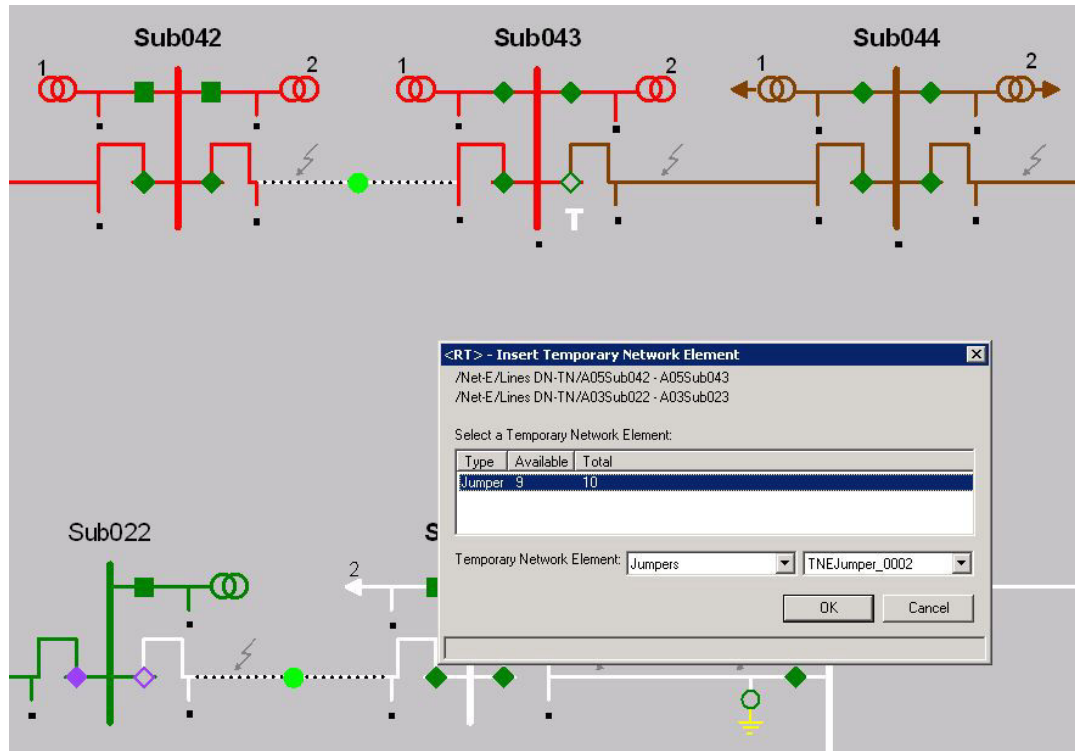


[rguios02-020113-02.tif, 1, en_US]

Figure 15-5 Context Menu

The context menu is displayed.

- Select **Insert TNE**.
- The **Insert Temporary Network Element** dialog is displayed.
- You can specify the position of the **Jumper** on the selected line by moving the green position marker.
- Select **Jumper** from the list.

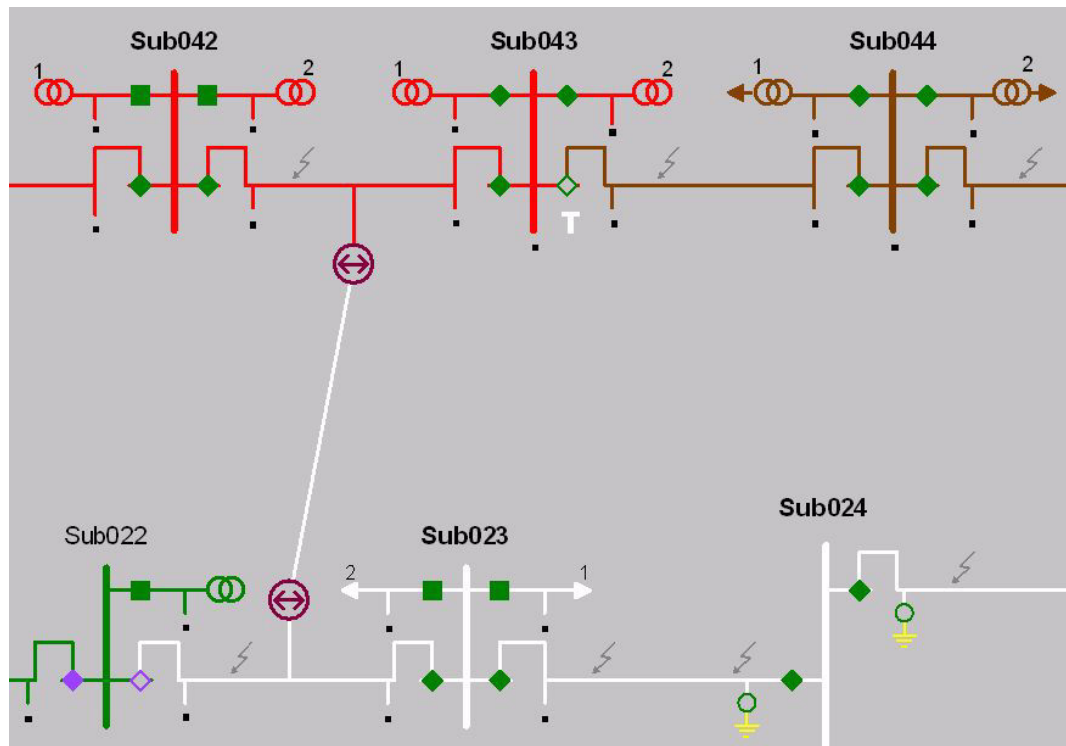


[rguios03-020113-03.tif, 1, en_US]

Figure 15-6 Select Jumper

- Select an element from the list of available jumpers.
By default, the first available element is selected.

- Click **OK**.
The selected **Jumper** is inserted on the selected lines.



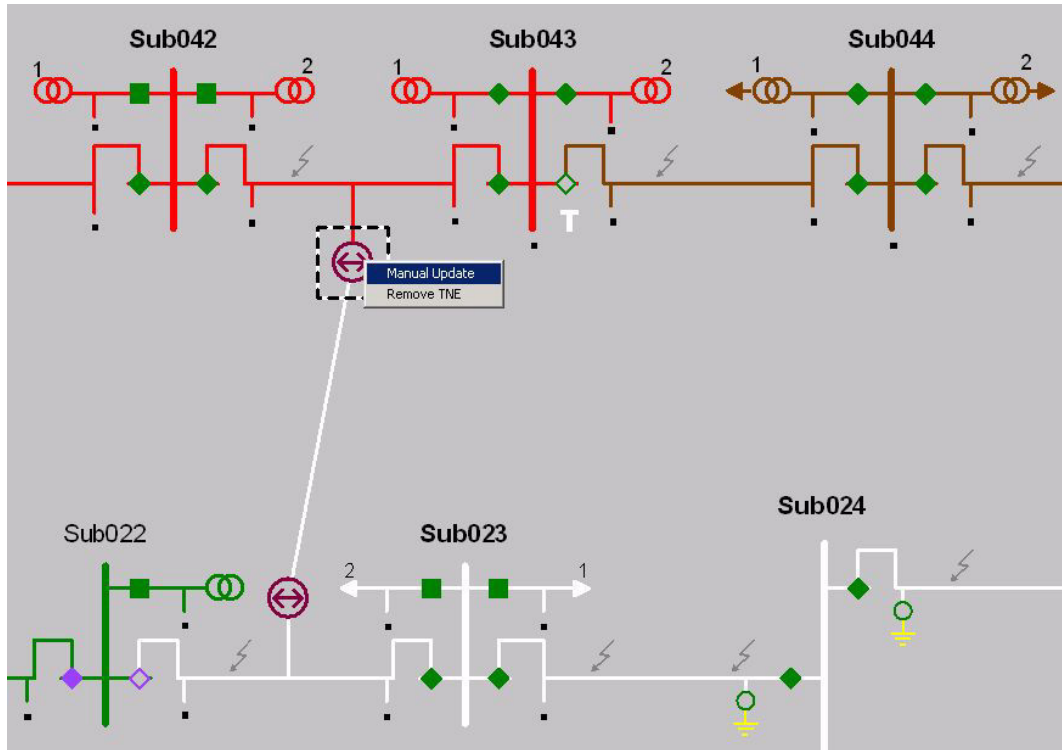
[rguios04-020113-04.tif, 1, en_US]

Figure 15-7 Jumper Inserted

15.2.2.2 Activating a Jumper

The goal of this task is to activate a **Jumper** in the single-line diagram.

- From the worldmap, select one side of the **Jumper** to activate.
- Right-click on the selected side of the **Jumper**.



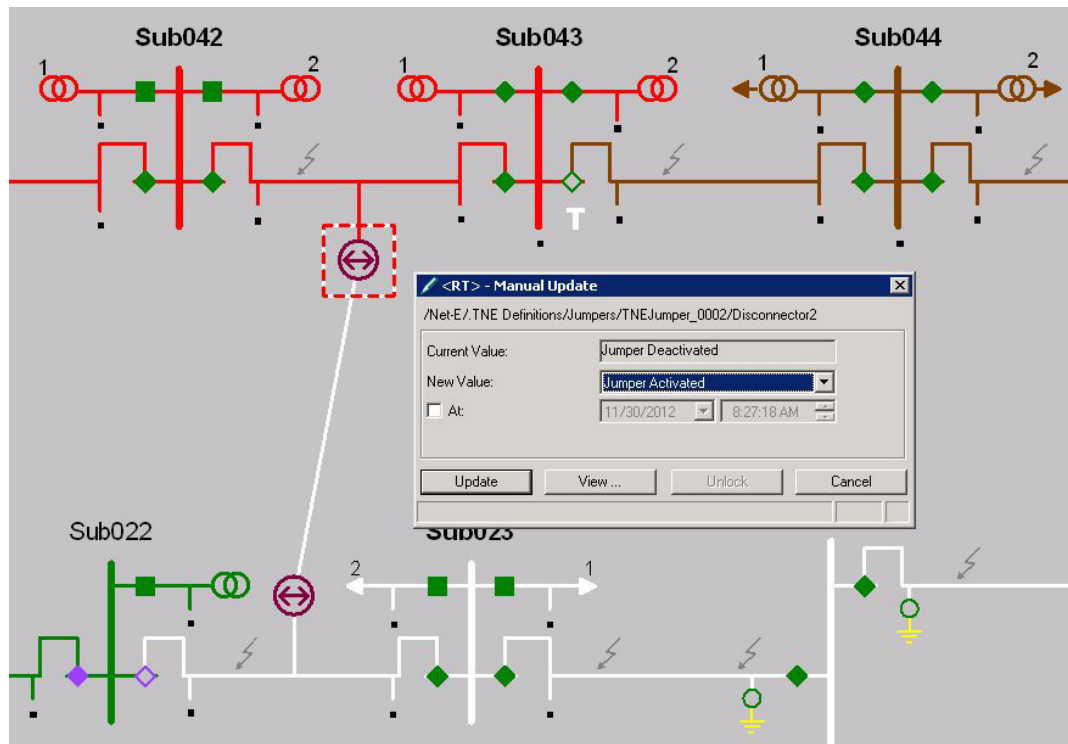
[rguios05-020113-05.tif, 1, en_US]

Figure 15-8 Context Menu

The context menu is displayed.

- Select **Manual Update**.

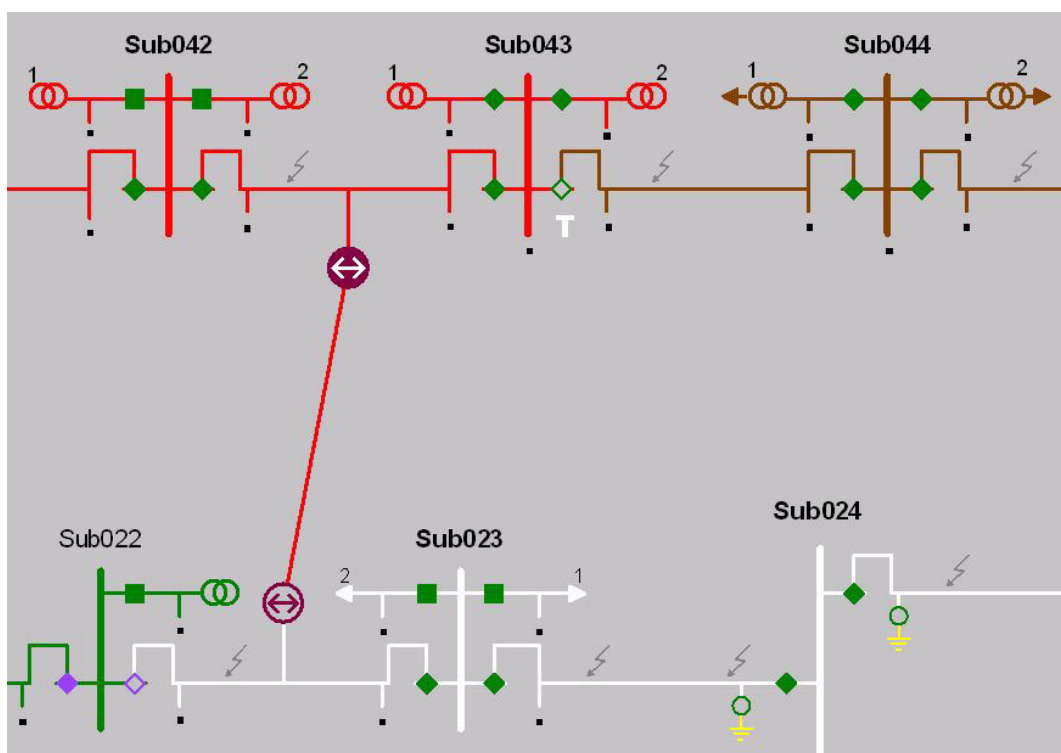
- The **Manual Update** dialog is displayed and **Jumper activated** is proposed as new value for the state of the Jumper.



[rguios40-030113-32.tif, 1, en_US]

Figure 15-9 Manual Update Dialog

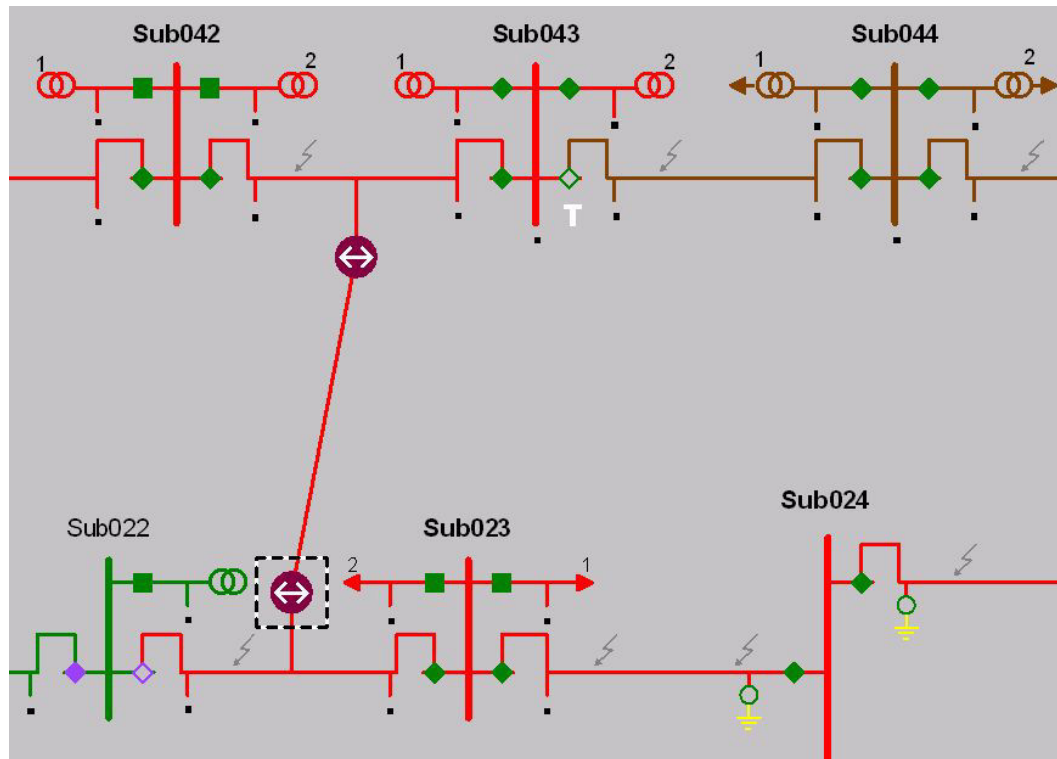
- Click **Update** to activate the side of the **Jumper**.
One side of the **Jumper** is activated.



[rguios07-020113-07.tif, 1, en_US]

Figure 15-10 One Side of the Jumper is Activated

Similarly you can activate the other side of the **Jumper**.



[rguios08-020113-08.tif, 1, en_US]

Figure 15-11 Activated Jumper



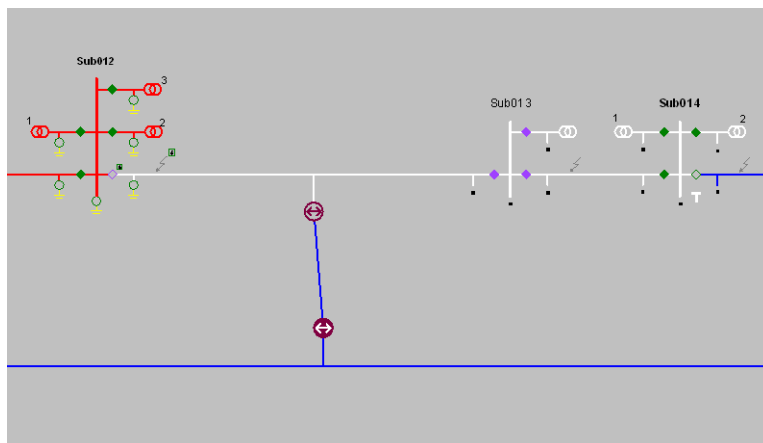
NOTE

When a new **Jumper** is inserted, by default the **Jumper** is deactivated.

15.2.2.3 Deactivating a Jumper

The goal of this task is to deactivate a **Jumper** in the single-line diagram.

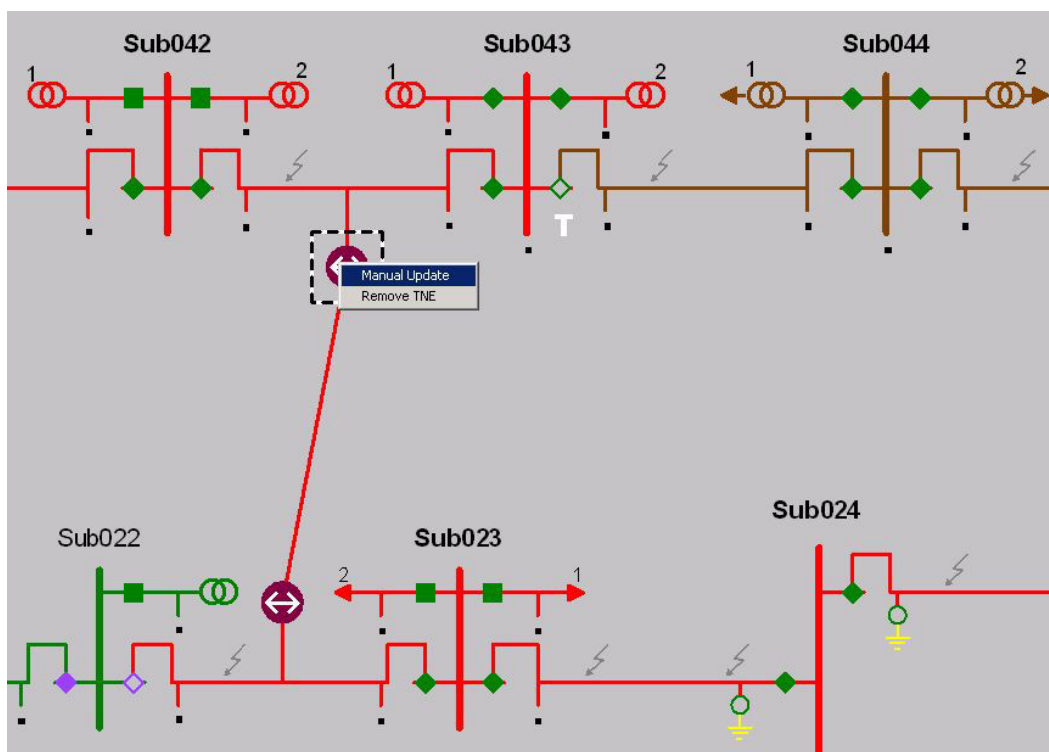
- From the worldmap, select one side of the **Jumper** to deactivate.



[rguios12-050811-01.tif, 1, en_US]

Figure 15-12 Select One Side of the Jumper to Deactivate

- Right-click on the **Jumper**.



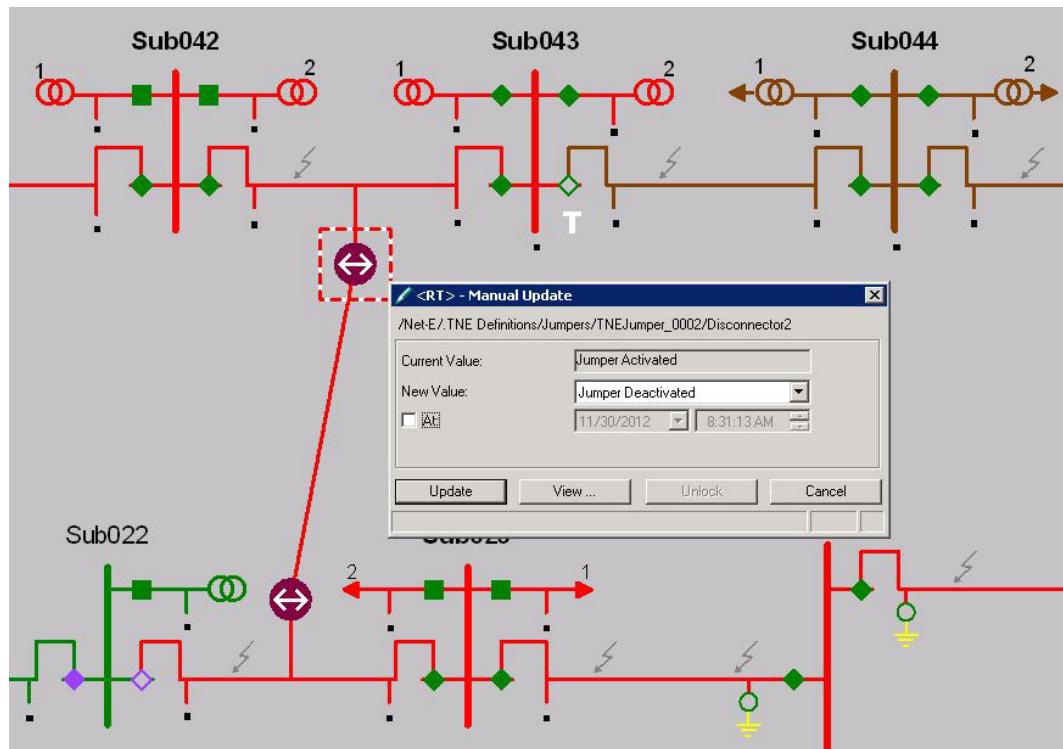
[rguios09-020113-09.tif, 1, en_US]

Figure 15-13 Context Menu

The context menu is displayed.

- Select **Manual Update**.

- The **Manual Update** dialog is displayed and **Jumper deactivated** is proposed as new value for the state of the **Jumper**.



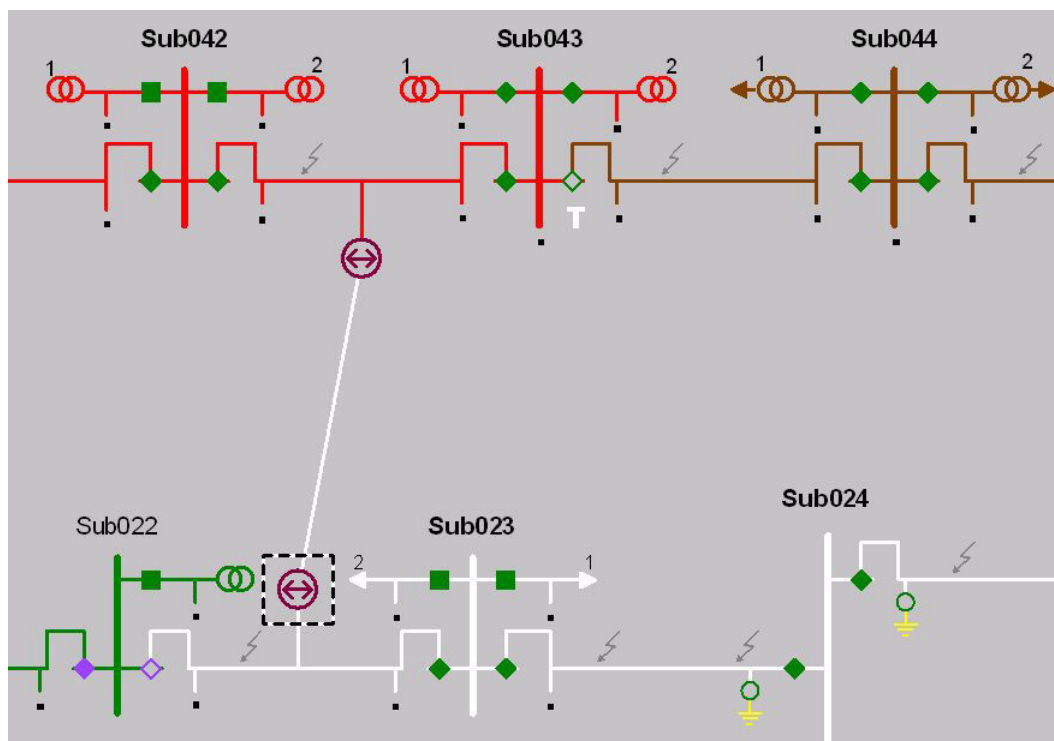
[rguios10-030113-10.tif, 1, en_US]

Figure 15-14 Manual Update Dialog

- Click **Update** to deactivate one side of the **Jumper**.

One side of the **Jumper** is deactivated.

If you want to deactivate the complete **Jumper**, perform this again for the remaining side.



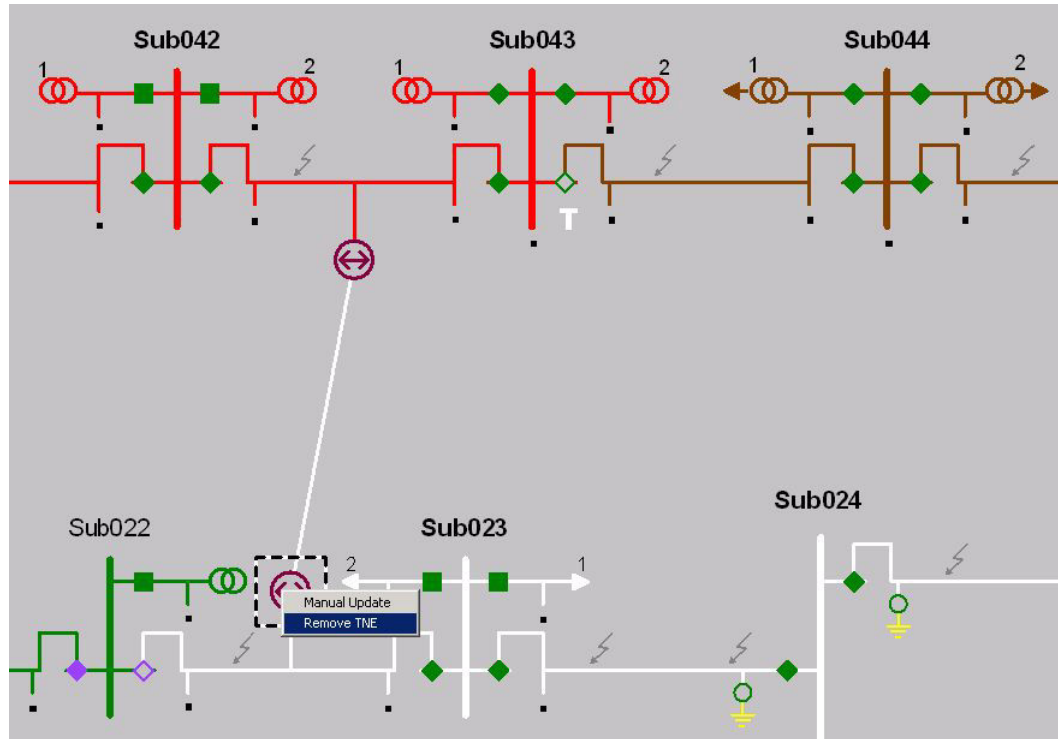
[rguios11-030113-11.tif, 1, en_US]

Figure 15-15 Jumper Deactivated on Both the Sides

15.2.2.4 Removing a Jumper

The goal of this task is to remove a **Jumper** in the single-line diagram.

- From the worldmap, select one of the elements of the **Jumper** to remove.
- Right-click on the **Jumper**.



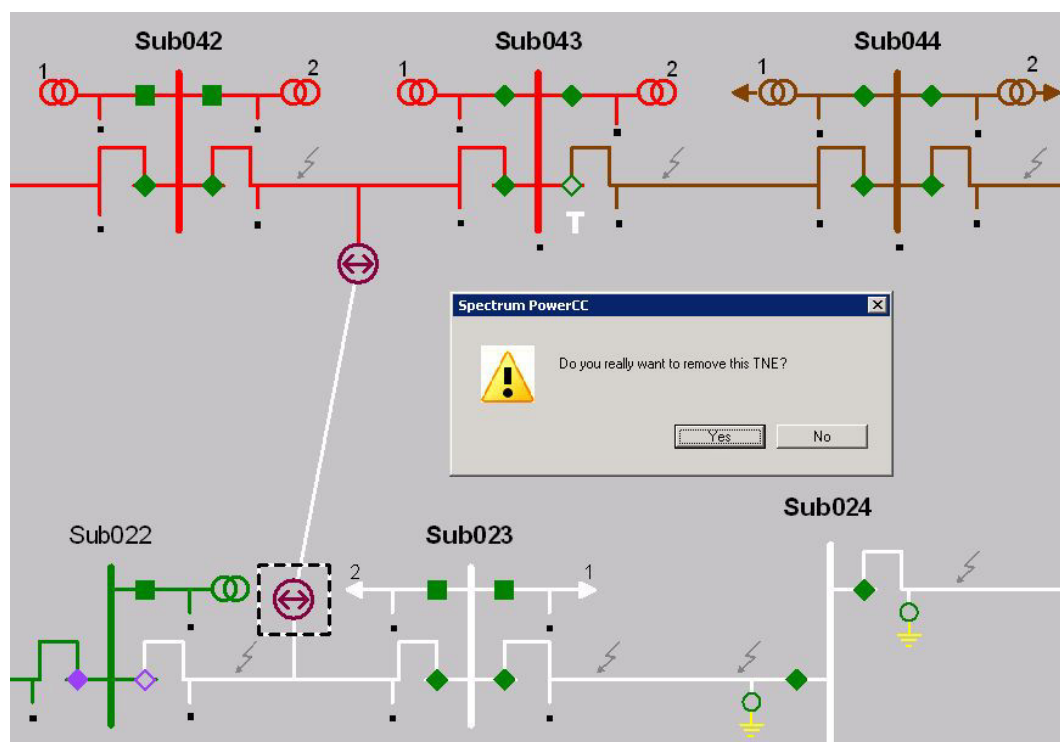
[rguios12-030113-12.tif, 1, en_US]

Figure 15-16 Context Menu

The context menu is displayed.

- Select **Remove TNE**.

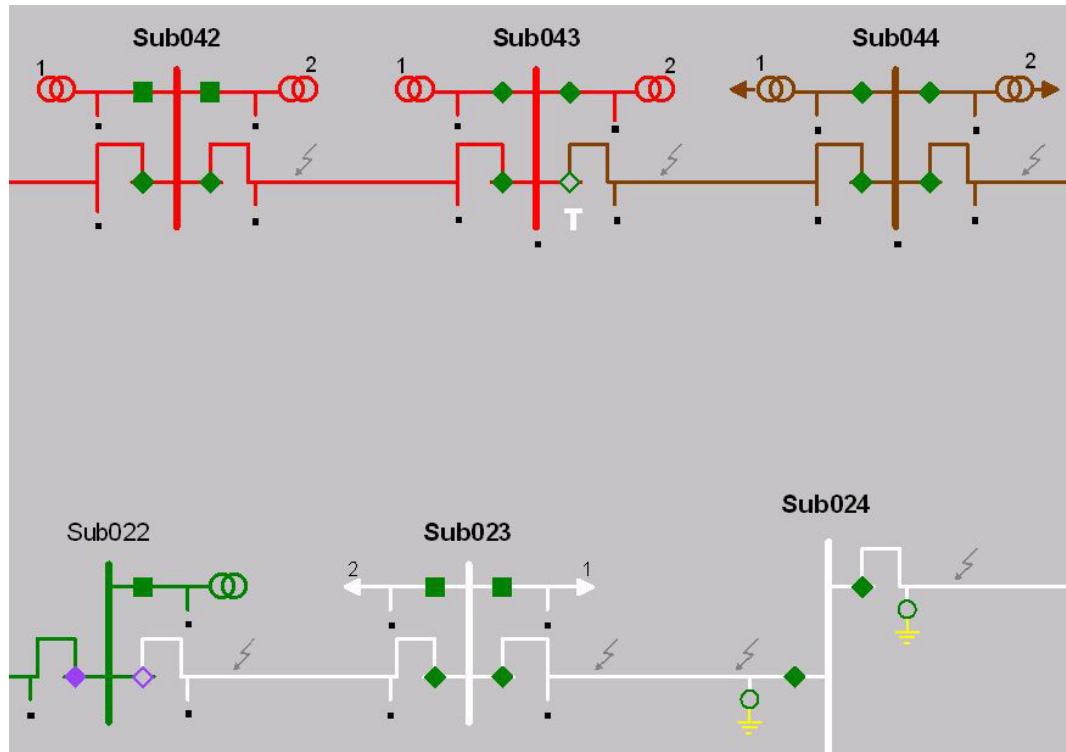
The Spectrum Power 5 dialog is displayed, that prompts for the confirmation to remove the **Jumper**.



[rguios13-030113-13.tif, 1, en_US]

Figure 15-17 Confirmation Dialog

- Click **Yes**.
The **Jumper** is removed.



[rguios14-030113-14.tif, 1, en_US]

Figure 15-18 Jumper Removed

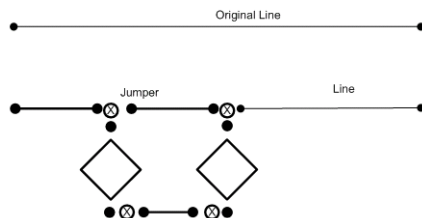


NOTE

A jumper can only be removed, after it is deactivated.

Restriction on Jumper

Jumper Placed Parallel to the Line Segment

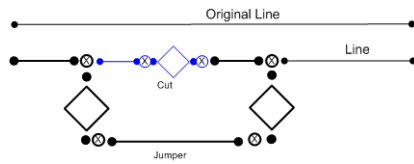


[ruios053-130511-01.tif, 1, en_US]

Figure 15-19 Jumper Placed Parallel to the Line Segment

If a **Jumper** is placed parallel to a line segment, it will have both the ends connected to same line segment. This way of inserting a **Jumper** is not allowed in Spectrum Power 5.

Jumper Placed Across a Cut



[ruios054-130511-01.tif, 1, en_US]

Figure 15-20 Jumper Placed Across a Cut

If a line segment is **Cut** first and then a **Jumper** is placed across the **Cut**. This is a valid scenario. But while removing, if **Cut** is removed first then the **Jumper** is in parallel to a line segment which is not allowed. Hence, Spectrum Power 5 will not allow **Cut** being removed earlier than the **Jumper** in this scenario. So the correct order of removing TNEs in the above scenario is, remove the **Jumper** first and then remove the **Cut**.

The same is true for inserting a **Jumper** across a **Ground**, although this is not really a use case.



NOTE

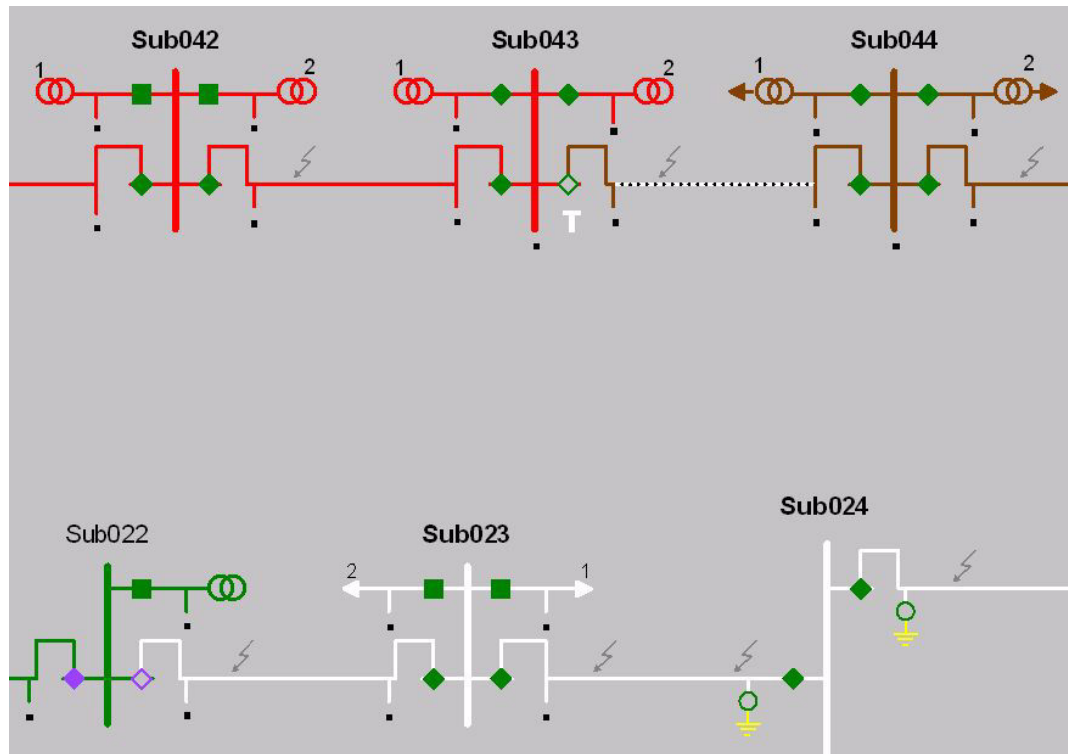
While removing the TNEs, it is important to consider the order of removal.

15.2.3 Cut

15.2.3.1 Inserting a Cut

The goal of this task is to insert a cut in the single-line diagram.

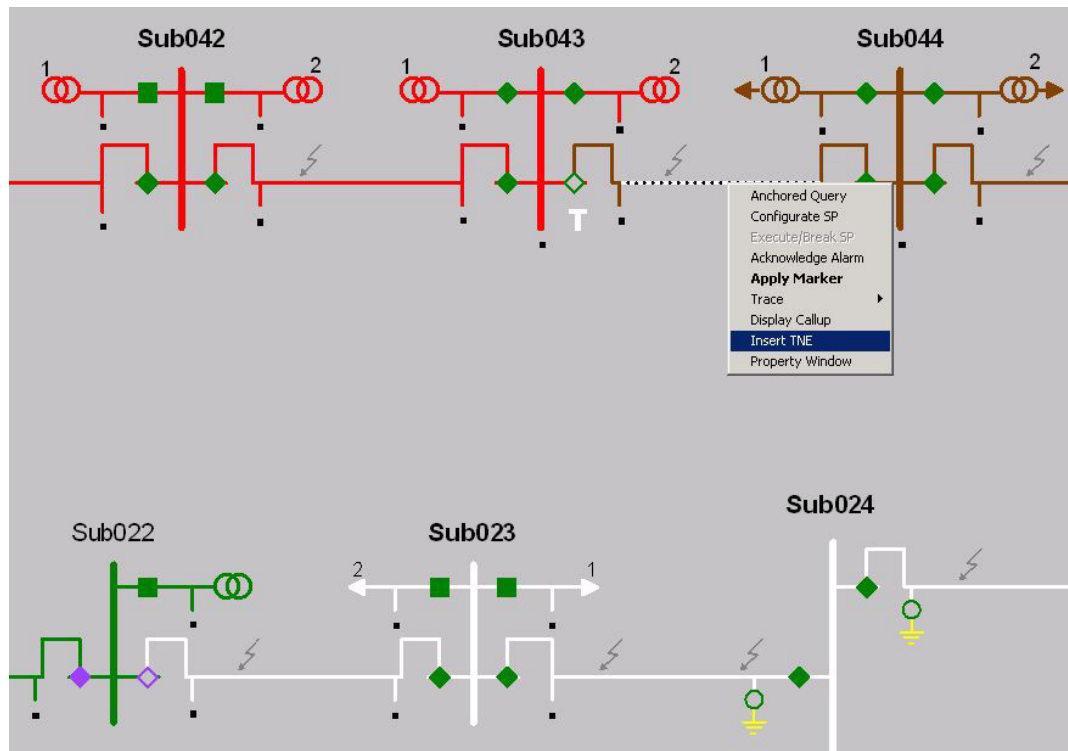
- From the worldmap, select a line to insert the **Cut**.



[rguios15-030113-15.tif, 1, en_US]

Figure 15-21 Line Selected to Insert Cut

- Right-click on one of the selected line.

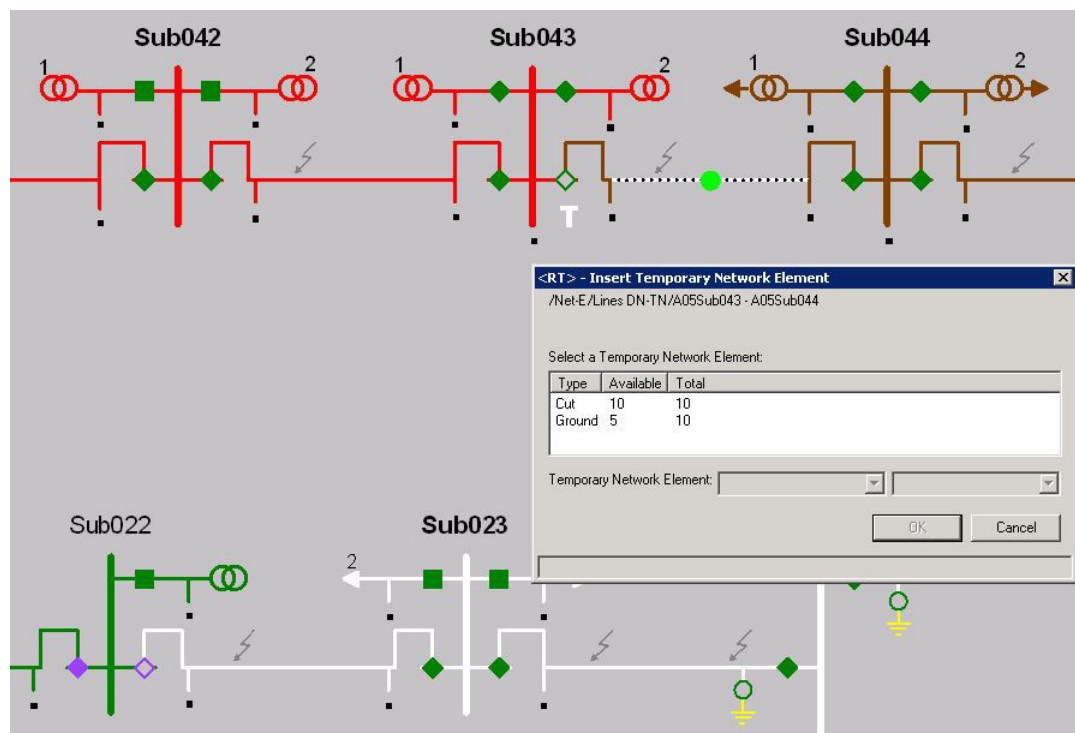


[rguios16-030113-16.tif, 1, en_US]

Figure 15-22 Context Menu

The context menu is displayed.

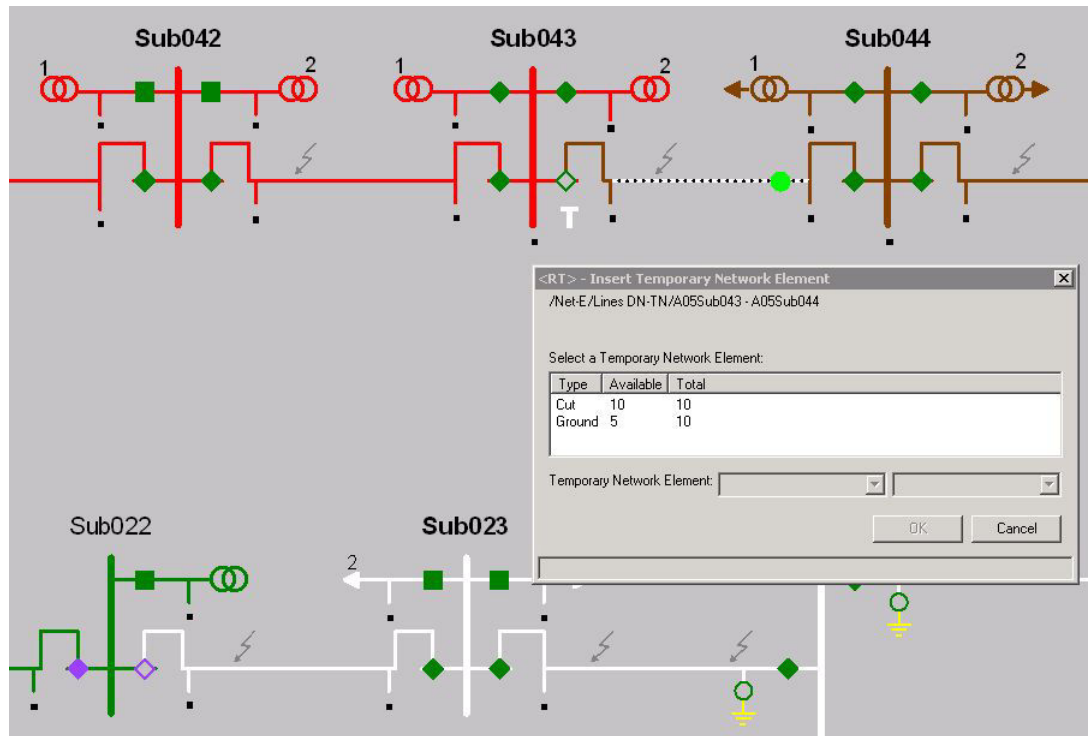
- Select **Insert TNE**.
- The **Insert Temporary Network Element** dialog is displayed.



[rguios17-030113-17.tif, 1, en_US]

Figure 15-23 Insert Temporary Network Element Dialog

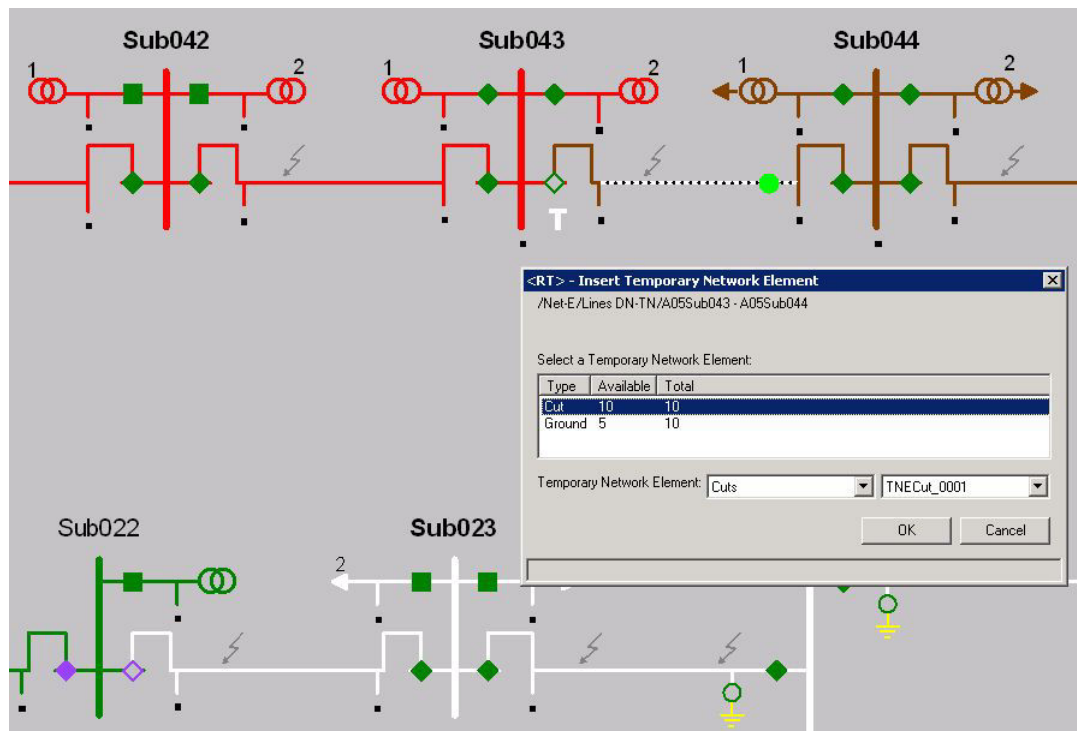
- You can specify the position of the **Cut** on the selected line by moving the green position marker.



[rguios18-030113-18.tif, 1, en_US]

Figure 15-24 Placing the Position Marker

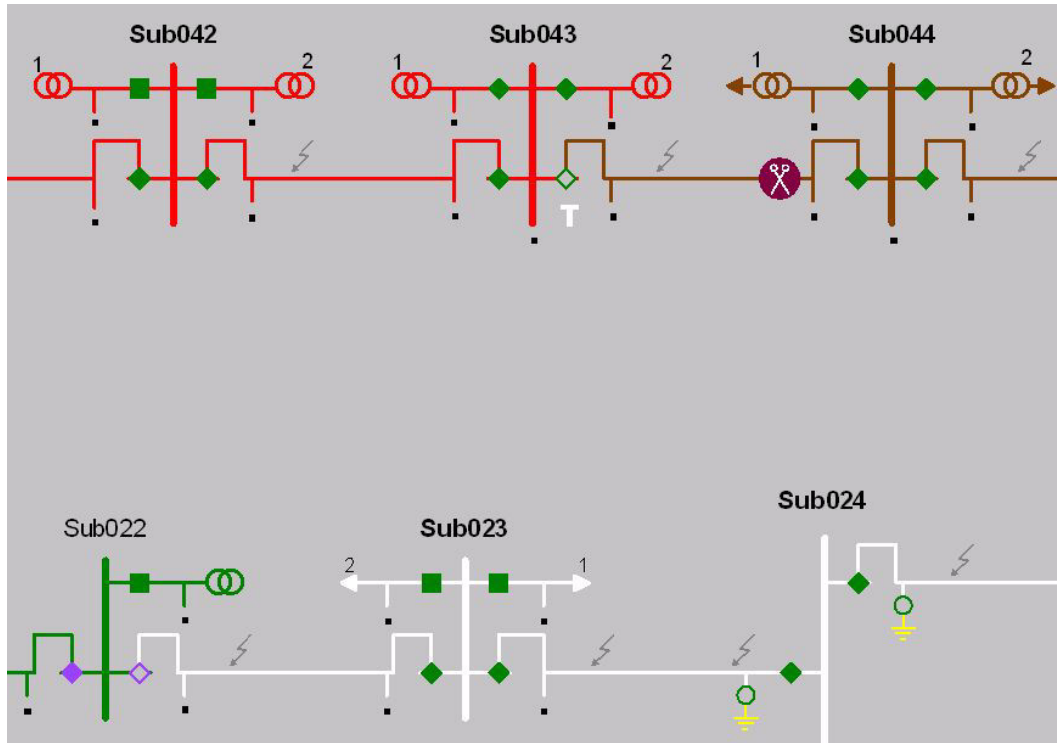
- Select **Cut** from the list.



[rguios20-030113-20.tif, 1, en_US]

Figure 15-25 Select Cut

- Click **OK**.
- The **Cut** is inserted on the selected line.



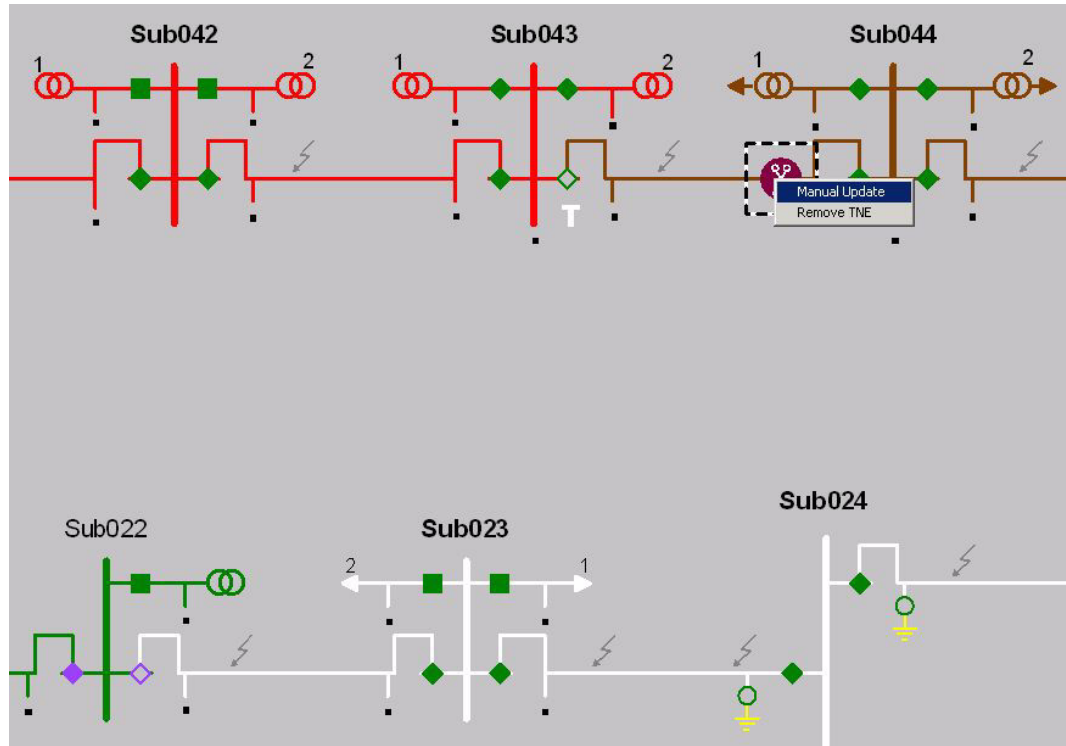
[rguios21-030113-21.tif, 1, en_US]

Figure 15-26 Cut Inserted

15.2.3.2 Activating a Cut

The goal of this task is to activate a **Cut** in the single-line diagram.

- From the worldmap, select the **Cut** to activate.
- Right-click on the **Cut**.



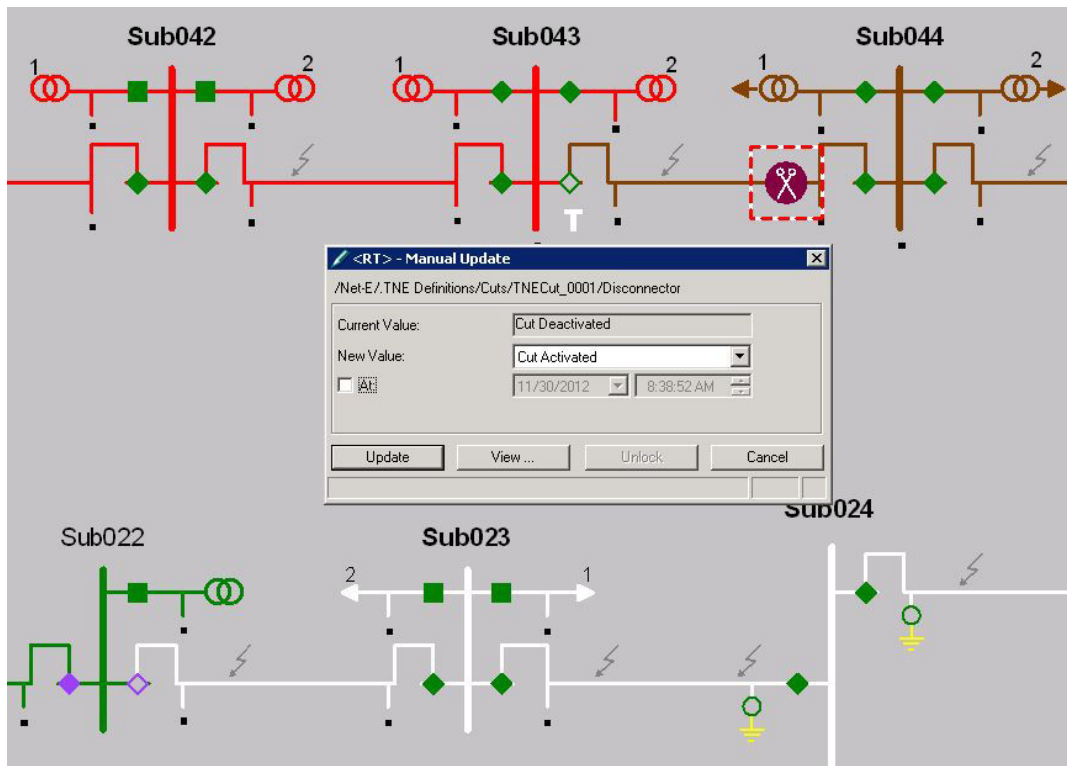
[rguios22-030113-22.tif, 1, en_US]

Figure 15-27 Context Menu

The context menu is displayed.

- Select **Manual Update**.

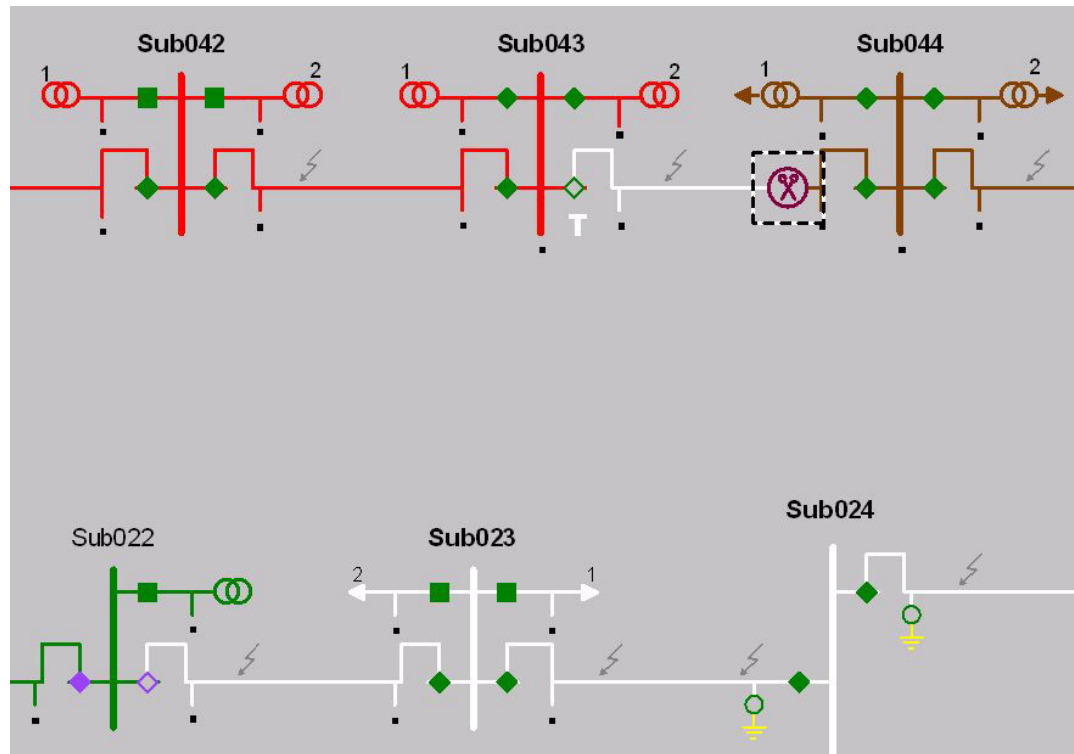
- The **Manual Update** dialog is displayed and **Cut Activated** is proposed as new value for the state of the cut.



[rguios23-030113-23.tif, 1, en_US]

Figure 15-28 Manual Update Dialog

- Click **Update** to activate the cut.



[rguios24-030113-24.tif, 1, en_US]

Figure 15-29 Cut Activated



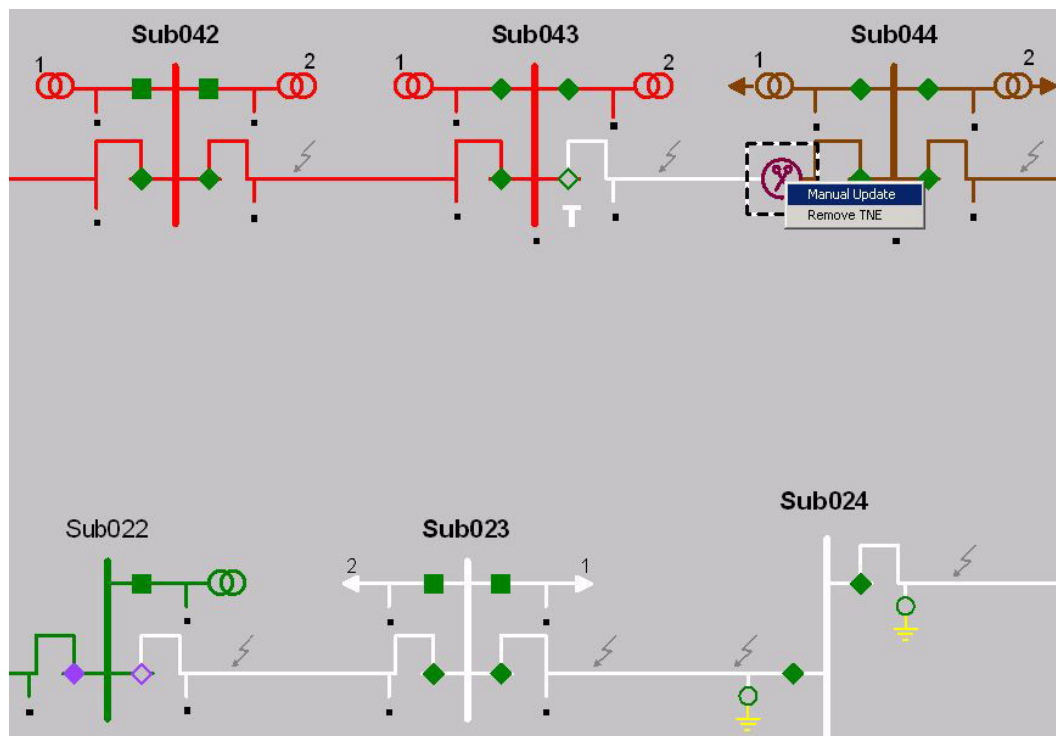
NOTE

When a new **Cut** is inserted, by default the **Cut** is deactivated.

15.2.3.3 Deactivating a Cut

The goal of this task is to deactivate a **Cut** in the single-line diagram.

- From the worldmap, select the **Cut** to deactivate.
- Right-click on the **Cut**.



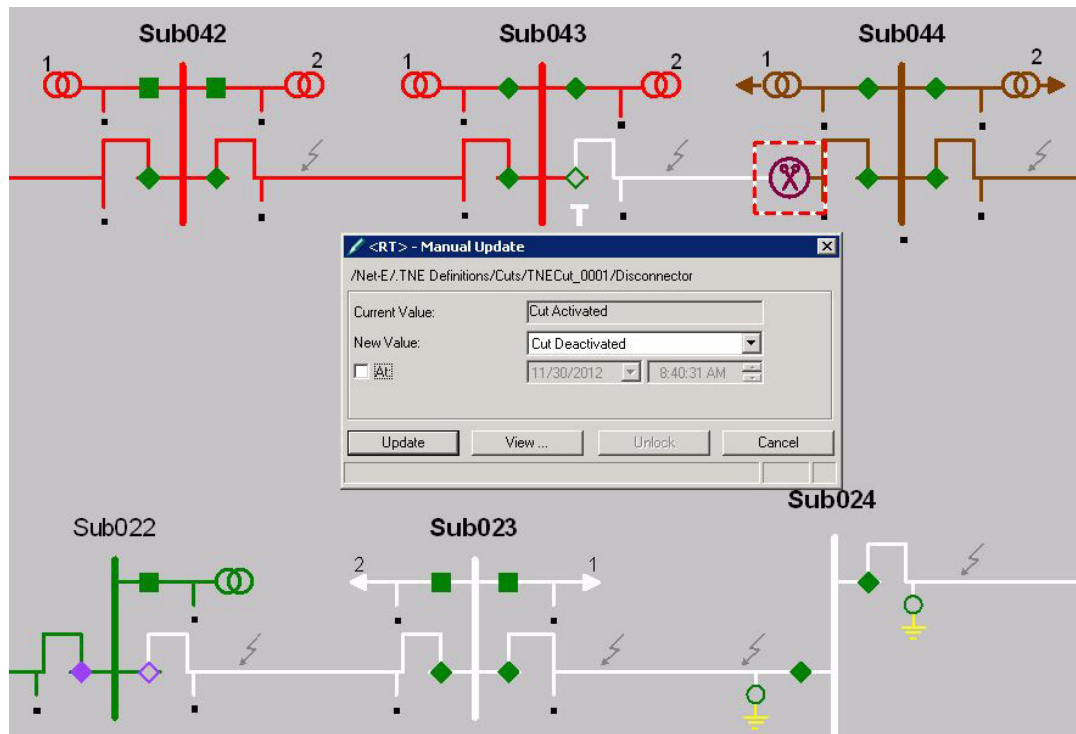
[rguios25-030113-25.tif, 1, en_US]

Figure 15-30 Context Menu

The context menu is displayed.

- Select **Manual Update**.

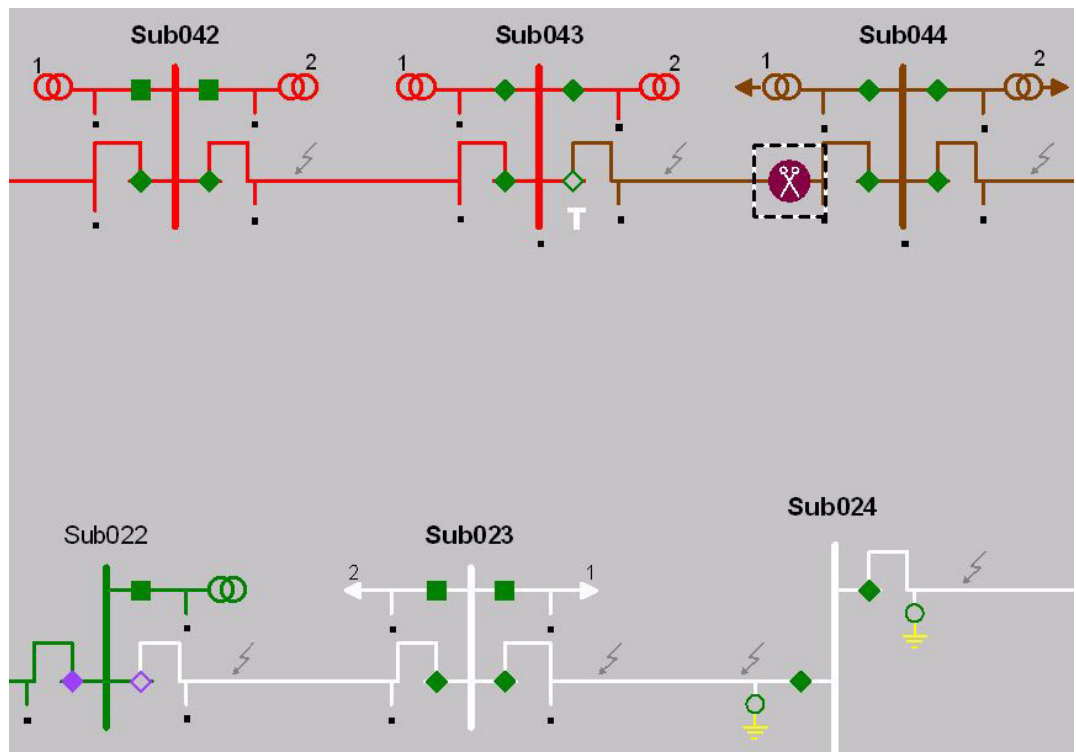
- The **Manual Update** dialog is displayed and **Cut Deactivated** is proposed as new value for the state of the cut.



[rguios26-030113-26.tif, 1, en_US]

Figure 15-31 Manual Update Dialog

- Click **Update** to deactivate the cut.



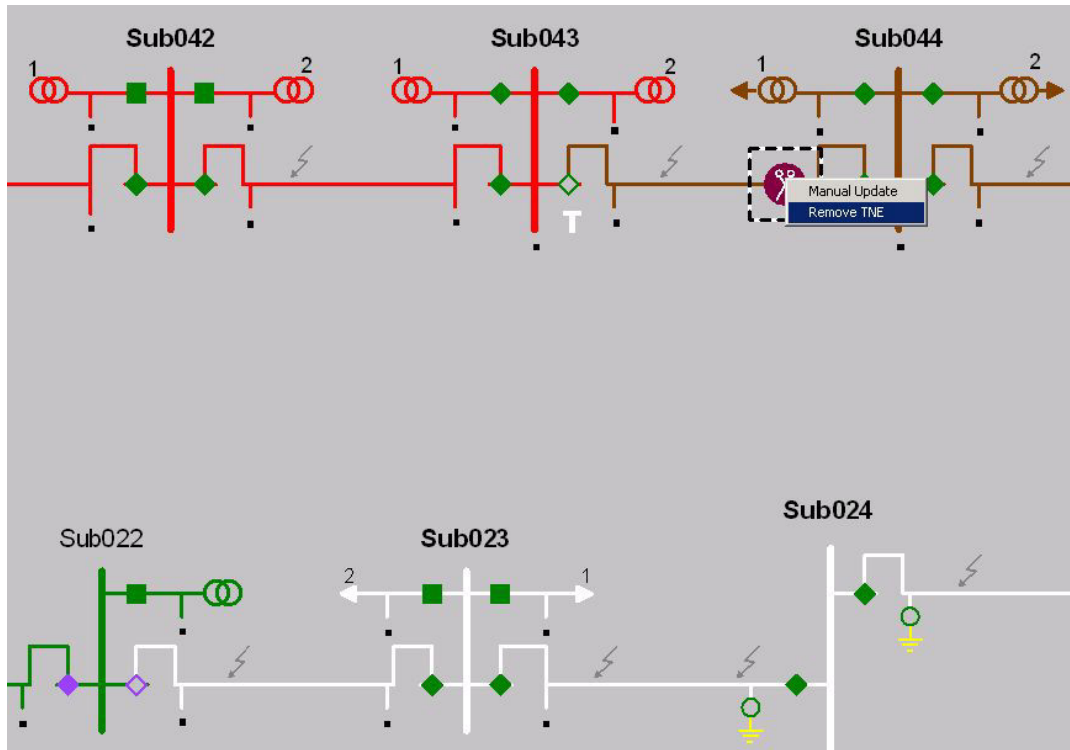
[rguios27-030113-27.tif, 1, en_US]

Figure 15-32 Cut Deactivated

15.2.3.4 Removing a Cut

The goal of this task is to remove a **Cut** in the single-line diagram.

- From the worldmap, select the **Cut** to remove.

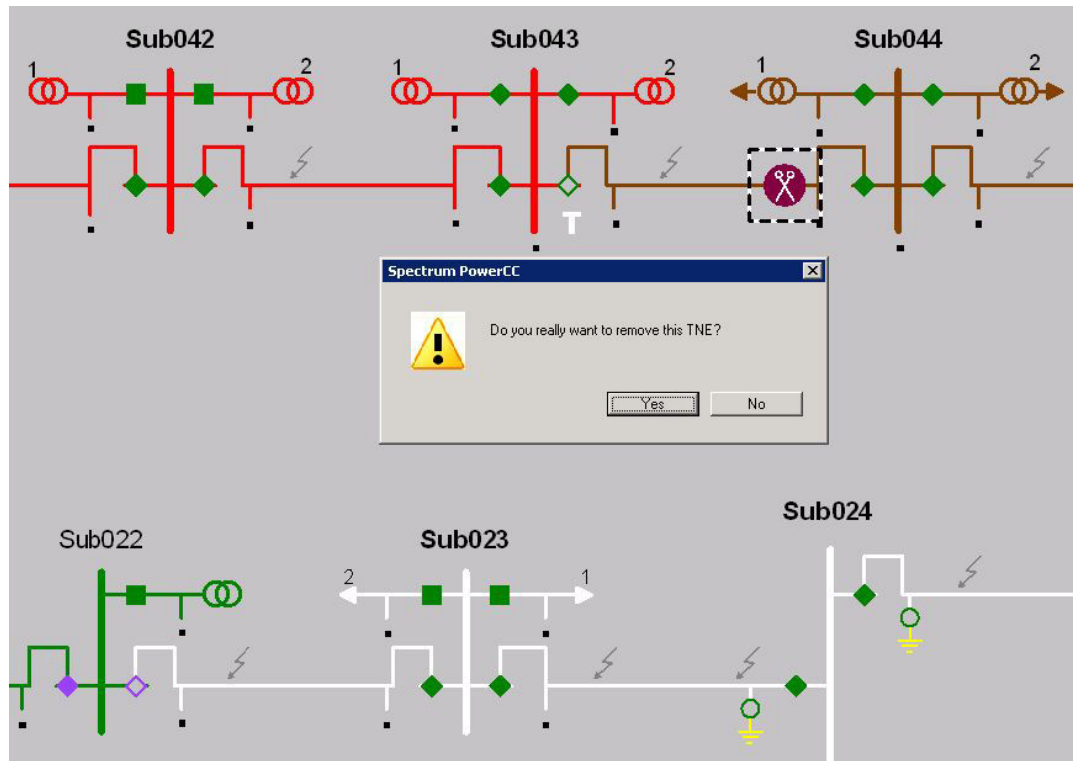


[rguios28-030113-28.tif, 1, en_US]

Figure 15-33 Select Cut

- Right-click on the **Cut**.
The context menu is displayed.
- Select **Remove TNE**.

- The Spectrum Power 5 dialog is displayed, that prompts for the confirmation to remove the **Cut**.



[rguios29-030113-29.tif, 1, en_US]

Figure 15-34 Confirmation Dialog

- Click **Yes**.
The **Cut** is removed.



NOTE

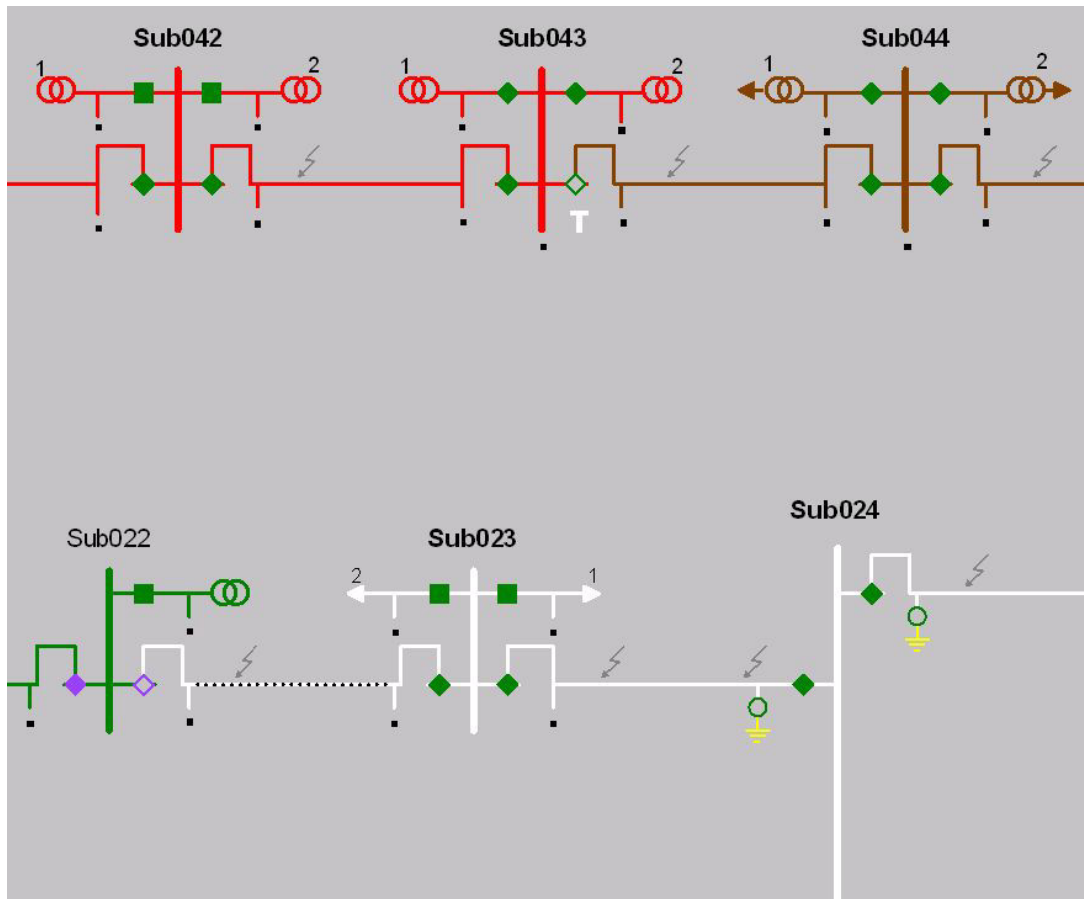
A cut can only be removed, after it is deactivated.

15.2.4 Ground

15.2.4.1 Inserting a Ground

The goal of this task is to insert a **Ground** in the single-line diagram.

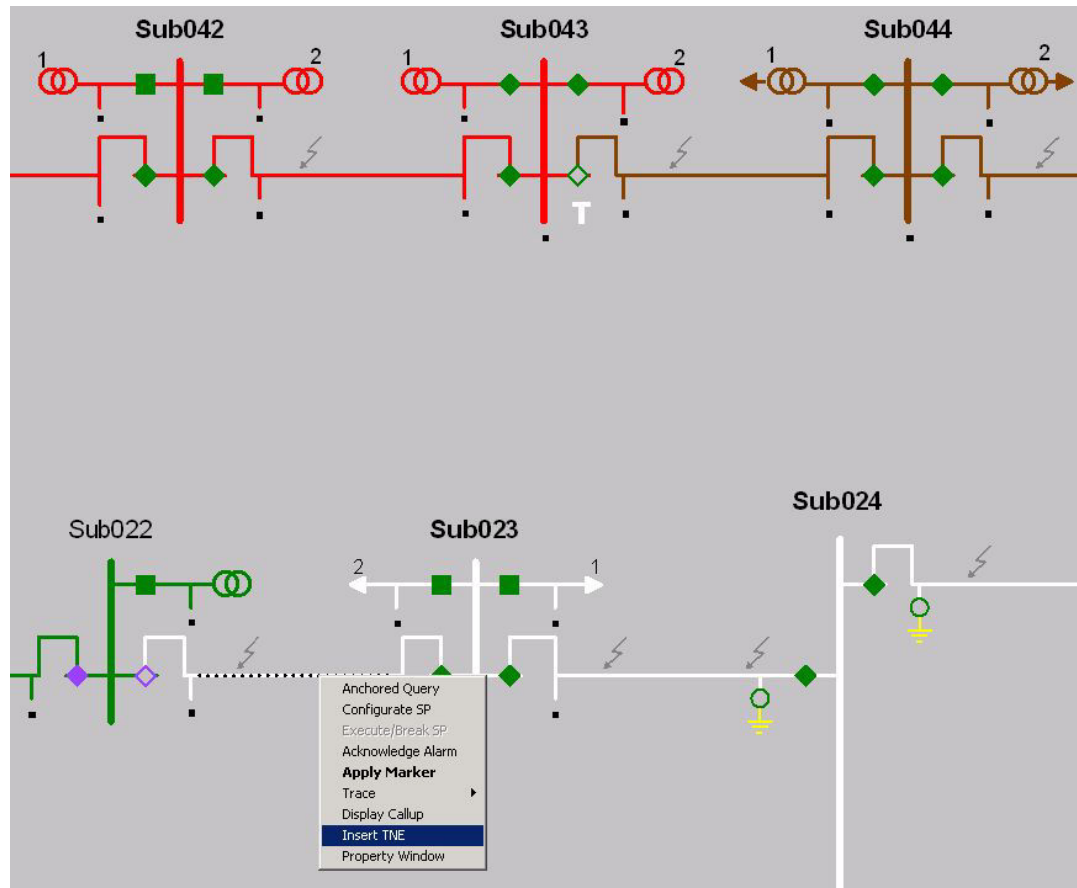
- From the worldmap, select a line or busbar segment to insert the **Ground**.



[rguios30-030113-30.tif, 1, en_US]

Figure 15-35 Line Selected to Insert Ground

- Right-click on the line.



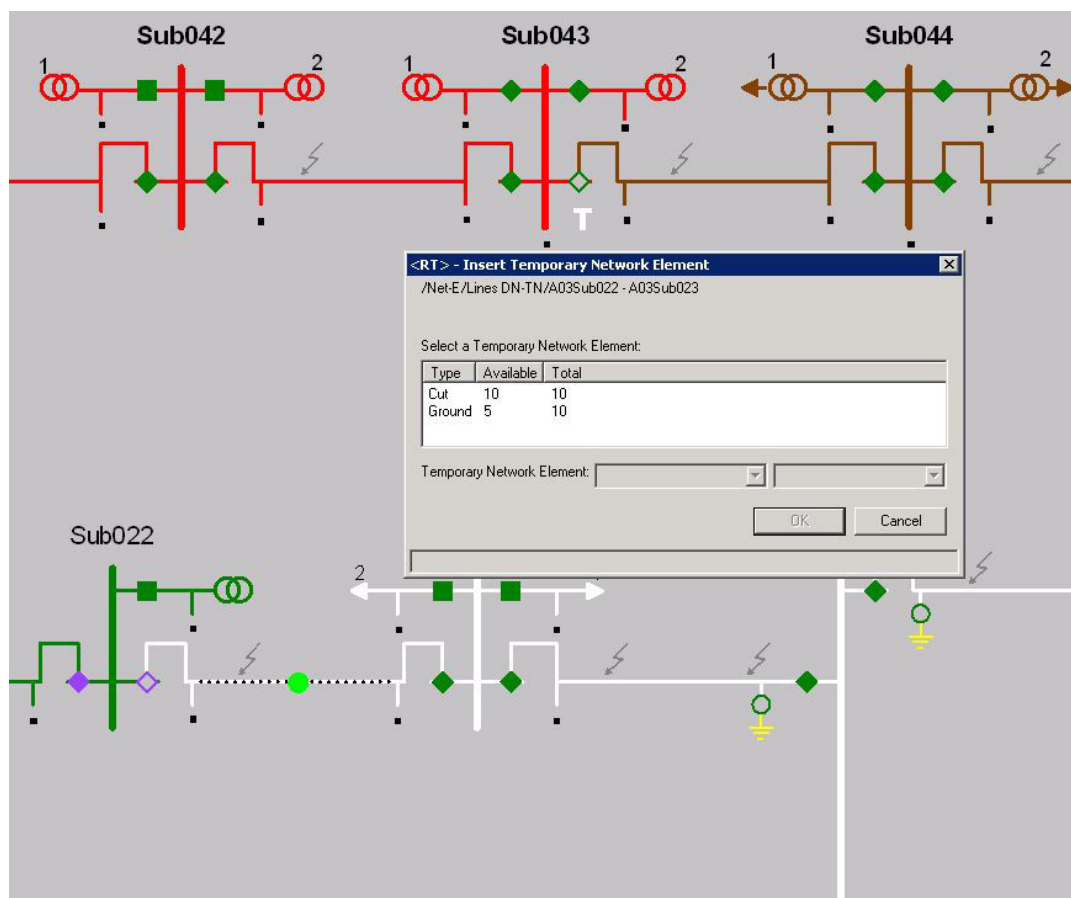
[rguios31-030113-31.tif, 1, en_US]

Figure 15-36 Context Menu

The context menu is displayed.

- Select **Insert TNE**.

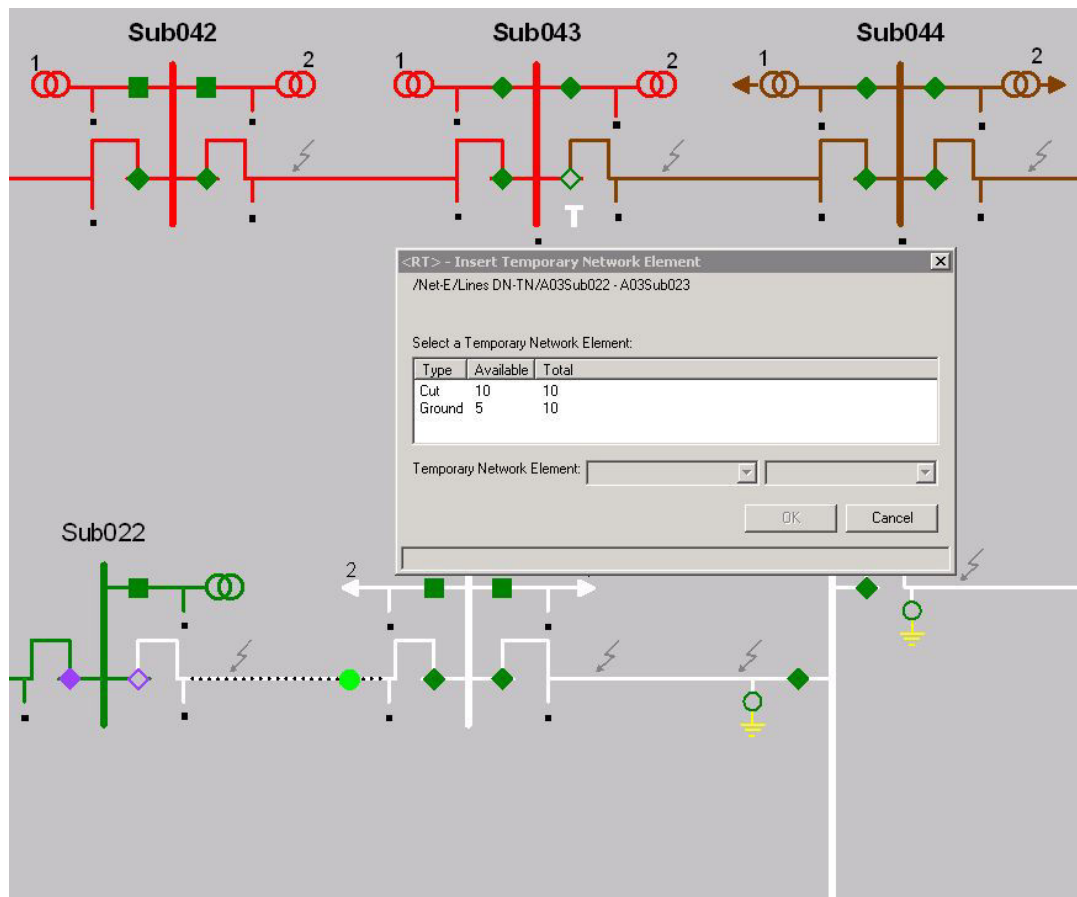
- The **Insert Temporary Network Element** dialog is displayed.



[rguios32-030113-32.tif, 1, en_US]

Figure 15-37 Insert Temporary Network Element Dialog

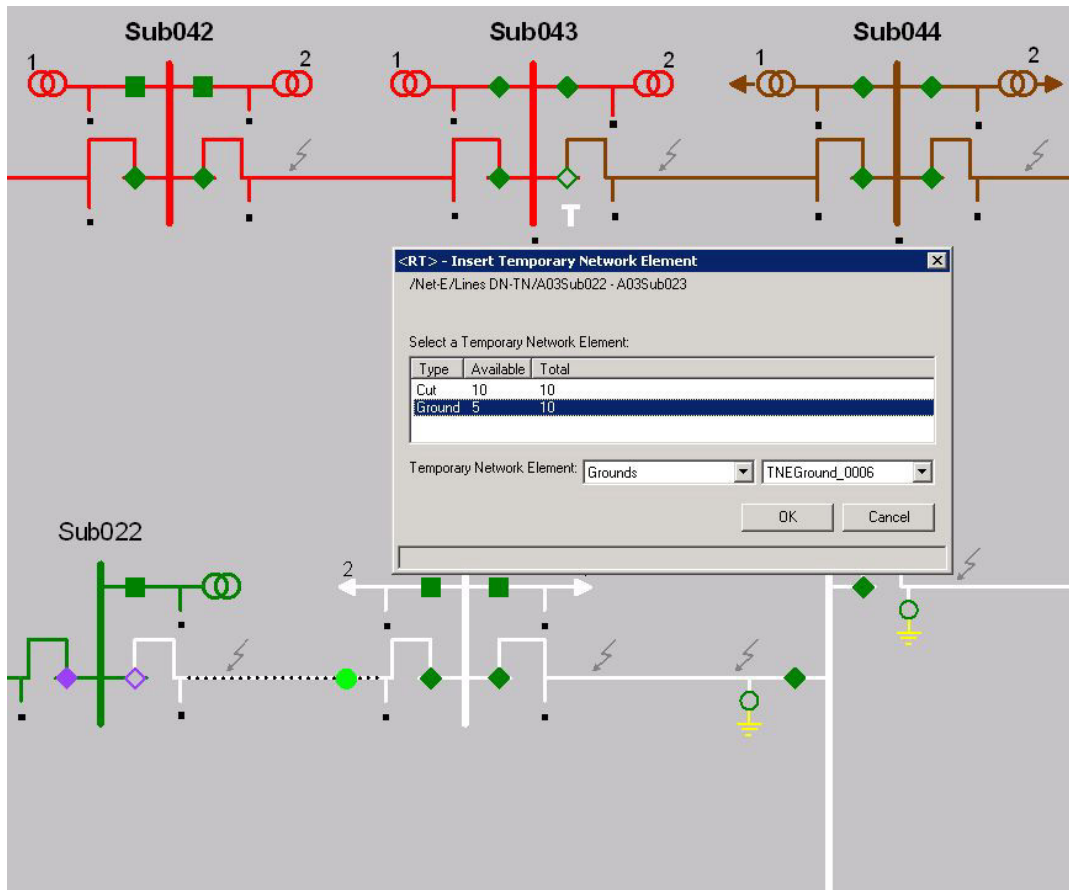
- You can specify the position of the **Ground** on the selected line by moving the green position marker.



[rguios33-030113-33.tif, 1, en_US]

Figure 15-38 Place the Position Marker

- Select **Ground** from the list.

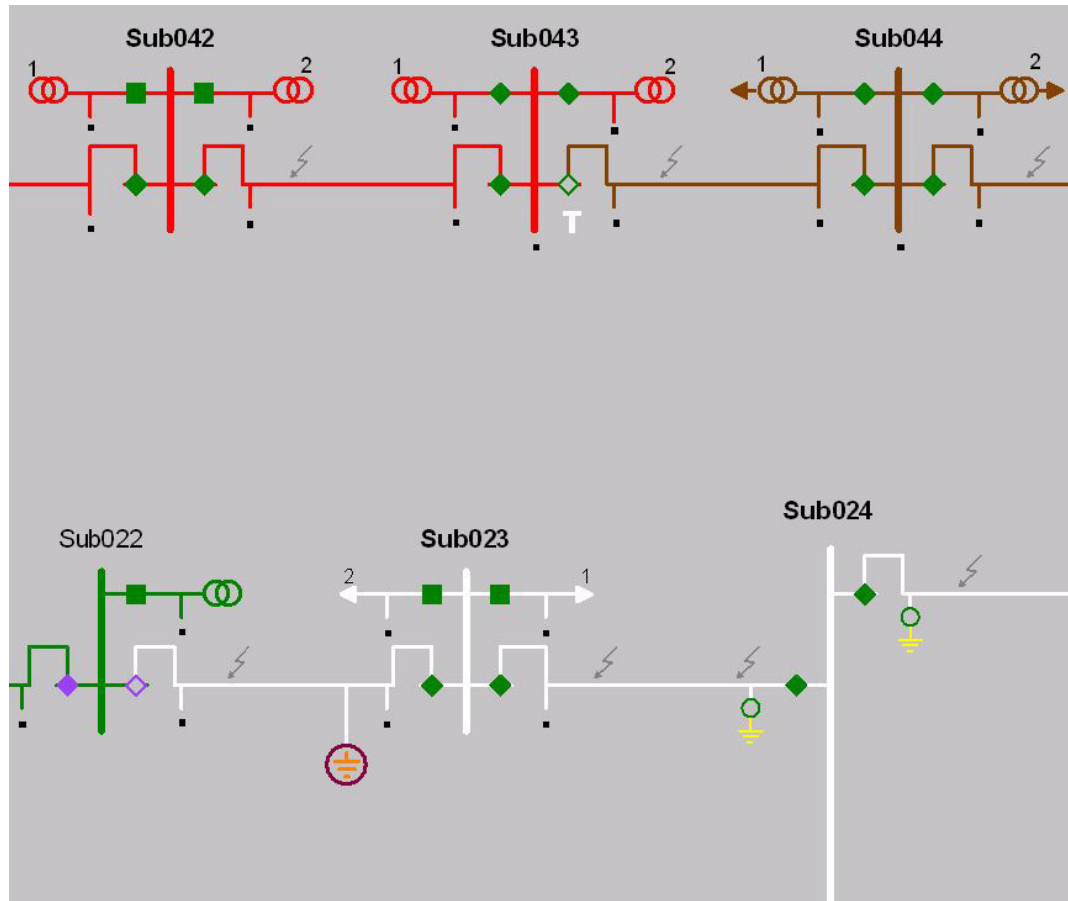


[rguios34-030113-34.tif, 1, en_US]

Figure 15-39 Select Ground

- Select an element from the list of available grounds.
By default, the first available element is selected.

- Click **OK**.
The selected **Ground** is inserted on the selected line.



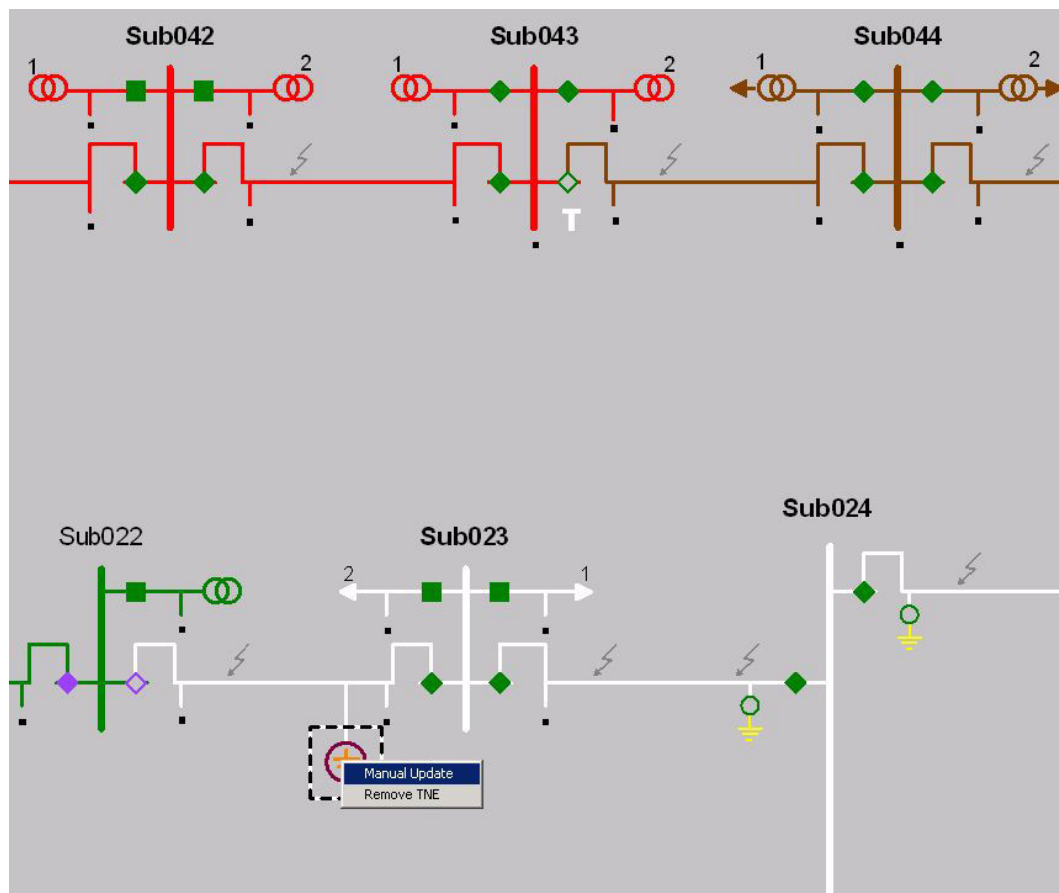
[rguios35-030113-35.tif, 1, en_US]

Figure 15-40 Inserted Ground

15.2.4.2 Activating a Ground

The goal of this task is to activate a **Ground** in the single-line diagram.

- From the worldmap, select the **Ground** to activate.
- Right-click on the **Ground**.



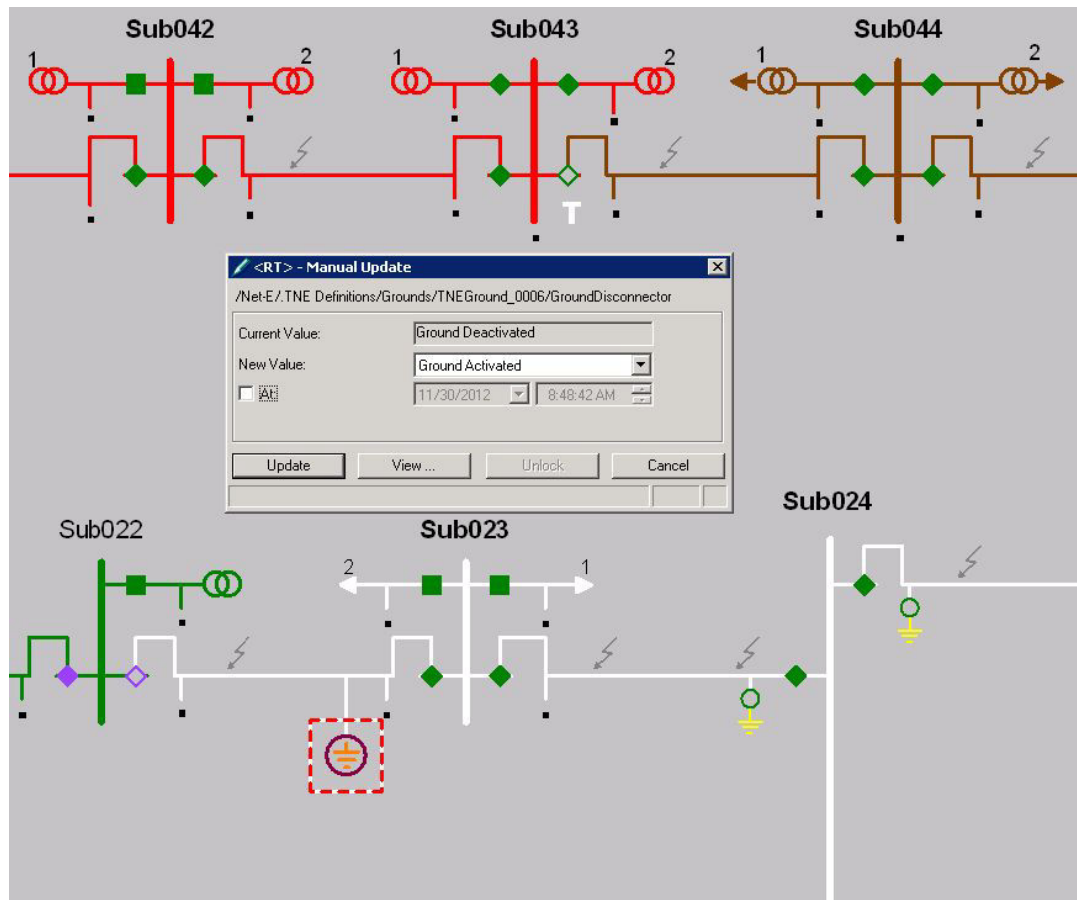
[rguios36-030113-36.tif, 1, en_US]

Figure 15-41 Context Menu

The context menu is displayed.

- Select **Manual Update**.

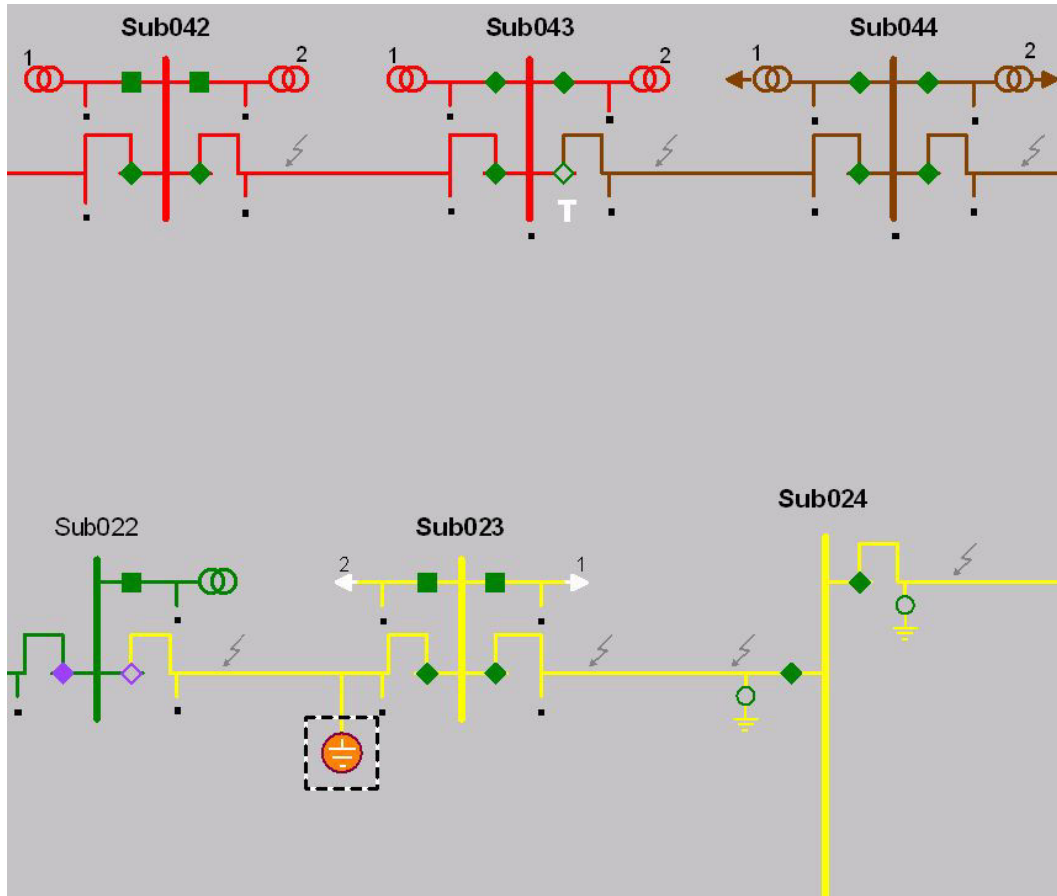
- The **Manual Update** dialog is displayed and **Ground Activated** is proposed as new value for the state of the ground.



[rguios37-030113-37.tif, 1, en_US]

Figure 15-42 Manual Update Dialog

- Click **Update** to activate the ground.



[rguios38-030113-38.tif, 1, en_US]

Figure 15-43 Ground Activated



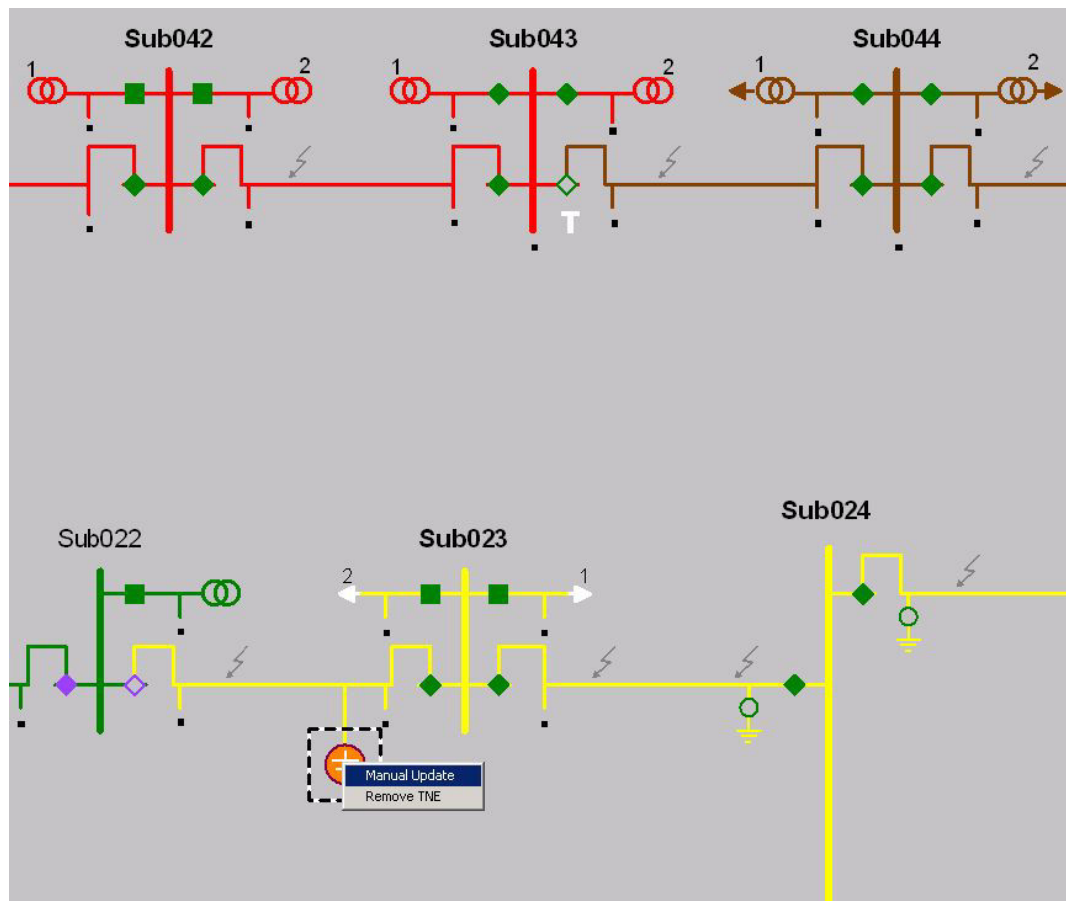
NOTE

When a new **Ground** is inserted, by default the **Ground** is deactivated.

15.2.4.3 Deactivating a Ground

The goal of this task is to deactivate a **Ground** in the single-line diagram.

- From the worldmap, select the **Ground** to deactivate.

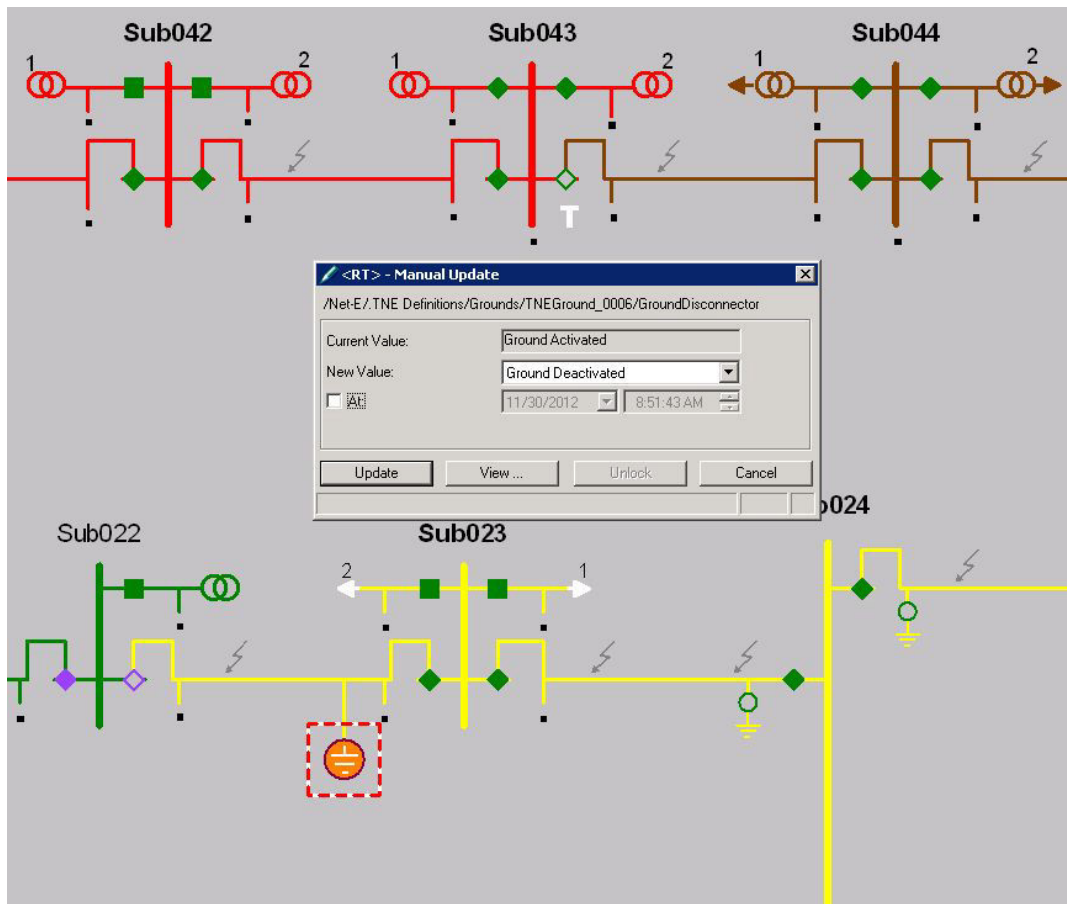


[rguios39-030113-39.tif, 1, en_US]

Figure 15-44 Ground to Deactivate

- Right-click on the **Ground**.
The context menu is displayed.
- Select **Manual Update**.

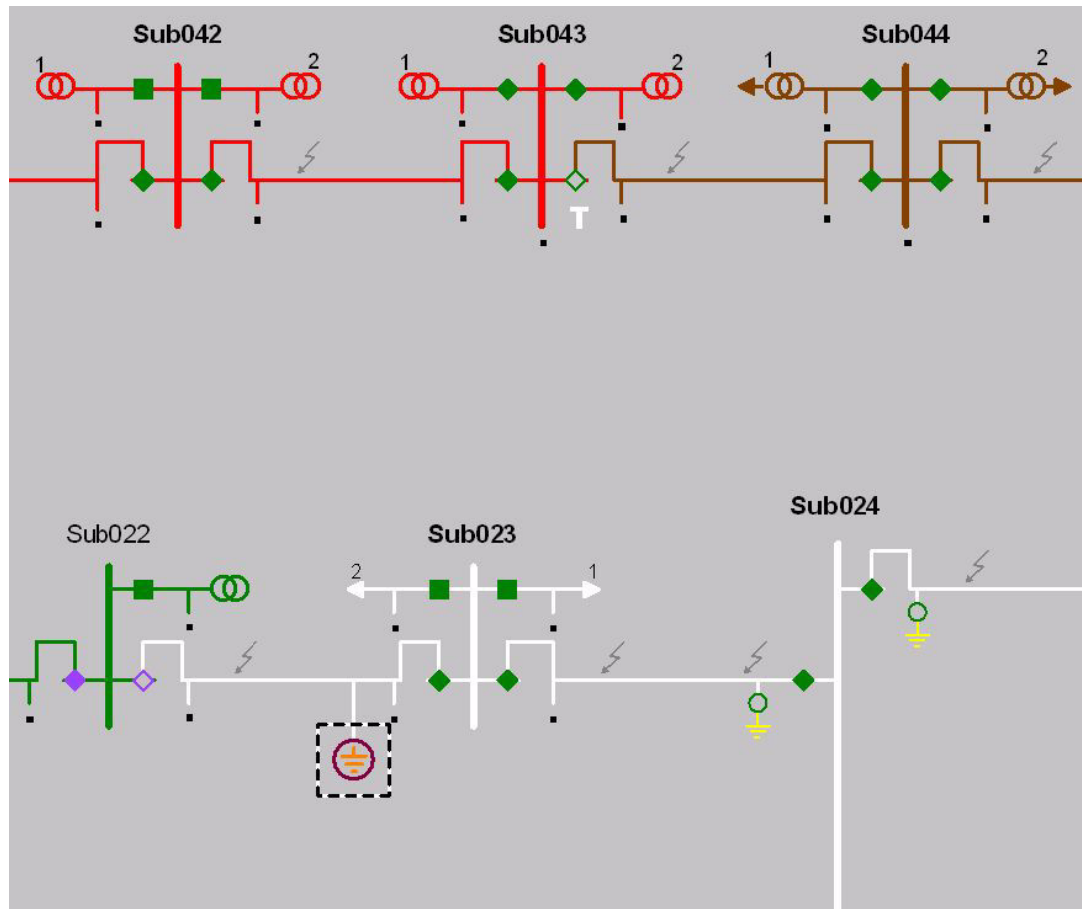
- The **Manual Update** dialog is displayed and **Ground Deactivated** is proposed as new value for the state of the ground.



[rguios40-030113-40.tif, 1, en_US]

Figure 15-45 Manual Update Dialog

- Click **Update** in order to deactivate the ground.



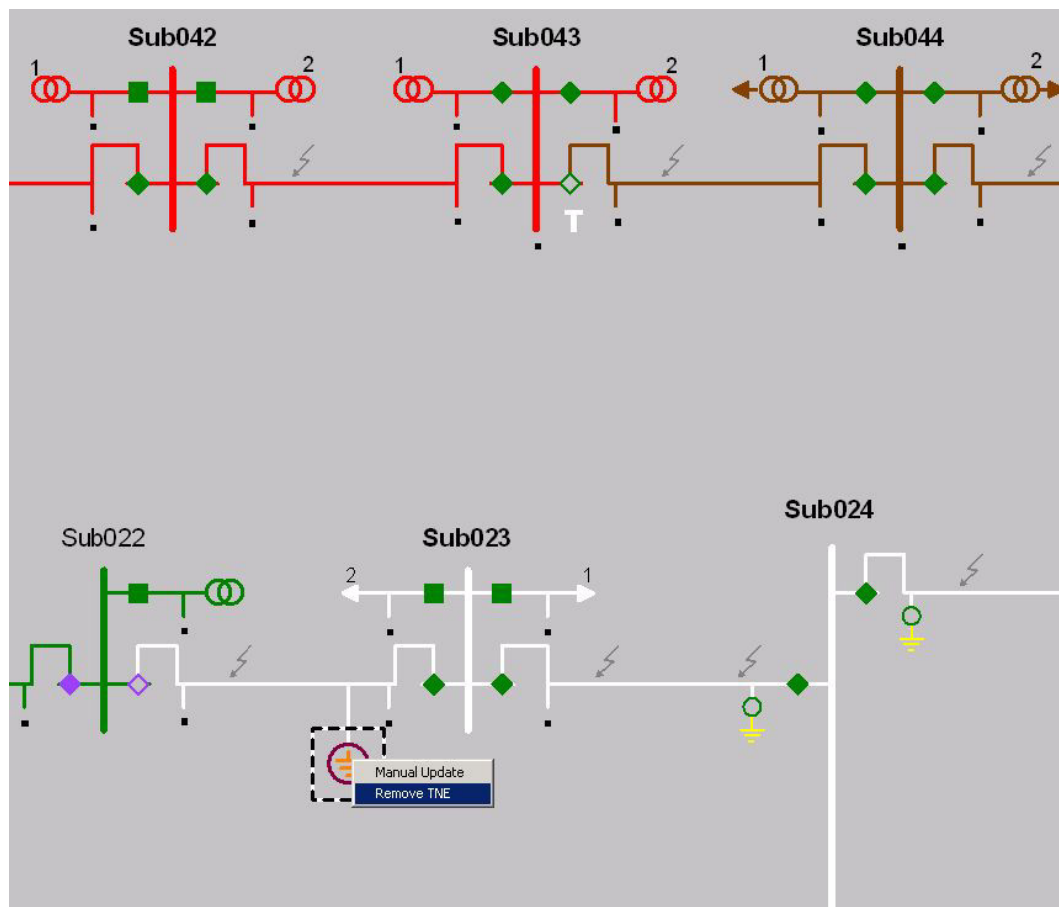
[rguios41-030113-41.tif, 1, en_US]

Figure 15-46 Ground Deactivated

15.2.4.4 Removing a Ground

The goal of this task is to remove a **Ground** in the single-line diagram.

- From the worldmap, select the **Ground** to remove.
- Right-click on the **Ground**.



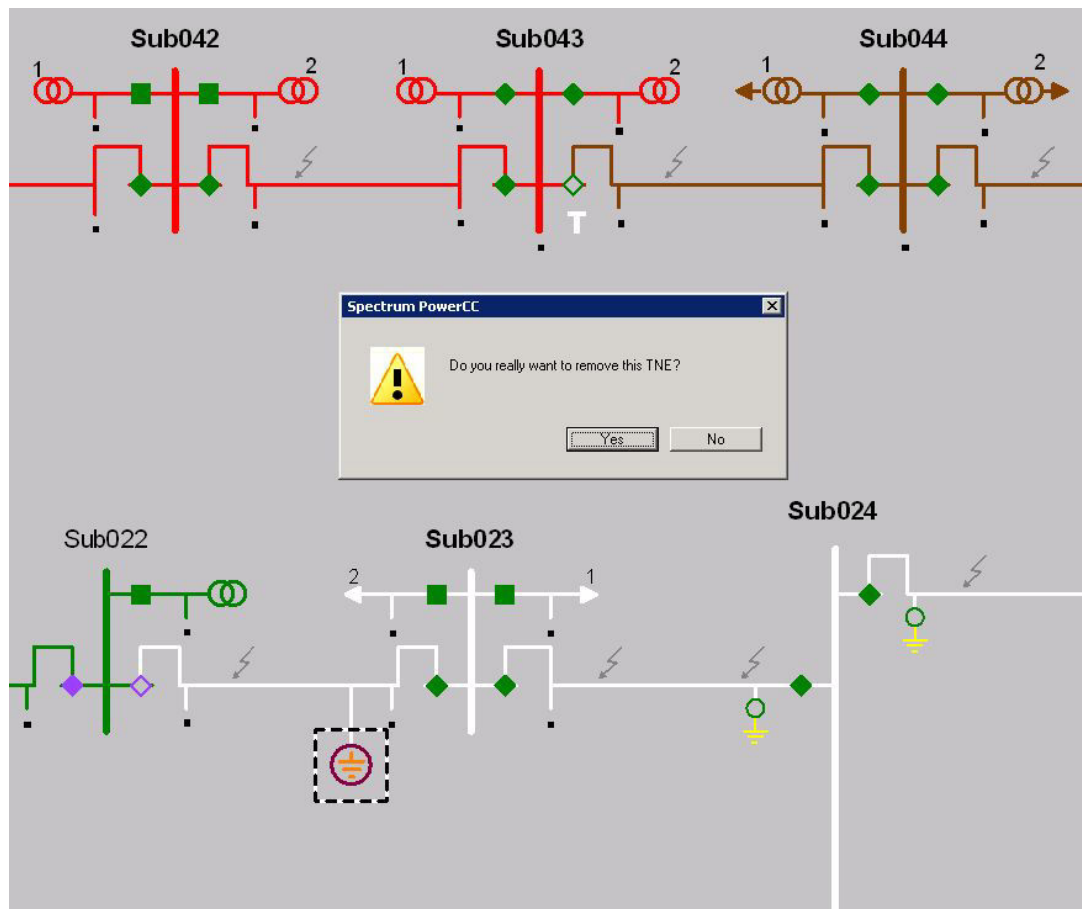
[rguios42-030113-42.tif, 1, en_US]

Figure 15-47 Context Menu

The context menu is displayed.

- Select **Remove TNE**.

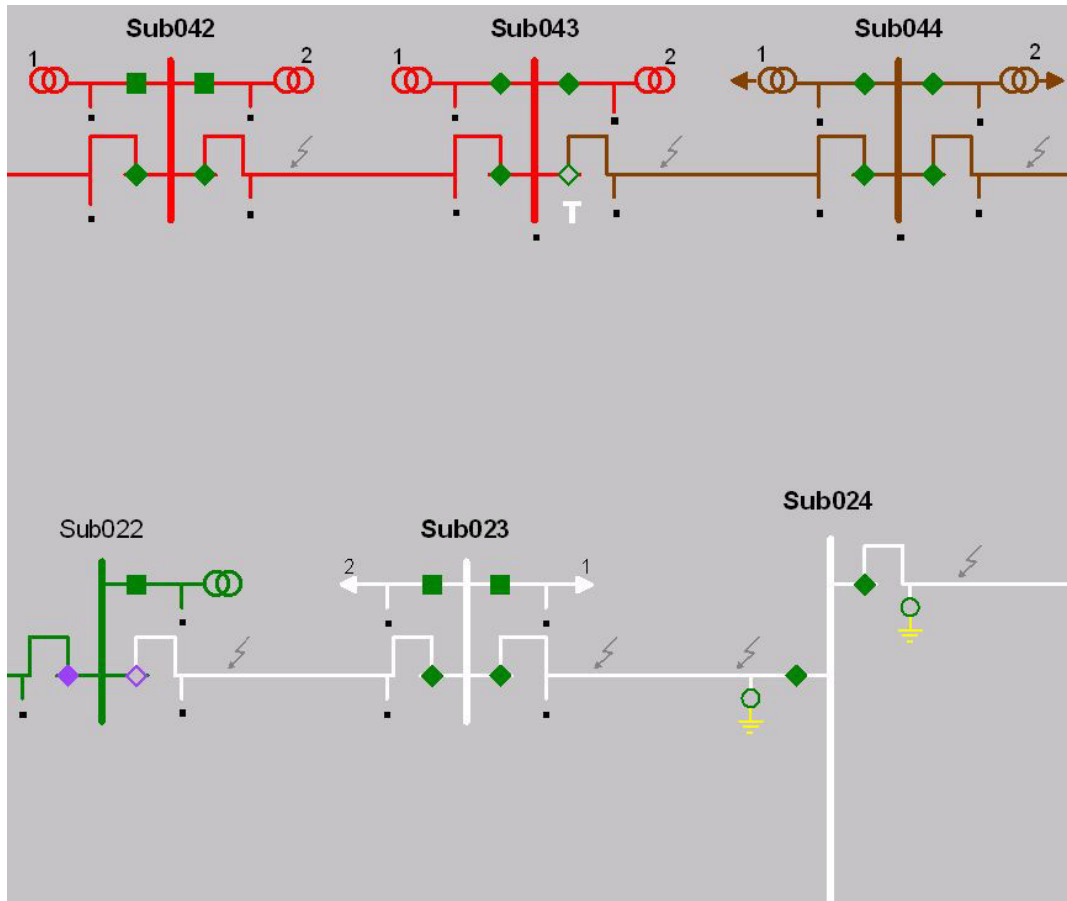
- The Spectrum Power 5 dialog is displayed, that prompts for the confirmation to remove the **Ground**.



[rguios43-030113-43.tif, 1, en_US]

Figure 15-48 Confirmation Dialog

- Click **Yes**.
The **Ground** is removed.



[rguios44-030113-44.tif, 1, en_US]

Figure 15-49 Ground Removed



NOTE

A Ground can only be removed after it is deactivated.

15.2.5 Monitoring the TNEs

Alarmlists

The system creates messages, for the insertion/removal and the activation/deactivation of a TNE. The reporting is done on the level of the parent equipment, that is, the line or the busbar. For jumpers the reporting is done for each parent equipment separately.

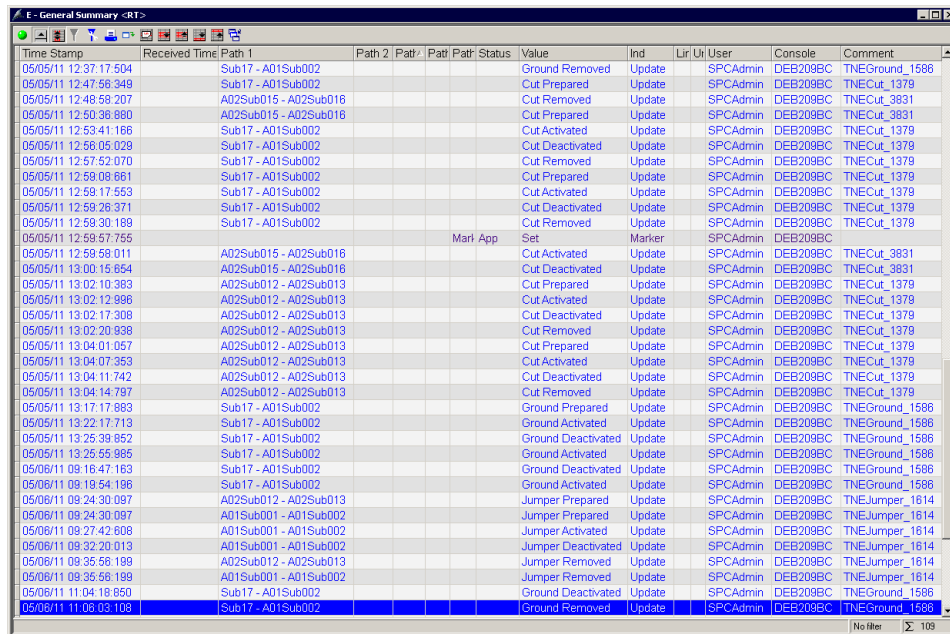
The following types of events are recorded:

- Messages for TNE-Insertion**
- Messages for TNE-Removal**
- Messages for TNE-Activation/Deactivation**

General Summary

The goal of this task is to view the status of the TNE and to view corresponding single-line diagram.

- From the PAT, click .
- The **General Summary** is displayed.

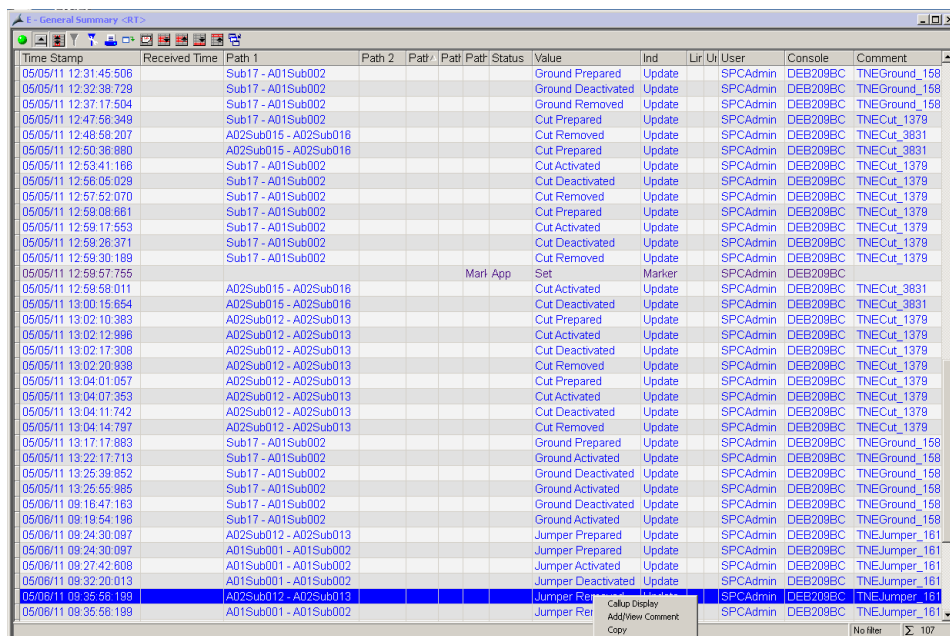


Time Stamp	Received Time	Path 1	Path 2	Pat1	Pat2	Pat3	Status	Value	Ind	Lir	U	User	Console	Comment
05/05/11 12:37:17.504		Sub17 - A01Sub002						Ground Removed	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/05/11 12:47:56.349		Sub17 - A01Sub002						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:48:58.207		A02Sub015 - A02Sub016						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 12:50:36.880		A02Sub015 - A02Sub016						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 12:53:41.166		Sub17 - A01Sub002						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:56:05.029		Sub17 - A01Sub002						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:57:52.070		Sub17 - A01Sub002						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:58:08.661		Sub17 - A01Sub002						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:17.553		Sub17 - A01Sub002						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:26.371		Sub17 - A01Sub002						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:30.188		Sub17 - A01Sub002						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:57.755							Mari App	Set	Marker			SPCAdmin	DEB209BC	
05/05/11 12:59:58.011		A02Sub015 - A02Sub016						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 13:00:15.654		A02Sub015 - A02Sub016						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 13:02:10.383		A02Sub012 - A02Sub013						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:02:12.986		A02Sub012 - A02Sub013						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:02:17.308		A02Sub012 - A02Sub013						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:02:20.938		A02Sub012 - A02Sub013						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:01.057		A02Sub012 - A02Sub013						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:07.353		A02Sub012 - A02Sub013						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:11.742		A02Sub012 - A02Sub013						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:14.797		A02Sub012 - A02Sub013						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:17:17.883		Sub17 - A01Sub002						Ground Prepared	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/05/11 13:22:17.713		Sub17 - A01Sub002						Ground Activated	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/05/11 13:25:39.852		Sub17 - A01Sub002						Ground Deactivated	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/05/11 13:25:55.985		Sub17 - A01Sub002						Ground Activated	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/06/11 09:16:47.163		Sub17 - A01Sub002						Ground Deactivated	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/06/11 09:16:54.196		Sub17 - A01Sub002						Ground Activated	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/06/11 09:24:30.097		A02Sub012 - A02Sub013						Jumper Prepared	Update			SPCAdmin	DEB209BC	TNEJumper_1614
05/06/11 09:24:30.097		A01Sub001 - A01Sub002						Jumper Prepared	Update			SPCAdmin	DEB209BC	TNEJumper_1614
05/06/11 09:27:42.608		A01Sub001 - A01Sub002						Jumper Activated	Update			SPCAdmin	DEB209BC	TNEJumper_1614
05/06/11 09:32:20.013		A01Sub001 - A01Sub002						Jumper Deactivated	Update			SPCAdmin	DEB209BC	TNEJumper_1614
05/06/11 09:35:56.199		A02Sub012 - A02Sub013						Jumper Removed	Update			SPCAdmin	DEB209BC	TNEJumper_1614
05/06/11 09:35:56.199		A01Sub001 - A01Sub002						Jumper Removed	Update			SPCAdmin	DEB209BC	TNEJumper_1614
05/06/11 11:04:18.850		Sub17 - A01Sub002						Ground Deactivated	Update			SPCAdmin	DEB209BC	TNEGround_1586
05/06/11 11:06:03.108		Sub17 - A01Sub002						Ground Removed	Update			SPCAdmin	DEB209BC	TNEGround_1586

[uios041-130511-01.tif, 1, en_US]

Figure 15-50 General Summary

- Right-click on a row.



Time Stamp	Received Time	Path 1	Path 2	Pat1	Pat2	Pat3	Status	Value	Ind	Lir	U	User	Console	Comment
05/05/11 12:31:45.506		Sub17 - A01Sub002						Ground Prepared	Update			SPCAdmin	DEB209BC	TNEGround_158
05/05/11 12:32:38.729		Sub17 - A01Sub002						Ground Deactivated	Update			SPCAdmin	DEB209BC	TNEGround_158
05/05/11 12:37:17.504		Sub17 - A01Sub002						Ground Removed	Update			SPCAdmin	DEB209BC	TNEGround_158
05/05/11 12:47:56.349		Sub17 - A01Sub002						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:48:58.207		A02Sub015 - A02Sub016						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 12:50:36.880		A02Sub015 - A02Sub016						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 12:53:41.166		Sub17 - A01Sub002						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:56:05.029		Sub17 - A01Sub002						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:57:52.070		Sub17 - A01Sub002						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:58:08.661		Sub17 - A01Sub002						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:17.553		Sub17 - A01Sub002						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:26.371		Sub17 - A01Sub002						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:30.188		Sub17 - A01Sub002						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 12:59:57.755							Mari App	Set	Marker			SPCAdmin	DEB209BC	
05/05/11 12:59:58.011		A02Sub015 - A02Sub016						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 13:00:15.654		A02Sub015 - A02Sub016						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_3831
05/05/11 13:02:10.383		A02Sub012 - A02Sub013						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:02:12.986		A02Sub012 - A02Sub013						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:02:17.308		A02Sub012 - A02Sub013						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:02:20.938		A02Sub012 - A02Sub013						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:01.057		A02Sub012 - A02Sub013						Cut Prepared	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:07.353		A02Sub012 - A02Sub013						Cut Activated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:11.742		A02Sub012 - A02Sub013						Cut Deactivated	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:04:14.797		A02Sub012 - A02Sub013						Cut Removed	Update			SPCAdmin	DEB209BC	TNECut_1379
05/05/11 13:17:17.883		Sub17 - A01Sub002						Ground Prepared	Update			SPCAdmin	DEB209BC	TNEGround_158
05/05/11 13:22:17.713		Sub17 - A01Sub002						Ground Activated	Update			SPCAdmin	DEB209BC	TNEGround_158
05/05/11 13:25:39.852		Sub17 - A01Sub002						Ground Deactivated	Update			SPCAdmin	DEB209BC	TNEGround_158
05/05/11 13:25:55.985		Sub17 - A01Sub002						Ground Activated	Update			SPCAdmin	DEB209BC	TNEGround_158
05/06/11 09:16:47.163		Sub17 - A01Sub002						Ground Deactivated	Update			SPCAdmin	DEB209BC	TNEGround_158
05/06/11 09:16:54.196		Sub17 - A01Sub002						Ground Activated	Update			SPCAdmin	DEB209BC	TNEGround_158
05/06/11 09:24:30.097		A02Sub012 - A02Sub013						Jumper Prepared	Update			SPCAdmin	DEB209BC	TNEJumper_161
05/06/11 09:24:30.097		A01Sub001 - A01Sub002						Jumper Prepared	Update			SPCAdmin	DEB209BC	TNEJumper_161
05/06/11 09:27:42.608		A01Sub001 - A01Sub002						Jumper Activated	Update			SPCAdmin	DEB209BC	TNEJumper_161
05/06/11 09:32:20.013		A01Sub001 - A01Sub002						Jumper Deactivated	Update			SPCAdmin	DEB209BC	TNEJumper_161
05/06/11 09:35:56.199		A02Sub012 - A02Sub013						Jumper Re	Update			SPCAdmin	DEB209BC	TNEJumper_161
05/06/11 09:35:56.199		A01Sub001 - A01Sub002						Jumper Re	Update			SPCAdmin	DEB209BC	TNEJumper_161

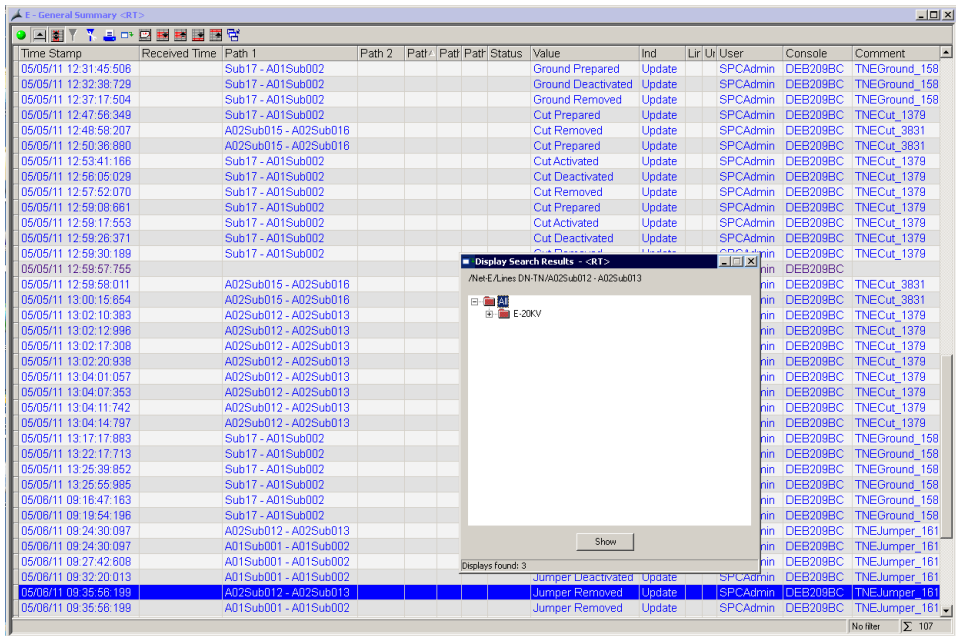
[uios042-130511-01.tif, 1, en_US]

Figure 15-51 Context Menu

The context menu is displayed.

- Select **Callup Display**.

The **Display Search Results** dialog is displayed.



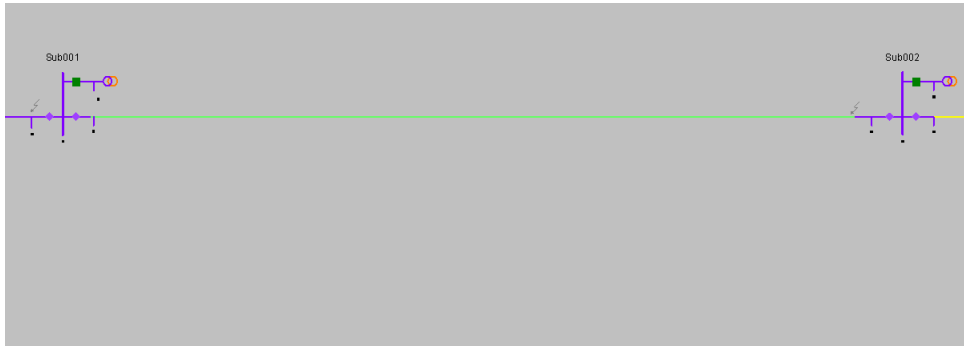
[ruios043-130511-01.tif, 1, en_US]

Figure 15-52 Display Search Results Dialog

- Select the single-line diagram from the tree view and click **Show**.

It displays the corresponding single-line diagram.

In case of more than one worldmap indicated, select the desired worldmap (view).



[ruios044-130511-01.tif, 1, en_US]

Figure 15-53 Single-line Diagram

Runtime Explorer

The **TNE View** of the **Runtime Explorer** will always show all currently inserted TNEs (independent of their state). Sorting is possible for all view columns.



NOTE

Filtering is not possible.

The following View columns are available:

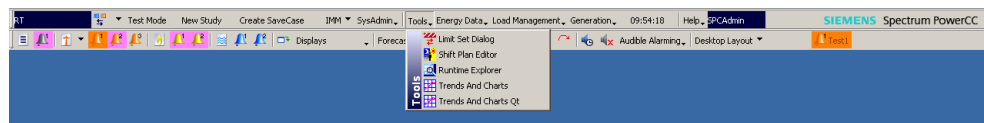
Table 15-2 Runtime Explorer TNE View Attributes

Column	Description
Name	Name of the TNE
Type	Type of the TNE
Connected side 1	Parent equipment path (can be a TNE again)
Connected side 2	Second Parent equipment path (only used for jumpers)
Status side 1	Status of the TNE switch 1 (Deactivated/Activated)
Status side 2	Status of the TNE switch 2 (Deactivated/Activated)
Timestamp	Date and time when the TNE was inserted.
Operator	Operator who has inserted the TNE.
Console	Console from where the TNE was inserted.

Viewing the TNEs

The goal of this task is to view the status of the TNE and to view corresponding single-line diagram using **Runtime Explorer**.

- From the PAT, select **Tools -> Runtime Explorer**.



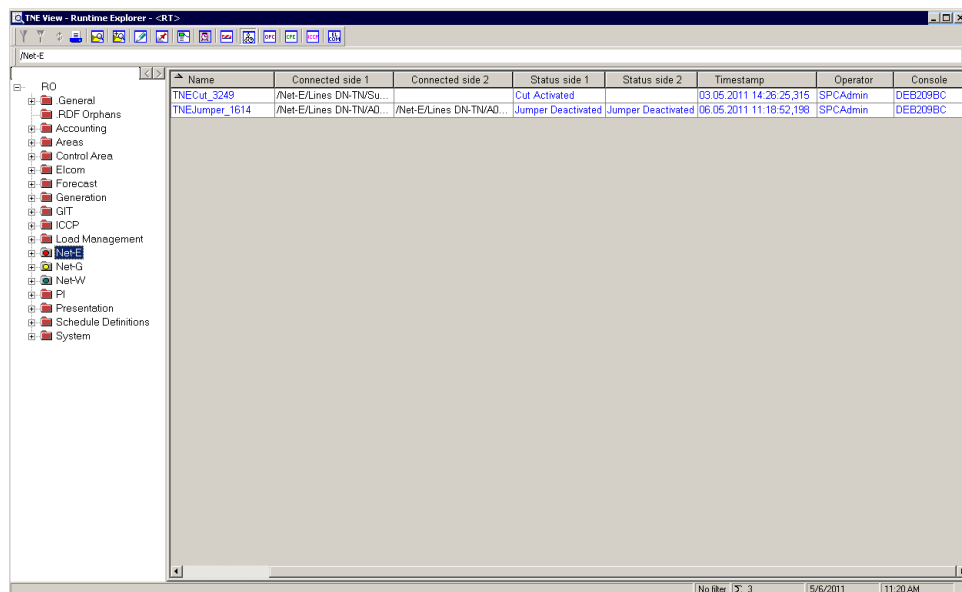
[ruios045-130511-01.tif, 1, en_US]

Figure 15-54 Power 5 Application Toolbar - Runtime Explorer

The **Runtime Explorer** dialog opens.

- Select **TNE View**.

This view displays the all the TNEs and their status in the tabular view.



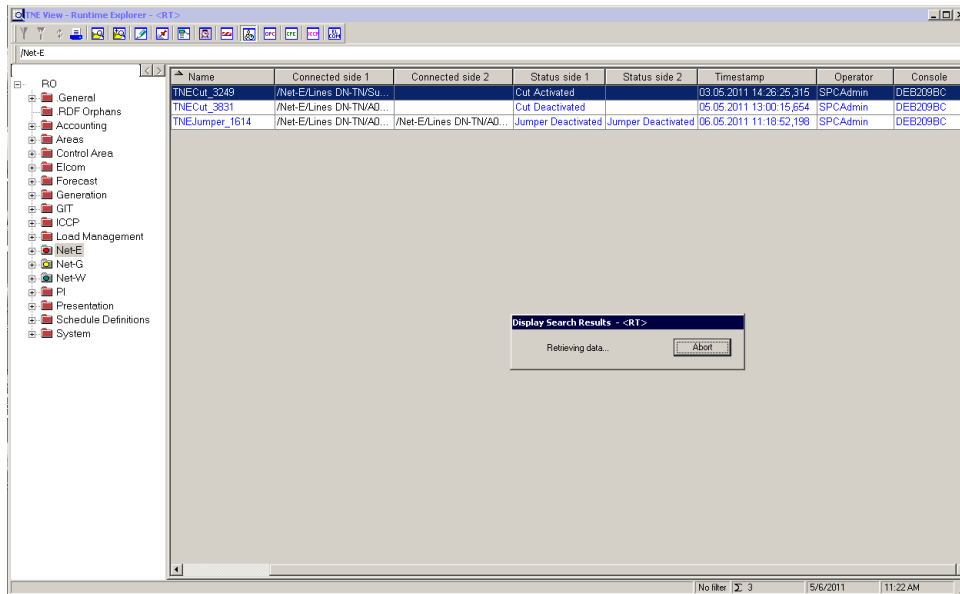
[ruios046-130511-01.tif, 1, en_US]

Figure 15-55 Runtime Explorer

- Right-click on any row.
The context menu is displayed.

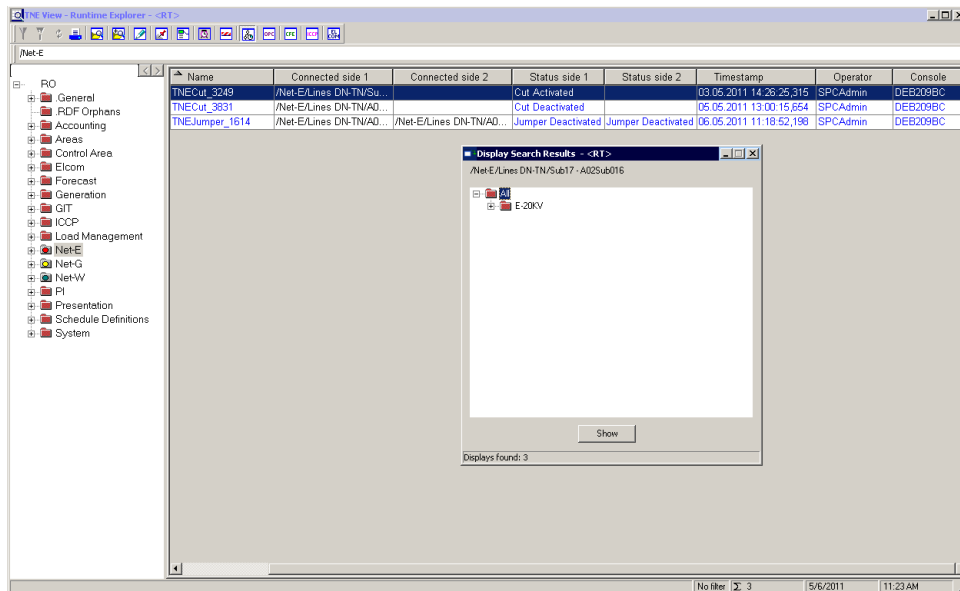
- Select **Display Callup**.

It starts retrieving the data and a **Display Search Results** dialog is displayed.



[ruios047-130511-01.tif, 1, en_US]

Figure 15-56 Callup Display - Retrieving Data



[ruios048-130511-01.tif, 1, en_US]

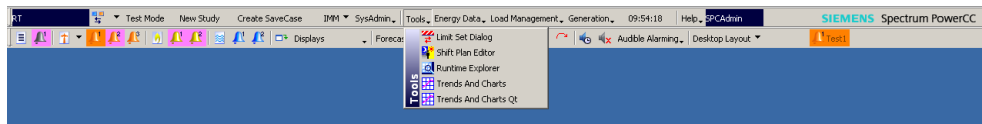
Figure 15-57 Display Search Results

- Select the single-line diagram from the tree view and click **Show**.
The corresponding single-line diagram is displayed.

Removal of TNEs

The goal of this task is to remove the TNEs using **Runtime Explorer**.

- From the PAT, select **Tools -> Runtime Explorer**.



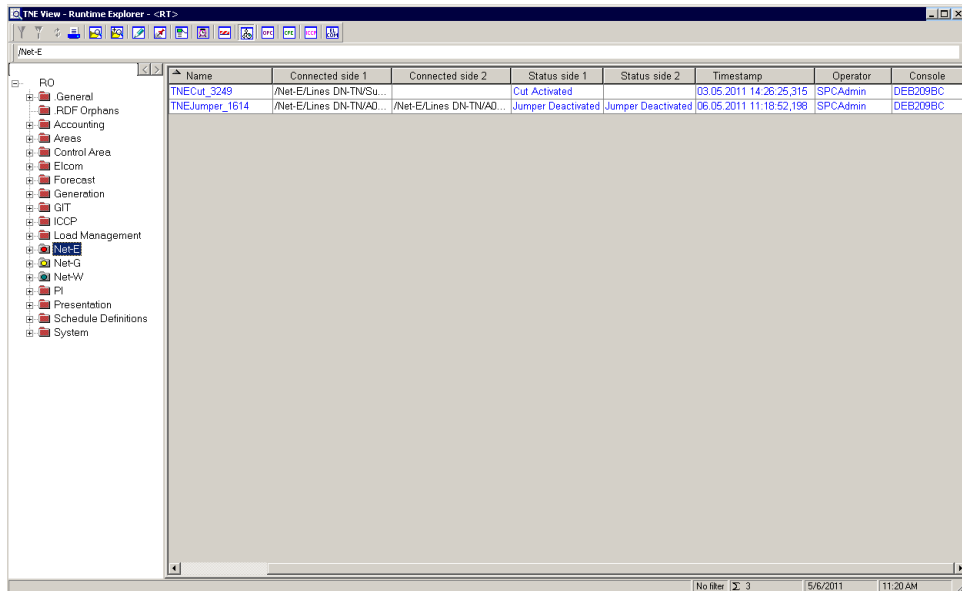
[ruis045-130511-01.tif, 1, en_US]

Figure 15-58 Power 5 Application Toolbar - Runtime Explorer

The **Runtime Explorer** dialog opens.

- Select **TNE View**.

This view displays the all the TNEs and their status in the tabular view.



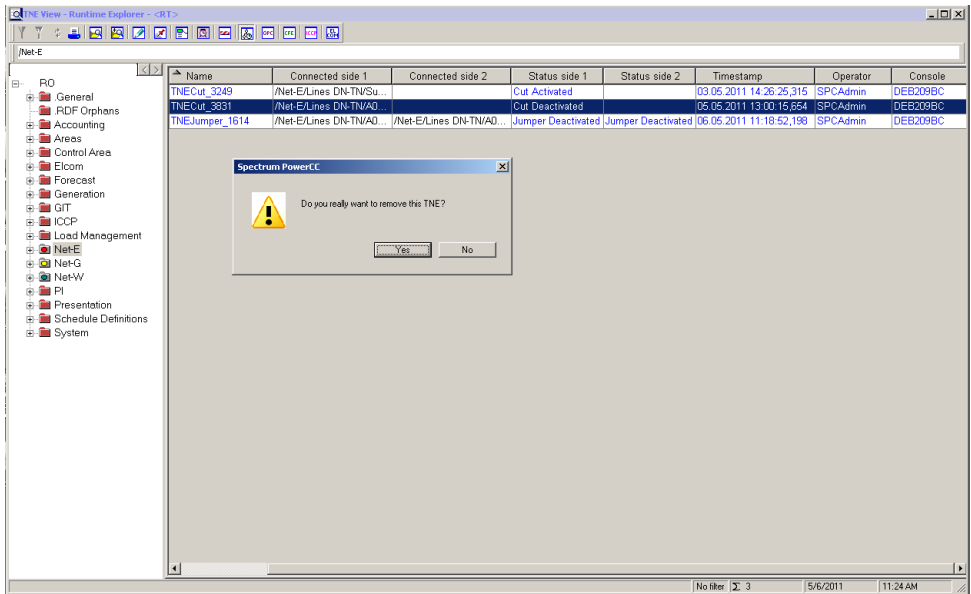
[ruis046-130511-01.tif, 1, en_US]

Figure 15-59 Runtime Explorer

- Right-click on any row.
The context menu is displayed.

- Select **Remove TNE**.

The Spectrum Power 5 dialog is displayed, that prompts for the confirmation to remove the TNE.



[ruios049-130511-01.tif, 1, en_US]

Figure 15-60 Confirmation Dialog

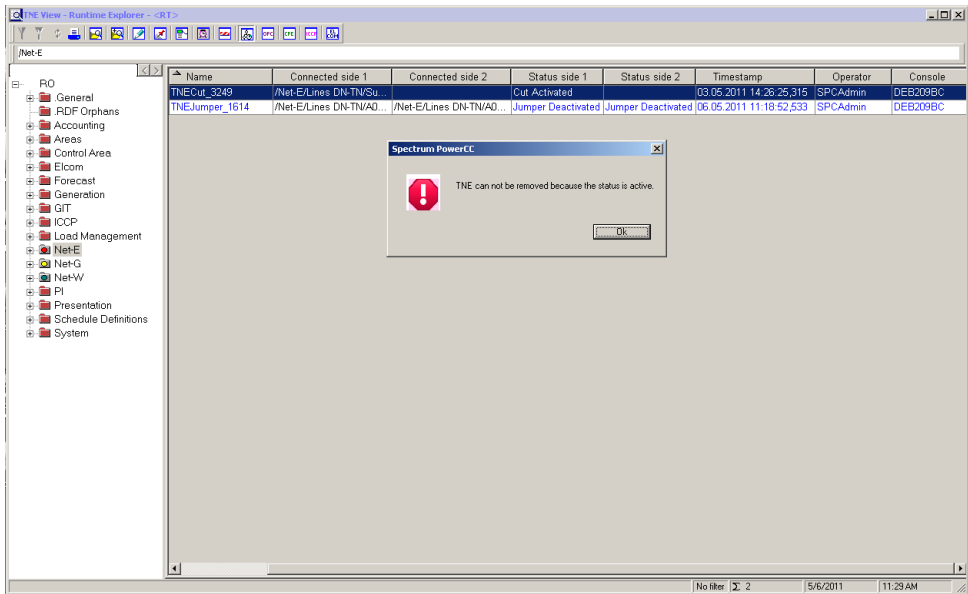
- Click **Yes**.

The corresponding TNE is removed.



NOTE

To remove any TNE, it has to be deactivated first.
For example, if you try to remove the **Cut** whose status is activated, it displays an error message.



[ruios051-130511-01.tif, 1, en_US]

Figure 15-61 Error Message

15.3 User Interface Description

15.3.1 User Roles Using TNE Elements

The following user rights for TNE activities are provided:

- InsertTneAccess: The user can insert a TNE
- ActivateTneAccess: The user can activate or deactivate a TNE
- RemoveTneAccess: The user can remove a TNE

